STUDIO OPERATING MANUAL & TECHNICAL REFERENCE · VERSION 0.4.2

Cave Recorder (caverec)

Multitrack Rehearsal Recording, Live Studio Tracking, Multi-Camera Video, Phase Alignment, Console Mixing & Cave Companion Mobile

```
CH 01–18 [##########..] –3.2 dBFS | RETRO 30M | PHASE +0.02 ms | 48.0 kHz INT
```

Manual Structure

PART I : STUDIO VISION, SETUP & END-TO-END WORKFLOWS

Ch. 1: Overview & Studio Philosophy · Ch. 2: Hardware Setup, Audio Connections & First Session · Ch. 3: Typical Studio & Rehearsal Workflows

PART II : MIX CONSOLE, BUILT-IN PROCESSORS, PHASE ALIGNMENT & PLUGINS

Ch. 4: Mix Console, Signal Flow & Channel Strips · Ch. 5: Built-In Processors, Phase Alignment & External Plugins (CLAP & AU)

PART III : WAVEFORM TIMELINE, SESSION REEL & CLOUD COLLABORATION

Ch. 6: Waveform Timeline, Pass Lanes, Markers & Ranges · Ch. 7: Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync

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PART I | CHAPTER 1

Overview & Studio Philosophy

The Human Musicianship Manifesto, the Anti-Bloat Curated Console Philosophy, independent t=0 Takes, 30-minute retroactive memory, live stereo Prints, track-level Overdub Passes, sample-accurate 50 FPS multi-camera studio video, and the Cave Companion Mobile ecosystem.

1.1 Why Cave Recorder Exists: The Human Musicianship Manifesto

Cave Recorder (`caverec`) is a multitrack studio recorder, live mixing console, automatic phase-alignment processor, and band collaboration workstation purpose-built for rehearsal rooms, live tracking spaces, and project studios.

Modern music production has drifted toward sterile isolation. In an era of AI-generated tracks and hyper-commercial pop production, performances are routinely recorded one person at a time to a rigid click track, snapped to a mathematical bar/beat grid, and stripped of pitch and timing micro-variations. That pipeline works for synthetic programmed audio, but it destroys the soul of music that depends on human interaction in a shared room.

Across jazz, folk, stoner rock, doom, psychedelic and space rock, krautrock, blues, soul, and punk, the magic of a great record lives in the unrepeatable human moment: the shared eye contact before a dynamic breakdown, the collective tempo push of a rhythm section leaning into a chorus, the natural acoustic bleed of drums and cranked amplifiers breathing in the same room, and the spontaneous twenty-minute jam that nobody planned.

Cave Recorder takes a clear philosophical stand:

1. **Human Musicianship First, Zero Grid or Pitch-Correction Bloat:** Cave Recorder intentionally omits MIDI sequencers, piano rolls, beat grids, tempo-warping elastic audio, and robotic vocal pitch correction. Time is measured in real physical seconds, milliseconds, and sample frames (`00:00:00.000`) so the band's natural tempo fluctuations and groove breathe without a visual metronome dictating the performance.

2. **Technology as an Invisible Studio Assistant, Never a Replacement:** Historically, capturing a full band live in one room with the punch and clarity of classic analog studio records required a dedicated recording engineer, an isolated control room, and two days spent moving microphone stands by millimeters to eliminate phase cancellation and amplifier buzz. Self-recording bands in rehearsal caves and basements rarely have a second person at the desk or hours to spend on technical setup. Cave Recorder acts as an invisible studio engineer that handles rolling 30-minute memory capture, one-click `-18 dBFS` gain staging, multicore spectral phase and polarity alignment, bleed-aware drum and amplifier gating, and non-destructive mistake surgery automatically. Musicians stay focused on their instruments and each other.
3. **Curated Analog-Inspired Console Instead of 90 Confusing Plugins:** General-purpose workstations overwhelm musicians with dozens of overlapping legacy plugins, floating windows, and arcane parameter matrices. Cave Recorder provides a single, deeply integrated studio console and rack suite where every processor opens on a musical sweet spot for immediate results, while exposing deep physical controls underneath when an experienced mix engineer wants surgical precision.

Traditional digital audio workstations are modeled around a single linear timeline designed for solo production at a desk. When a full band plugs into an 18-channel audio interface for a three-hour rehearsal or live tracking session, a single continuous timeline creates immediate studio bottlenecks: finding a specific song run-through requires scrolling through hours of waveforms, spontaneous jams are lost when nobody remembers to press Record, bandmates wait days for someone to bounce rough mixes, and multi-microphone phase cancellation hollows out the drum kit and bass rig.

Cave Recorder solves four foundational studio and rehearsal problems:

1.1.1 Independent Zero-Based Takes Instead of a 3-Hour Linear Timeline

In a band rehearsal or tracking session, musicians play multiple run-throughs of songs alongside spontaneous improvisations. In Cave Recorder, every time you press Record (**R**) with Overdub Mode turned off, the transport creates a brand-new, self-contained **Take** (`Take 001`, `Take 002`, `Take 003`, ...) that starts at its own zeroed timeline origin (`00:00:00.000`, sample `0`).

Two different Takes are never appended end-to-end down a multi-hour timeline and are never stacked on top of one another in the same timeline view. Instead, switching the active Take in the Top Transport Deck (`< TAKE 001 >`) or via **ALT** + **[** and **ALT** + **]** immediately switches the Waveform Timeline and Mix Console playback to that song's `00:00:00.000` origin while keeping the room's microphone assignments, channel names, subgroup routing, and console mix intact.

1.1.2 30-Minute Retroactive Capture Memory (RAM & RETRO)

The best moments in a rehearsal room often happen when the transport is stopped while the band is warming up or exploring a groove between songs. Whenever the audio interface is active and the green `RAM` indicator is lit in the Top Transport Deck, Cave Recorder continuously retains the last 30 minutes of live multi-channel audio across all record-armed channels in dedicated system memory.

Pressing the RETRO button in the Top Deck (or pressing **SHIFT** + **R**) immediately rescues the rolling 30-minute capture memory to disk as a new numbered Take (Take<NNN>), complete with individual pre-fader multitrack stems and a reference stereo file.

1.1.3 Simultaneous Live Console Print and Raw Multitrack Stems

During every Take, Cave Recorder writes two distinct sets of audio files to disk in real time:

1. **Raw Pre-Fader Multitrack Stems (rec/):** Every record-armed input channel (CH01 through CH18) writes a discrete, uncompressed 32-bit floating-point mono .wav stem tapped directly after input gain trim and prior to channel insert processing and fader moves. This guarantees clip-free archival stems for later mixing or overdubbing.
2. **Live Stereo Master Bus Print (mix/):** Simultaneously, when 2-TRK is armed on the MST MASTER BUS, a dedicated 2-track recorder tapped at the post-limiter output of the Master Bus captures the live stereo console mix (Print001.wav) with automatic Plugin Delay Compensation (PDC) latency trimming, including all channel EQ, VCA compression, tape saturation, subgroup processing, and auxiliary reverb and delay returns.

With a single click on FLAC PRINT or MP3 PRINT in the Session Reel (or automatically upon cloud upload when AUTO-FLAC ON UPLOAD or AUTO-MP3 ON UPLOAD is enabled), Cave Recorder encodes the live stereo Print to 24-bit lossless .flac and/or 320 kbps CBR .mp3 (with a Xing/Info seek header for direct playback inside the Google Drive web browser player) and links it into the project-wide _Mixes/ folder. Band members walk out of rehearsal with finished stereo reference mixes ready for phone, browser, or car playback, while the engineer retains pristine 32-bit floating-point raw stems.

1.1.4 Console-First Layout and Automatic Multi-Mic Phase & Polarity Alignment

Cave Recorder eliminates nested plugin windows and hidden routing matrices in favor of a fixed, dark-chassis studio console interface. Input gain trim with one-click CALIBRATE ALL TRIMS, polarity inversion (Ø), per-channel sub-sample delay (Φ) paired with a zero-slot 5-stage allpass **Phase Align Network**, an 8-slot reorderable processor rack (12-band parametric Dynamic EQ with dual-trace spectrum analyzer, multi-topology VCA/FET/Opto compressor, bleed- and harmonicity-aware noise gate, vocal rider, de-esser, transient shaper, micro-pitch doubler, 3-mode delay, 4-pole resonant ladder filter, studio plate reverb, 4-mode modulation, tape saturation, CLAP .clap plugins, and native macOS AUv3/AUv2 Audio Unit effects), stereo Aux Sends (RVB and DLY), Subgroup Bus routing (SUB 1..16), and 100 mm logarithmic faders are always accessible directly on the console faceplate.

When recording a drum kit or multi-miked guitar and bass cabinets, acoustic arrival-time differences and cabinet filter dispersion between close microphones, overheads, DI boxes, and room microphones cause comb filtering and 6 to 12 dB of low-frequency cancellation. Clicking ALIGN ALL in the Top Deck aligns acoustic arrival times to sub-sample accuracy (us / smp), while the two-tier **Polarity & Phase Maximizer** solves both instant broadband polarity inversions (Mode 1, completing in under 50 ms) and deep multicore frequency-dependent allpass phase rotation (Mode 2, 30 s to 30 min anytime solver with loudness-matched Candidates A, B, C, D auditioning) without moving a single microphone stand.

1.2 The Four-Tier Recording Hierarchy

Cave Recorder organizes every recording into a strict four-tier studio hierarchy:

```
Project ---> Session ---> Take (t = 0) ---> Pass (Rec001 / Rec002-OD)
```

On disk, this hierarchy maps cleanly to human-readable folders and self-describing filenames inside your Cave Recorder Library Root:

```
<LibraryRoot>/
├── .caverec/
│   └── projects.json                    # Library project registry
└── GF/                                  # 1. PROJECT folder (Tag: GF)
    ├── .caverec_project.json            # Cloud sync manifest
    ├── _Mixes/                          # Project Prints, Audio & Video
Mixdowns
    │   ├── GF_20261020-1_Take001_Print001_1947.flac
    │   ├── GF_20261020-1_Take001_Print001_1947.mp3
    │   ├── GF_20261020-1_Take001_Mix001_2130.flac
    │   ├── GF_20261020-1_Take001_Mix001_2130.mp3
    │   └── GF_20261020-1_Take001_VideoMix001_2135.mp4
    └── GF_20261020-1_Rehearsal/         # 2. SESSION folder
        ├── GF_20261020-1_session.json   # Console, routing & markers
        ├── snapshots/                   # Point-in-time Snapshots
        │   ├── GF_20261020-1_Snap001_20261020T195012Z.json
        │   └── GF_20261020-1_Snap002_20261020T211500Z.json
        ├── Take001_1947_SongA/          # 3. TAKE 001 (t = 0)
        │   ├── GF_20261020-1_Take001_take.json
        │   ├── rec/                     # 4. PASS files (Base + OD)
        │   │   ├── GF_20261020-1_Take001_01-KickIn_Rec001_1947.wav
        │   │   ├── GF_20261020-1_Take001_02-SnrTp_Rec001_1947.wav
        │   │   ├── GF_20261020-1_Take001_04-GtrAmp_Rec001_1947.wav
        │   │   └── GF_20261020-1_Take001_04-GtrAmp_Rec002-OD_2015.wav
        │   ├── video/                   # Multi-Camera 50/25 FPS CFR Video &
Sidecar Index
        │   │   ├── GF_20261020-1_Take001_Cam01-Drums_1947.mp4
        │   │   ├── GF_20261020-1_Take001_Cam01-Drums_1947.vididx
        │   │   ├── GF_20261020-1_Take001_Cam02-RoomMobile_1947.mp4
        │   │   └── GF_20261020-1_Take001_Cam02-RoomMobile_1947.vididx
        │   ├── mix/                     # Live Prints, Audio & Split-Screen
Video Mixdowns
        │   │   ├── GF_20261020-1_Take001_Print001_1947.wav
        │   │   ├── GF_20261020-1_Take001_Print001_1947.flac
        │   │   ├── GF_20261020-1_Take001_Print001_1947.mp3
        │   │   ├── GF_20261020-1_Take001_Mix001_2130.flac
        │   │   ├── GF_20261020-1_Take001_Mix001_2130.mp3
        │   │   └── GF_20261020-1_Take001_VideoMix001_2135.mp4
        │   └── bounce/                  # Offline Stem Bounces
        │       └── GF_20261020-1_Take001_01-KickIn_Bounce001_2135.wav
        └── Take002_1958_ImprovJam/      # 3. TAKE 002 (t = 0)
            └── ...
```

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior
Project	Band, artist, or musical venture	<LibraryRoot>/<ProjectTag>/	Top-level workspace (for example, Grottfolk with tag GF, or Miscellaneous with tag MISC). Contains all Sessions for that band plus a unified _Mixes/ folder for rapid listening and cloud sharing. Managed via the PROJECT modal.
Session	Single rehearsal night, tracking day, or studio gathering	<ProjectTag>/<SessionID>_<Tag>/	Defines the room's physical microphone setup, channel strip names, input routing, subgroup buses, and console processing shared across every Take recorded that day. Session dates roll over at **08:00 local time** so late-night recordings after midnight stay grouped with the same evening session.

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior
Take	Complete multitrack song run-through or jam (t = 0)	Take<NNN>_<Stamp>[_<TakeTag>]/	An independent multitrack recording starting at 00:00:00.000. Pressing Record (**R**) when Overdub Mode (OD) is off always creates a new numbered Take (Take 001, Take 002, ...) and switches the active timeline to that Take.
Base Pass (Rec001)	Initial track-level recording inside a Take	rec/*_<CC>-<Name>_Rec001_<Stamp>.wav	The original 32-bit float pre-fader mono stem captured on an armed channel strip (PASS 01) when the Take was first recorded from 00:00:00.000.
Overdub Pass (Rec002-OD)	Layered track-level recording inside an existing Take	rec/*_<CC>-<Name>_Rec<MMM>-OD_<Stamp>.wav	Additional track-level performance (OD 02, OD 03, ...) captured inside the currently active Take when Overdub Mode (OD, **O**) is latched on the transport bar.

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior
Pass Lanes	Expandable track sub-lanes in Waveform Timeline	Stored in *_Take<NNN>_take.json	Visual lanes (PASS 01 Base Pass and OD 02, OD 03 Overdub Passes) expanded in WAVEFORM TIMELINE (**CMD** + **2** / **CTRL** + **2**) with **E** (and collapsed with **SHIFT** + **E**) for auditioning (Use PASS 01 / ★ OD 02), muting (M), and removing (Del) passes within the active Take.
Video Angle & .vididx Index	Sample-locked studio or mobile camera recording inside a Take	video/*_Take<NNN>_Cam<CC>-<Name>_<Stamp>.<mp4|vididx>	Constant Frame Rate (50 FPS or 25 FPS) camera angle recorded from connected desktop USB/ HDMI cameras or wireless **Cave Companion** Android and iOS mobile camera nodes, paired with a binary .vididx frame-to-sample index file locked to the 48.0 kHz audio clock.

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior
Print (Print<NNN>)	Real-time post-Master-Bus stereo recording	mix/ *_Take<NNN>_Print<NNN>_<Stamp>.<wav|flac|mp3>	Captured live alongside the raw stems while tracking a Take when 2-TRK is armed on the Master Bus. Converting a Print to 24-bit .flac (FLAC PRINT) and/or 320 kbps CBR .mp3 (MP3 PRINT) links it into <ProjectTag>/_Mixes/ for immediate band playback without waiting for an offline render.
Mixdown (Mix<NNN> / VideoMix<NNN>)	Offline Master Bus stereo audio or split-screen video render	mix/ *_Take<NNN>_Mix<NNN>_<Stamp>.<wav|flac|mp3> / VideoMix<NNN>.mp4	Faster-than-real-time stereo Master Bus audio render (FLAC / MP3) or multi-camera split-screen .mp4 video mixdown (Single, Side-by-Side Crop, Side-by-Side 16:9, 3-Cam Band, Quad 2x2) of a Take linked into <ProjectTag>/_Mixes/.
Stem Bounce (Bounce<NNN>)	Offline post-processor track or subgroup render	bounce/ *_Take<NNN>_<Target>_Bounce<NNN>.wav	Offline render of an individual channel strip or Subgroup Bus capturing all insert processing and phase alignment for delivery to external remixing workflows.

Studio Entity	Scope	On-Disk Pattern	Purpose & Studio Behavior
Snapshot (Snap<NNN>)	Immutable console state capture	snapshots/ <SessionID>_Snap<NNN>_<UTC>.json	Point-in-time capture of the entire Mix Console state (faders, pans, EQ bands, compressor thresholds, routing, and active Pass Lane selections), created automatically after every recording pass or manually via SNAP MANUAL (**CMD** + **S** / **CTRL** + **S**).
Live Mix (LIVE MIX)	Automatic live console backup during Snapshot audition	Held in working session memory	Whenever you audition a historical Snapshot in the Session Reel, Cave Recorder preserves your current working console state in LIVE MIX so you can toggle A/B LIVE MIX or click RETURN TO LIVE MIX (**BACKSPACE**) without losing a single fader move.

DISTINGUISHING SESSION-LEVEL TAKES FROM TRACK-LEVEL PASSES

Cave Recorder maintains a strict separation between **Takes** and **Passes**. A **Take** (Take 001, Take 002) is an entire song or jam recorded across the studio room starting at 00:00:00.000, selected in the Top Transport Deck or SESSION REEL (**CMD** + **4** / **CTRL** + **4**). A **Pass** (PASS 01 Base Pass or OD 02 Overdub Pass) is an individual performance recorded on a single channel strip inside the currently active Take, displayed inside expandable **Pass Lanes** in WAVEFORM TIMELINE (**CMD** + **2** / **CTRL** + **2**).

1.3 Tour of the Studio Workspace

Cave Recorder opens into a dark studio console workspace divided into three horizontal tiers: the persistent **Top Transport & Session Header Deck**, the **Upper Inspector Wells**, and the **Lower Console Channel & Bus Strips**.



Figure 1.1: *The Cave Recorder Mix Console workspace showing the Top Transport Deck, Upper Inspector Wells, and Lower Channel Strips.*

1. **Top Transport & Session Header Deck (Top):** Remains pinned at the top of the screen across all seven workspace views (MIX CONSOLE through PLUGINS and KEYMAP). Houses project and session selection, Google Drive cloud sync status, 30-minute retroactive memory controls, main transport buttons, the active Take selector, high-precision digital timecode and sample counters, global multi-mic phase alignment controls, and latency diagnostics.
2. **Upper Inspector Wells (Middle):** Displays four side-by-side real-time visualizers and editors for the currently selected channel strip or Subgroup Bus:
   - **8-Slot Reorderable Processor Rack & Tape Controls (Left Well):** Shows slots 1 through 8 (1:CLAP, 2:12B EQ, 3:VCA CMP, 4:AUX SND, ...), per-slot bypass (B), up/down slot reordering arrows, a dedicated TAPE saturator enable/bypass toggle with analog tape parameters (DRV, WRM, BIAS, HC, OUT, MIX), SWAP EQ/CMP, and channel RESET.
   - **12-Band Parametric EQ & Dual-Trace RTA Spectrum Analyzer (Center-Left Well):** Interactive Bode magnitude curve with draggable filter nodes (BELL, LSHF, HSHF, HPF, TILT) superimposed over a pre-EQ and post-EQ real-time spectrum trace (20 Hz to 20 kHz).
   - **VCA Compressor & Gain Reduction Transfer Curve (Center-Right Well):** Displays the compressor knee and transfer curve (-48 dB to 0 dB), input/output threshold markers, sidechain high-pass filter (SC HPF), ratio presets (2:1, 4:1, 8:1), parallel blend (MIX 50%), and real-time Gain Reduction (GR) metering.
   - **Upper Mini-Timeline & Multi-Tier Waveform Overview (Right Well):** Displays the composite waveform of the active Take with playhead tracking, PASSES expansion shortcut, LOCK PH/CUR, -> CUR, and zoom controls (ZOOM +, ZOOM -, ZOOM 1:1).

3. **Lower Console Strips (Bottom):** Houses up to 18 Input Channel Strips (CH01..CH18), stereo Subgroup Bus Strips (DRUM BUS, INST BUS), dedicated stereo Aux Return Strips (AUX 1 REVERB with zero-latency convolution IR and AUX 2 TAPE DLY with capstan tape echo), and the MST MASTER BUS strip with Master Tape, Master Compressor, brickwall Limiter, live 2-TRK Print record arm, independent FLAC and MP3 format toggles, and one-click BOUNCE offline Mixdown render controls.

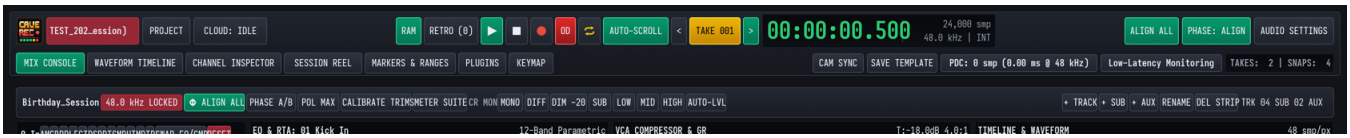


Figure 1.2: The Top Transport and Session Header Deck.

1.3.1 Top Deck Row 1: Session Identity, Transport, Timecode & Phase Controls

The top row of the header deck is organized into three functional clusters from left to right:

Cluster	Control / Badge	Keyboard Shortcut	Studio Function
Left: Session & Cloud	Active Session Pill (TEST_202...ession))	Status Indicator	Displays the currently loaded Session ID and Session Name. Highlights in studio burgundy when a live session is active on disk.
Left: Session & Cloud	PROJECT	**CMD** + **O** (macOS) / **CTRL** + **O** (Win/Linux)	Opens the Project and Session Browser modal to switch Projects, create a new Session from a template, link a remote shared project, or open a recent Session.
Left: Session & Cloud	CLOUD: IDLE	Mouse Click	Opens the Cloud Sync Quick Controls popover (AUTO-FLAC ON UPLOAD, AUTO-MP3 ON UPLOAD, SYNC SESSION FULL, SYNC SESSION MIXES) and displays real-time Google Drive transfer status.
Center: Transport & Take	RAM	Mouse Click	Enables or disables the continuous 30-minute retroactive capture memory across all armed input channels. Glows emerald green when active.

Cluster	Control / Badge	Keyboard Shortcut	Studio Function
Center: Transport & Take	RETRO (0)	**SHIFT** + **R**	Flushes the rolling 30-minute retroactive capture memory to disk as a new numbered Take and increments the rescued-take counter.
Center: Transport & Take	Play (▶)	**SPACE**	Starts or pauses playback of the active Take from the current playhead position. Glows green while transport is rolling.
Center: Transport & Take	Stop (■)	**SHIFT** + **SPACE**	Stops playback or active recording immediately and finalizes open audio files and post-recording Snapshots.
Center: Transport & Take	Record (●)	**R**	Starts or stops multitrack recording. Creates a new t = 0 Take (Take<NNN>) when OD is off, or records a new Overdub Pass (Rec<MMM>-OD) inside the active Take when OD is latched.
Center: Transport & Take	OD (Overdub Mode)	**O**	Latches Overdub Mode (glows red when active). While latched, pressing Record (**R**) records new track-level Overdub Passes on armed channels while playing back existing tracks of the active Take.
Center: Transport & Take	Cycle / Loop (⇄)	**C**	Toggles continuous loop playback and loop overdubbing across the active Take range.
Center: Transport & Take	AUTO-SCROLL	Mouse Click	Keeps the Waveform Timeline viewport following the advancing playhead during playback and recording.
Center: Transport & Take	< TAKE 001 >	**ALT** + **[** / **ALT** + **]**	Steps backward or forward through the Session's non-deleted Takes, immediately loading that Take's 00:00:00.000 timeline and Pass Lanes.

Cluster	Control / Badge	Keyboard Shortcut	Studio Function
Center: Transport & Take	00:00:00.500 / 24,000 smp	Readout	High-contrast emerald LED timecode readout (HH:MM:SS.mmm) paired with the exact sample frame position (24,000 smp) within the active Take.
Center: Transport & Take	48.0 kHz I INT	Readout	Displays the active hardware sample rate (44.1, 48.0, 88.2, or 96.0 kHz) and digital clock lock state (INT).
Right: Phase & Hardware	ALIGN ALL	Mouse Click (Bindable in KEYMAP)	Runs automatic multi-microphone arrival-time and polarity correlation across all active channels relative to the reference channel.
Right: Phase & Hardware	PHASE: ALIGN	Mouse Click	Toggles global phase alignment processing between active (PHASE: ALIGN) and bypassed (PHASE: BYP) for instant A/B listening.
Right: Phase & Hardware	AUDIO SETTINGS	**CMD** + **,** (macOS) / **CTRL** + **,** (Win/Linux)	Opens the Audio Device & Latency Settings modal to configure interfaces, sample rate, buffer size, and hardware loopback calibration.

1.3.2 Top Deck Row 2: Workspace View Tabs, Templates & Latency Diagnostics

The second row of the Top Deck provides direct navigation across Cave Recorder's seven workspace views on the left and session template and latency management on the right:

Control / View Tab	Keyboard Shortcut	Workspace Purpose
MIX CONSOLE	**CMD** + **1** (macOS) / **CTRL** + **1** (Win/Linux)	Full studio mixing console with Upper Inspector Wells, 18 input strips, Subgroup Buses, Aux Returns, and Master Bus. Detailed in Chapter 4: Mix Console, Signal Flow & Channel Strips.

Control / View Tab	Keyboard Shortcut	Workspace Purpose
WAVEFORM TIMELINE	**CMD** + **2** (macOS) / **CTRL** + **2** (Win/Linux)	Multi-track 2D zoomable waveform editor, expandable **Pass Lanes** for overdub comping, marker flags, loop ranges, and Take splitting. Detailed in Chapter 6: Waveform Timeline, Pass Lanes, Markers & Ranges.
CHANNEL INSPECTOR	**CMD** + **3** (macOS) / **CTRL** + **3** (Win/Linux)	Dedicated full-screen channel and Subgroup Bus workbench with expanded 12-band EQ, Dual-Trace RTA, VCA Compressor, Gate, Tape Saturation, and CLAP / Audio Unit plugin editors. Detailed in Chapter 5: Built-In Processors, Phase Alignment & External Plugins (CLAP & AU).
SESSION REEL	**CMD** + **4** (macOS) / **CTRL** + **4** (Win/Linux)	Session dashboard containing the Takes Catalog, one-click FLAC and MP3 Print encoding, offline Mixdown and Stem Bounce rendering, undo history, Snapshots, Live Mix A/B comparison, and Cloud Sync controls. Detailed in Chapter 7: Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync.
MARKERS & RANGES	**CMD** + **5** (macOS) / **CTRL** + **5** (Win/Linux)	Sortable tabular manager for all Take markers (M001..) and loop/export ranges (R001..), with one-click playhead jump (GO TO) and range-to-Take extraction (NEW TAKE). Detailed in Chapter 6: Waveform Timeline, Pass Lanes, Markers & Ranges.
PLUGINS	**CMD** + **7** (macOS) / **CTRL** + **7** (Win/Linux)	Dedicated **CLAP & macOS Audio Unit Plugin Manager** (PLUGIN MANAGER & INDEXED CATALOG (CLAP & AU)) for incremental directory scanning (RESCAN DIRECTORIES), custom .clap search folders, status and format filtering, and missing Ghost Slot relinking (LOCATE...). Detailed in Section 5.7 in Chapter 5.

Control / View Tab	Keyboard Shortcut	Workspace Purpose
KEYMAP	**CMD** + **6** (macOS) / **CTRL** + **6** (Win/Linux)	Interactive keyboard shortcut reference and custom keybinding editor with live conflict detection across all 59 studio actions. Detailed in Chapter 8: Action, Mouse Gesture & Keyboard Shortcut Reference.
SAVE TEMPLATE	Mouse Click	Opens the Save Mix Console as Template popover to store current channel names, routing, and processor racks as a reusable Session Template (excluding audio takes and undo history).
PDC: 0 smp (0.00 ms @ 48 kHz)	Mouse Click	Opens the Latency & Plugin Delay Compensation (PDC) Transparency Breakdown popover showing per-track and per-bus sample compensation and roundtrip I/O offsets.
Low-Latency Monitoring	Mouse Click	Bypasses high-latency lookahead or external plugins on record-armed monitoring paths during overdubbing so musicians hear zero-lag foldback.
TAKES: 2 | SNAPS: 4	Mouse Click	Live tally of active Takes and saved console Snapshots in the current Session. Clicking opens the Session Reel.

1.4 The Anti-Bloat Curated Console Philosophy

General-purpose workstations and large commercial plugin bundles often ship with 70 to 90 separate effects plugins: five nearly identical equalizers, eight redundant compressors named after vintage circuit board revisions, four separate de-essers, and dozens of single-purpose utility modules. In a fast-moving band rehearsal or self-engineered tracking session, scrolling through huge plugin menus and juggling floating windows breaks musical momentum and causes decision fatigue.

Cave Recorder replaces plugin sprawl with a **Curated Native Console & Rack Suite** built around five engineering pillars:

1.4.1 Pillar 1: Fewer, Deeper Multi-Mode Instruments Instead of Redundant Clones

Rather than cluttering the rack with dozens of single-purpose plugins, Cave Recorder consolidates classic studio hardware topologies into focused, multi-mode processors:

- **One 3-Topology Compressor (VCA, FET, Opto):** Switches cleanly between feed-forward RMS bus glue (VCA), ultra-fast discrete transistor transient limiting with odd-harmonic grit (FET), and program-dependent dual-stage electroluminescent optical leveling (Opto).
- **One Context-Aware 3-Variant Gate (Expander, Drum Bleed, Amp Dynamic):** Combines a smooth downward vocal/room expander, a differential sidechain tom/snare bleed gate with automatic fundamental lock ([LRN]), and a pitch-harmonicity and spectral-entropy amplifier gate built for high-gain stoner, doom, and psychedelic guitar rigs.
- **One 3-Mode Echo & Delay (Capstan Tape, BBD Analog, Pristine Digital):** Covers saturating magnetic tape echo, dark bucket-brigade analog delay with compander breathe and clock noise, and clean stereo rhythmic delay with dynamic ducking.
- **One 4-Mode Studio Modulator (Through-Zero Flanger, Dimension Chorus, Optical Phaser, Harmonic Tremolo):** Covers tape-style through-zero flanging, motionless stereo dimension widening, 4/6/8/12-stage photocell phasing, and dual-band crossover harmonic tremolo.

1.4.2 Pillar 2: Immediate Musical Sweet Spots with Deep Custom Power Underneath

Every native processor in Cave Recorder is tuned so that engaging it on a channel strip immediately improves the sound without requiring five minutes of parameter tweaking:

- **One-Click Calibration & Auto-Setup:** Clicking CALIBRATE ALL TRIMS sets every channel's input gain to the -18 dBFS RMS console sweet spot with a -6 dBFS peak ceiling. Clicking [LRN] on the Drum Bleed gate listens to a single tom hit and automatically locks the resonant key filter to that drum's fundamental pitch (55 Hz to 350 Hz, default 110 Hz). Engaging Vocal Rider with Breath / Bleed Freeze holds the fader steady during pauses so drum bleed is never boosted between vocal lines.
- **Full Surgical Depth When Needed:** Beneath these immediate defaults, mix engineers have full access to per-band Dynamic EQ thresholds, 16-sample smoothed filter trajectories, sidechain Tilt EQ, smooth gate ballistics, 5-stage allpass frequency/Q coordinates, and loudness-matched AUTO-LVL bypass auditioning.

1.4.3 Pillar 3: Solving Real Live-Room Physics (Phase Dispersion, Bleed & High-Gain Sustain)

Plugins designed for isolated bedroom overdubs fall apart when an entire band plays loud in an untreated room. Cave Recorder's processors are engineered specifically for the acoustic realities of live band tracking:

- **Multi-Mic Phase Cancellation & Allpass Dispersion:** Combining a bass DI with a miked ported bass cabinet, or blending 8 microphones across a drum kit, introduces both acoustic time-of-flight delays and frequency-dependent phase rotation across cabinet ports and crossover networks. Every channel strip includes a zero-slot top-of-strip **Phase Align Network** driven by the two-mode **Polarity & Phase Maximizer** (Mode 1 instant binary polarity solver in under 50 ms, and Mode 2 multicore spectral allpass solver).

- **Snare-into-Tom Spill & Cymbal Wash:** The Drum Bleed gate uses a differential key detector (comparing the target tom against a snare or overhead reference) and a dynamic decay low-pass filter (20 kHz sweeping down to 250 Hz to 800 Hz, default 380 Hz) so tom hits ring out naturally without sucking bursts of hi-hat and cymbal wash into the mix.
- **Stoner, Doom & Psychedelic High-Gain Guitar Dynamics:** Standard threshold noise gates destroy quiet clean fingerpicking (-34 dBFS) and chop off intentional harmonic amplifier feedback bloom, or else sit wide open while stacked fuzz pedals roar at -22 dBFS. The Amp Dynamic gate measures autocorrelation pitch periodicity ($H \in [0, 1]$) and spectral flatness so whispered clean passages and 1.5 s to 5.0 s feedback sustain pass untouched while 60 Hz hum and fuzz hiss clamp shut in 15 ms to 40 ms (default 18 ms to 22 ms) on a palm-mute stop.

1.4.4 Pillar 4: Non-Destructive Live-Take Salvage and Mistake Surgery

When a band captures a once-in-a-lifetime live groove, you cannot ask them to play the exact same improvisation again just because a snare rimshot clipped an preamp converter for 12 samples, a vintage combo amp buzzed at 60 Hz, or the guitarist hit a wrong chord in bar 42.

Cave Recorder treats live-take rescue as a core console workflow rather than an external audio restoration chore. Directly inside the Waveform Timeline and Channel Inspector, you can reconstruct clipped ADC peaks ([REPAIR CLIPPING] using smooth spline and spectral waveform interpolation), notch out 12 harmonics of 50/60 Hz power-line hum without dulling guitar tone, comp a single bar from an Overdub Pass ([USE OD 02 IN RANGE]), graft a clean chord from another bar ([PATCH FROM SOURCE...]), or apply a spectral spot-heal ([SMOOTH OVER ARTIFACT]) while leaving the original 32-bit floating-point recording on disk completely untouched.

1.4.5 Pillar 5: Strict Separation Between Master Bus Printing and Control Room Monitoring

In a rehearsal or tracking room, engineers constantly need to check mono fold-down compatibility (MONO), solo the stereo side difference (DIFF), isolate the sub-bass (SUB < 90 Hz) or midrange (MID 350 Hz - 3.5 kHz) across studio monitors, dim the speakers (DIM -20 dB) to talk to the band, or A/B a heavy processing chain at matched loudness (AUTO-LVL).

In Cave Recorder, the **Control Room Monitor Bar** is wired strictly *downstream* of the Master Bus 2-TRK live Print tap and offline BOUNCE engine. You can solo frequency bands, flip to mono, or dim the control room speakers right in the middle of recording a live Take without ever altering or interrupting the stereo Print001.wav being written for the band.

Live-Room Studio Challenge	Why Traditional DAWs Struggle	Cave Recorder Curated Console Solution
Hollow drums and thin bass from multi-mic phase cancellation	Requires manual sample nudging or expensive external phase plugins inserted on every track	Top-of-strip zero-slot **Phase Align Network** + ALIGN ALL sub-sample delay + Mode 1 (< 50 ms) & Mode 2 multicore spectral allpass solver

Live-Room Studio Challenge	Why Traditional DAWs Struggle	Cave Recorder Curated Console Solution
Fuzz/tube amp hum vs. quiet clean picking and feedback sustain	Amplitude noise gates choke quiet clean notes and chop off harmonic feedback tails	Amp Dynamic gate using pitch harmonicity and spectral flatness (Feedback Bloom Lock + 18 ms Palm-Mute Clamp + HF Tame)
Snare and cymbal spill triggering rack and floor tom gates	Standard gates false-trigger on loud snare hits and let harsh cymbal wash through on tom decay	Drum Bleed gate with differential key rejection, [LRN] fundamental auto-detection, and dynamic decay low-pass filter
Inconsistent input trim levels across 18 live mics	Slow manual trim riding before rehearsal can start	One-click CALIBRATE ALL TRIMS (Gain Utility) normalizing channels to -18 dBFS RMS with a -6 dBFS peak ceiling
Vocalist steps back from mic or drum bleed fills vocal pauses	Compressor boosts room bleed during pauses unless automated by hand	Vocal Rider (±6 dB zero-latency leveler) with speech-weighted detector and **Breath / Bleed Freeze Gate**
Clipped drum transient, 60 Hz amp buzz, or 1-bar chord flub in a magic live Take	Requires exporting stems to external restoration editors or re-recording the whole song	Non-destructive [REPAIR CLIPPING], 12-Harmonic Hum Canceller, Range Comping, [PATCH FROM SOURCE...], and [SMOOTH OVER ARTIFACT]
Louder-sounds-better bias and accidental speaker-check printing	Monitor EQ/mono plugins on the Master Bus accidentally get printed into the mixdown file	**Control Room Monitor Bar** (SUB/LOW/MID/HIGH, MONO, DIFF, DIM) wired strictly post-Print/post-Bounce + AUTO-LVL loudness-matched bypass

PART I | CHAPTER 2

Hardware Setup, Audio Connections & First Session

Connecting multi-channel USB and ADAT interfaces, automatic studio interface selection, macOS first-launch setup and Microphone/Camera permissions, locking digital clocks, calibrating roundtrip loopback latency, connecting 50 FPS / 25 FPS studio cameras with the 3-Second Optical & Audio Flash Latency Calibration modal, and creating your first Project and Session.

2.1 Connecting Multi-Channel Audio Interfaces (USB, Thunderbolt & ADAT)

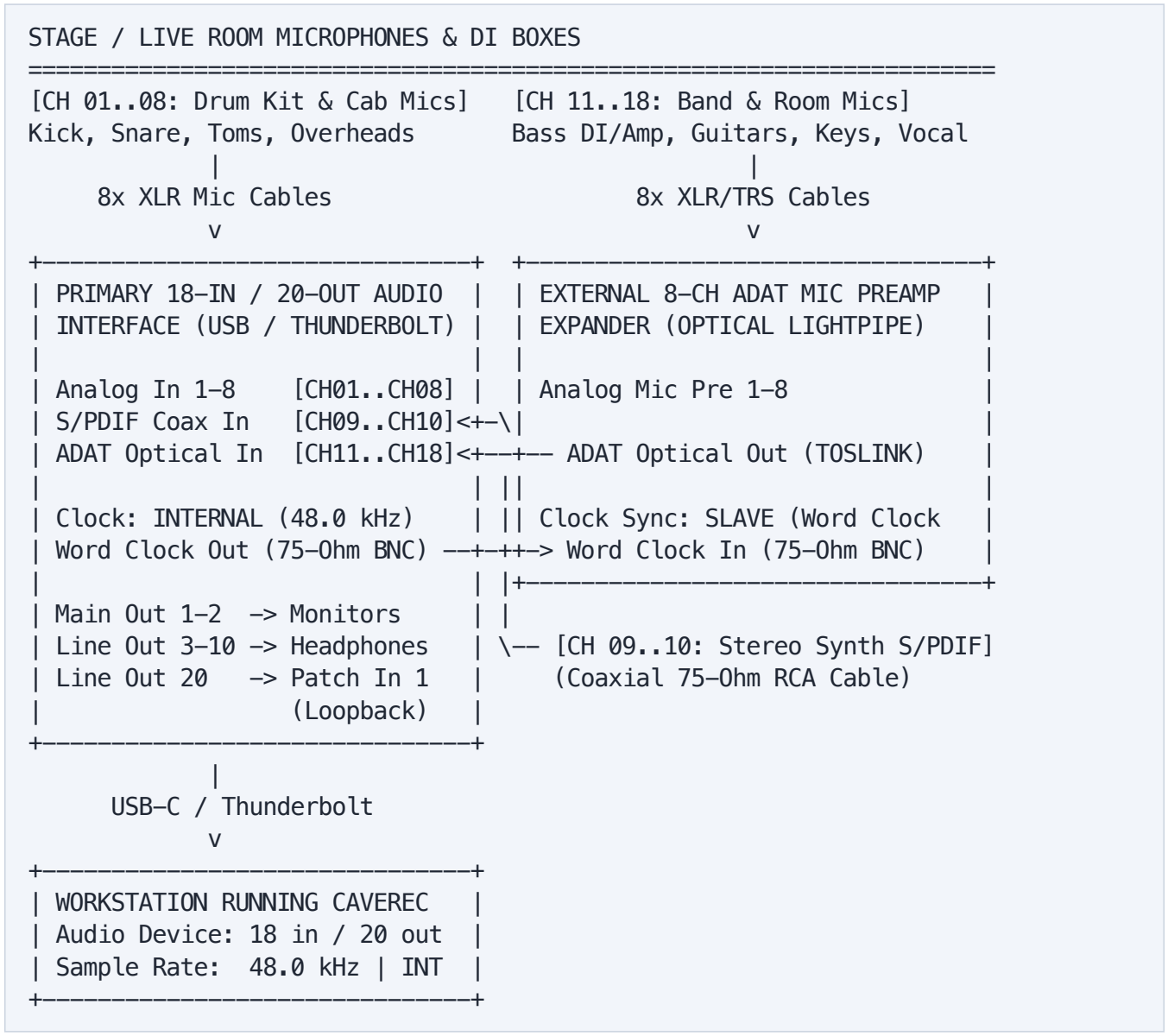
Cave Recorder supports up to 32 simultaneous mono or stereo-linked input tracks (`CH01` through `CH32`), making it an ideal match for rackmount 18-in / 20-out or 32-channel USB and Thunderbolt studio interfaces paired with 8-channel ADAT optical microphone preamp expanders.

A standard 18-channel rehearsal and tracking rig combines three physical input groups into a single unified audio device:

1. **Inputs 1-8 (Primary Interface Onboard Mic/Line Preamps):** Eight XLR/TRS combo preamps on the main rack unit, typically dedicated to close drum microphones, guitar cabinet mics, rack and floor toms, and stereo drum overheads.
2. **Inputs 9-10 (Coaxial S/PDIF or Auxiliary Line Return):** Two digital coaxial S/PDIF channels or line inputs used for stereo hardware synthesizers, stage pianos, or drum machines (`SpdifSynthL` and `SpdifSynthR`).
3. **Inputs 11-18 (8-Channel ADAT Optical Preamp Expander):** Eight additional microphone and instrument preamps fed from an external 8-channel ADAT expander into the primary interface's `ADAT OPTICAL IN` port over a TOSLINK fiber-optic cable, dedicated to `BassDi`, `BassAmp`, `GuitarL`, `GuitarR`, `KeysL`, `KeysR`, `LeadVocal`, and `RoomAmbience`.

2.1.1 Studio Signal & Clock Wiring Diagram

The diagram below illustrates the physical audio, optical ADAT, Word Clock, studio monitor, and loopback calibration connections for an 18-channel tracking room:



2.1.2 Recommended 18-Channel Full-Band Patch List

Connecting instruments in a consistent console order allows you to reuse your saved Cave Recorder Session Templates (SAVE TEMPLATE) across every rehearsal without re-patching:

Input & Strip	Typical Studio Source & Microphone	+48V Phantom	Subgroup Bus	Gain-Staging Target
IN 01 (CH01 01KickIn)	Kick Drum Inside (Dynamic or boundary mic)	Off (On for boundary)	SUB 1 (DRUM BUS)	Peaks at -6 dBFS on hard hits
IN 02 (CH02 02SnrTp)	Snare Drum Top (Dynamic cardioid, align Φ + Ø)	Off	SUB 1 (DRUM BUS)	Peaks at -6 dBFS on rimshots
IN 03 (CH03 03OvrhdL)	Mono Drum Overhead or Snare Bottom (align Φ)	On / Off	SUB 1 (DRUM BUS)	Peaks at -8 dBFS

Input & Strip	Typical Studio Source & Microphone	+48V Phantom	Subgroup Bus	Gain-Staging Target
IN 04 (CH04 04GtrAmp)	Close Guitar Cabinet Mic or Hi-Hat Condenser	Off / On	MST (MASTER BUS)	Average -18 dBFS RMS
IN 05 (CH05 TomRack)	Rack Tom Clip-On Dynamic	Off	SUB 1 (DRUM BUS)	Peaks at -6 dBFS on fills
IN 06 (CH06 TomFloor)	Floor Tom Dynamic	Off	SUB 2 / SUB 1	Peaks at -6 dBFS on fills
IN 07 (CH07 OverheadL)	Drum Overhead Left (Matched condenser pair)	On	SUB 1 (DRUM BUS)	Peaks at -8 dBFS on crashes
IN 08 (CH08 OverheadR)	Drum Overhead Right (Matched condenser pair)	On	SUB 2 / SUB 1	Peaks at -8 dBFS on crashes
IN 09 (CH09 SpdifSynthL)	Stereo Digital Synth Left (Coaxial S/PDIF L)	Off	SUB 1 / SUB 2	Average -18 dBFS RMS
IN 10 (CH10 SpdifSynthR)	Stereo Digital Synth Right (Coaxial S/PDIF R)	Off	SUB 2 (INST BUS)	Average -18 dBFS RMS
IN 11 (CH11 BassDi)	Bass Guitar Active/ Passive DI Box (Line/ Inst)	On (Active DI)	SUB 1 / SUB 2	Average -18 dBFS, peaks -8 dBFS
IN 12 (CH12 BassAmp)	Bass Cabinet Dynamic or FET Condenser (align Φ to CH11)	Off / On	SUB 2 (INST BUS)	Average -18 dBFS, peaks -8 dBFS
IN 13 (CH13 GuitarL)	Rhythm Guitar Amp Close Dynamic (Cardioid Dynamic / Ribbon)	Off	SUB 1 / SUB 2	Average -18 dBFS RMS
IN 14 (CH14 GuitarR)	Lead Guitar Amp Close Dynamic or Second Cab Mic	Off	SUB 2 (INST BUS)	Average -18 dBFS RMS
IN 15 (CH15 KeysL)	Stage Piano / Keyboard Stereo DI Left	Off	SUB 2 (INST BUS)	Average -18 dBFS RMS
IN 16 (CH16 KeysR)	Stage Piano / Keyboard Stereo DI Right	Off	SUB 2 (INST BUS)	Average -18 dBFS RMS
IN 17 (CH17 LeadVocal)	Lead Vocal Dynamic or Large-Diaphragm Condenser	Off / On	MST (MASTER BUS)	Peaks at -6 dBFS on loud chorus
IN 18 (CH18 RoomAmbience)	Live Room Condenser or Ribbon (Never +48V on passive ribbon)	On / Off	SUB 1 (DRUM BUS)	Average -20 dBFS RMS

2.1.3 Digital Clock Synchronization (INT vs. External ADAT / Word Clock)

Whenever two or more digital audio devices convert audio simultaneously (such as a primary USB interface and an external 8-channel ADAT optical preamp), every converter in the chain must lock to a single master sample clock. If both units run on their own internal crystal oscillators, slight clock drift causes periodic clicks, pops, and high-frequency distortion across the expanded channels.

Follow one of these two studio clocking topologies:

1. **Topology A: Primary Interface as Master (INT), Expander Slaved via Word Clock or Bidirectional ADAT (Recommended):**
   - Set your primary audio interface clock source to **Internal (INT)** at 48.0 kHz (displayed in the Cave Recorder Top Deck as 48.0 kHz | INT and in the console session bar as 48.0 kHz LOCKED).
   - Run a **75-ohm BNC Word Clock cable** from the primary interface WORD CLOCK OUT to the ADAT expander WORD CLOCK IN (switching the expander's rear clock selector to WC IN with 75-ohm termination active), or run a second TOSLINK cable from the primary interface ADAT OUT to the expander ADAT IN and set the expander clock selector to ADAT IN.
2. **Topology B: ADAT Expander as Master, Primary Interface Slaved to ADAT:**
   - If your external 8-channel preamp has only an ADAT OUT port and no WORD CLOCK IN or ADAT IN port, set the external preamp to INTERNAL 48 kHz and set your primary audio interface clock source to **External ADAT** in its hardware control panel.

> **PREVENTING DIGITAL CLOCK CLICKS AND SAMPLE RATE MISMATCHES**
> Standard single-wire ADAT Lightpipe carries 8 discrete audio channels at 44.1 kHz or 48.0 kHz. At 88.2 kHz or 96.0 kHz (S/MUX mode), a single optical cable carries only 4 channels. For full 18-channel band recording with an 8-channel ADAT expander, operate both hardware units and Cave Recorder at 48000 Hz (48.0 kHz) and confirm the red 48.0 kHz LOCKED badge is lit in the console header before tracking.

2.1.4 First-Launch Setup on macOS (Developer ID Signing, Notarization & Ad-Hoc Fallback)

The macOS release of Cave Recorder (caverec-v0.4.2-macos-arm64.zip / Cave Recorder.app) supports both **Apple Developer ID Application signing and notarization** and **ad-hoc signed local builds**:

- **Notarized Developer ID Releases:** Signed and notarized with Apple for CoreAudio multi-channel microphone and audio interface input, multi-camera studio video capture, local Wi-Fi companion and cloud networking, and third-party CLAP and Audio Unit plugin hosting, so **Cave Recorder.app** opens directly from /Applications without Gatekeeper warnings.
- **Ad-Hoc Signed Builds (Gatekeeper Quarantine Clearance):** When opening an ad-hoc signed build downloaded from a web browser on macOS for the first time, macOS Gatekeeper places a quarantine flag on the bundle and displays an unverified developer prompt. Clear the quarantine flag using either of these two methods before your first studio session:

- **Method A: macOS System Settings (Open Anyway):**
- Double-click **Cave Recorder.app** in /Applications and click **Done** (or **Cancel**) on the initial Gatekeeper prompt.
- Open **System Settings -> Privacy & Security**, scroll down to the **Security** section, and click **Open Anyway** next to "Cave Recorder.app" was blocked from use because it is not from an identified developer.
- Authenticate with Touch ID or your macOS administrator password and confirm **Open**.
- **Method B: Terminal Quarantine Removal (xattr -cr):**
- Open the macOS **Terminal** application and run xattr -cr "/Applications/Cave Recorder.app".
- Launch **Cave Recorder.app** normally from /Applications or Spotlight, and click **Allow** when macOS prompts for microphone, audio interface, and camera access.

2.2 Configuring Audio Device Settings (AUDIO SETTINGS)

Click the AUDIO SETTINGS button at the top right of the Top Transport Deck (or press **CMD** + **,** on macOS / **CTRL** + **,** on Windows and Linux) to open the **Audio Device & Latency Settings** modal (MODAL: AUDIO DEVICE & LATENCY SETTINGS).

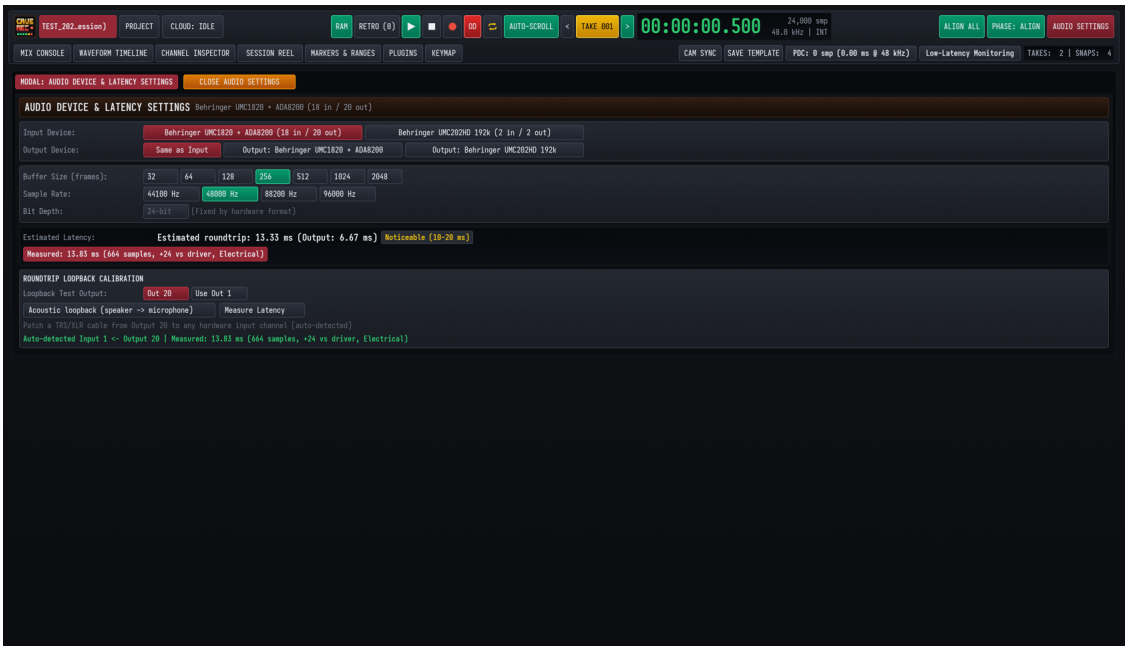


Figure 2.1: *The Audio Device Settings modal showing input/output device selection, sample rate, buffer size, and hardware loopback calibration.*

2.2.1 Selecting Input and Output Devices & macOS Microphone Permission Diagnostics

- **Automatic Multi-Channel Studio Interface Selection:** On startup, Cave Recorder automatically prioritizes connected multi-channel studio audio interfaces (such as 18-channel USB/ADAT interfaces) over 2-channel built-in laptop microphones while keeping the input and output sample clocks unified on the same hardware device.

- **Input Device:** Displays all active CoreAudio (macOS), ASIO/WASAPI (Windows), or ALSA (Linux) recording devices detected on the workstation, along with their available channel counts (for example, Studio USB + ADAT Interface (18 in / 20 out) or Desktop 2-Channel USB Interface (2 in / 2 out)). Click the desired interface button to bind Cave Recorder's input engine.
- **Output Device:** Select Same as Input whenever your studio monitors and headphone amplifiers are connected to the same multi-channel interface as your microphones. Using a single unified hardware device guarantees that input and output converters share the same sample clock and ensures deterministic roundtrip loopback calibration. Separate output devices remain selectable when monitoring through an independent desktop DAC.
- **Built-In macOS Microphone Permission Status & Recovery Guidance:** On macOS, CoreAudio delivers silent (0.0) input buffers across all hardware channels whenever macOS Privacy & Security blocks microphone access. Cave Recorder continuously checks the operating system Microphone permission status (Authorized, Not Determined, Restricted, or Denied) inside AUDIO SETTINGS:
  - **One-Click Permission Request (Not Determined):** If macOS has not yet prompted for audio input access, Cave Recorder triggers the system Microphone privacy prompt automatically (or on demand from AUDIO SETTINGS) so you can click **Allow** in one step.
  - **Denied or Stale Permission Recovery:** If microphone access was previously denied in **System Settings -> Privacy & Security -> Microphone**, or if a live CoreAudio input stream delivers 100% digital silence across all channels after updating Cave Recorder.app, AUDIO SETTINGS displays a clear status banner instructing you to toggle **Cave Recorder** off and back on inside **System Settings -> Privacy & Security -> Microphone** and restart the app.

2.2.2 Buffer Size (frames), Sample Rate & Bit Depth

- **Buffer Size (frames):** Selectable from 32, 64, 128, 256, 512, 1024, or 2048 sample frames per hardware buffer cycle.
  - **Low Buffer Sizes (32, 64, or 128 frames):** Minimize live input-to-output foldback delay when a vocalist or guitarist is monitoring their own performance through Cave Recorder's built-in console EQ, VCA compressor, and AUX 1 REVERB.
  - **Medium Buffer Sizes (256 or 512 frames):** Provide generous processor headroom during 18-channel full-band tracking when musicians monitor via the audio interface's direct hardware mixer, or during mixing sessions with multiple third-party CLAP plugins.
- **Sample Rate:** Selectable between 44100 Hz, 48000 Hz, 88200 Hz, and 96000 Hz. Use 48000 Hz (48.0 kHz) for 18-channel USB/ADAT studio rigs and video-synchronized live sessions.
- **Bit Depth:** Shows the hardware converter transport word length (24-bit (Fixed by hardware format)). Regardless of the hardware interface's 24-bit converter format, Cave Recorder processes all internal signal paths and writes all on-disk multitrack stem files (rec/*.wav) in uncompressed 32-bit floating-point format.

Buffer Size (frames)	One-Way Buffer @ 48.0 kHz	Driver Roundtrip (incl. Converter/USB Safety)	Recommended Studio Scenario
32 frames	0.67 ms	~4.0 - 5.5 ms	Solo vocal or guitar tracking through software effects on fast workstations
64 frames	1.33 ms	~5.5 - 7.5 ms	Live overdubbing (OD) with software cue monitoring and Low-Latency Monitoring
128 frames	2.67 ms	~8.0 - 10.5 ms	General multi-track tracking and rehearsal capture
256 frames	5.33 ms	~13.3 - 14.0 ms	18-channel band rehearsals using hardware direct monitoring
512 frames	10.67 ms	~24.0 - 26.0 ms	Full console mixing, Subgroup Bus processing, and heavy CLAP plugin chains
1024 / 2048 frames	21.33 / 42.67 ms	~45.0 - 90.0 ms	Mastering, offline Stem Bounce preparation, and high-channel-count archival playback

2.2.3 30-Minute Retroactive Capture Memory (RAM)

While your audio interface is active, the green RAM button in the Top Transport Deck controls Cave Recorder's continuous 30-minute retroactive capture memory. When RAM is enabled:

- Every record-armed channel strip (R engaged) continuously writes incoming 32-bit float audio into a dedicated 30-minute rolling capture memory even while the main transport is stopped.
- At 48.0 kHz, 30 minutes of 32-bit audio requires approximately 330 MB of system memory per armed channel (roughly 5.9 GB when all 18 channels are armed simultaneously).
- If you are working on a laptop with limited memory and do not need retroactive jam rescue, click RAM in the Top Deck to disable background capture memory.

2.3 Calibrating Roundtrip Loopback Latency (Measure Latency)

When a musician records an Overdub Pass (Rec002-OD) while listening to playback of a Base Pass (Rec001), the backing track travels out through the digital-to-analog converter (DAC) to the musician's headphones, and the musician's new performance travels back in through the analog-to-digital converter (ADC) and USB or Thunderbolt bus.

Audio device drivers only report nominal buffer sizes (Estimated roundtrip: 13.33 ms, or 640 samples at 256 frames / 48.0 kHz). They do not account for the additional samples introduced by hardware oversampling filters, optical ADAT connections, and USB or Thunderbolt transport stages (which typically add 16 to 64 extra samples). Without hardware loopback calibration, every overdubbed track lands slightly late relative to the drums.

Cave Recorder includes a built-in hardware impulse loopback analyzer (ROUNDTRIP LOOPBACK CALIBRATION) that measures the exact physical sample delay from output jack to input jack:

1. **Lower Monitor Volume:** Turn down your studio monitor controller and headphone amplifiers completely so the calibration test impulse does not play through loudspeakers.
2. **Patch a Physical Loopback Cable:** Connect a balanced 1/4-inch TRS or XLR patch cable from your interface's dedicated loopback test output (Out 20 or Out 1) directly into any hardware analog input channel (such as Input 1). Ensure +48V phantom power is turned **off** on that input channel and set its physical preamp gain knob near unity (roughly 9 to 10 o'clock).
3. **Select the Loopback Test Output:** Open AUDIO SETTINGS (**CMD** + **,** / **CTRL** + **,**) and locate the **ROUNDTRIP LOOPBACK CALIBRATION** panel at the bottom. Choose Out 20 (convenient on 20-output interfaces where Output 20 can stay permanently patched to an unused line input) or click Use Out 1 if patching from main Output 1. (For acoustic room delay tests through a speaker and microphone, toggle Acoustic loopback (speaker -> microphone)).
4. **Run the Impulse Measurement:** Click Measure Latency. Cave Recorder emits a short, high-precision measurement impulse out the selected output port, automatically detects which hardware input channel received the return signal, and correlates the exact sample arrival offset.
5. **Verify the Measured Offset:** Inspect the green confirmation readout at the bottom of the modal and the red badge under Estimated Latency: Auto-detected Input 1 <- Output 20 | Measured: 13.83 ms (664 samples, +24 vs driver, Electrical)
6. **Close Audio Settings:** Click CLOSE AUDIO SETTINGS at the top left of the modal.

> **HOW CAVE RECORDER APPLIES YOUR MEASURED LOOPBACK OFFSET**
> Whenever you record an Overdub Pass (Rec002-OD) with OD enabled, Cave Recorder automatically tags and shifts the newly recorded Pass Lane by the negative of the total roundtrip offset (Overdub Record-Offset (G + R): -664 smp, combining any active Plugin Delay Compensation G with the calibrated hardware roundtrip R). You can inspect the exact sample offset on every Take row in SESSION REEL (**CMD** + **4** / **CTRL** + **4**) and on every Overdub Pass Lane in WAVEFORM TIMELINE (**CMD** + **2** / **CTRL** + **2**), or click RE-APPLY MEASURED OFFSET in the Session Reel to update earlier Takes after recalibrating.

2.4 Creating Your First Project and Session (PROJECT Modal)

Press **CMD** + **O** on macOS (**CTRL** + **O** on Windows and Linux) or click the PROJECT button in the Top Transport Deck to open the **Project and Session Browser** modal.

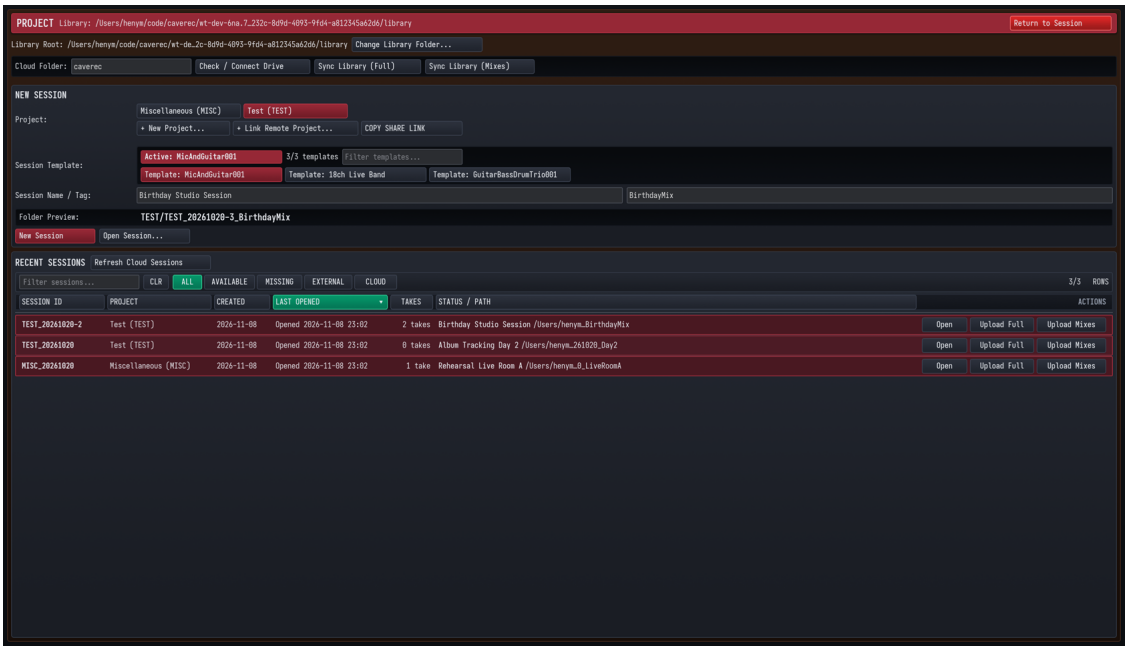


Figure 2.2: *The Project and Session Browser modal showing Projects, Templates, and Session history.*

2.4.1 Configuring the Library Root and Cloud Folder

At the top of the PROJECT modal, verify your storage paths before creating your first session:

1. **Library Root:** Displays the master local disk directory where all Projects, Sessions, and Takes are stored. Click Change Library Folder... to point Cave Recorder at a fast internal NVMe SSD or dedicated external Thunderbolt recording drive.
2. **Cloud Folder:** Displays the configured Google Drive folder name (for example, caverec). Click Check / Connect Drive to verify cloud credentials, Sync Library (Full) to synchronize complete session multitracks across machines, or Sync Library (Mixes) to synchronize only the lightweight <ProjectTag>/_Mixes/ reference folders.

2.4.2 Creating a New Project or Linking a Bandmate's Remote Project

Every Session belongs to a **Project** (such as a band name, album project, or client ensemble) identified by an uppercase 1-to-8 character **Project Tag** (GF, TEST, MISC).

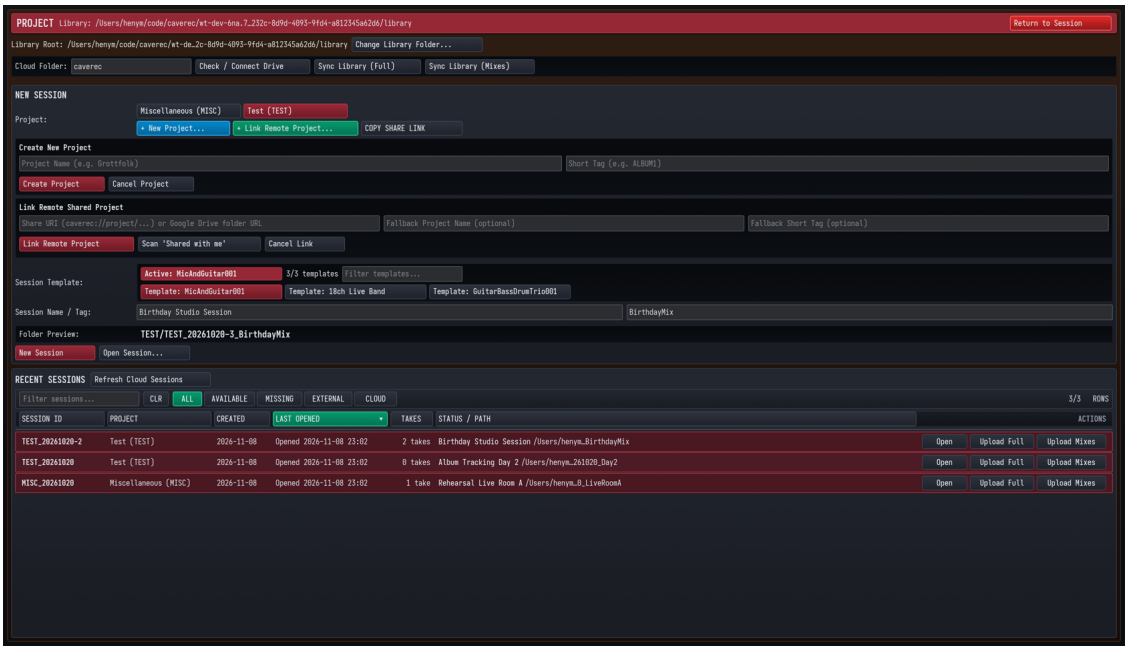


Figure 2.3: *Creating a new Project and Session in the Project Browser modal.*

- **To create a new local Project:**
  - In the **NEW SESSION** panel, click `+ New Project...` below the Project selector pills.
  - Enter the **Project Name** (for example, `Grottfolk`) and a short alphanumeric **Project Tag** consisting of 1 to 8 uppercase letters or digits (for example, `GF`).
  - Click `Create Project`. Cave Recorder creates `<LibraryRoot>/GF/` along with `<LibraryRoot>/GF/_Mixes/` and selects the new Project pill.
- **To link a collaborator's shared Google Drive Project:**
  - Click `+ Link Remote Project...` to expand the **Link Remote Shared Project** form.
  - Paste a `caverec://project/...` share URI (copied from a bandmate's `COPY SHARE LINK` button) or a Google Drive folder URL, or click `Scan 'Shared with me'` to auto-discover Cave Recorder projects shared with your Google account.
  - Click `Link Remote Project`.

2.4.3 Selecting a Session Template and Starting a New Session

Once your Project pill is selected (highlighted in burgundy):

1. **Choose a Session Template:** In the `Session Template` row, select a starting console layout:
   - **`Template: MicAndGuitar001` (Built-In Default Template):** Pre-selected on startup, `MicAndGuitar001` provides a ready-to-record 2-channel singer-songwriter and acoustic rehearsal setup with **0 Subgroup Buses** and both stereo auxiliary returns (`AUX 1 REVERB` Plate Reverb and `AUX 2 TAPE DLY` Tape Delay) active:
   - **Channel 1 (`Mic`, routed from `IN 01` to `MASTER BUS`, Record Armed `R`):** Pre-configured with an active `80 Hz` Butterworth high-pass filter (`HPF`) on Band 1 and an enabled `-8.0 dB` Post-Fader (`PST`) send to `AUX 1 REVERB` (`EMT 140 Plate`).
   - **Channel 2 (`Guitar`, routed from `IN 02` to `MASTER BUS`, Record Armed `R`):** Pre-configured with an active `80 Hz` Butterworth high-pass filter (`HPF`) on Band 1 and unity `0.0 dB` fader gain.

- **Template: GuitarBassDrumTrio001:** Built-in 11-channel band template with three stereo Subgroup Buses (Guitar Bus, Drum Bus, and parallel Drum Crush Bus with 8:1 VCA compression), stereo guitars (Guitar L / Guitar R), Bass DI, Room Mic, 4-mic drum kit (Kick, Snare Top, Overhead L, Overhead R), and pre-assigned guitar/vocal overdub channels.
- **Template: 18ch Live Band:** Full 18-channel band console (01KickIn through RoomAmbience) with DRUM BUS, INST BUS, EQ high-pass filters, and VCA compressor defaults.
- Any custom studio setup you save via SAVE TEMPLATE in the Top Deck appears automatically in this list and can be filtered via the Filter templates... search box.

2. **Enter Session Name and Session Tag:** Type a descriptive **Session Name** in the left field (for example, Birthday Studio Session) and a short filesystem-friendly **Session Tag** in the right field (for example, BirthdayMix).
3. **Verify the Folder Preview:** Inspect the Folder Preview line (TEST/TEST_20261020-3_BirthdayMix). Cave Recorder automatically combines the Project Tag, the session date (using the 08:00 local time rollover rule), the daily session sequence number, and your Session Tag.
4. **Launch the Session:** Click New Session to initialize the session folder on disk and enter the MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**). To reopen an existing session instead, locate it in the **RECENT SESSIONS** table at the bottom (filtering by ALL, AVAILABLE, MISSING, EXTERNAL, or CLOUD) and click Open.

2.4.4 Pre-Session Track Naming, Input Routing & Gain Staging

Before the band starts playing the first song, perform a 60-second console check in MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**):

1. **Verify Hardware Input Routing (IN 01 v):** On each channel strip, confirm the input selector dropdown (IN 01 v, IN 02 v, …) matches the physical preamp jack where that microphone or DI box is plugged in.
2. **Name Each Strip Clearly:** Double-click any channel strip header (01KickIn, 02SnrTp, …) to rename it. Clear strip names propagate automatically into every recorded .wav stem filename (01-KickIn_Rec001.wav) and Waveform Timeline lane.
3. **Set Analog Preamp Gain and Digital Input Trim (TRM):** Have the band play at full rehearsal volume:
   - Adjust your audio interface's physical analog preamp knobs so sustained signals average around **-18 dBFS RMS** (in the healthy green zone of the channel meter) and the loudest drum hits peak around **-6 dBFS** (in the lower amber zone, well below 0.0 dBFS hardware clipping).
   - Use each strip's digital TRM slider (-24.0 dB to +24.0 dB) to fine-tune the level feeding the 8-slot processor rack and recorded pre-fader stem.
4. **Engage Record Arm (R) and Input Monitoring (I):** Click R on every active input strip so it lights red, and confirm the Top Deck RAM indicator is lit green so 30-minute retroactive memory begins capturing immediately.
5. **Save Your Room Template:** Click SAVE TEMPLATE in Row 2 of the Top Deck, enter a template name for your room setup, and click SAVE AS TEMPLATE so future sessions load this exact configuration in one click.

2.5 Connecting Studio Cameras & 3-Second Camera Latency Calibration

Cave Recorder includes a sample-accurate **Multi-Camera Studio Video Suite** that captures multiple USB, UVC, HDMI-to-USB capture-card, built-in studio cameras, and wireless **Cave Companion** mobile camera nodes directly into <TakeFolder>/video/ alongside your 32-bit float multitrack audio stems.

2.5.1 Constant Frame Rate (50 FPS & 25 FPS) Integer Audio-Clock Lock & 50 Hz Anti-Flicker

Consumer video recorders and webcams frequently operate at variable frame rates (VFR) or fractional NTSC cadences (29.97 FPS or 59.94 FPS) that produce non-integer audio sample counts per video frame (1601.6 samples/frame at 48.0 kHz) and cause rolling-shutter banding under 50 Hz studio LED lighting. Cave Recorder locks every studio camera to a strict **Constant Frame Rate (CFR)** cadence tied to the 48.0 kHz audio interface clock:

Video Capture Mode	Frame Duration	Exact 48.0 kHz Audio Cadence	Encoding & GOP Profile	Studio Use Case
1080p50 / 720p50 CFR (Default)	20.000 ms	**960 samples/frame** (exact integer)	H.264 .mp4, **Zero B-Frames**, 0.5 s (25-frame) closed GOP + .vididx index	High-motion drum cams, close fretboard angles, and full-band rehearsal video with zero 50 Hz mains lighting flicker
1080p25 / 720p25 CFR	40.000 ms	**1920 samples/frame** (exact integer)	H.264 .mp4, **Zero B-Frames**, 0.5 s (12/13-frame) closed GOP + .vididx index	Wide room cameras, lower-light rehearsal spaces, or extended multi-hour jam sessions

Because every video frame maps to an exact integer number of 48.0 kHz audio samples (960 samples at 50 FPS or 1920 samples at 25 FPS) and is encoded with **zero bidirectional B-frames** and a short **0.5 s keyframe interval (GOP)**, timeline scrubbing in WAVEFORM TIMELINE (**CMD / CTRL** + **2**) snaps instantaneously to any drum transient without multi-second decoder lag.

2.5.2 Running the 3-Second 4-Pulse Camera Latency Calibration Modal

Even when a camera's frame cadence is locked to 50 FPS (20.0 ms per frame), USB video capture cards, HDMI converters, and CMOS rolling-shutter image sensors introduce a fixed pipeline delay (typically 35 ms to 120 ms) between the physical moment a drumstick strikes a snare head and the moment that decoded video frame arrives in workstation memory.

To measure and compensate for this optical pipeline delay to **sub-frame microsecond accuracy** without a mechanical clapperboard, Cave Recorder provides the **3-Second Camera Latency Calibration Modal** (CALIBRATE CAMERA LATENCY):

1. **Position the Camera, Select a Calibration Mode & Open Calibration:** Open the camera settings panel (CAM SYNC), click **CALIBRATE CAMERA LATENCY**, and select one of the three calibration modes in the mode selector bar:
   - **4-PULSE VERNIER FLASH (External / Companion Cameras Pointed at the Display or Rear LED Torch):** Emits a 4-pulse quarter-frame staggered screen/LED flash + studio monitor click sequence for automated sub-frame optical latency measurement.
   - **FULL-WINDOW WHITE FLASH (Built-In Laptop Webcams Facing the Musician):** Temporarily replaces the entire modal with a full-window #FFFFFF maximum-luminance white flash during each 80 ms pulse so a built-in laptop camera (such as a MacBook FaceTime HD Camera) facing away from the screen captures the reflected room/face illumination jump.
   - **INTERACTIVE CLAPPER / HAND CLAP (Visual Hand-Clap Filmstrip Inspector):** Records a 3-second audio-video capture while you clap your hands once in front of the camera lens, automatically locates the acoustic clap transient peak, and opens an interactive -5..+5 frame-stepping filmstrip inspector ([◀ PREV FRAME] / [NEXT FRAME ▶]) around the impulse so you can select the exact frame of physical hand contact and save the calibrated camera latency offset.
2. **Execute the 3-Second (34,320-Sample) Calibration Sequence:** Click **START 3-SECOND CALIBRATION**. In 4-Pulse Vernier or Full-Window White Flash mode, Cave Recorder runs an automated 4-phase optical and acoustic measurement spanning 34,320 samples (715 ms active measurement window inside the 3-second sequence) at 48.0 kHz:
   - **Vernier Staggered Optical Flash + Studio Monitor Click:** Cave Recorder emits **4 synchronized calibration pulses** staggered at quarter-frame (5.0 ms / 240-sample) Vernier offsets relative to the 20.0 ms (50 FPS) camera frame grid. Each pulse pairs an **80 ms full-white screen flash** (or high-intensity mobile LED Torch flash) with a crisp **acoustic calibration click** routed through the studio monitors into an active live-room microphone.
   - **Direct-Path Acoustic Onset Detector:** The audio analyzer identifies the exact direct-path acoustic arrival sample of each monitor click while rejecting late room reflections, establishing the true physical acoustic timestamp on the 48.0 kHz audio clock.
   - **Partial-Exposure Luminance Weighting:** Because a CMOS sensor integrating light over a 20.0 ms frame window captures a partial-exposure brightness ramp on the exact frame where the 80 ms flash turns on, Cave Recorder weights the rising luminance fraction across the 4 Vernier-staggered pulses to solve the optical arrival time with **sub-frame microsecond precision** (finer than a single 20.0 ms video frame).
3. **Automatic Per-Camera Calibration Storage:** Once the 4-pulse sequence completes, the modal displays the measured optical offset (for example, Measured Camera Latency: +64.25 ms (+64,250 us | 3.21 frames @ 50 FPS)) and stores the calibrated latency offset in the session by hardware camera identifier. Every subsequent Take recorded with that camera automatically stamps its .vididx frame index with the calibrated microsecond offset so drumstick impacts and guitar strums lock visually to the audio waveform.

PART I | CHAPTER 3

Typical Studio & Rehearsal Workflows

Step-by-step guides for band rehearsal capture, 30-minute retroactive memory rescue, multi-mic phase alignment and Polarity Maximizer optimization, overdub comping, live-take salvage, multi-camera rehearsal video capture with Cave Companion wireless mobile camera nodes and split-screen mixdowns, and band mix sharing.

This chapter walks through seven end-to-end studio workflows that cover the day-to-day operation of Cave Recorder: capturing a full-band rehearsal with instant stereo reference mixes, rescuing an unrecorded jam from the 30-minute retroactive memory, phase-aligning a multi-microphone drum kit and guitar rig, tracking and comping layered overdubs in Pass Lanes, mixing with non-destructive Snapshots and Google Drive synchronization, salvaging a live band Take with non-destructive mistake surgery, and capturing multi-camera 50 FPS studio and wireless **Cave Companion** mobile video with split-screen .mp4 and .fcpxml v1.9 multicam exports.

3.1 Workflow 1: Full-Band Rehearsal Capture & Instant Reference Prints

Studio Scenario: A four- or five-piece band gathers for a two-hour evening rehearsal. You want to record every song run-through across 12 to 18 channels without slowing down the room, mark strong sections on the fly, and hand every band member compact 24-bit .flac and 320 kbps .mp3 stereo mixes before they finish packing up their gear.

3.1.1 Rehearsal Capture Procedure

1. **Launch the Session from Your Rehearsal Template:**
   - Press **CMD** + **O** (macOS) or **CTRL** + **O** (Windows/Linux) to open the PROJECT modal.
   - Select your band's Project pill, click Template: 18ch Live Band (or your custom room template), type a Session Name (Tuesday Night Rehearsal) and Session Tag (Rehearsal), and click New Session.
2. **Arm All Active Input Channels:**
   - Press **CMD** + **1** (macOS) or **CTRL** + **1** (Windows/Linux) to view MIX CONSOLE.
   - Confirm that CH01 through CH18 (or all active mic and DI channels) have their red Record Arm (R) buttons engaged (click R on each strip, or select a strip and press **SHIFT** + **A**).
   - Verify that Overdub Mode (OD) in the Top Transport Deck is **off** (unlit) so each recording starts a fresh t = 0 Take.
3. **Dial a Quick Monitor and Live Print Balance:**
   - While the band warms up, verify input levels average around -18 dBFS RMS with peaks around -6 dBFS.

- Set channel faders and stereo pans (OverheadL/OverheadR, GuitarL/GuitarR, KeysL/KeysR), blend in AUX 1 REVERB (RVB) on snare and vocals, and engage TAPE saturation, the brickwall LIM (CEIL -0.3dB), and the 2-TRK live Print button on the MST MASTER BUS strip. Because the live stereo **Print** records directly from the Master Bus output whenever 2-TRK is armed, this monitor balance becomes your immediate reference mix.

4. **Start Recording Song 1 (Take 001):**
   - Just before the drummer counts in, press **R** (or click Record ● in the Top Deck).
   - Cave Recorder creates Take 001 at 00:00:00.000 and simultaneously writes raw 32-bit float pre-fader stems (Rec001) into Take001/rec/ and the live stereo Master Bus mix (Print001.wav) into Take001/mix/.
5. **Drop Cue Markers on the Fly (M):**
   - Whenever a chorus groove locks in, a bridge arrangement changes, or a solo stands out during the performance, press **M** on the keyboard.
   - Cave Recorder drops a timestamped standalone marker (M005, M006, ...) at the exact live playhead sample without interrupting recording.
6. **Stop at the End of the Song (SPACE):**
   - When the final chord rings out, press **SPACE** (or **SHIFT** + **SPACE**).
   - Cave Recorder closes all open audio files for Take 001 and automatically captures a post-recording console Snapshot (Auto-Save (Post-Recording)).
7. **Capture Subsequent Songs as Independent Takes (Take 002, Take 003, ...):**
   - When the band counts in the next run-through or the next song, simply press **R** again.
   - Cave Recorder automatically increments to Take 002 with a brand-new 00:00:00.000 timeline origin. Repeat **R**, play the song, and press **SPACE** for every song of the night.
8. **Tag Keeper Takes and Convert Live Prints to .flac and .mp3 (FLAC PRINT / MP3 PRINT / FLAC ALL PRINTS / MP3 ALL PRINTS):**
   - At the end of rehearsal, press **CMD** + **4** (macOS) or **CTRL** + **4** (Windows/Linux) to open the SESSION REEL.
   - In the **TAKES CATALOG**, click EDIT TAKE 001 to assign a descriptive song tag (such as NewSong_Run1 or ChorusKeeper).
   - Click FLAC PRINT 001 or MP3 PRINT 001 on individual Takes, or click FLAC ALL PRINTS and MP3 ALL PRINTS in the Session Reel header bar to batch-encode every live stereo .wav Print to 24-bit lossless .flac and 320 kbps CBR .mp3 (with a Xing/Info seek header) and link both into <ProjectTag>/_Mixes/. (If you adjusted faders after recording and want an updated offline render instead of the live Print, toggle FLAC and/or MP3 on the MST MASTER BUS strip and click BOUNCE to render an offline Mix001.<flac|mp3> Mixdown).
9. **Sync Reference Mixes to Bandmates' Phones and Browsers:**
   - Click UPLOAD SES MIXES in the Session Reel header (or open the Top Deck CLOUD popover and click SYNC SESSION MIXES).
   - Cave Recorder uploads the compact .flac and .mp3 reference mixes to Google Drive in seconds so every band member can stream .mp3 mixes directly inside the Google Drive web browser preview player or download lossless .flac masters on the drive home.

> **AUTOMATIC FLAC AND MP3 ENCODING ON CLOUD UPLOAD**
> Enable the AUTO-FLAC ON UPLOAD and AUTO-MP3 ON UPLOAD toggles in the SESSION REEL (**CMD** + **4** / **CTRL** + **4**) or Top Deck CLOUD popover. Whenever you click UPLOAD SES MIXES or UPLOAD SES FULL, Cave Recorder automatically converts any uncompressed .wav Prints into 24-bit .flac and/or 320 kbps CBR .mp3 files and populates <ProjectTag>/_Mixes/ prior to uploading.

3.2 Workflow 2: Rescuing an Unrecorded Jam with 30-Minute Retroactive Capture (RETRO)

Studio Scenario: Between rehearsed songs, the transport is stopped while the guitarist tunes. The drummer and bassist start an improvised groove, the rest of the band joins in, and eight minutes later they finish the best song idea of the night. Nobody pressed Record.

Because Cave Recorder continuously maintains a rolling 30-minute retroactive capture memory across all armed inputs whenever RAM is active, the entire multi-channel performance is still in system memory waiting to be saved.

3.2.1 Retroactive Rescue Procedure

1. **Keep RAM Active and Channels Armed During Rehearsal:**
   - At the start of your session, verify that the RAM button in the Top Transport Deck is illuminated green and all live studio channels (CH01..CH18) have their red R (Record Arm) buttons engaged.
   - Even while the transport is stopped (▪), Cave Recorder continuously retains the most recent 30 minutes of incoming 32-bit float audio from every armed channel in memory.
2. **Trigger Retroactive Capture (RETRO):**
   - As soon as the unrecorded jam finishes (or even while the band is still playing), click the RETRO (0) button in the Top Transport Deck, or press **SHIFT** + **R** .
3. **Verify the Rescued Take:**
   - Cave Recorder immediately flushes the rolling multi-channel capture memory to disk as a new numbered Take (Take<NNN>), increments the Top Deck counter badge to RETRO (1), builds the multi-tier waveform overview, and switches the active console and timeline to the rescued Take.
4. **Trim Pre-Jam Room Chatter into a Clean Song Take (SPLIT TAKE):**
   - Press **CMD** + **2** (macOS) or **CTRL** + **2** (Windows/Linux) to switch to WAVEFORM TIMELINE.
   - Scrub or zoom (ZOOM IN (+) or **=** , or vertical/horizontal drag on the timeline ruler) to find the exact moment the drummer counted in the jam, and click to place the playhead just before the first beat.
   - Click SPLIT TAKE in the Waveform Timeline toolbar (or press **M** to drop a marker, open MARKERS & RANGES with **CMD** + **5** / **CTRL** + **5** , and click NEW TAKE -> END on that marker row).
   - Cave Recorder extracts the jam from that marker forward into its own clean 00:00:00.000 Take and lets you move the earlier tuning Take to the session trash via TOMB TAKE 001 (or permanently remove an unwanted retroactive Take via PURGE RETRO 001) in SESSION REEL (**CMD** + **4** / **CTRL** + **4**).

3.3 Workflow 3: Multi-Mic Drum & Guitar Cab Phase Alignment (ALIGN ALL & Φ)

Studio Scenario: You are tracking a full band across 18 channels (CH01 01KickIn, CH02 02SnrTp, CH03 03OvrhdL, CH04 04GtrAmp, CH05 TomRack, CH06 TomFloor, CH07 OverheadL, CH08 OverheadR, CH09 SpdifSynthL, CH10 SpdifSynthR, CH11 BassDi, CH12 BassAmp, CH13 GuitarL, CH14 GuitarR, through CH18 RoomAmbience). Because acoustic wavefronts from the kick and snare travel several feet through the air before reaching the overhead and room microphones, and because a bass amp microphone lags behind a direct Bass DI box, summing the raw channels causes phase cancellation that thins out low-end punch and smears transient focus.



Figure 3.1: Mix Console configured for multi-channel band tracking and phase alignment.

3.3.1 Multi-Mic Phase Alignment Procedure

1. **Record a 10-to-15 Second Full-Kit & Band Reference Pass:**
   - Arm all drum and multi-miked instrument channels (R) in MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**).
   - Press **R** and have the drummer play a 10-to-15 second groove that hits the kick, snare, toms, hi-hat, and cymbals clearly while the bassist and guitarist play locked eighth notes.
   - Press **SPACE** to stop recording.
2. **Verify the Timing Reference Channel (0us):**
   - Look at the phase delay readout directly beneath each channel strip's top identifier badge (CH01 01KickIn shows -2.4dB 0us).
   - By default, CH01 (01KickIn or Overhead L) serves as the 0us acoustic timing anchor against which correlated channels are measured.
3. **Run Automatic Multi-Mic Phase Alignment (ALIGN ALL):**
   - Click the green ALIGN ALL button in the Top Transport Deck (or Φ ALIGN ALL in the session toolbar bar).

- To align only a single channel strip (such as aligning CH12 BassAmp to CH11 BassDi), select that strip and click its individual Φ button. Both Trigger Global Multi-Mic Phase Align and Trigger Phase Align on Inspected Track can also be bound to custom keyboard shortcuts in KEYMAP (**CMD** + **6** / **CTRL** + **6**).
- To manually flip 180-degree polarity (Ø) on the currently inspected channel strip at any time, click Ø or press **SHIFT** + **P** while in MIX CONSOLE.

4. **Inspect Sub-Sample Delay (us) and Polarity (Ø) Corrections:**
   - Cave Recorder cross-correlates the recorded multi-channel waveforms and engages the green Φ phase button and blue Ø polarity button where needed:
   - On CH02 02SnrTp, notice the readout updates to +1.5dB 420us with both polarity inversion (Ø blue) and phase delay (Φ green) engaged to lock the snare transient in phase with the kick and overheads.
   - On CH03 03OvrhdL, notice the readout updates to +0.0dB 1850us with Φ engaged to compensate for the acoustic propagation distance between the close mics and overheads.
5. **A/B Audition the Phase Alignment (PHASE: ALIGN vs. PHASE: BYP):**
   - Press **SPACE** to play back the drum recording through your studio monitors.
   - Click PHASE: ALIGN in the Top Deck (or PHASE A/B in the session sub-header) to toggle between PHASE: ALIGN (active) and PHASE: BYP (bypassed).
   - Listen to the low-frequency weight of the kick drum, the body of the snare, and the phantom center image of the kit snap into focus when PHASE: ALIGN is engaged. All subsequent Takes in the Session inherit these aligned delay and polarity settings automatically.
6. **Run the Two-Mode Polarity & Phase Maximizer (Mode 1 Instant Binary & Mode 2 Multicore Allpass Solver):**
   - For an immediate < 50 ms broadband polarity check across connected acoustic groups (such as all drum mics in Group 1 and a bass DI + mic'd bass cabinet pair in Group 2), open the **Polarity & Phase Maximizer** and trigger **Mode 1: Instant Binary Polarity Maximizer**, selecting **Low-End Weight (30-200 Hz Punch)**, **Broadband Loudness (ITU-R BS.1770-4 K-Weighted LUFS)**, **Transient Crest & Impact (60 Hz - 6 kHz + Top-K Crest Re-Scoring)**, or **Anti-Comb Spectral Smoothness**. Your original manual Ø, delay, and allpass settings are automatically preserved for A/B comparison.
   - When mixing a dispersive ported bass cabinet (bass DI + mic'd cabinet) or an 8-mic drum kit where simple time delay cannot align both 60 Hz sub-bass resonance and 2.5 kHz stick attack, switch to **Mode 2: Deep Multicore Spectral Phase & Allpass Solver** (30 s to 30 min anytime budget).
   - Toggle **[GAIN-MATCH AUDITION]** on, click between **[CANDIDATE A: PURE Ø - ZERO DISPERSION]** (optimal 180° polarity flips only, 0 allpass stages, 0.0 us time shift), **[CANDIDATE B: TIGHT LOW-END LOCK - MINIMAL POLES]** (polarity plus at most 1st-order LF rotator and 1 low-frequency biquad allpass stage <= 250 Hz, group delay < 2.5 ms), **[CANDIDATE C: DEEP MULTI-BAND ALLPASS]** (polarity plus up to 5-stage allpass cascade, group delay <= 6.0 ms), and **[CANDIDATE D: FULL WAVEFRONT + ALLPASS]** (causal microsecond time-of-flight alignment 0..25,000 us on unlocked spot mics plus polarity and sparse allpass cascade) with click-free 10 ms crossfades, and click **[APPLY SELECTED TO CONSOLE]** to program the top-of-strip zero-slot 5-stage **Phase Align Network** across the console.

3.4 Workflow 4: Layered Guitar & Vocal Overdubs with Pass Lane Comping (OD)

Studio Scenario: The band has tracked a solid rhythm section performance in Take 001. Now the guitarist or lead vocalist wants to loop the chorus section, record three alternate solo or vocal passes (Rec002-OD, Rec003-OD, Rec004-OD) inside Take 001, and select the best performance using **Pass Lanes**.

3.4.1 Overdub Tracking & Pass Lane Comping Procedure

1. **Select the Target Take (TAKE 001):**
   - Use < TAKE 001 > in the Top Deck (**ALT** + **[** / **ALT** + **]**) or click LOAD TAKE 001 in SESSION REEL (**CMD** + **4** / **CTRL** + **4**) so the rhythm Take is loaded at 00:00:00.000.
2. **Disarm Rhythm Tracks and Arm Only the Overdub Channel:**
   - In MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**), click the red R buttons on the drum and bass strips (CH01..CH03, CH05..CH12) to disarm them so they play back their PASS 01 (Rec001) stems from disk.
   - On the target overdub channel (for example, CH04 04GtrAmp or CH17 LeadVocal), engage both R (Record Arm, or press **SHIFT** + **A** while the strip is selected in MIX CONSOLE) and I (Input Monitor, or press **SHIFT** + **I**) so the performer hears their live input blended with playback.
3. **Latch Overdub Mode (OD) and Low-Latency Monitoring:**
   - Click OD in the Top Transport Deck (or press **O**) so the OD button illuminates bright red.
   - Click Low-Latency Monitoring in Row 2 of the Top Deck so any high-latency plugins on non-armed buses are bypassed in the live monitoring path.
4. **Define the Section Loop Range (K, L & ⇄):**
   - Press **CMD** + **2** (macOS) or **CTRL** + **2** (Windows/Linux) to open WAVEFORM TIMELINE.
   - Place the playhead two bars before the solo or chorus entrance and press **K** (or click RANGE START (K)).
   - Place the playhead a bar after the section ends and press **L** (or click RANGE END (L)).
   - Press **C** (or click Cycle/Loop ⇄ in the Top Deck) to enable loop playback across the range.
5. **Record Multiple Overdub Passes (OD 02, OD 03, OD 04):**
   - Press **R** to start overdub recording inside Take 001. While the rhythm tracks play back, Cave Recorder captures the new performance onto CH04 as Rec002-OD (displayed as OD 02) and automatically applies your calibrated roundtrip loopback offset (-640 smp (-13.3ms)).
   - Press **SPACE** at the end of the pass, return to the start of the range, and press **R** again to capture OD 03 (Rec003-OD) and OD 04 (Rec004-OD).
6. **Expand Pass Lanes in the Waveform Timeline (E):**
   - In WAVEFORM TIMELINE (**CMD** + **2** / **CTRL** + **2**), press **E** (or click EXPAND ALL PASSES in the timeline toolbar, or click the red pass-count pill v (2) on an individual track header) to expand the track's **Pass Lanes**. Press **SHIFT** + **E** (or click COLLAPSE ALL PASSES) at any time to fold them back into a single compact track row.

- The top slice of the expanded track displays PASS 01 (BASE PASS), followed by stacked slices for each recorded Overdub Pass (OD 02, OD 03, OD 04, each badged OVERDUB with its sample span Span: 0..48000 smp and net record-offset -640 smp (-13.3ms)). Notice that the Mix Console channel strip badge also updates to show OD 02 in amber when an Overdub Pass is active.

7. **Audition and Select Active Pass Lanes (Use PASS 01 / ★ OD 02):**
   - The currently selected pass on a track is badged ACTIVE COMP LANE and displays a green star button (★ PASS 01 or ★ OD 02), while every other pass is badged STANDBY COMP LANE with a clickable Use PASS 01 or Use OD 02 button.
   - While looping playback (**SPACE**), click Use PASS 01, Use OD 02, or Use OD 03 on any Pass Lane row to promote that pass immediately as the active playback lane for the track.
   - To silence a pass temporarily without changing selection, click M (Mute) on that Pass Lane row, or click Del to remove a flawed pass from the Take.

3.5 Workflow 5: Mixing, Non-Destructive Snapshots & Band Cloud Sharing

Studio Scenario: Tracking and overdub selection are complete. Now you want to shape a polished mix using Subgroup Buses and Aux Sends, save several mix variations as **Snapshots**, compare a vocal-up mix against an earlier Snapshot using **Live Mix (LIVE MIX)**, render an offline **Mixdown** and **Stem Bounces**, and synchronize the project to Google Drive.

3.5.1 Bus Mixing, Snapshot A/B & Cloud Sync Procedure

1. **Organize Channels into Subgroup Buses (DRUM BUS & INST BUS):**
   - In MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**), click the output routing selector (OUT: SUB 1 v) on all drum channels (CH01..CH08, CH18) to route them to SUB 1 (DRUM BUS).
   - Route synths, bass, guitars, and keys (CH09..CH16) to OUT: SUB 2 v (INST BUS), and route lead vocals (CH17) directly to OUT: MST v (MASTER BUS).
2. **Apply Bus Processing and Spatial Aux Sends (RVB & DLY):**
   - Click SEL on the DRUM BUS strip to load its 8-slot processor rack into the Upper Inspector Wells.
   - In the **12-Band Parametric EQ** well (2:12B EQ), sculpt a low-shelf boost around 60 Hz and a gentle presence lift at 4.5 kHz while watching the pre/post Dual-Trace RTA spectrum.
   - In the **VCA Compressor** well (3:VCA CMP), select a 4:1 ratio, engage the sidechain high-pass filter (SC HPF), and set MIX 50% for punchy New York-style parallel drum compression.
   - Use each channel's RVB and DLY send sliders (toggling PST for post-fader send operation) to feed AUX 1 REVERB (EMT 140 Plate zero-latency convolution reverb) and AUX 2 TAPE DLY (CAPSTAN TAPE echo with WOW 18%, DMP 4.2k, and PP ping-pong stereo widening).
3. **Save Named Mix Variations as Manual Snapshots (SNAP MANUAL):**
   - Press **CMD** + **S** on macOS (**CTRL** + **S** on Windows and Linux) from any workspace view, or press **CMD** + **4** / **CTRL** + **4** to open SESSION REEL and click SNAP MANUAL in the HISTORY toolbar, to capture your baseline mix as Snap 004 [MANUAL].

- Return to MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**) and try a more aggressive mix variation (for example, pushing DRIVE +4.5dB on the Master Tape saturator and riding the lead guitar +2.0 dB).

4. **A/B Compare Historical Snapshots with Your Live Mix (A/B LIVE MIX):**
   - In SESSION REEL (**CMD** + **4** / **CTRL** + **4**), scroll to the **SNAPSHOTS** table and click AUDITION SNAP 001 (or AUDITION SNAP 004).
   - Cave Recorder automatically backs up your live, unsaved console adjustments into **Live Mix (LIVE MIX)** and displays the AUDITION MODE: SNAP 001 control bar.
   - While playback is rolling, click A/B LIVE MIX repeatedly to toggle instantaneously between the historical Snapshot and your live working mix with zero audio dropouts.
   - Click RETURN TO LIVE MIX (or press **BACKSPACE**) to restore your live working mix, or click LOAD SNAPSHOT if you prefer the historical Snapshot and want to adopt it as the new working session state.
5. **Render Offline Mixdowns and Stem Bounces:**
   - When your mix is finished, select FLAC (24-bit lossless), MP3 (320 kbps CBR), or both on the MST MASTER BUS strip in MIX CONSOLE (**CMD** + **1** / **CTRL** + **1**) and click BOUNCE to render an offline stereo **Mixdown** (Mix001.flac / Mix001.mp3) linked directly into <ProjectTag>/_Mixes/, or trigger offline per-track **Stem Bounces** (Bounce001.wav) for archival delivery.
6. **Synchronize to Google Drive (UPLOAD SES MIXES / UPLOAD SES FULL):**
   - Click UPLOAD SES MIXES in SESSION REEL (**CMD** + **4** / **CTRL** + **4**) to push all stereo .flac and .mp3 Prints and Mixdowns to Google Drive for rapid band review.
   - Click UPLOAD SES FULL to synchronize the entire session folder (including 32-bit float rec/ stems, snapshots/, and session history) so remote bandmates who have linked the project can open the full multitrack session on their own studio workstations.

3.6 Workflow 6: Live-Room Gain Calibration, Bleed Gating & Rescuing a Magic Band Take

Studio Scenario: A stoner/psych-rock four-piece has just played an electrifying 9-minute live run-through in Take 003 with everyone in the same room. The groove and collective tempo push are unrepeatable, so re-recording the song is out of the question. However, the raw tracking session has five real-world rehearsal-room flaws:

1. Preamplifier gains were set unevenly during warm-up.
2. Loud snare rimshots are bleeding into the rack and floor tom microphones, while stacked fuzz pedals on CH13 GuitarL hum at -22 dBFS during stops even though the guitarist plays a whisper-quiet -34 dBFS clean fingerpicked bridge and holds a 4-second harmonic feedback bloom at the outro.
3. One monster snare rimshot in the climax clipped the hardware ADC converter for 16 samples.
4. The vintage bass amp picked up 60 Hz mains buzz.
5. The guitarist hit one wrong chord in bar 42 of Chorus 2.

3.6.1 Step-by-Step Live-Room Cleanup & Mistake Surgery Procedure

1. **One-Click Console Gain Calibration (CALIBRATE ALL TRIMS):**

- Open MIX CONSOLE (**CMD** / **CTRL** + **1**) and click **CALIBRATE ALL TRIMS** in the Gain Utility processor. Cave Recorder scans the active Pass stems of Take 003 and sets every channel's input trim so average playing level hits -18.0 dBFS RMS with a -6.0 dBFS peak ceiling clamp.

2. **Isolate Toms with Drum Bleed ([LRN] & Dynamic Decay LPF):**
   - Select CH05 TomRack, assign the **Noise Gate** in its processor rack, and switch the gate mode to **Drum Bleed**.
   - Solo a single rack tom hit on the timeline and click **[LRN]**. The resonant key filter (55 Hz to 350 Hz, default 110 Hz, Q = 4.0) locks onto the drum's 118 Hz fundamental pitch while the differential detector rejects snare and hi-hat spill. As each tom hit rings out, the dynamic decay low-pass filter sweeps from 20 kHz down to 250 Hz..800 Hz (default 380 Hz), removing harsh cymbal wash from the tom tail. Repeat on CH06 TomFloor.
3. **Tame Fuzz Roar While Preserving Clean Picking & Feedback (Amp Dynamic):**
   - Select CH13 GuitarL and switch its Noise Gate to **Amp Dynamic**.
   - During the -34 dBFS clean fingerpicked bridge, the autocorrelation pitch periodicity detector (H > 0.65) automatically engages **Whisper Clean Lock**, keeping every quiet note wide open without manual automation.
   - During the outro's 4-second harmonic feedback swell, **Feedback Bloom Lock** (H > 0.80, release 1.5 s..5.0 s, default 2.5 s) holds the gate open until the guitarist slams the final palm-mute stop, where **Palm-Mute Clamp** silences the 60 Hz fuzz roar in 15 ms..40 ms (default 18 ms..22 ms) along a click-free S-curve. Enable **HF Tame** to shelf down > 3.5 kHz fuzz fizz on fading chord tails.
4. **Level the Live Vocal Without Boosting Drum Spill (Vocal Rider + De-Esser):**
   - On CH17 LeadVocal, enable **Vocal Rider** (Target -18 dBFS, Range ±6 dB) with **Breath / Bleed Freeze** set to -46 dBFS so the fader rides quiet verses and loud choruses smoothly but freezes whenever the singer steps back from the mic between lines. Follow with **De-Esser** to catch 120 Hz P/B plosives and 6.5 kHz sibilance before the Opto compressor.
5. **Reconstruct the Clipped Snare Rimshot ([REPAIR CLIPPING]):**
   - Switch to WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**), select CH02 02SnrTp, bracket the clipped rimshot with K and L, and click **[REPAIR CLIPPING]**. Cave Recorder's spectral waveform interpolation reconstructs the missing 16-sample transient peak cleanly.
6. **Strip 60 Hz Mains Buzz (Hum Canceller) & Graft the Clean Bar 42 Chord ([PATCH FROM SOURCE...]):**
   - On CH12 BassAmp, enable the **60 Hz 12-Harmonic Comb Hum Canceller** (45 Hz..65 Hz fundamental, Q = 20..100, default Q = 60) to notch out 60 Hz .. 720 Hz transformer buzz without thinning out the bass cabinet low-mids.
   - On CH13 GuitarL, bracket the 1-bar wrong chord in Chorus 2 (K and L), click **[PATCH FROM SOURCE...]**, and point to the matching bar in Chorus 1. Cave Recorder micro-aligns the phase and transient correlation within ±50 ms and crossfades the clean chord in place while leaving the original Rec001.wav stem untouched on disk.
7. **Check Translation on the Control Room Monitor Bar & Bounce:**
   - Audition the mix through the **Control Room Monitor Bar** (SUB, MID + MONO, and AUTO-LVL bypass) while verifying that the Master Bus **Broadcast Metering Suite** shows -11.5 LUFS Short-Term, -0.5 dBTP True Peak, and a punchy 8.5 dB PSR crest factor. Click BOUNCE with FLAC and MP3 enabled to publish the rescued live performance to _Mixes/.

3.7 Workflow 7: Multi-Camera Rehearsal Video Capture, Cave Companion Mobile Camera Nodes & Split-Screen Video Mixdowns

Studio Scenario: Your band wants to capture a live rehearsal performance on video from three angles (a desktop USB/HDMI overhead drum camera, plus two bandmates' phones mounted on mic stands as wireless `1080p50` **Cave Companion** camera nodes), trigger recording remotely from across the room, and export both a 3-camera split-screen `.mp4` video with console-mixed audio and a Final Cut `.fcpxml` v1.9 multicam timeline before leaving the studio.

3.7.1 Step-by-Step Multi-Camera & Mobile Companion Procedure

1. **Pair Bandmates' Phones in the `PAIR MOBILE COMPANION` Modal:**
   - Open the **Pair Mobile Companion** modal on the desktop workstation. Cave Recorder displays a `caverec://pair` QR code (containing the workstation's local Wi-Fi studio address, a secure pairing token, and the optional Google Drive project folder ID) alongside a 6-digit PIN fallback and an app download QR code for **`caverec-v0.4.2-android-arm64.apk`** (Android) and **`caverec-v0.4.2-ios-arm64.ipa`** (iOS).
   - Each bandmate scans the pairing QR code inside the **Cave Companion** app. Assign **`Full Control + Camera`** to the drummer's phone (so the drummer can start and stop Takes from behind the kit) and **`Camera + Player`** to the guitarist's phone.
2. **Verify `50 FPS` CFR Lock & Calibrate Camera Latency (`3-Second` 4-Pulse Flash):**
   - Arm the desktop USB/HDMI drum camera (`Cam01-Drums`) at `1080p50` CFR (`960` audio samples/frame at `48.0 kHz`). If calibrating the camera for the first time, run the **3-Second Camera Latency Calibration** modal (`4-pulse 80 ms` screen flash + studio monitor click) so Cave Recorder stores its exact sub-frame optical latency.
   - On both phones, switch to the **Wireless `1080p50` CFR Mobile Camera Node** tab. Confirm the green **Clock Lock Badge** shows `< 0.5 ms` wireless clock lock and live 16-channel peak meters. For first-time phone optical calibration, run the phone's **Rear LED Torch 4-Pulse Calibration** so its hardware sensor offset is recorded.
3. **Trigger Multi-Track Audio + Multi-Camera Video Recording Remotely:**
   - From behind the drum kit, the drummer taps **Record (●)** on the Cave Companion Remote Transport screen.
   - Desktop Cave Recorder creates `Take 001` at `00:00:00.000`, illuminates the red **Recording Tally Border** across the desktop Video Monitor and both mobile camera screens, and records 18 channels of 32-bit float audio stems (`rec/*.wav`), the live Master Bus stereo Print (`mix/Print001.wav`), and sample-locked `50 FPS` CFR video streams (`video/*.mp4` + `.vididx` frame indices).
4. **Automatic Wireless Camera Ingest When Recording Stops:**
   - When the song ends, the drummer taps **Stop (▪)**.
   - Both phones immediately finalize their local `1080p50` `.mp4` and `.vididx` index files and transfer them automatically over local Wi-Fi (with integrity verification and automatic resume) into `<TakeFolder>/video/` on the desktop workstation.
   - As each camera transfer completes, Cave Recorder automatically ingests the new angle into `Take 001`'s **Video Timeline Strip** keyframe-snapped thumbnail filmstrip in `WAVEFORM TIMELINE` (**`CMD`** / **`CTRL`** + **`2`**). If any angle needs a fine visual tweak, drag its **Camera Trim Offset** slider while watching the real-time **Video Monitor**.

5. **Render a Split-Screen `.mp4` Video Mixdown & `.fcpxml` v1.9 Multicam Timeline:**
   - Open the **Export Video Mixdown & Multicam FCPXML** modal from SESSION REEL (**CMD** / **CTRL** + **4**).
   - Select the **3-Cam Band** layout (or **Side-by-Side Crop** with horizontal camera pan adjustments, **Side-by-Side 16:9**, or **Quad 2x2**) and click **RENDER VIDEO MIXDOWN (.MP4)**. Cave Recorder's split-screen video compositor and hardware video encoder render `<TakeFolder>/mix/GF_20261020-1_Take001_VideoMix001.mp4` (combining 50 FPS video with AAC-LC stereo console audio) alongside gapless 48.0 kHz .flac and .mp3 audio prints, linking them into `<ProjectTag>/_Mixes/`.
   - Click **EXPORT MULTICAM FCPXML (.FCPXML)** to write both a stacked split-screen timeline (`<adjust-crop>`, `<adjust-transform>`, and `<timeMap>` clock-drift tables) and a native `<mc-clip>` multi-camera switching timeline into `<TakeFolder>/export/`.
6. **Review Multi-Angle Video & Audio Prints on Mobile (Take Player & Google Drive Sync):**
   - Bandmates open the **Multi-Angle & Multi-Print Take Player** in the Cave Companion app (or pull from **Google Drive Cloud Sync**, which automatically skips already-uploaded files and resumes interrupted transfers) to switch live between all three camera angles (Cam01-Drums, Cam02-GtrMobile, Cam03-RoomMobile) and .flac/.mp3 audio prints with sample-locked audio-video playback.

PART II | CHAPTER 4

Mix Console, Signal Flow & Channel Strips

Dynamic console topology (up to 32 mono or stereo-linked input tracks, 16 Subgroup Buses, and 8 Aux Returns), automatic trim calibration, top-of-strip phase alignment, multi-instance reorderable 8-slot racks, Channel Inspector large processor views with interactive XY control pads, Control Room Monitor Bar, Broadcast LUFS/dBTP/PSR and Spectral Phase Metering, and precision fader operation.

The Cave Recorder **Mix Console** (`CMD` + `1` on macOS, `CTRL` + `1` on Windows and Linux) models an in-line large-format studio recording and mixing desk. Every input channel, subgroup bus, stereo auxiliary return, and master bus stage is visible on a single dark-chassis surface without hidden routing matrices or nested sub-menus.

This chapter details the end-to-end audio signal path from physical preamplifiers to disk stems and main monitors, the seven functional zones of an Input Track channel strip, the architecture of Subgroup Buses, Aux Returns, and the Master Bus, and the mouse and keyboard modifiers used for precision console mixing.

4.1 Console Architecture & End-to-End Signal Flow

Understanding where audio is tapped for multitrack disk capture versus monitor and bus mixing is fundamental to operating Cave Recorder with confidence during live band tracking.


CAVE RECORDER END-TO-END CONSOLE SIGNAL FLOW
PHYSICAL AUDIO INPUT (IN 01 .. IN 18)
[Input Trim (TRM: -24.0 to +24.0 dB / CALIBRATE ALL TRIMS -18 dBFS)]
[Polarity Invert (Ø: 180°)] -> [Sub-Sample Delay (Φ: 0..50,000 us)]
[Zero-Slot Phase Align Network (5-Stage Allpass Rotator, 10 ms X-Fade)]
(Armed [R] + Record)
(When Input Mon [I] is OFF)
RAW PRE-FADER STEM RECORDER
<Take>/rec/..._Rec001.wav
(Pre-inserts & pre-fader!)
DISK PASS LANE PLAYBACK READER
Active Pass: PASS 01 or OD 02..N
(Plus Range Comp & Spot-Heal)
<-- AUX TAP: [PRE] (Pre-Insert / Pre-Fader)
8-SLOT MULTI-INSTANCE REORDERABLE PROCESSING RACK (Slots 1..8)
* Input TAPE Saturation, Gain Utility & Transient Shaper
* 12-Band Dynamic EQ (2:12B EQ) <-- AUX TAP: [EQ] (Post-EQ)
* VCA/FET/Opto Comp, 3-Mode Gate (Expander/Drum Bleed/Amp Dyn)
* Amp Rig (HEAD ON/BYP + .nam) & Standalone Cab IR (.wav/.nam)
* Vocal Rider, De-Esser, MicroPitch, Delay, Ladder, Plate, Mod
* Movable Aux Send Tap Slot (4:AUX SND), CLAP & AU Plugins
[Channel Fader (-inf to +12.0 dB), Pan (PAN) & Mute/Solo (M/S)]
(OUT: SUB 1 / SUB 2 / MST)
(Aux Send Bus Summing)
--> AUX TAP: [PST] (Default)
STEREO SUBGROUPS (SUB 1..16)
DRUM BUS, INST BUS
* Trim (TRM), LINK & SEL
* 8-Slot SUB RACK & Sends
* Balance (BAL), M/S, Fader
STEREO AUXILIARY RETURNS
* AUX 1 REVERB (EMT 140 Plate)
LC, Pre-Delay (PD), Decay (DCY)
* AUX 2 TAPE DLY (3-Mode Delay)
WOW, DMP, Time (TM), FB, PP
(Summing into Master Bus)
STEREO MASTER BUS (MST MASTER BUS)
1. Master Tape (TAPE: DRIVE 0.0..+24.0 dB, WARM 0..100%)
2. Master Bus Glue Compressor (COMP: Shared Threshold THR)
3. Brickwall Limiter (LIM: THR -48..0 dB, CEIL -6.0..0.0 dB)
4. Master Fader (-inf..+12.0 dB), Mute (M) & Broadcast Metering
(When [2-TRK] Enabled)
[CONCURRENT LIVE STEREO PRINT WAV]
(Plus [FLAC]/[MP3] + [BOUNCE] -> _Mixes/)
(Strictly Post-Print & Post-Bounce!)
CONTROL ROOM MONITOR BAR (SPEAKER CHECK & LOUDNESS A/B)

4.1.1 Dual-Path Capture: Raw Pre-Insert Stems Plus Live Processed Prints

In traditional rehearsal spaces and project studios, engineers face a dilemma when tracking a full band:

1. **The Safe But Uninspiring Approach:** Record flat tracks with zero processing so nothing can be ruined by an overly aggressive compressor or EQ setting, leaving the band with dry, unmixed rough bounces at the end of the night.
2. **The Inspiring But Risky Approach:** Print heavy compression, EQ, and tape saturation directly onto the multitrack stems, risking permanent distortion or pumping if a guitarist boosts a pedal mid-song.

Cave Recorder resolves this dilemma through a split capture topology:

- **Raw Pre-Fader Multitrack Stems (<TakeFolder>/rec/):** Every armed Input Track (R) writes its 32-bit floating-point WAV file (Rec001 Base Pass or Rec002-OD Overdub Pass) immediately after Input Trim (TRM), Polarity (Ø), and Phase Alignment (Φ), **before** the 8-slot insert rack and channel fader. You can dial in aggressive 8:1 drum compression, heavy tape drive, high-pass filters, or external CLAP and macOS Audio Unit plugins on the console during rehearsal knowing the underlying multitrack stem on disk remains 100% clean and unclipped.
- **Concurrent Live Stereo Master Print (<TakeFolder>/mix/):** Simultaneously, while the transport is recording a Take, the Master Bus (2-TRK enabled) captures the complete post-fader, post-subgroup, post-reverb, and post-limiter stereo mix in real time to a Print<NNN> WAV file with automatic Plugin Delay Compensation (PDC) latency trimming. Selecting FLAC and/or MP3 and clicking BOUNCE on the Master Bus renders a 24-bit .flac mix file, a 320 kbps CBR .mp3 mix file, or both simultaneously and links them into the Project's _Mixes/ folder for immediate band sharing and cloud sync, while you retain every raw multitrack stem for later remixing or offline stem bouncing (covered in Chapter 7: Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync).

> **INPUT TRIM (TRM), POLARITY (Ø), AND PHASE (Φ) FEED THE STEM RECORDER**
> Because TRM, Ø, and Φ sit upstream of the multitrack stem tap, calibrating preamp gain mismatches and aligning multi-mic drum phase at the top of the channel strip ensures your recorded stems are already phase-coherent and properly gain-staged on disk. All downstream insert slots (TAPE, 12B EQ, VCA CMP, CLAP), Aux Sends, Faders, and Subgroup processors remain non-destructive during playback and remixing.

4.2 Input Track Channel Strips & Dynamic Console Topology (0..32 Tracks)

Cave Recorder supports a dynamic console topology of **0 to 32 mono or stereo-linked Input Track channel strips (CH01 through CH32)** alongside **0 to 16 stereo Subgroup Buses (SUB 1 through SUB 16)**, **0 to 8 stereo Auxiliary Return Buses (AUX 1 through AUX 8)**, and the **Master Bus (MST MASTER BUS)**:

- **Adding & Removing Strips (+ TRK, + SUB, + AUX & Strip X):** Click + TRK in the console toolbar to append a new Input Track (up to 32 tracks), + SUB to add a stereo Subgroup Bus (up to 16 subgroups), or + AUX to add a stereo Aux Return bus (up to 8 Aux returns). Removing a track, subgroup, or Aux return is fully reversible via UNDO (CMD / CTRL + Z), preserving its complete strip settings, routing, and Pass Lane state in the undo history.
- **Inline Scribble-Strip Renaming:** Click any Track, Subgroup, or Aux Return nameplate header (01KickIn, DRUM BUS, AUX 1 REVERB) to edit its name inline and press ENTER .
- **Mono & Stereo-Linked Input Channels (IN 01 vs. IN 01-02):** Each Input Track can be bound to a single mono hardware input (IN 01 through IN 32) or configured as a **Stereo-Linked Input Pair (IN 01-02, IN 03-04, ...)** for stereo synthesizers, stage pianos, drum overheads, or stereo room microphones. Stereo-linked tracks record 2-channel interleaved Broadcast Wave .wav stems and process left and right channels together through a single stereo channel strip.

Each Input Track strip is organized vertically from top to bottom in strict signal-flow order.



Figure 4.1: *Input Track Channel Strips showing input routing, trim, phase badges, insert slot summaries, aux sends, output bus routing, and 40-segment meters.*

4.2.1 Input Track Control Reference Table

Strip Zone	UI Control / Badge	Range / States	Default	Primary Studio Function
1. Header	CH01 .. CH18 Badge & Nameplate	Custom text label (e.g., 01KickIn)	CHNN	Identifies channel number and scribble-strip name. Clicking selects the strip for inspection.

Strip Zone	UI Control / Badge	Range / States	Default	Primary Studio Function
1. Header	Level & Delay Readout	-inf..+12.0dB, 0us..50000us	+0.0dB 0us	Displays current fader setting and active sub-sample phase delay offset in microseconds (us).
2. Input & Phase	Physical Input (IN 01 v)	IN 01..IN 18, NONE (PLAYBACK)	IN NN	Assigns physical hardware audio input channel or sets the strip to pure disk playback.
2. Input & Phase	Polarity Invert (Ø)	Off / Active (cyan, **SHIFT** + **P**)	Off	Flips signal polarity by 180° on the inspected strip to correct underside snare mics or inverted cables.
2. Input & Phase	Phase Align (Φ)	Off (dark) / Active (emerald)	Off	Engages sub-sample acoustic delay compensation (0.00 ms to +50.00 ms) calculated by ALIGN ALL.
2. Input & Phase	Input Trim (TRM)	-24.0 dB to +24.0 dB	+0.0 dB	Adjusts pre-insert digital input gain before disk stem capture and the 8-slot rack.
3. Rack & Pass	Rack Slot Summary	0 OF 8 SLOTS .. 8 OF 8 SLOTS	3 OF 8 SLOTS	Shows active processor count in the channel's 8-slot insert rack.
3. Rack & Pass	Active Pass Badge	PASS 01, OD 02, OD 03..	PASS 01	Displays which Pass Lane is currently selected for playback inside the active Take.
4. Aux Sends	Reverb Send (RVB)	-inf (-60.0 dB) to +12.0 dB	-12.0 dB	Controls send level from this channel to the stereo AUX 1 REVERB bus.
4. Aux Sends	Delay Send (DLY)	-inf (-60.0 dB) to +12.0 dB	-18.0 dB	Controls send level from this channel to the stereo AUX 2 TAPE DLY bus.
4. Aux Sends	Send Tap Selector	PRE / EQ / PST	PST	Selects pre-insert (PRE), post-EQ/pre-fader (EQ), or post-fader (PST) send pickoff point.

Strip Zone	UI Control / Badge	Range / States	Default	Primary Studio Function
5. Routing & Pan	Output Bus (OUT: ... v)	SUB 1..SUB 16, MST	OUT: MST v	Routes post-fader channel output to one or more Subgroup Buses and/or the Master Bus.
5. Routing & Pan	Stereo Pan (PAN)	L 100% .. C .. R 100%	C (Center)	Positions the channel across the stereo field using a constant-power panning taper.
6. Switch Bank	Mute (M)	Off / Muted (**SHIFT** + **M**)	Off	Silences post-fader channel audio and PST aux sends without stopping pre-fader recording.
6. Switch Bank	Solo (S)	Off / Soloed (**SHIFT** + **S**)	Off	Isolates soloed channels in the control room monitor path without affecting multitrack recording.
6. Switch Bank	Record Arm (R)	Unarmed / Armed (red, **SHIFT** + **A**)	Armed on active inputs	Arms the channel for disk stem capture and rolling 30-minute retroactive memory capture.
6. Switch Bank	Input Monitor (I)	Auto Playback / Live Input (**SHIFT** + **I**)	Off	Forces live physical input monitoring through the channel strip even during transport playback.
7. Fader & Meter	Channel Fader & 40-Seg Meter	-inf (-60.0 dB) to +12.0 dB	+0.0 dB	Controls post-insert mix level alongside a 40-segment Peak/RMS bargraph with peak hold.

4.2.2 Zone 1 & Zone 2: Track Header, Input Routing, Polarity, Phase & Trim

At the top of each Input Track strip (CH01 through CH18), the **Track Header** displays the channel badge (CH01 highlighted in emerald when selected) and the custom scribble-strip name (01KickIn, 02SnrTp, 03OvrhdL, 04GtrAmp). A crimson outline borders the entire selected channel strip to show at a glance which track is currently loaded into the four Upper Console Wells (on the Mix Console view, **CMD** / **CTRL** + **1**) and the full-screen CHANNEL INSPECTOR view (**CMD** / **CTRL** + **3**). Immediately beneath the track name, a compact dual telemetry line reports the current fader gain and the sub-sample phase delay offset in microseconds (such as -2.4dB 0us on CH01, +1.5dB 420us on CH02, and +0.0dB 1850us on CH03).



Figure 4.2: *Selecting a physical audio input and routing buses on the Mix Console.*

Clicking the **Physical Input Selector** (IN 01 v) opens the CH NN PHYSICAL INPUT popover menu, allowing you to bind the channel strip to any physical hardware input (IN 01 through IN 18) on your connected USB, Thunderbolt, or ADAT audio interface, or select NONE (PLAYBACK) for dedicated mixdown or virtual instrument return tracks.

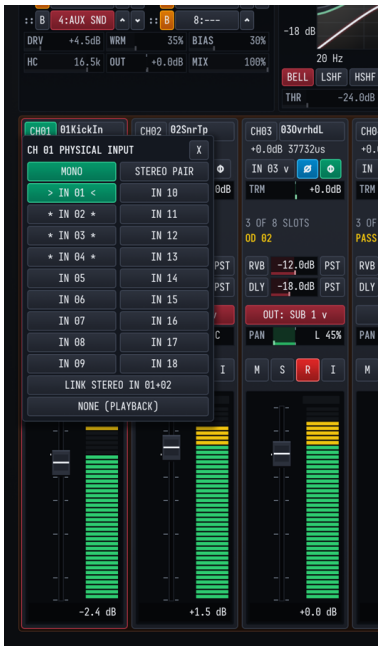


Figure 4.3: *The Input Routing selector dropdown menu.*

Directly to the right of the input dropdown sit two phase-conditioning switches paired with a zero-slot top-of-strip allpass phase rotator:

- **Ø (Polarity Invert, SHIFT + P on the inspected strip):** Reverses the positive and negative waveform excursions by 180°. It illuminates cyan when active (for example, on CH02 02SnrTp when phase-aligning a snare microphone against overheads or a bottom snare mic, or after running the Mode 1 Instant Binary Polarity Maximizer).

- **Φ (Sub-Sample Phase Alignment & Zero-Slot Phase Align Network):** Illuminates emerald when automatic or manual multi-mic phase alignment is active on the channel. In addition to applying a sub-sample time-of-flight delay between 0.00 ms and +50.00 ms (0 us to 50,000 us), every channel strip contains a built-in, zero-slot 5-stage **Phase Align Network** (one 1st-order 20-500 Hz low-frequency rotator plus four 2nd-order 20-12,000 Hz biquad allpass stages with a clickless 10 ms dual-branch preset crossfade) driven by the Mode 2 Deep Multicore Spectral Phase & Allpass Solver. Because the Phase Align Network lives at the top of the channel strip alongside Ø and Φ, all eight insert slots remain completely free for creative mixing. See Chapter 5: Built-In Processors, Phase Alignment & External Plugins (CLAP & AU) for the complete multi-mic alignment and Polarity Maximizer workflow.

Below the input and phase row, the horizontal **Input Trim (TRM)** bar adjusts pre-insert gain from -24.0 dB to +24.0 dB around a center 0.0 dB detent. Use TRM (or click **CALIBRATE ALL TRIMS** in the Gain Utility processor or console header to automatically calibrate all active channels to -18 dBFS RMS with a -6 dBFS peak ceiling in one step) so that raw stems hit disk and analog-modeled insert processors in their optimal dynamic sweet spot.

4.2.3 Zone 3 & Zone 4: Multi-Instance Insert Rack Summary, Active Pass Badge & Aux Sends

Zone 3 provides instant status feedback for both multi-instance insert processing and track-level Pass selection:

- **N OF 8 SLOTS (Multi-Instance Reorderable 8-Slot Rack):** Reports how many of the channel's eight reorderable processor slots are currently populated (for example, 3 OF 8 SLOTS when 12B EQ, VCA CMP, and AUX SND are active). Any built-in studio processor (12-Band Dynamic EQ, VCA/FET/Opto Compressor, Tape Saturation, Context-Aware Gate, Guitar/Bass Amp Rig, standalone Cab IR Speaker Cabinet & Linear IR Loader, De-Esser, Vocal Rider, Transient Shaper, Ladder Filter, Modulation, Micro-Pitch Doubler, Gain Utility, or Signal Repair) or external CLAP / macOS Audio Unit plugin can be inserted in **multiple slots simultaneously** on the same channel or Subgroup Bus. For example, you can place a surgical 12-Band Dynamic EQ in Slot 1 before an Amp Rig head (HEAD ON/BYP) in Slot 2, insert stereo Modulation in Slot 3, follow it with a standalone Cab IR speaker cabinet in Slot 4, and add a post-cabinet 12-Band Dynamic EQ and dual FET + Opto compressors in Slots 5 through 7, with independent per-slot parameter state, bypass, and real-time telemetry in the CHANNEL INSPECTOR.
- **Active Pass Badge (PASS 01 / OD 02):** Displays in gold which track-level **Pass** is currently active inside the loaded Session **Take**. PASS 01 indicates the original Rec001 Base Pass captured when the Take began at 00:00:00.000, while OD 02, OD 03, and higher indicate subsequent track-level Overdub Passes (Rec002-OD, Rec003-OD) recorded inside that same Take.

> **TAKES VS. PASSES ON THE CHANNEL STRIP**
> Never confuse a session-level **Take** (TAKE 001 in the Top Deck transport bar, representing an entire song performance starting at 00:00:00.000) with a track-level **Pass** (PASS 01 or OD 02 on the channel strip, representing an individual recording pass on that single track within the active Take). Expanding and comping **Pass Lanes** is covered in Chapter 6: Waveform Timeline, Pass Lanes, Markers & Ranges.

Zone 4 houses the **Banked Stereo Auxiliary Sends (AUX 1..8)**:

- **RVB / AUX 1 & DLY / AUX 2 (Plus Banked Sends AUX 3..8):** Feeds the console's stereo Aux Return buses (AUX 1 REVERB, AUX 2 TAPE DLY, and up to six additional stereo Aux Return buses AUX 3 through AUX 8), adjustable from -inf (-60.0 dB) to +12.0 dB by dragging horizontally across the crimson send bar. When a session uses more than two Aux Return buses, Zone 4 displays banked send selectors on the channel strip while the CHANNEL INSPECTOR (**CMD** / **CTRL** + **3**) exposes the full 8-bus Aux Send matrix simultaneously.
- **3-Way Send Tap Selector (PRE / EQ / PST):** Clicking the tap button to the right of any send cycles through three pickoff points in the channel signal path:
  - **PRE (Pre-Insert /Pre-Fader):** Taps the signal immediately after TRM, Ø, and Φ, unaffected by channel EQ, compression, fader moves, or channel Mute (M). Ideal for feeding a dedicated musician headphone cue mix.
  - **EQ (Post-EQ /Pre-Fader):** Taps the signal immediately after the 12B EQ slot (so low-cut filters and resonant notch cuts clean up the send) but before the VCA compressor and channel fader.
  - **PST (Post-Fader, Default):** Taps the signal after the channel insert rack, fader, and Mute switch so that riding the channel fader up or down naturally scales its wet reverb and delay contribution in proportion.

4.2.4 Zone 5, Zone 6 & Zone 7: Output Routing, Switch Bank, Fader & 40-Segment Meter

Zone 5 controls where the channel's post-fader stereo image is summed:

- **Output Bus Selector (OUT: SUB 1 v, OUT: SUB 2 v, OUT: MST v):** Assigns the channel to a stereo Subgroup Bus (illuminated in crimson when routed to a Subgroup such as OUT: SUB 1 v for DRUM BUS) or directly to the stereo Master Bus (OUT: MST v). In the expanded CHANNEL INSPECTOR view (**CMD** / **CTRL** + **3**), clicking the Output Bus selector opens the multi-bus routing popover, where a single track or subgroup can route to multiple Subgroup Buses and the Master Bus simultaneously for parallel bus processing.
- **Stereo Pan Bar (PAN):** Positions the signal between left (L 100%), center (C), and right (R 100%) using a studio -3.0 dB constant-power panning law so a signal panned hard left or right maintains the same perceived loudness as when centered.

Zone 6 provides the four primary console switches (M, S, R, I). All four switches can be clicked directly on any strip or toggled via **SHIFT** shortcuts on the currently inspected channel (leaving unmodified single keys free for global transport and marker actions):

1. **M (Mute, SHIFT + M on inspected strip):** Silences the post-fader output and PST aux sends. Unmodified M drops a cue marker on the timeline without muting audio, and muting a channel on the Mix Console **never** interrupts pre-fader multitrack stem recording on an armed track.
2. **S (Solo, SHIFT + S on inspected strip):** Engages Solo-in-Place on the monitor bus so you can inspect an individual microphone or subgroup in stereo context while keeping multitrack recording untouched.
3. **R (Record Arm, illuminates red, SHIFT + A on inspected track):** Arms the track for multitrack WAV capture when transport Record (R) is engaged, and includes the channel in the rolling 30-minute retroactive memory capture (RAM / RETRO, triggered via SHIFT + R). Even when software Input Monitoring (I) is off, any record-armed track (R) continuously feeds live physical input signal to its 40-segment LED meter and the Dual-Trace Pre/Post-EQ RTA so you can verify preamp headroom and spectral balance during zero-latency hardware direct-monitoring sessions without doubling the monitor signal in the headphones.
4. **I (Input Monitor, SHIFT + I on inspected track):** Switches the channel strip's audible monitor source between automatic disk playback of the active Pass Lane (PASS 01 / OD 02) and live audible monitoring of the physical hardware input (IN 01..IN 18) through the channel's insert rack and fader. Enable I when rehearsing parts through console effects or checking microphone placement while playing back an existing Take.

Zone 7 contains the **Logarithmic Studio Fader** and **40-Segment Peak/RMS Meter**:

- **Fader Cap, Scale & Direct Numeric Entry:** Provides smooth logarithmic resolution around the 0.0 dB unity gain mark, ranging from +12.0 dB down to -inf (-60.0 dB), with a dedicated numeric dB readout at the bottom of the meter well (-2.4 dB, +1.5 dB, +0.0 dB). Clicking the numeric dB readout below any fader opens an inline text entry box where you can type an exact decibel value (such as -3.5 or 0) and press ENTER.
- **40-Segment LED Bargraph:** Combines instantaneous digital sample peak detection (including a floating yellow peak-hold segment and red 0.0 dBFS overload latch) with continuous RMS loudness ballistics:
  - **Emerald Segments (-48 dB to -12 dB):** Nominal tracking and mixing operating zone.
  - **Amber Segments (-12 dB to -3 dB):** Healthy transient peak zone.
  - **Red Cap (-3 dB to 0.0 dBFS):** Digital full-scale peak warning. Click CLR CLIP on the Master Bus or in the Channel Inspector (or press X in the Channel Inspector) to clear latched clip indicators.

4.3 Subgroup Buses, Aux Returns & Master Bus

To the right of the Input Track strips sit the console's stereo summing and return strips: **Subgroup Buses** (DRUM BUS, INST BUS, expandable from 0 to 16 stereo subgroups via + SUB), **Stereo Aux Returns** (AUX 1 REVERB and AUX 2 TAPE DLY by default, expandable from 0 to 8 stereo Aux Return bays via + AUX), and the **Master Bus** (MST MASTER BUS).

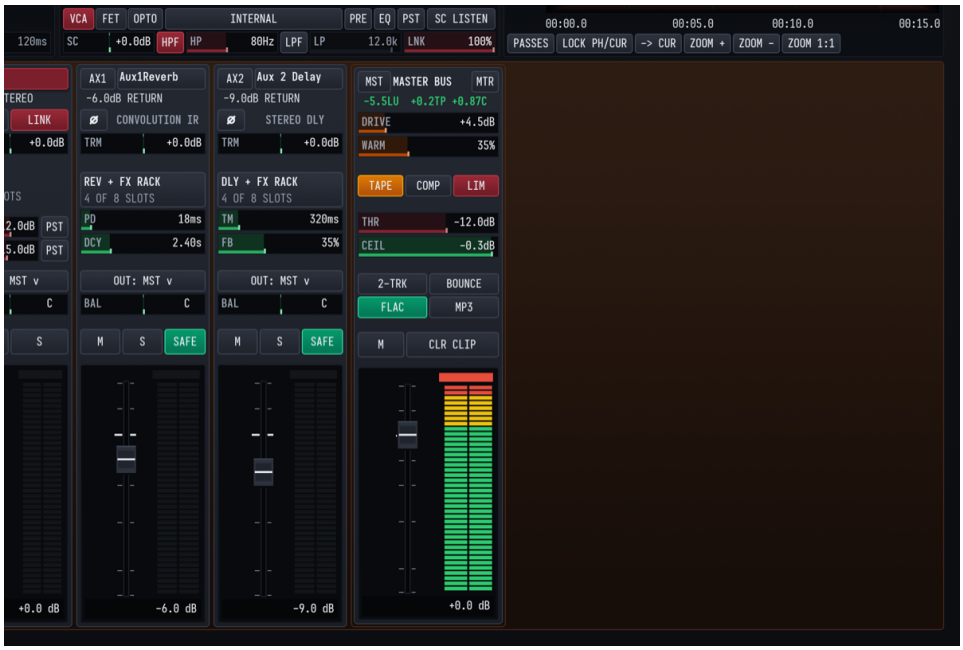


Figure 4.4: *Subgroup Buses, Stereo Aux Returns, and the Master Bus strip on the right side of the Mix Console.*

4.3.1 Stereo Subgroup Buses (DRUM BUS / SUB 1, INST BUS / SUB 2 .. SUB 16)

Subgroup Buses let you sum related Input Tracks (such as all drum microphones into DRUM BUS, or guitars and keys into INST BUS) so you can process the entire group through a shared 8-slot insert rack and control its overall level with a single stereo fader.

Each Subgroup Bus strip provides:

- **Crimson Bus Header & Stereo Readout:** Displays the custom bus name (DRUM BUS, INST BUS) and current stereo fader gain (+0.0dB STEREO).
- **SEL (Select Bus for Inspection):** Clicking SEL loads the Subgroup Bus into the four Upper Console Wells (**CMD** / **CTRL** + **1**) and the CHANNEL INSPECTOR view (**CMD** / **CTRL** + **3**). Every Cave Recorder Subgroup Bus owns a full **8-Slot Reorderable SUB RACK** with its own Tape Saturation stage, 12-Band Parametric EQ with Dual-Trace RTA, Analog VCA Bus Compressor, and third-party CLAP and macOS Audio Unit (AUv3 / AUv2) plugin slots.
- **LINK (Stereo Dynamics & Gain Link):** Illuminates crimson when enabled, locking left and right detector sidechains and gain stages across the stereo bus so off-center floor tom or guitar hits never skew the stereo center image.
- **Subgroup Trim (TRM, -24.0 dB to +24.0 dB):** Gain-stages the summed multi-channel signal before it enters the Subgroup's 8-slot SUB RACK. If summing eight hot drum tracks drives the subgroup bus compressor too hard, pull back TRM on DRUM BUS rather than disturbing your individual drum fader balance.
- **SUB RACK Status (3 OF 8 SLOTS):** Indicates how many processors are active in the subgroup's insert rack.
- **Subgroup Aux Sends (RVB & DLY):** Sends the entire post-rack subgroup mix to AUX 1 REVERB (default -12.0 dB) or AUX 2 TAPE DLY (default -15.0 dB, range -inf to +12.0 dB) to place a whole drum kit or horn section into the same acoustic room.

- **Output Bus Selector (OUT: MST v) & Stereo Balance (BAL):** Routes the subgroup to the MASTER BUS (or cascades into downstream higher-numbered subgroups via the multi-bus routing popover in the Channel Inspector) and adjusts left/right stereo balance (L 100%..C..R 100%).
- **Bus Mute (M), Solo (S), Fader (-inf to +12.0 dB) & 40-Segment Meter:** Controls final subgroup level and solo isolation. Soloing a Subgroup Bus automatically solos its contributing post-fader input channels in the monitor path.

4.3.2 Dedicated Stereo Aux Returns (AUX 1 REVERB & AUX 2 TAPE DLY)

Cave Recorder includes two hardwired, zero-latency stereo effect return strips so you never have to sacrifice Input Track slots or configure third-party plugins just to set up studio plate reverb and tape slapback during a session:

Return Strip	Header & Engine Readouts	Interactive Strip Drag Bars	Strip Switches & Routing
AUX 1 REVERB	EMT 140 Plate impulse response name, LC 90Hz input low-cut readout (20 Hz to 1000 Hz), and FFT CONV IR / ZERO-LATENCY engine status.	**PD (Pre-Delay):** 0 ms to 200 ms (e.g., 18ms). **DCY (Decay Time):** 0.20 s to 12.00 s (e.g., 2.40s), dynamically rebuilding and swapping the stereo plate impulse response in real time.	OUT: MST v routing, WET 100% readout, Mute (M), WET button (toggles 100% full wet vs. 50% blend), Return Fader (-6.0 dB), and 40-segment meter.
AUX 2 TAPE DLY	CAPSTAN TAPE model header, WOW 18% DMP 4.2k modulation and high-cut damping readout (500 Hz to 20.0 kHz), and TAPE ECHO / PP 60% width status.	**TM (Delay Time):** 1 ms to 2000 ms (e.g., 320ms). **FB (Feedback):** 0% to 95% strip drag bar (98% engine max, e.g., 35%).	OUT: MST v routing, FB 35% readout, Mute (M), PP Ping-Pong stereo bounce button, Return Fader (-9.0 dB), and 40-segment meter.

For complete acoustic modeling details and studio tuning recipes for AUX 1 REVERB and AUX 2 TAPE DLY, see Section 5.5 in Chapter 5.

4.3.3 The Stereo Master Bus (MST MASTER BUS)

The rightmost strip on the Mix Console is the **Master Bus (MST MASTER BUS)**, which sums all Subgroup Buses, direct Input Tracks, and Stereo Aux Returns into the final stereo program bus feeding your control room monitors and the concurrent **Live Stereo Print Recorder**.

```
+-------------------------------------------------------------------+
|                  MST MASTER BUS PROCESSING CHAIN                  |
+-------------------------------------------------------------------+
  Summed Stereo Bus (Subgroups + Direct Tracks + Aux Returns)
         |
         v
  [1] MASTER TAPE SATURATION ([TAPE] Toggle, Amber when Active)
      * DRIVE: 0.0 dB to +24.0 dB (default +4.5 dB)
      * WARM:  0% to 100% harmonic density (default 35%)
         |
         v
  [2] MASTER BUS GLUE COMPRESSOR ([COMP] Toggle)
      * Stereo VCA bus compression controlled by THR (-48.0..0.0 dB)
         |
         v
  [3] LOOKAHEAD BRICKWALL LIMITER ([LIM] Toggle, Crimson when Active)
      * THR (Threshold): -48.0 dB to 0.0 dB (default -12.0 dB)
      * CEIL (True-Peak Ceiling): -6.0 dB to 0.0 dB (default -0.3 dB)
         |
         v
  [4] MASTER FADER (-inf to +12.0 dB) & MASTER MUTE ([M])
         |
         +---> Main Hardware Monitor Outputs (Channels 1 & 2)
         +---> [2-TRK] Live Stereo Print WAV + [FLAC]/[MP3] + [BOUNCE]
```

Every control on the MST MASTER BUS strip is designed for immediate tactile operation during tracking and mixing:

- **DRIVE & WARM Drag Bars:** Adjust the input drive (0.0 dB to +24.0 dB, shown at +4.5dB) and analog transformer/oxide warmth (0% to 100%, shown at 35%) of the Master Tape stage.
- **TAPE, COMP & LIM Processor Toggles:** Independently engage or bypass the Master Tape stage (TAPE, amber), Master Bus Glue VCA Compressor (COMP), and Lookahead Brickwall Limiter (LIM, crimson).
- **THR (Threshold) & CEIL (Brickwall Ceiling) Drag Bars:** THR (-48.0 dB to 0.0 dB, default -12.0dB) sets the compression and limiting threshold, while CEIL (-6.0 dB to 0.0 dB, default -0.3dB) locks the absolute digital output ceiling so live rehearsal Print files and offline Mixdown renders never clip inter-sample converter reconstructions.
- **2-TRK (Live 2-Track Print Toggle):** When enabled, every transport Record pass automatically records the post-Master-Bus stereo mix in real time to <TakeFolder>/mix/<SessionID>_Take<NNN>_Print<PPP>_<Stamp>.wav with automatic Plugin Delay Compensation (PDC) latency trimming so the printed stereo master aligns sample-accurately with the recorded multitrack stems.

- **FLAC, MP3 & BOUNCE (Master Mixdown Format Toggles & Trigger):** The FLAC and MP3 buttons independently toggle 24-bit lossless .flac and 320 kbps CBR .mp3 export formats (either or both can be enabled at once). Clicking BOUNCE renders a faster-than-realtime master mixdown of the active Take (<SessionID>_Take<NNN>_Mix<MMM>_<Stamp>.<flac|mp3>, with a Xing/Info seek header on .mp3 files for instant scrubbing in the Google Drive web player) and links the output files into <ProjectTag>/_Mixes/ for immediate band listening and Google Drive synchronization.
- **CLR CLIP (Clear Clip Latches):** Resets latched red peak-overload indicators across the Master Bus and all channel meters.
- **Master Mute (M), Master Fader (-inf to +12.0 dB) & Stereo 40-Segment Meter:** Provides final control room level control and high-resolution peak/RMS metering.

4.3.4 Control Room Monitor Bar & AUTO-LVL Loudness-Matched Bypass

Below the Master Bus and inside the Master/Monitor section sits the **Control Room Monitor Bar**. Crucially, the Control Room Monitor Bar is wired **strictly downstream** of the Master Bus 2-TRK live stereo Print tap and the offline BOUNCE mixdown engine. You can engage any speaker-check filter, fold the monitors to mono, or dim the control room volume while a live Take is recording without ever affecting the Print001.wav file being written to disk.

Monitor Bar Control	Acoustic / DSP Specification	Studio Mixing Purpose
MONO (M⊕)	Unity-gain (L + R) × 0.5 sum (-6.02 dB per channel)	Verifies single-speaker translation (club PA systems, smart speakers, and phone playback) and immediately reveals any hollow comb filtering or out-of-phase stereo guitars.
DIFF (L-R)	(L - R) × 0.7071 (-3.01 dB) stereo side difference	Solos only the stereo side information (panned guitars, wide overheads, dimension chorus, and plate reverb tails) while canceling dead-center kick, snare, bass, and lead vocals.
SUB Band Solo	4th-order (24 dB/oct) Linkwitz-Riley (LR4) low-pass < 90 Hz	Isolates sub-bass fundamentals so you can hear whether the kick drum and bass guitar are fighting each other or rumbling out of phase below 90 Hz.
LOW Band Solo	LR4 band-pass 90 Hz - 350 Hz	Isolates the lower-midrange warmth and mud zone (snare body, rhythm guitar palm-mutes, bass upper harmonics, and room boom).
MID Band Solo	LR4 band-pass 350 Hz - 3.5 kHz	Isolates the core midrange where vocal intelligibility, snare crack, and guitar attack live, emulating a classic small single-cone studio cube speaker check.

Monitor Bar Control	Acoustic / DSP Specification	Studio Mixing Purpose
HIGH Band Solo	LR4 high-pass > 3.5 kHz	Isolates cymbal air, vocal sibilance, pick transient click, and high-gain amplifier fizz.
DIM (-20 dB)	Smooth -20.0 dB default control room attenuation (-40.0 dB to -6.0 dB configurable)	Drops studio monitor volume by 20 dB for low-level balance checking (where vocal-to-snare balance is easiest to judge) or talking with bandmates between songs.
AUTO-LVL Bypass	Input vs. output RMS tracker (300 ms integration window, gated below -60 dBFS, clamped to ±18.0 dB safety range)	Automatically matches the output loudness of a processor or entire 8-slot rack to its pre-insert input loudness when toggling bypass (B), defeating the human ear's "louder sounds better" psychoacoustic bias.

> **CHECKING MIX TRANSLATION WITH MID + MONO**
> Engaging **MID (350 Hz - 3.5 kHz)** and **MONO (M⊕)** together emulates a single-driver midrange studio cube speaker. If your lead vocal, snare drum, bass harmonics, and lead guitar are all clearly audible and balanced in MID + MONO, your mix will translate reliably across everything from car stereos and laptops to festival PA systems.

4.3.5 Broadcast & Phase Studio Metering Suite (Meter Probe & Console Metering Deck)

In addition to per-strip 40-segment Peak/RMS bargraphs, Cave Recorder provides a comprehensive **Broadcast Loudness, Crest-Factor & Spectral Phase Metering Suite** accessible on the Master Bus metering deck and insertable at any point in a channel or subgroup signal path via the **Meter Probe** rack processor:

1. **ITU-R BS.1770-4 K-Weighted Loudness (LUFS & LRA):**
   - Cascaded two-stage K-weighting filter (Stage 1: +4.0 dB high-shelf head-acoustics pre-filter at 1681.97 Hz, Stage 2: 2nd-order RLB high-pass filter at 38.13 Hz).
   - **M (Momentary Loudness, 400 ms sliding window):** Tracks fast phrase-level loudness swings.
   - **S (Short-Term Loudness, 3.0 s sliding window):** Primary reference for balancing chorus-to-verse intensity.
   - **I (Integrated Loudness, Dual-Gate -70 LUFS absolute + -10 LU relative):** Measures overall song loudness from start to finish with a one-click reset button.
   - **LRA (Loudness Range in LU):** Measures the macroscopic dynamic spread between quiet verses and loud climaxes (10th to 95th percentile of short-term loudness).
2. **4x Oversampled True-Peak Meter (dBTP):**
   - Uses a 4x oversampled FIR interpolator (48 taps, 12 taps per phase) to reconstruct the continuous analog waveform between digital samples and catch inter-sample clipping that standard 0.0 dBFS sample-peak meters miss. Keep Master Bus CEIL at -0.3 dBTP to -1.0 dBTP when encoding FLAC and 320 kbps MP3 files so consumer DACs and streaming transcoders never distort.

3. **Crest Factor & Micro-Dynamic Ballistics (PSR & PLR):**
   - Continuously computes **Peak-to-Short-Term Loudness Ratio (PSR = TruePeak - LUFS_S)** and **Peak-to-Integrated Loudness Ratio (PLR = TruePeak - LUFS_I)** to show at a glance whether your Master Bus Limiter is preserving drum punch or crushing the life out of the band:
   - **PSR > 10.0 LU (Emerald, OPEN / DYNAMIC):** Uncompressed acoustic, jazz, folk, and dynamic live-room recordings with full transient snap.
   - **PSR 6.0 to 10.0 LU (7 dB to 9 dB target, Gold, PUNCHY):** Competitive modern rock and heavy live-band loudness that still retains visceral kick and snare punch.
   - **PSR < 6.0 LU (Crimson, CRUSHED Warning):** Heavy brickwall limiting is squashing drum transients flat and inducing listener ear fatigue. Back off LIM THR or raise subgroup drum dynamics.
4. **Stereo Lissajous Goniometer & 32-Band Spectral Phase Scope (-1.0 to +1.0):**
   - Displays a real-time 45°-rotated Mid/Side Lissajous vectorscope alongside a vertical **32-Band Log-Spaced Spectral Phase Correlation Display** driven by 32 2nd-order state-variable bandpass filters (20 Hz to 20 kHz, Q = 2.2).
   - Each frequency band from 20 Hz to 20 kHz plots its normalized left-right cross-correlation $C(b) \in [-1.0, +1.0]$. Points near +1.0 (center spine) are tightly mono-compatible, +0.2 to +0.8 indicate healthy stereo width, and excursions toward -1.0 glow amber/red to pinpoint the exact frequency band where out-of-phase overheads, room mics, or stereo widening processors will cancel out when summed to mono.
5. **31-Band 1/3-Octave ANSI/ISO RTA with Visual Spectral Tilt (-4.5 dB/oct Studio & -3.0 dB/oct Pink):**
   - Offers selectable **Peak/RMS**, **Fast (125 ms) / Slow (1.0 s)** ballistics, and three visual reference tilt slopes: **-4.5 dB/oct (Modern Studio Reference)**, **-3.0 dB/oct (Pink Noise Reference)**, and **0.0 dB/oct (Flat)**. In -4.5 dB/oct Studio mode, a well-balanced full-band mix appears as a near-horizontal line across the analyzer, making booming 120 Hz room resonances or harsh 4 kHz cymbal build-ups effortless to spot visually.

4.4 Precision Mouse & Keyboard Modifiers for Faders, Knobs & 2D XY Pads

All faders, horizontal value bars (TRM, RVB, DLY, PAN, BAL, PD, DCY, TM, FB, DRIVE, WARM, THR, CEIL), tiered rotary knobs, interactive EQ/compressor graph nodes, and **2D XY Control Pads** across the Mix Console and Channel Inspector Large Processor Views share a unified set of studio mouse gestures and modifier keys.

Interaction / Gesture	macOS Modifier	Windows / Linux Modifier	Console Behavior
Standard Adjustment	Click + Drag (Vertical on faders/knobs, Horizontal on bars, 2D on XY pads)	Click + Drag	Smoothly sweeps the parameter (or paired X/Y parameters on a 2D XY pad) across its studio operating curve.

Interaction / Gesture	macOS Modifier	Windows / Linux Modifier	Console Behavior
10x Fine Precision Adjustment	**SHIFT** + Click + Drag	**SHIFT** + Click + Drag	Engages 10x fine-gear resolution (0.1 dB steps on faders, trim, and thresholds, 0.1 Hz on low-frequency EQ bands, 0.01 on Q, 1% on mix/pan bars, and 10x fine gearing on 2D XY control pads without cursor jumping).
Instant Reset to Unity / Default	Double-Click or **OPT** + Click	Double-Click or **ALT** + Click	Resets any fader or trim to 0.0 dB unity, any PAN or BAL bar to center (C), and any EQ node, compressor knob, or 2D XY pad puck to its neutral default.
Click-to-Type Numeric dB Entry	Click numeric dB readout below fader, type value, press **ENTER**	Click numeric dB readout below fader, type value, press **ENTER**	Opens an inline numeric entry field at the bottom of the fader well so you can type an exact level (e.g., -2.5 or 0.0) and press **ENTER** (or **ESC** to cancel).
Stepped Wheel Adjustment	Mouse Scroll Wheel	Mouse Scroll Wheel	Increments or decrements the hovered fader, bar, or knob in calibrated steps. Hovering over an EQ node and scrolling adjusts filter bandwidth (Q).
Live Rack Slot Reordering	Drag :: Handle or click ^ / v	Drag :: Handle or click ^ / v	Moves a processor slot up or down inside the 8-slot channel or subgroup rack in real time with clickless crossfading and zero audio dropouts.
Quick Slot Bypass	Click [B] next to any slot	Click [B] next to any slot	Bypasses that individual rack slot (illuminates amber when bypassed) for instant A/B comparison without unloading the processor.

HOLDING SHIFT MID-DRAG FOR SURGICAL SWEET SPOTS

You do not need to press **SHIFT** before clicking a control. Start dragging a fader, rotary knob, EQ node, compressor threshold, or 2D XY pad puck normally to get close to the target sound, then press and hold **SHIFT** without releasing the mouse button to drop directly into 10x fine-tuning mode with zero position jump.

PART II | CHAPTER 5

Built-In Processors, Phase Alignment & External Plugins (CLAP & AU)

Channel Inspector Large Processor Views, interactive XY control pads, and collapsible Mini-Timeline, 12-Band Dynamic EQ, VCA/FET/Opto Compressors, Drum Bleed and Stoner/Doom Amp Dynamic Gates, Guitar and Bass Amp Rig with 2x/4x oversampling, HEAD ON/BYP, and native .nam Neural Amp Capture (Standard, Mastering, and Turbo training modes), standalone Speaker Cabinet and Impulse Response Loader, Vocal Rider, De-Esser, Transient Shaper, Micro-Pitch Doubler, 3-Mode Delay, 4-Pole Ladder Filter, Studio Plate Reverb, 4-Mode Modulation, Polarity & Phase Maximizer, Audio Restoration, and CLAP and macOS Audio Unit plugins.

Every Input Track (CH01 through CH32) and every stereo Subgroup Bus (SUB 1 through SUB 16) in Cave Recorder includes a zero-slot top-of-strip **5-Stage Allpass Phase Align Network** and a dedicated, multi-instance reorderable **8-Slot Processing Rack** powered by Cave Recorder's **Curated Native Console & Rack Suite** (12-Band Dynamic EQ, VCA/FET/Opto Compressor, Tape Saturation, Guitar & Bass Amp Rig with HEAD ON/BYP and native .nam Neural Amp Capture, standalone Cab IR Speaker Cabinet & Linear Impulse Response Loader, Vocal De-Esser, Context-Aware Gate, Signal Repair, Vocal Rider, Transient Shaper, Ladder Filter, Studio Plate Reverb, 4-Mode Modulation, Micro-Pitch Doubler, Gain Utility, and Aux Send Tap) alongside cross-platform third-party **CLAP (.clap)** plugins (on macOS, Windows, and Linux), native **macOS Audio Unit (AUv3 and AUv2)** audio effects ('aufx' and 'aumf'), automatic multi-mic phase and polarity alignment, and up to eight dedicated stereo Auxiliary Returns (AUX 1 through AUX 8).

This chapter covers the dual-view inspection workflow (Upper Console Wells vs. full-screen CHANNEL INSPECTOR with Large Processor Views, 2D XY Control Pads, and Collapsible Mini-Timeline), **Tape Saturation (TAPE), Gain Utility & Differential Transient Shaper**, the **12-Band Dynamic Parametric EQ & Dual-Trace Spectrum Analyzer (EQ & RTA), Vocal Rider, Split-Band De-Esser & Micro-Pitch Doubler, Multi-Topology VCA / FET / Opto Compressors (with External Sidechain Key Routing), the Context-Aware 3-Variant Noise Gate (Expander, Drum Bleed, Amp Dynamic), Guitar/Bass Amp Rig ([CIR] Tube Models, HEAD ON/BYP & [NAM] Neural Amp Capture) & Standalone Cab IR Speaker Cabinet / Linear IR Loader, Stereo Convolution Reverb (AUX 1 REVERB), 3-Mode Delay (Capstan Tape, BBD Analog, Pristine Digital), 4-Pole Non-Linear Ladder Filter, Studio Plate Reverb & 4-Mode Modulation, Automatic Multi-Mic Phase Alignment (ALIGN ALL) & the Two-Mode Multicore Polarity & Phase Maximizer (Mode 1 & Mode 2), CLAP & macOS Audio Unit Plugin Hosting (1:CLAP, PLUGINS Manager, PLUGIN GUI, Ghost Slots, PDC & Low-Latency Monitoring)**, and **Non-Destructive Audio Restoration (ADC De-Clipper & 12-Harmonic Hum Canceller)**.

5.1 Upper Console Wells vs. Dedicated CHANNEL INSPECTOR View

Cave Recorder provides two complementary workspaces for viewing and shaping the processors on any Input Track or Subgroup Bus:

1. **The Four Upper Console Wells (MIX CONSOLE View, CMD / CTRL + 1):** Docked permanently along the upper deck of the Mix Console so you can adjust inserts, EQ curves, and compression while keeping all channel faders and bus meters in view.
2. **The Dedicated CHANNEL INSPECTOR View (CMD / CTRL + 3):** Expands the selected channel or subgroup across the entire screen with a high-resolution Dual-Trace Real-Time Spectrum Analyzer, dedicated rotary parameter knobs, multi-bus output routing popovers, and a scrolling Gain Reduction history trace.



Figure 5.1: *The four Upper Inspector Wells at the top of the Mix Console.*

5.1.1 The Four Upper Console Wells (MIX CONSOLE)

Clicking any Input Track header (CH01..CH18) or clicking the SEL button on any Subgroup Bus (DRUM BUS, INST BUS) immediately binds the four Upper Console Wells to that strip:

Upper Well	Header Label	Core Controls & Displays
Well 1 (Far Left)	<Track/Bus Name> (e.g., 01 Kick In or DRUM BUS)	Quick action buttons (TAPE, SWAP EQ/CMP, RESET), the **8-Slot Reorderable Rack** (Slots 1..8 with :: drag handles, B bypass switches, and ^ / v move arrows), and the 6-parameter **Tape Saturation** bar bank (DRV, WRM, BIAS, HC, OUT, MIX).
Well 2 (Center-Left)	EQ & RTA: <Track Name> (12-Band EQ)	Interactive 20 Hz to 20 kHz parametric response graph (-18 dB and +18 dB reference gridlines with full -24 dB to +24 dB band range), draggable band nodes, real-time spectrum trace, and filter shape pills (BELL, LSHF, HSHF, HPF, TILT).
Well 3 (Center-Right)	VCA COMPRESSOR & GR (T:-18.0dB 4.0:1)	Interactive transfer curve (-48 dB to 0 dB), threshold and knee handles, live red Gain Reduction (GR) shading, and quick-setting pills (SC HPF, 2:1, 4:1, 8:1, MIX 50%).

Upper Well	Header Label	Core Controls & Displays
Well 4 (Far Right)	WAVEFORM TIMELINE (48 smp/px)	Multi-resolution scrolling waveform monitor for the selected channel with Pass Lane and zoom triggers (PASSES, LOCK PH/CUR, -> CUR, ZOOM +, ZOOM -, ZOOM 1:1).

5.1.2 Full-Screen CHANNEL INSPECTOR View, Large Processor Decks, 2D XY Pads & Collapsible Mini-Timeline (CMD / CTRL + 3)

Pressing **CMD** + **3** (macOS) or **CTRL** + **3** (Windows/Linux) switches to the dedicated **CHANNEL INSPECTOR** view. Along the top selector bar, single-click pills (01 Kick In, 02 Snare Top, 03 Overhead L, 04 Guitar Amp, DRUM BUS, INST BUS, AUX 1, MST) let you step across every Input Track, Subgroup Bus, Aux Return, and Master Bus strip in the session without returning to the Mix Console.



Figure 5.2: *The full-screen Channel Inspector view for an Input Track.*

In the CHANNEL INSPECTOR view, the left column anchors a full-height channel strip (complete with active Pass selector * PASS 01 on Input Tracks or STEREO SUBGROUP / SUBGROUP BUS badges on Subgroups, Output Bus selector, M, S, REC, MON, Ø, Φ, TRIM, rotary PAN, full-scale calibrated meter with +12 dB to -48 dB markings, and CLR CLIP), while the right-hand instrument workspace is governed by the top **Signal-Flow Chain & Layout Mode Bar**:

- **Signal-Flow Chain Bar & Three Workspace Layout Modes (SPLIT, CHAIN LANE, EQ+CMP):**
  - **Interactive Signal-Flow Chain Pills (Slots 1..8):** Displays the active processors across the strip's 8 rack slots in exact top-to-bottom signal evaluation order. Clicking any slot pill in the chain bar immediately focuses that slot's dedicated **Large Processor View** and telemetry (even when multiple instances of the same processor type are loaded in different slots).

  - **SPLIT Mode (Default Side-by-Side Workbench):** Stacks EQ & DUAL-TRACE RTA and VCA COMPRESSOR & GR TRACE on the left (11:9 split) while displaying the focused slot's full-height **Large Processor Card** on the right.
  - **CHAIN LANE Mode (Horizontally Scrollable Device Chain):** Displays every active processor slot across the strip side-by-side in signal-flow order as 580 px-wide hardware module cards so you can tweak an entire multi-slot chain (Gate -> Amp Rig -> Modulation -> Cab IR -> EQ -> Compressor) simultaneously.
  - **EQ+CMP Mode:** Collapses the right-hand Large Processor Card so the 12-Band Parametric EQ and Compressor graphs span the full width of the inspector.
- **Full-Height Large Processor Views & Interactive 2D XY Control Pads:** Every built-in processor (STUDIO TAPE SATURATION, CONTEXT-AWARE EXPANDER GATE, VOCAL DE-ESSER & PLOSIVE, AUTOMATIC VOCAL RIDER, TRANSIENT SHAPER & FOCUS, GUITAR / BASS AMP RIG & NAM, SPEAKER CABINET & IR LOADER, LADDER FILTER, ANALOG MODULATION SUITE, STEREO MICRO-PITCH DOUBLER, HUM & DE-CLIP REPAIR, GAIN & M/S UTILITY), the stereo Aux Return engines (AUX RETURN STEREO REVERB, AUX RETURN STEREO DELAY), the Master Bus Brickwall Limiter (MASTER TRUE-PEAK LIMITER), and external plugin slots (EXTERNAL CLAP PLUGIN SLOT) feature a dedicated three-zone hardware faceplate:
  - **Zone A (Interactive 2D XY Control Pad & Real-Time Telemetry Bridge):** Combines a real-time transfer curve, envelope trace, or spectral response contour with a draggable **2D XY Control Pad** that sweeps two paired studio parameters simultaneously (such as DRIVE × WARMTH on Tape Saturation, ATTACK × SUSTAIN on Transient Shaper, CUTOFF × RESONANCE on Ladder Filter, MIC HPF × MIC LPF on Amp Rig and Cab IR, or THRESHOLD × CEILING on the Master True-Peak Limiter), flanked by a dedicated input/output/attenuation LED telemetry bridge. Hold **SHIFT** while dragging the XY puck for 10x fine precision without cursor jumps, or Double-Click (**OPT** / **ALT** + Click) to reset both axes to unity defaults.
  - **Zone B (Hero Rotary Control Bay):** Large-diameter **Hero** and **Standard** rotary knobs for the processor's primary sonic controls.
  - **Zone C (Topology Mode Bank, Sidechain & Output Trim Sub-Panel):** Direct mode selector buttons, sidechain filters, and output trim/wet-dry mix controls.
- **EQ & DUAL-TRACE RTA** & **VCA COMPRESSOR & GR TRACE:** Provide direct band-select buttons for all 12 EQ bands (B1 through B12), all seven filter shape buttons (BELL, LSHF, HSHF, HPF, LPF, NTCH, TILT), a continuous scrolling Gain Reduction (GR Trace) timeline, and dedicated rotary knobs (FREQ, GAIN, Q, THRESH, RATIO, ATTACK, RELEASE, MAKEUP, MIX).
- **Multi-Bus Output Routing Popover:** Clicking the green OUT: MST or crimson OUT: SUB 1 button in the left inspector strip opens the OUTPUT BUSES popover ([x] MASTER BUS, [ ] SUB 2: INST BUS), allowing any track or subgroup to feed multiple buses in parallel.
- **Collapsible Single-Track Mini-Timeline (HIDE TL / SHOW TL):** Below the processor decks, the inspector waveform strip provides quick Pass Lane actions for the active Take (PASSES or **E** to expand Pass Lanes, PASS 01 to select the active pass, AUD P1 to audition, MUTE P1 to mute, and DEL P1 to deactivate). Clicking **HIDE TL** in the top-right corner of the Chain Bar (or in the mini-timeline header) collapses the bottom waveform strip down to a compact 28 px summary bar (SHOW TL) so the upper EQ, Compressor, and Large Processor Views reclaim the full vertical height of your display.

- **Dedicated Inspector Keyboard Shortcuts:** While inspecting any strip, press **H** to toggle the compressor Sidechain High-Pass Filter (SC HPF), **X** to clear the channel peak overload latch (CLR CLIP), **E** to expand or collapse Pass Lanes (PASSES), or **SHIFT** + **M** / **S** / **A** / **I** / **P** to toggle Mute, Solo, Record Arm, Input Monitor, or Polarity (Ø).



Figure 5.3: The Channel Inspector view switched to inspect a Subgroup Bus.

5.1.3 Multi-Instance 8-Slot Rack, Reordering, SWAP EQ/CMP & RESET

In Well 1 (01 Kick In or DRUM BUS) and in the left rack column of CHANNEL INSPECTOR, the eight processing slots (1 through 8) execute sequentially from Slot 1 down to Slot 8:

- **Multi-Instance Slot Assignment (SLOT N PROCESSOR Overlay):** Clicking any slot pill (1 through 8) opens the floating **SLOT N PROCESSOR** selector menu. Any built-in zero-latency studio processor (12-Band Dynamic EQ, VCA Compressor, Tape Saturation, Vocal De-Esser, Context-Aware Gate, Signal Repair, Guitar/Bass Amp Rig, Speaker Cabinet & IR, 2P/4P Ladder LPF, Vocal Rider, Transient Shaper, Micro-Pitch Doubler, Gain & Polarity Utility, Studio Modulation, or Aux Send Tap) or external **CLAP / AU Plugin** can be assigned to **any number of slots on the same strip**. Each slot maintains its own independent parameter state, bypass switch (B), and real-time metering telemetry. Clicking a slot in the rack or in the Inspector Chain Bar focuses that specific slot instance for editing.
- **Dragging :: or Clicking ^ / v:** Reorders any processor (1:CLAP, 2:12B EQ, 3:VCA CMP, 4:AUX SND, 5:AMP RIG, 6:CAB IR) to an earlier or later slot in real time without audio clicks or dropouts. Moving 4:AUX SND above 3:VCA CMP, for example, taps the channel's Aux Sends before compression even when the send is set to rack-tap mode.
- **SWAP EQ/CMP Button:** Instant one-click macro in the top-left rack header that swaps the slot positions of the 12B EQ and VCA CMP processors. Use SWAP EQ/CMP to audition whether filtering low-end mud **before** the compressor detector (EQ -> CMP) or shaping tonal balance **after** dynamic peak control (CMP -> EQ) sounds more natural on a vocal, bass, or drum bus.

- **B (Slot Bypass) & RESET:** Clicking B next to any slot (or on the 12B EQ / VCA CMP badge) bypasses that single processor in place (illuminated amber). Clicking the crimson RESET button returns all built-in processors on the selected strip to their flat factory defaults.

5.2 Tape Saturation & Input Drive (TAPE)

Located in Well 1 beneath the 8-slot rack, the built-in **Tape Saturation (TAPE)** processor models the non-linear magnetic hysteresis, transformer core warmth, bias asymmetry, and head-bump high-frequency roll-off of a calibrated analog multitrack tape machine running at 15 or 30 IPS.

Click the TAPE pill in the Well 1 header to engage or bypass the tape stage on the selected Input Track or Subgroup Bus.

5.2.1 Tape Saturation Parameter Reference

Parameter	UI Label	Range	Default	Acoustic & Signal Effect
Input Drive	DRV	0.0 dB to +24.0 dB	+4.5 dB	Pushes signal level into the magnetic saturation transfer curve, smoothly soft-clipping fast transient spikes while increasing harmonic density.
Harmonic Warmth	WRM	0% to 100%	35%	Controls the low-mid transformer head-bump and low-order (2nd and 3rd) harmonic enrichment between 80 Hz and 350 Hz.
Tape Bias	BIAS	0% to 100%	30%	Adjusts virtual AC bias current. Lower values (10% to 25%) emphasize gritty odd-order harmonics, while higher values (40% to 75%) produce smooth, warm even-order compression.
High-Cut Roll-Off	HC	4.0 kHz to 20.0 kHz	16.5 kHz	Simulates analog tape high-frequency self-erasure and smooths out harsh cymbal sizzle, close-mic brass bite, or digital guitar amp fizz.
Output Trim	OUT	-12.0 dB to +12.0 dB	+0.0 dB	Compensates for loudness gained from DRV so you can A/B the TAPE button at matched perceived volume.
Parallel Wet/Dry Mix	MIX	0% to 100%	100%	Blends the dry uncompressed input signal with the saturated tape path directly inside the processor.

5.2.2 Studio Tape Recipes for Tracking & Mixing

Source Material	DRV	WRM	BIAS	HC	OUT	MIX	Sonic Goal
Snare Top (02SnrTp)	+6.5 dB	40%	35%	14.5 kHz	-2.5 dB	100%	Rounds off paper-thin stick transients by 3 dB to 4 dB on the peak meter while thickening the 200 Hz shell fundamental.
Bass DI Harmonic Weight	+9.0 dB	65%	25%	8.5 kHz	-4.0 dB	70%	Generates audible upper harmonics (200 Hz to 900 Hz) on a clean DI bass so the note pitch cuts through small speakers while keeping 30% clean sub-fundamental via MIX.
Drum Bus Glue (DRUM BUS)	+3.5 dB	35%	50%	17.5 kHz	-1.5 dB	100%	Cohesively binds kick, snare, and overheads together and absorbs micro-transients before the signal reaches the Subgroup VCA Compressor.
Harsh Guitar Cab Ribbon/Dynamic	+4.0 dB	50%	60%	11.0 kHz	-1.5 dB	100%	Softens 4 kHz to 8 kHz pick harshness and adds spongy analog midrange push.

5.2.3 Gain Utility: One-Click Trim Calibration (CALIBRATE ALL TRIMS), Elliptical Bass-Mono & Allpass Dispersion

In a self-recorded rehearsal or tracking session, microphone preamplifiers are often set hastily while musicians are warming up, leaving some channels peaking near -1 dBFS while others sit down at -32 dBFS. Feeding uncalibrated levels into analog-modeled compressors, tape saturators, and gates forces you to fight every threshold knob on the console.

The **Gain Utility** processor solves gain staging and stereo-field conditioning at the top of the rack:

Parameter / Action	Range / Specification	Default	Studio Application

Parameter / Action	Range / Specification	Default	Studio Application
CALIBRATE ALL TRIMS	One-click console-wide auto-trim (-18.0 dBFS RMS target, -6.0 dBFS peak ceiling clamp)	Action Button	Analyzes active Pass audio across the Take and sets every channel's input trim so average energy sits at the -18.0 dBFS analog console sweet spot without any transient exceeding -6.0 dBFS.
Coarse Gain & Fine Trim	Coarse Gain: -36.0 dB to +36.0 dB (5 ms smoothed ramp) Fine Trim: -3.0 dB to +3.0 dB (0.05 dB steps)	0.0 dB / 0.00 dB	Clean digital level staging prior to downstream dynamics or saturation, with 0.05 dB vernier precision for exact A/B gain matching.
Stereo Pan (Pan)	100L .. C .. 100R (equal-power -3.0 dB center law)	C (0.0)	Positions the pre-insert or post-insert signal across the stereo field inside the rack.
Independent L/R Polarity	ØL, ØR (180° per channel)	Off	Corrects out-of-phase stereo cables or Mid/Side matrix inversions independently on left and right channels.
Channel Routing Mode	Stereo, Mono Sum, Swap L/R, Left Only, Right Only, Side Diff Only	Stereo	Collapses a stereo input to mono, swaps reversed overhead channels, isolates one side of a dual-mic guitar cab, or solos the stereo difference (L - R) component.
Mid/Side Width	0% (Mono) to 200% (Super-Wide)	100%	Scales the Side ((L - R) * 0.5) signal relative to Mid ((L + R) * 0.5) to narrow an overly wide keyboard patch or expand a stereo bus.
5 Hz DC Offset Blocker	1st-order high-pass (R = 0.99935 @ 48 kHz)	Off	Removes asymmetrical DC voltage offsets from vintage preamps or fuzz pedals that would otherwise cause clicks on edits and rob compressor headroom.

Parameter / Action	Range / Specification	Default	Studio Application
Elliptical Bass-Mono (20..300 Hz)	1st-order high-pass filter on the Side (S) channel only (Off or 20 Hz to 300 Hz)	Off (120 Hz cutoff)	Folds stereo sub-bass energy below the cutoff frequency to dead-center mono (classic vinyl mastering "elliptical" filter) so stereo synth bass, chorus pedals, and room rumble drive both studio monitors in unison.

*(Note: Zero-slot 5-stage allpass phase dispersion and rotation is built directly into the top of every channel strip via the dedicated **Phase Align Network** alongside Ø and Φ, as detailed in Section 5.6.4.)*

5.2.4 Differential Envelope Transient Shaper

Unlike compressors that depend on an absolute decibel threshold (and therefore react differently to quiet ghost notes versus loud rimshots), the built-in **Transient Shaper** uses a **level-independent differential envelope detector**. It compares a fast envelope follower (E_fast, 1.0 ms attack / 25.0 ms release) against a slow envelope follower (E_slow, 25.0 ms attack / 150.0 ms release) in the logarithmic decibel domain:

```
Δ_dB(n) = 20 log10(E_fast(n) + ε) − 20 log10(E_slow(n) + ε)
```

Because the difference Δ_dB(n) depends only on the *rate of change* of the envelope and not its absolute volume, every ghost note and loud accent receives consistent transient shaping:

- **Attack (-12.0 dB to +12.0 dB):** Positive values amplify the initial 1 ms to 20 ms stick crack, kick beater click, or acoustic guitar pick attack. Negative values soften overly spiky transients or push close mics deeper into the room.
- **Sustain (-24.0 dB to +24.0 dB):** Positive values bring up the decaying drum shell ring, room ambience, or guitar chord bloom after the initial attack has passed. Negative values tighten boomy rehearsal rooms and shorten long tom or kick tails cleanly without hard gate chatter.
- **Detector Focus Band (20 Hz to 16,000 Hz):** Focuses the differential detector on a specific frequency zone (for example, 2.5 kHz to trigger only on snare wire snap while ignoring low-frequency kick bleed).
- **Soft Clip Output Guard:** Engages a zero-overshoot tanh saturator on the output stage so boosting Attack by +8 dB adds visceral drum impact without clipping downstream buses.

5.3 12-Band Parametric EQ & Dual-Trace Spectrum Analyzer (EQ & RTA)

Each Input Track and Subgroup Bus features a precision **12-Band Parametric Equalizer (2:12B EQ)** paired with a **Dual-Trace High-Resolution Real-Time Spectrum Analyzer (EQ & DUAL-TRACE RTA)**. Designed for zero-zipper-noise live adjustment, the equalizer handles everything from steep sub-bass rumble filtering and surgical snare-ring notch removal to broad, musical spectral tilt shaping.

5.3.1 12-Band EQ Parameter & Filter Shape Reference

Control / Filter Shape	UI Badge / Knob	Range / Behavior	Default	Studio Application
Band Selectors	B1 .. B12	12 independent bands	B1..B5 active	Selects which band (B1 through B12) is bound to the filter shape pills and right-hand rotary knobs (FREQ, GAIN, Q).
Bell (Peaking)	BELL	Proportional-Q peaking/dipping	Default on mid bands	Boosts or cuts a targeted frequency band around FREQ with bandwidth controlled by Q.
Low Shelf	LSHF	Shelving below corner FREQ	Band B2	Boosts or attenuates all frequencies below FREQ (e.g., adding 80 Hz kick weight or thinning muddy acoustic guitar low-mids).
High Shelf	HSHF	Shelving above corner FREQ	Band B5	Adds 10 kHz vocal and overhead air or shelves down overly bright cymbal bleed.
High-Pass Filter (Low-Cut)	HPF	Steep low-cut below FREQ	Band B1	Removes stage floor rumble, air-conditioning hum, and plosive energy from vocals, guitars, and overheads.
Low-Pass Filter (High-Cut)	LPF	Steep high-cut above FREQ	Optional	Rolls off ultrasonic hiss, guitar amp buzz above 7 kHz, or hi-hat bleed in kick and tom microphones.
Surgical Notch	NTCH	Deep narrow band-reject	Optional	Eliminates annoying metallic snare drum overtones, acoustic guitar body wolf notes, or 50 Hz/60 Hz mains hum without dulling adjacent frequencies.

Control / Filter Shape	UI Badge / Knob	Range / Behavior	Default	Studio Application
Spectral Tilt	TILT	Linear dB/octave fulcrum tilt	Optional	Pivots the entire frequency spectrum around FREQ (default 1 kHz). Positive gain brightens highs while simultaneously attenuating lows, and negative gain warms lows while softening highs.
Center / Corner Frequency	FREQ	20.0 Hz to 20,000 Hz	Band-dependent	Sets the target frequency for the selected band (B1..B12).
Band Gain	GAIN	-24.0 dB to +24.0 dB	+0.0 dB	Sets boost or attenuation depth for BELL, LSHF, HSHF, and TILT shapes (plotted against -18 dB, 0 dB, and +18 dB horizontal reference lines).
Bandwidth / Resonance	Q	0.10 (wide) to 18.00 (narrow)	1.00	Controls filter sharpness on BELL/NTCH bands or corner resonance/slope steepness on HPF, LPF, LSHF, and HSHF.

5.3.2 Operating Interactive EQ Nodes & Reading the Dual-Trace RTA

You can shape EQ either via the right-hand rotary knobs (FREQ, GAIN, Q) in the CHANNEL INSPECTOR view (**CMD** / **CTRL** + **3**) or by manipulating the colored band nodes directly inside the EQ & RTA graph on both the Mix Console and Channel Inspector:

- **Click + Drag Node Horizontally / Vertically:** Sweeps FREQ (20 Hz to 20 kHz) and GAIN (-24.0 dB to +24.0 dB) simultaneously.
- **Mouse Scroll Wheel (or SHIFT + Vertical Drag) Over a Node:** Narrows or widens the filter Q (0.10 to 18.00).
- **SHIFT + Drag Node:** Engages 10x fine precision (0.1 dB gain steps and sub-Hz frequency resolution) for locking onto exact room resonances.
- **Double-Click Node (or OPT / ALT + Click):** Resets the band's GAIN to 0.0 dB.

Behind the white composite EQ curve, the **Dual-Trace High-Resolution Real-Time Spectrum Analyzer (DUAL-TRACE RTA)** renders two live acoustic traces simultaneously:

1. **Pre-EQ Input Spectrum (Darker Shaded Trace):** Shows the raw frequency energy entering the 12B EQ slot.
2. **Post-EQ Output Spectrum (Bright Emerald Trace):** Shows the resulting frequency energy leaving the 12B EQ slot after all active bands are applied.

Because the high-resolution analyzer resolves narrow intervals down into the lowest bass octaves (20 Hz to 250 Hz), you can visually distinguish between the 55 Hz fundamental of a kick drum and the 82 Hz low E string of a bass guitar, place a narrow BELL or NTCH node directly on a ringing 480 Hz snare overtone spike, and watch the emerald post-EQ trace flatten out in real time. On any record-armed track (R), the Dual-Trace RTA remains active even when software Input Monitoring (I) is turned off, allowing you to inspect live microphone spectrum and dial in high-pass or notch filters during zero-latency hardware direct-monitoring sessions.

5.3.3 Per-Band Dynamic EQ (Threshold-Driven Resonance & Mud Control)

In a live rehearsal or tracking room, a problematic frequency is rarely constant. A rhythm guitar cabinet might sound balanced during open chords but boom violently at 165 Hz only on chugged palm-mutes. A lead vocalist might sound warm in the verse but turn piercing at 3.2 kHz only when belting the chorus. Cutting 165 Hz or 3.2 kHz by -6 dB with a static EQ fixes the loud peaks at the cost of leaving the instrument thin and lifeless for the rest of the song.

To solve this, any of the 12 parametric bands (B1 through B12) in 12B EQ can be switched into **Dynamic EQ Mode**:

Dynamic Band Parameter	Range	Default	Acoustic Behavior
DYN (Dynamic Enable)	Off (Static) / Active	Off	Activates envelope-driven gain modulation on the selected EQ band (BELL, LSHF, or HSHF).
DYN THRESH (Threshold)	-60.0 dB to 0.0 dB	-24.0 dB	Sets the sidechain bandpass level at which the band begins moving from its static GAIN toward DYN RANGE.
DYN RANGE (Max Excursion)	-18.0 dB to +18.0 dB	-6.0 dB	Sets the maximum dynamic cut (negative values) or dynamic upward expansion boost (positive values) applied when the band's frequency energy exceeds DYN THRESH.
DYN ATTACK	0.5 ms to 100.0 ms	8.0 ms	Controls how fast the filter node dips or boosts once the narrow-band sidechain detector crosses threshold.
DYN RELEASE	5.0 ms to 1000.0 ms	80.0 ms	Controls how smoothly the EQ band returns to its resting gain after the resonance or boom subsides.

Each active dynamic band runs its own dedicated state-variable bandpass sidechain detector tuned to that band's exact FREQ and Q, updating the audio filter coefficients smoothly every 16 samples. On the EQ & RTA graph, a shaded dynamic excursion halo around the node shows the real-time decibel deflection on every beat.

5.3.4 Vocal Rider: Automatic Zero-Latency Vocal Leveler with Breath & Bleed Freeze Gate

When a singer performs live in the same room as a loud band, their distance from the microphone capsule naturally shifts between intimate verses and full-voice choruses. Feeding those 12 dB level swings directly into a compressor forces the compressor into heavy -12 dB gain reduction on loud notes while leaving quiet lines buried. Worse, if you try to use a standard automatic gain rider in a live rehearsal room, the rider mistakes the pause between vocal lines for a "quiet vocal" and cranks the fader up by +6 dB, flooding the mix with loud snare and cymbal bleed.

Cave Recorder's **Vocal Rider** is engineered specifically for live-room vocals, bass, and solo instruments:

- **Target Level (-36.0 dBFS to -6.0 dBFS, default -18.0 dBFS) & Range (±6.0 dB):** Smoothly rides the pre-compressor channel level by up to +6.0 dB on quiet phrases and -6.0 dB on loud phrases with zero lookahead latency, delivering a consistent -18 dBFS vocal into your downstream compressor.
- **Speech-Weighted Sidechain Detector (250 Hz to 4.0 kHz):** Filters out low-frequency kick/bass thump below 250 Hz and hi-hat sizzle above 4.0 kHz so the gain rider reacts strictly to fundamental vocal body and presence.
- **Breath & Bleed Freeze Gate (Freeze Threshold, -70.0 dBFS to -24.0 dBFS, default -48.0 dBFS):** Whenever the speech-weighted vocal energy drops below Freeze Threshold (such as when the singer steps back or takes a breath between phrases), the rider immediately **freezes its gain fader** (or glides gently toward 0.0 dB unity) so live drum and amplifier room spill is never boosted into the mix.

5.3.5 Split-Band De-Esser & Dynamic Low-Shelf Plosive Arrestor

Close vocal microphones in rehearsal and live tracking rooms suffer from two distinct transient faults at opposite ends of the spectrum: low-frequency P and B breath blasts (plosives) that thump the subwoofers, and piercing high-frequency S, Sh, and T sibilance (4.5 kHz to 9.0 kHz) that turns harsh when pushed through vocal saturation or plate reverb.

The built-in **De-Esser** solves both problems in a single zero-latency processor:

1. **Stage 1: Dynamic Low-Shelf Plosive Arrestor (120 Hz):** An 80 Hz low-pass transient detector monitors sub-bass air blasts. Normal vocal chest resonance (110 Hz to 250 Hz) passes through completely untouched, but the instant a P or B microphone pop hits below 80 Hz, a fast (0.5 ms attack, 60 ms release) dynamic low-shelf filter at 120 Hz attenuates the low end by up to -18.0 dB, removing the thump without thinning out the singer's voice.
2. **Stage 2: Split-Band Sibilance Suppressor (2.0 kHz to 16.0 kHz, default 6.5 kHz, up to -24.0 dB):** Unlike wideband de-essers that duck the entire vocal whenever an S occurs (causing an unnatural "lisp"), the split-band detector isolates only the sibilant frequency band (Sibilance Freq adjustable from 2.0 kHz to 16.0 kHz with default 6.5 kHz, Max Reduction 0.0 dB to -24.0 dB with default -8.0 dB) and compresses only the high-frequency shelf while leaving the 200 Hz to 3.0 kHz vocal vowel core at 100% volume. Engaging SC LISTEN solos the isolated sibilance detector so you can sweep Sibilance Freq to the exact whistle pitch of the singer's S consonants.

5.3.6 Haas-Free Stereo Micro-Pitch Doubler

A classic studio trick for widening a single mono rhythm guitar, synthesizer lead, or backing vocal is the "Haas delay" (delaying the right channel by 15 ms relative to the left). Unfortunately, summing a static Haas delay to mono (MONO M⊕) creates severe comb-filtering notches that make the instrument sound hollow and phasey on single-speaker PA systems or phones.

The **Micro-Pitch Doubler** generates lush stereo width that remains 100% solid when folded to mono:

- **Opposite-Polarity Dual Pitch Shifters (Detune: 1.0 to 30.0 cents, default ±9.0 cents):** Pitch-shifts the Left wet voice slightly downward (-Detune cents) and the Right wet voice slightly upward (+Detune cents) using dual overlapping Hann-windowed grain buffers driven by a 0.75 Hz quadrature LFO (90° stereo phase offset).
- **Decorrelated Pre-Delay (5.0 ms to 40.0 ms, default 14.0 ms L / 19.0 ms R) & High-Cut Warmth (1.5 kHz to 16.0 kHz):** Because the delayed voices are continuously pitch-decorrelated and rolled off in the extreme highs (High Cut), they fuse with the dry center channel to create a wide stereo halo without producing static comb-filter notches in MONO.

5.4 Analog VCA Compressor & Gain Reduction Meter (VCA COMPRESSOR & GR)

The **Analog VCA Compressor (3:VCA CMP)** models a classic studio voltage-controlled amplifier dynamics processor (inspired by large-format British console bus compressors and fast American VCA limiters). It combines punchy transient control with a built-in **Sidechain High-Pass Filter (SC HPF, shortcut H in the Channel Inspector)** and **New York Parallel Mix (MIX)** stage.

5.4.1 VCA Compressor Parameter Reference

Parameter / Switch	UI Label	Range / Presets	Default	Studio Function
Threshold	THRESH	-60.0 dB to 0.0 dB	-18.0 dB	Sets the input level (dBFS) above which gain reduction begins. Drag the yellow threshold node directly on the transfer graph or turn the THRESH knob.

Parameter / Switch	UI Label	Range / Presets	Default	Studio Function
Ratio	RATIO (2:1, 4:1, 8:1, 20:1)	1.0:1 to 20.0:1	4.0:1	Determines how strongly signal excursions above THRESH are attenuated. Use the quick-ratio buttons (2:1 gentle leveling, 4:1 studio punch, 8:1 aggressive control, 20:1 peak limiting) or sweep continuously with the RATIO knob.
Attack Time	ATTACK	0.1 ms to 100.0 ms	10.0 ms	Controls how quickly the VCA gain cell clamps down once a transient crosses THRESH. Slow attacks (15 ms to 35 ms) let drum stick snap pass through uncompressed, while fast attacks (0.5 ms to 3 ms) tame sharp peaks.
Release Time	RELEASE	10.0 ms to 1000.0 ms	100.0 ms	Controls how fast gain recovers to 0.0 dB after the signal drops below THRESH. Set release so gain reduction recovers smoothly in time with the song's tempo without audible pumping.
Makeup Gain	MAKEUP	0.0 dB to +24.0 dB	+0.0 dB	Restores output level to match the uncompressed peak or perceived loudness.
Parallel Wet/ Dry Mix	MIX (MIX 100%, MIX 50%)	0% to 100%	100%	Blends the uncompressed dry signal with the compressed VCA output. Click MIX 50% for instant 50/50 New York-style parallel compression.

Parameter / Switch	UI Label	Range / Presets	Default	Studio Function
Sidechain High-Pass	SC HPF (`H` in Inspector)	Off / Active (crimson, 90 Hz HPF)	Active on drums/bus	Inserts a 90 Hz high-pass filter into the compressor's detector circuit so heavy kick drum sub-bass energy does not trigger excessive gain reduction across the rest of the frequency spectrum.
Stereo Sidechain Link	LINK	Off / Linked (crimson)	Linked	Links left and right level detectors on stereo channels and Subgroup Buses so both channels compress identically and preserve stereo imaging.

5.4.2 Interactive Transfer Curve & Real-Time Gain Reduction (GR) Trace

The VCA COMPRESSOR & GR graph displays input level (-48 dB to 0 dB) on the horizontal axis against output level (-24 dB to 0 dB) on the vertical axis:

- **Interactive Threshold & Knee Nodes:** Drag the highlighted gold node along the diagonal line to set THRESH, and drag the upper slope handle (or click 2:1, 4:1, 8:1, 20:1) to tilt the compression ratio above threshold.
- **Scrolling Gain Reduction (GR Trace):** As audio passes through the compressor, a shaded crimson trace sweeps across the graph showing instantaneous and historical gain reduction depth in decibels. Watching the GR Trace alongside the waveform makes it easy to dial ATTACK and RELEASE so the compressor breathes cleanly between snare hits or vocal phrases.

5.4.3 Studio Starting Settings for Common Sources

Source / Bus	RATIO	ATTACK	RELEASE	SC HPF	MIX	Target GR	Why It Works
Kick In & Snare Top	4:1	18.0 ms	80.0 ms	Off	100%	-3 to -6 dB	The 18 ms attack lets the initial beater click and stick crack pass through at full impact before clamping the drum decay, adding punch and tightening sustain.

Source / Bus	RATIO	ATTACK	RELEASE	SC HPF	MIX	Target GR	Why It Works
Bass Guitar (DI or Amp)	4:1 or 8:1	12.0 ms	140.0 ms	**On** (SC HPF)	80%	-4 to -7 dB	Engaging SC HPF prevents the low E (41 Hz) and B (31 Hz) strings from over-triggering the VCA cell, keeping low and high notes at equal volume across the fretboard.
Lead & Backing Vocals	2:1 or 4:1	6.0 ms	120.0 ms	**On** (SC HPF)	100%	-3 to -5 dB	Catches loud chorus phrases smoothly while keeping intimate verses intelligible above the band without requiring constant fader riding.
Drum Bus (DRUM BUS) Parallel Crush	8:1 or 20:1	10.0 ms	75.0 ms	**On** (SC HPF)	50% (MIX 50%)	-8 to -12 dB	Heavy compression brings up room ambience and snare sustain, while MIX 50% retains 100% of the dry, natural drum transients in parallel.

5.4.4 Multi-Topology Compressor Modes (VCA, FET, Opto) & Sidechain Spectral Tilt

Instead of cluttering your plugin menu with separate vintage compressor emulations, Cave Recorder's compressor provides a 3-way hardware topology selector (VCA | FET | Opto) alongside a continuous **Sidechain Spectral Tilt** filter (-6.0 dB/oct to +6.0 dB/oct around 1.0 kHz):

Topology Mode	Detector & Transfer Characteristic	Harmonic & Ballistic Signature	Ideal Studio Sources
VCA (Voltage-Controlled Amplifier)	Feed-forward true RMS detector with precision logarithmic decibel computer and selectable hard/soft knee.	Ultra-clean, predictable transient control and cohesive stereo bus glue without coloring the midrange.	DRUM BUS, INST BUS, MST MASTER BUS, tight kick and snare shaping, and percussive acoustic guitars.
FET (Field-Effect Transistor)	Ultra-fast peak-sensing discrete transistor circuit (Attack down to 0.02 ms, Release 5 ms to 500 ms) with high-ratio (4:1, 8:1, 12:1, 20:1) transfer curve.	Aggressive transient grab with subtle odd-order (3rd harmonic) discrete FET saturation that pushes tracks forward in a dense mix.	Snare crack enhancement, explosive crushed room microphones, aggressive rock/punk vocals, and picked bass guitar.

Topology Mode	Detector & Transfer Characteristic	Harmonic & Ballistic Signature	Ideal Studio Sources
Opto (Optical Electroluminescent Cell)	Program-dependent photocell attenuator (10 ms soft onset) with dual-stage release memory (fast 60 ms initial recovery for the first 50% of gain reduction, transitioning to a slow 400 ms to 1.5 s decay for sustained compression) and fixed ~3:1 gentle leveling slope.	Zero pumping, velvety even-order (2nd harmonic) warmth, and natural phrase-to-phrase leveling.	Lead vocals, fretless or fingerstyle bass, pedal steel, horns, strings, and organ.

- **Sidechain Spectral Tilt (SC TILT, -6.0 dB/oct to +6.0 dB/oct):** Tilts the frequency sensitivity of the compressor detector around 1.0 kHz. Positive tilt (+3.0 to +6.0 dB/oct) makes the compressor react more strongly to bright 2 kHz - 6 kHz vocal belt notes and guitar pick transients while ignoring low-end body, whereas negative tilt makes the detector clamp down on low-mid weight.
- **External Sidechain Key Routing (SC SRC / KEY INPUT) & Sidechain Low-Pass Filter (SC LPF):** Any built-in dynamics or filter slot (VCA/FET/Opto Compressor, Context-Aware Gate, Ladder Filter) or third-party plugin slot can receive its detector key signal from **INTERNAL (Self)** or from any external **Input Track**, **Subgroup Bus**, or **Aux Return** strip in the session. Each external sidechain route provides a selectable tap point (PRE-INS, POST-EQ, PRE-FDR, or POST-FDR), a dedicated -24.0 dB to +24.0 dB sidechain input gain trim, and an optional compressor **Sidechain Low-Pass Filter (SC LPF, 200 Hz to 20.0 kHz, default 12.0 kHz)** so you can duck a bass DI or fuzz guitar cleanly from the fundamental thump of the kick drum (80 Hz to 250 Hz) without false-triggering on high-frequency beater click. Cave Recorder automatically validates the routing graph to prevent feedback cycles and applies slot-level Plugin Delay Compensation across all sidechain taps.

5.4.5 Context-Aware 3-Variant Noise Gate (Expander, Drum Bleed, Amp Dynamic)

Standard threshold noise gates are notorious for ruining live band recordings: on drums they false-trigger on loud snare hits and let bursts of harsh cymbal wash through on every tom decay, while on high-gain guitar amplifiers they either chop off quiet fingerpicked notes and intentional feedback sustain or sit wide open while fuzz pedals hum at -22 dBFS.

Cave Recorder replaces conventional gating with a **Context-Aware 3-Variant Noise Gate** tailored to the physics of the source instrument:

```
+=====================================================================+
|          CONTEXT-AWARE 3-VARIANT STUDIO NOISE GATE MODES            |
+=====================================================================+
  [1] EXPANDER (Vocal, Acoustic & Room Cleanup)
      * Smooth Downward Ratio (1.2:1 .. 10:1) + Dual-Threshold Hysteresis
      * Gently lowers background floor by -6..-18 dB without hard cutoffs
  ---------------------------------------------------------------------
  [2] DRUM BLEED (Rack Tom, Floor Tom & Kick / Snare Isolation)
      * Differential Detector: E_key (Resonant Fundamental) - alpha * E_ref
      * One-Click [LRN] Auto-Detects Drum Fundamental Pitch (55..350 Hz)
      * 1.5 ms Lookahead + < 0.25 ms Attack preserves 100% of stick snap
      * Dynamic Decay Low-Pass Filter sweeps 20 kHz -> 250..800 Hz (380 Hz)
        so hi-hat and cymbal bleed vanish during the tom ring-out!
  ---------------------------------------------------------------------
  [3] AMP DYNAMIC (Stoner, Doom, Psych & High-Gain Guitar / Bass Rigs)
      * Autocorrelation Pitch Periodicity (H in [0, 1]) + Spectral Flatness
      * Whisper Clean Lock: Pitched notes (H > 0.65) stay open at -34 dBFS
      * Feedback Bloom Lock: Harmonic sustain (H > 0.80) holds 1.5..5.0 s
      * Palm-Mute Clamp: Chug stops clamp 60 Hz hum & fuzz hiss in 15..40 ms
        along a click-free S-Curve + Split-Band [HF Tame]
+=====================================================================+
```

Variant 1: `Expander` (Smooth Downward Expander)

Designed for lead vocals, horns, and acoustic instruments where a hard gate cut would sound unnatural. Below Open Threshold (with independent Close Threshold hysteresis to prevent chatter), the expander attenuates the floor smoothly according to Ratio (1.2:1 to 10.0:1) down to a musical Range / Floor ceiling (for example, -12.0 dB), tucking headphone bleed and stage shuffle down into the mix without ever dropping to dead digital silence.

Variant 2: `Drum Bleed` (Differential Tom/Snare Bleed Gate with [LRN] & Dynamic Decay LPF)

When a loud snare rimshot sits 40 cm from a rack tom microphone, the snare spill inside the tom mic is often nearly as loud in broad decibels as a medium tom fill. Drum Bleed separates the target drum from adjacent spill through four specialized stages:

1. **Resonant Sidechain Fundamental Filter & One-Click [LRN] Fundamental Lock:** A high-Q (Q = 4.0) resonant bandpass filter focuses the detector on the specific fundamental pitch of the drum shell (55 Hz to 350 Hz, default 110 Hz). Clicking **[LRN]** while the drummer hits the tom (or while playing back a single tom hit on the timeline) runs a parabolic spectral peak detector that automatically locks Key Freq to that exact drum shell resonance (for example, 110 Hz on a 12" rack tom or 78 Hz on a 16" floor tom).
2. **Differential Spill Rejection (`E_key - alpha * E_ref`):** Compares the energy inside the drum's fundamental bandpass filter against high-frequency snare wire and cymbal reference energy (> 800 Hz). Even a massive snare rimshot or crash cymbal burst has high E_ref relative to E_key, so the differential detector rejects the snare hit and keeps the tom gate shut.
3. **`1.5 ms` Lookahead & Ultra-Fast `< 0.25 ms` Attack:** Opens the gate slightly ahead of the physical stick impact so the initial 4 kHz stick transient is never dulled or clicked.

4. **Envelope-Driven Dynamic Decay Low-Pass Filter (20 kHz down to 250..800 Hz, default 380 Hz):** When a tom is struck, the audio path opens at full 20 kHz bandwidth so the stick attack snaps clearly. As the tom decays over the next 40 ms to 300 ms, the drum's own high frequencies vanish quickly while its 80 Hz - 150 Hz fundamental keeps ringing. Drum Bleed automatically sweeps a smooth 12 dB/oct low-pass filter from 20 kHz down to an adjustable 250 Hz to 800 Hz cutoff (default 380 Hz) in lockstep with the tom decay envelope, filtering out 100% of the harsh hi-hat and cymbal bleed during the tom ring-out while letting the warm low-end drum resonance sustain naturally.

Variant 3: Amp Dynamic (Stoner, Doom & Psychedelic High-Gain Guitar & Bass Gate)

In stoner rock, doom, sludge, psychedelic rock, space rock, and blues-rock, guitarists and bassists rely on high-gain tube heads driven by cascaded fuzz, overdrive, and boost pedals. This creates an impossible problem for conventional amplitude-threshold noise gates:

- **The Idle Noise Floor is Loud (-22 dBFS):** When the guitarist stops playing, stacked fuzz and tube stages produce loud 50/60 Hz transformer hum, rectifier buzz, and wideband thermal hiss around -26 dBFS to -22 dBFS.
- **Clean Fingerpicking is Quieter than the Fuzz Floor (-34 dBFS):** When the guitarist rolls back their guitar volume knob or kicks off a fuzz pedal for a hypnotic clean psychedelic intro, the delicate fingerpicked notes sit down around -34 dBFS (12 dB *below* the loud fuzz noise floor). Any standard noise gate set to -20 dBFS to block fuzz hum will completely mute the clean intro!
- **Intentional Feedback Bloom vs. Tight Palm-Mute Stops:** At the end of a doom riff, the guitarist may hold a chord in front of the cabinet for 5 seconds, letting it morph into singing harmonic feedback (Feedback Bloom), and then suddenly slam a tight staccato palm-mute stop (Chug Stop) where the room must drop dead silent in 18 ms to 22 ms.

Amp Dynamic solves this by analyzing **signal physics rather than raw volume alone**, combining an **Autocorrelation Pitch Periodicity / Harmonicity Detector ($H \in [0, 1]$)** across the 55 Hz to 1,200 Hz guitar/bass fundamental range with a **Spectral Flatness / Entropy Measure ($SFM \in [0, 1]$)**:

Musical Playing State	Acoustic Signature (H & SFM)	Automatic Amp Dynamic Response
Quiet Clean Fingerpicking (-34 dBFS)	High pitch periodicity ($H > 0.65$), concentrated harmonic partials	**Whisper Clean Lock:** Lowers the effective gate threshold by up to 18 dB whenever coherent string periodicity is detected, keeping delicate -34 dBFS clean notes and long acoustic-style sustains wide open without touching a knob.

Musical Playing State	Acoustic Signature (H & SFM)	Automatic Amp Dynamic Response
Sustained Harmonic Amp Feedback	Very high periodicity (H > 0.80), near-sinusoidal low spectral flatness	**Feedback Bloom Lock:** Recognizes intentional harmonic cabinet feedback and stretches release hold to 1.5 s .. 5.0 s (1500..5000 ms, default 2.5 s) so singing feedback tails never choke or sputter out.
Abrupt Palm-Mute Riff Stop (Chug)	Rapid broadband energy drop (-12 dB in < 15 ms) + high-entropy pick/palm damping (SFM > 0.55, H < 0.25)	**15 ms .. 40 ms Palm-Mute Clamp (default 18 ms .. 22 ms):** Immediately overrides the slow sustain release and clamps 50/60 Hz hum and fuzz roar shut in 15 ms to 40 ms (default 18 ms to 22 ms) along a click-free S-curve.
Long Fading Chord Tail	Gradually declining harmonic energy with rising > 3.5 kHz fuzz hiss	**Split-Band HF Tame Dynamic De-Hisser:** Smoothly shelves down high-frequency white-noise fizz above 3.5 kHz as the chord fades while leaving the warm 100 Hz - 1.5 kHz tube sustain untouched.

5.4.6 Guitar & Bass Amp Rig ([CIR] Tube Circuit Models, HEAD ON/BYP & [NAM] Neural Amp Capture)

Recording a direct instrument (DI) feed alongside (or instead of) a loud stage amplifier gives a band complete freedom to re-voice guitar and bass tones after rehearsal, track late at night on headphones, or bring their exact personal tube heads and fuzz pedals inside the console without carrying heavy cabinets to every session.

Cave Recorder's **Amp Rig** processor combines an input calibration stage (dBu-referenced headroom Trim, INST/LINE pad, and one-click AUTO-CAL), an overdrive/fuzz stompbox boost stage (BOOST + DRV), two selectable amplifier engines (**[CIR] Physical Tube Circuit Models** vs. **[NAM] Native .nam Neural Amp Models**), a 4-control passive tone stack (BASS, MID, TREBLE, PRESENCE), power-amp transformer SAG, an independent **HEAD ON / HEAD BYP** amplifier head bypass toggle, and an integrated **Speaker Cabinet & Microphone stage** (8 modeled speaker cabinets plus external .wav/.nam linear impulse responses, 5 studio microphone capsules, continuous POS and AIR, and a CAB BYP toggle):

1. **[CIR] Mode (8 Physical Vacuum-Tube Amplifier Topologies with 2x / 4x Anti-Aliasing Oversampling):**
   - Models triode/pentode grid conduction, cathode bias shift, power-supply RC sag, and output transformer core hysteresis across 8 classic guitar and bass amplifier voicings: CLN65 (65 CLEAN 2x6L6 VIBRATO), JMP59 (59 JUMPED 4xEL34 LEAD), BRT80 (80 BRIT 3x12AX7 MASTER), CTR100 (CITRUS 4x12AX7 DIRTY), 5STG (5-STAGE 6L6 HIGH-GAIN), D-REC (MODERN OPEN-LOOP 5U4G), DM150 (73 ULTRALINEAR 4x6550), and BS300 (69 BAXANDALL 6x6550 BASS).

- Nonlinear tube and overdrive saturation stages run through 2x / 4x anti-aliasing oversampling so high-gain harmonics stay free of aliasing foldback across 44.1 kHz, 48.0 kHz, 88.2 kHz, and 96.0 kHz sessions.

2. **HEAD ON / HEAD BYP & CAB BYP (Independent Head and Cabinet Bypass):**
   - Toggling **HEAD BYP** bypasses the preamp, overdrive boost, tone stack, and power-amp stages while keeping Amp Rig's speaker cabinet and microphone section active (ideal when tracking a physical tube head's load-box line out that only needs speaker cabinet voicing).
   - Conversely, toggling **CAB BYP** inside Amp Rig bypasses its internal speaker cabinet so you can insert stereo rack processors (Modulation, 3-Mode Delay, or Studio Plate Reverb) in downstream rack slots and place a standalone **Cab IR** (SPEAKER CABINET & IR LOADER) processor after them (Amp Rig [CAB BYP] -> Stereo Chorus/Delay -> Cab IR).
3. **[NAM] Mode (Native .nam Neural Amp Real-Time Inference & Self-Contained Library):**
   - Loads open **.nam (Neural Amp Modeler)** files (A2, A1 Standard, and Feather/Lite architectures) directly into the channel strip with zero-latency real-time inference, anti-aliasing oversampling, and automatic sample-rate conversion when running sessions at 44.1 kHz, 48.0 kHz, 88.2 kHz, or 96.0 kHz.
   - **Two-Tier Self-Contained .nam Storage:** Imported or captured .nam profiles are indexed in your global studio library (<LibraryRoot>/.caverec/amp_captures/, along with optional return_wet.flac archives for later higher-quality re-training via **RE-TRAIN HQ (WET WAV)**) and are **automatically copied into <SessionFolder>/models/ and <TakeFolder>/models/** whenever assigned to a strip. Every Session and Take remains 100% self-contained when synced to Google Drive or opened on a bandmate's laptop without broken external file paths.
4. **3-Step Guided Hardware Amp & Pedal Capture Wizard (CAP) & Mastering Trainer:**
   - Clicking **CAP** in the Amp Rig inspector deck opens the **3-Step Hardware Neural Capture & Training Modal** (STEP 1: I/O & LOOPBACK CHECK -> STEP 2: SWEEP & TRAINER -> STEP 3: SANITY AUDIT & SAVE) so you can clone your own physical tube head, preamp, or fuzz/overdrive pedalboard:
   - **Step 1 (Hardware Patch, Preset Selection & Loopback Level Check):** Select the physical RE-AMP SEND output jack (OUT 01..18, feeding your amp/pedal input via a re-amp box) and MIC / DI RETURN input jack (IN 01..18, from your load box, attenuator line out, or cabinet microphone). Click the header preset selector button to cycle between the three training quality modes (Standard -> Mastering -> Turbo -> Standard):
   - **Standard (PRESET: STANDARD, Default):** Full studio capture using the complete 3-minute conditioning sweep and 150 training passes.
   - **Mastering (PRESET: MASTERING):** Maximum-accuracy mode using the full 3-minute conditioning sweep, up to 1,000 training passes, automatic best-checkpoint retention, adaptive step-size refinement on plateaus, and automatic early stopping when accuracy converges.
   - **Turbo (PRESET: TURBO):** Fast preview capture using a compact conditioning sweep and 60 training passes. Click **TEST LOOPBACK & LEVEL** to play a test burst through the hardware, verify return peak headroom (-18 to -3 dBFS) and signal-to-noise ratio (>= 36 dB), and lock sub-sample roundtrip latency and clock drift (PPM).

- **Step 2 (Conditioning Sweep, Live Convergence Telemetry & STOP & KEEP BEST):** Click **START CAPTURE & TRAINING** (with optional **DIM MONITORS (-20dB)**). Cave Recorder transmits the multi-frequency conditioning sweep, impulses, and dynamic transient bursts out the hardware send, captures and time-aligns the wet return, and runs hardware-accelerated model training in the background. A live 60 Hz convergence graph plots training and validation Error-to-Signal Ratio (ESR) across passes. At any point during training, you can click **STOP & KEEP BEST** to finish immediately while keeping the best validated model weights and running the post-training sanity audit (or click **CANCEL** to abort and discard the run).
- **Step 3 (Sanity Audit, Quality Badge, A/B Audition & Save to Library):** Upon completion, Cave Recorder validates the trained .nam model against automated studio quality gates (STUDIO < 1% ESR / GOOD 1-3% ESR quality badge, steady-state silence decay <= -70.0 dBFS, full-scale burst stability, and -18 LUFS calibrated output loudness metadata), lets you **A/B AUDITION** the trained neural model against the recorded hardware return, and saves the .nam file (plus the optional aligned wet .wav archive when **KEEP WET WAV** is enabled) directly into your Global Library and active Session/Take models/ folder.

5.4.7 Standalone Speaker Cabinet & Linear Impulse Response Processor (Cab IR & CAB IR Browser)

In addition to the cabinet stage built into Amp Rig, Cave Recorder provides a dedicated, reorderable **Cab IR (SPEAKER CABINET & IR LOADER)** rack processor (6:CAB IR). Inserting Cab IR in its own rack slot allows you to place stereo modulation (Dimension Chorus, Through-Zero Flanger, Optical Phaser, Harmonic Tremolo), stereo 3-Mode Delay, or Studio Plate Reverb **between** an amplifier head (Amp Rig with CAB BYP active) and the speaker cabinet, or to voice direct preamp DI tracks and external hardware load-box returns.

```
+=================================================================+
|     POST-HEAD STEREO EFFECTS LOOP INTO STANDALONE CABINET (Cab IR) |
+=================================================================+
  Slot 1: [Context-Aware Gate (Amp Dynamic)]
     |
     v
  Slot 2: [Amp Rig: HEAD ON + CAB BYP] (Tube Circuit or .nam Head Only)
     |
     v
  Slot 3: [Modulation / 3-Mode Delay] (Stereo FX Post-Head, Pre-Cab!)
     |
     v
  Slot 4: [Cab IR: SPEAKER CABINET & IR LOADER] (8 Models or .wav/.nam)
+=================================================================+
```

Both Cab IR and the cabinet section inside Amp Rig share two zero-latency speaker engines, five studio microphone models, an interactive MIC HPF × MIC LPF 2D XY spectral contour pad, and the **Three-Tier Portable Cabinet IR Library**:

1. **8 Modeled Physical Speaker Enclosures (Cabinet Impulse & Acoustic Resonance):**

- Synthesizes sealed and ported low-frequency cabinet resonance, speaker cone breakup modes, and multi-driver phase comb notches across 8 studio enclosures:
- **2x12 (OPEN 2x12 CERAMIC):** Open-back 2x12 combo loaded with bright American ceramic drivers (85 Hz open-back dipole resonance).
- **4x12 25 (CLOSED 4x12 25W):** Classic closed-back 4x12 loaded with warm 25W British ceramic drivers and early woody midrange breakup.
- **4x12 75 (CLOSED 4x12 75W):** Tight closed-back 4x12 loaded with 75W scooped-midrange ceramic drivers for articulate rhythm crunch.
- **4x12 BR (BIRCH 4x12 CABINET):** Heavy 18-ply birch closed-back 4x12 with thick low-mid projection and smooth high-end roll-off.
- **4x12 TG (TIGHT 4x12 CABINET):** Focused straight-front 4x12 tuned for fast, aggressive high-gain palm-mute transients.
- **4x12 OV (OVERSIZED 4x12 CAB):** Deep oversized closed-back 4x12 with extended 65 Hz sub-bass chest thump.
- **4x12 15 (PORTED 4x12 150W):** High-headroom 150W ported 4x12 enclosure built for drop-tuned stoner, doom, and sludge riffs.
- **8x10 (SEALED 8x10 BASS):** Infinite-baffle sealed 8x10 bass enclosure delivering fast, punchy low-end articulation without port chuffing.

2. **5 Studio Microphone Capsules & Continuous Placement Controls:**
   - Select between **DYN-C (CARDIOID DYNAMIC)**, **DYN-L (LARGE DYNAMIC)**, **RBN-8 (FIGURE-8 RIBBON)**, **CND-L (LARGE CONDENSER)**, and **DYN-B (BASS DYNAMIC)**.
   - Shape the mic placement continuously with **MIC POS** (0% bright center dust-cap to 100% warm outer cone edge, default 25%), **ROOM AIR** (0% to 100% early-reflection studio room ambiance), **LOW CUT** (20 Hz to 500 Hz 12 dB/oct high-pass filter, horizontal axis on the 2D XY pad), **HIGH CUT** (2.0 kHz to 20.0 kHz 12 dB/oct low-pass filter, vertical axis on the 2D XY pad), **180° Polarity Invert**, and **OUT TRIM** (-24.0 dB to +24.0 dB).
3. **Custom .wav & .nam ("Linear") Impulse Response Convolution (EXT IR):**
   - Supports mono and true-stereo linear impulse responses loaded from standard **.wav** files (16/24/32-bit PCM or 32-bit float) and **.nam "Linear"** FIR captures, evaluating up to 2048 FIR taps per channel (1024 taps at 48.0 kHz) with zero latency.
   - Impulse responses recorded at any sample rate are automatically resampled via a high-precision windowed-sinc interpolator to match the active session sample rate (44.1 kHz, 48.0 kHz, 88.2 kHz, or 96.0 kHz).
4. **Cabinet IR Browser Modal ([CAB IR]) & Three-Tier Portable Storage (Take -> Session -> Global):**
   - Clicking **[CAB IR]** in Amp Rig or Cab IR opens the **CABINET IR BROWSER & MODELED CABINET SELECTOR** modal. You can filter catalog entries via **ALL**, **.WAV**, **.NAM LIN**, and **STEREO** pills, sort by **CABINET IR NAME**, **FMT**, **SAMPLE RATE**, **CH**, or **TAPS**, and import external .wav or .nam files directly from disk.
   - Imported impulse responses are stored in your global library (<LibraryRoot>/.caverec/cab_irs/) and are **automatically replicated into <SessionFolder>/cab_irs/ and <TakeFolder>/cab_irs/** whenever assigned to a strip. When resolving an impulse response, Cave Recorder searches Take -> Session -> Global Library, ensuring every Session and Take opens with its exact speaker cabinet impulse responses intact across computers and Google Drive syncs.

5.5 Stereo Convolution Reverb (AUX 1 REVERB) & Stereo Tape Delay (AUX 2 TAPE DLY)

To keep CPU overhead near zero while providing authentic vintage studio ambience during live tracking and mixing, Cave Recorder provides two dedicated stereo Aux Return processors fed by the RVB and DLY send bars on every Input Track and Subgroup Bus.

5.5.1 AUX 1 REVERB: Zero-Latency Acoustic Convolution Reverb

AUX 1 REVERB uses a zero-latency real-time acoustic convolution engine (FFT CONV IR / ZERO-LATENCY) loaded with studio impulse responses, headed by the classic **EMT 140 Plate** alongside Studio Wood Room, Live Stone Chamber, and Vintage Spring Tank spaces.

Parameter	UI Label	Range	Default	Studio Function
Acoustic Space / IR	Header Model Name	EMT 140 Plate, Studio Room, Chamber, Spring	EMT 140 Plate	Selects the acoustic impulse response profile loaded into the stereo convolution engine.
Input Low-Cut Filter	LC (LC 90Hz)	20 Hz to 1000 Hz	90 Hz	Filters low-frequency rumble and kick/bass bleed out of the reverb input so the reverb tail stays airy and never muddies the low end of the mix.
Pre-Delay	PD	0 ms to 200 ms	18 ms	Inserts a short clean gap before the reverb tail begins. Setting PD between 18 ms and 45 ms preserves upfront dry vocal clarity while creating the illusion of a larger room (the Haas effect).
Decay Time	DCY	0.20 s to 12.00 s	2.40 s	Scales the RT60 reverberation decay envelope (0.8 s to 1.4 s for tight drum rooms, 1.8 s to 2.8 s for classic vocal/ snare plates, 3.5 s+ for ambient ballads) and dynamically rebuilds and hot-swaps the stereo studio plate impulse response in real time.

Parameter	UI Label	Range	Default	Studio Function
Wet Level Toggle	WET (WET 100%)	100% Full Wet / 50% Blend	100% (crimson)	Toggles the return strip between 100% pure wet bus operation and 50% attenuated wet level while keeping AUX 1 REVERB solo-safe.

5.5.2 AUX 2 TAPE DLY: Capstan Analog Tape Delay

AUX 2 TAPE DLY models a multi-head magnetic capstan tape echo unit (CAPSTAN TAPE / TAPE ECHO) complete with mechanical transport wow and flutter, oxide saturation, high-frequency head damping, and stereo ping-pong regeneration.

Parameter	UI Label	Range	Default	Studio Function
Capstan Wow & Flutter	WOW (WOW 18%)	0% to 100%	18%	Introduces organic capstan motor speed drift and subtle pitch modulation on echoes, widening vocals and guitar leads without synthetic chorus artifacts.
High-Frequency Damping	DMP (DMP 4.2k)	500 Hz to 20.0 kHz	4.2 kHz	Places a warm analog low-pass filter inside the feedback loop so each successive echo repeat becomes progressively darker and tucks neatly behind the dry performance.
Delay Time	TM	1 ms to 2000 ms	320 ms	Sets the echo interval between 10 ms doubling, 75 ms to 115 ms rockabilly/vocal slapback, and 250 ms to 600 ms rhythmic quarter-note or dotted-eighth throws.
Feedback Regeneration	FB	0% to 95% drag bar (98% max)	35%	Controls how much delayed output is fed back into the virtual record head (0% for a single slapback repeat, 25% to 45% for natural fading trails, and up to 95%/98% for sustained dub tape repeats without runaway clipping).
Ping-Pong Width & Toggle	PP (PP 60%)	Off / Active (0% to 100% width)	Active (60%)	Alternates echo repeats between the left and right channels at the selected stereo spread width (PP 60%) while keeping AUX 2 TAPE DLY solo-safe.

5.5.3 3-Mode Delay (Capstan Tape, BBD Analog, Pristine Digital) & Dynamic Ducking

Whether used on AUX 2 TAPE DLY or inserted directly into any channel or subgroup rack slot, Cave Recorder's delay processor offers three distinct physical delay architectures plus a built-in **Dynamic Ducking** envelope:

Delay Mode	Physical Circuit Model	Sonic & Spectral Behavior	Ideal Studio Uses
Capstan Tape	Multi-head magnetic tape transport with capstan wow and flutter (WOW), oxide saturation inside the feedback path, head-bump low-end warmth, and progressive high-cut (DMP).	Warm, evolving repeats that never clash with the dry vocal or guitar lead, capable of self-oscillating dub swells at FB > 90% without digital clipping.	Psychedelic guitar solos, dub/space-rock throws, rockabilly vocal slapback (85 ms), and horn section depth.
BBD Analog	Classic 4096-stage Bucket-Brigade Device (BBD) charge-transfer delay line (Delay Time 10 ms to 1000 ms, default 300 ms) with pre-emphasis/de-emphasis compander breathe, clock-rate-linked 24 dB/oct low-pass filters (4.8 kHz at short delays down to 1.6 kHz at long delays), subtle even-order warmth, and authentic -72 dBFS clock whine at long delay times.	Dark, murky, slightly gritty repeats that melt directly behind the dry instrument like a vintage analog pedalboard stompbox.	Stoner/doom lead guitar ambiance, krautrock motorik synth arpeggios, and atmospheric dark vocals.
Pristine Digital	Phase-accurate 32-bit floating-point stereo delay line with adjustable Low-Cut, High-Cut, Ping-Pong Width, and zero pitch drift.	Crystal-clear rhythmic repeats with exact transient definition.	Tight dotted-eighth rhythmic guitar patterns, stereo keyboard widening, and percussive echoes.

- **Built-In Dynamic Ducking (Ducking: 0.0 dB to -24.0 dB):** Monitors the dry input envelope and automatically attenuates the wet delay repeats while the vocalist or guitarist is actively playing a phrase. The moment the phrase ends, the ducking releases (180 ms) and the delay tail blooms up to full volume in the spaces between lines.

5.5.4 4-Pole Non-Linear Resonant Ladder Filter (with Envelope Follower & Sidechain)

The **Ladder Filter** models a classic transistor ladder low-pass filter with two selectable low-pass slopes (2P 12 dB/octave 2-pole low-pass and 4P 24 dB/octave 4-pole low-pass, default 4P) and per-stage analog saturation:

- **Cutoff (20 Hz to 16,000 Hz, default 1,200 Hz) & Resonance (0% to 100% Self-Oscillation):** Sweeps the resonant peak from deep sub-bass thunder to harmonic self-oscillation.

- **Class-A Input Drive (0.0 dB to +24.0 dB) & Bass Comp (0% to 100%, default 50%):** Standard 4-pole ladder filters lose low-end weight when high negative feedback (Resonance) is applied. Turning up Bass Comp restores the sub-bass fundamental before the feedback summer so bass guitars and synthesizers keep their chest-thumping low end even at aggressive resonance settings.
- **Bi-Directional Envelope Follower (Env Depth -5.0 to +5.0 octaves / -60 to +60 semitones), Fast / Smooth Response Speeds & External Sidechain:** Sweeps the filter cutoff upward or downward in proportion to input pluck dynamics with selectable **Fast** (3 ms attack / 60 ms release for snappy funk bass and guitar auto-wah) and **Smooth** (25 ms attack / 300 ms release for liquid psychedelic sweeps) detector ballistics, or locks onto an external kick-drum sidechain so a fuzz bass or synth drone pumps rhythmically with the drummer.

5.5.5 Algorithmic Studio Plate Reverb & 4-Mode Modulation Suite

When you want dedicated reverb or modulation inside an individual channel or Subgroup Bus rack slot alongside AUX 1 REVERB, Cave Recorder provides two zero-latency rack processors:

1. **Algorithmic Studio Plate Reverb (4-Stage Allpass Diffuser + 8-Line Feedback Delay Network):**
   - Uses four series allpass diffusers feeding an 8-channel Feedback Delay Network (FDN) with slow quadrature delay-tap modulation and independent two-band crossover damping (Bass Mult vs. High Damping).
   - Controls include Pre-Delay (0..150 ms, default 10 ms), Decay Time (0.4..8.0 s, default 2.2 s), High Damping (1.0..16.0 kHz, default 6.5 kHz), Bass Mult (0.4x..1.5x, default 0.9x), Diffusion (0%..100%, default 80%), Pre HPF (20..500 Hz, default 80 Hz), Pre LPF (2.0..20.0 kHz, default 12.0 kHz), Stereo Width (0%..150%, default 100%), and Wet/Dry Mix (0%..100%).
2. **4-Mode Studio Modulation (Through-Zero Flanger, Dimension Chorus, Optical Phaser, Harmonic Tremolo):**
   - **Through-Zero Flanger (Authentic Dual-Tape Through-Zero Flanging):** Delays the dry reference path by a fixed 2.5 ms offset while sweeping the modulated wet tape path from 0.05 ms to 4.95 ms with saturated feedback (-95% to +95%). As the modulated path crosses 2.5 ms, it passes straight through zero relative delay (t = 0), creating the unmistakable deep jet-whoosh null and phase inversion of two studio reel-to-reel tape machines pressed by thumb on the supply flange.
   - **Dimension Chorus (Motionless 3D Stereo Widener):** Models a classic 4-button studio dimension unit using dual triangle-wave BBD delay lines driven in quadrature with cross-blended anti-phase high-pass dry/wet matrixing. Expands mono guitars, keys, and vocals into a wide, motionless 3D stereo image without the seasick pitch warble of standard choruses.
   - **Optical Phaser (4, 6, 8, or 12-Stage Photocell Allpass Phaser):** Cascades 4, 6, 8, or 12 first-order allpass stages modulated by an asymmetric incandescent-lamp / photocell envelope (15 ms fast light-up attack vs. 120 ms logarithmic dark recovery) for liquid, swirling stoner/psych-rock guitar and electric piano movement.
   - **Harmonic Tremolo (Dual-Band Linkwitz-Riley Crossover Tremolo):** Splits the signal at a 400 Hz Linkwitz-Riley crossover and modulates the low band (< 400 Hz) and high band (> 400 Hz) 180° out of phase. While treble rises, bass dips (and vice versa), creating a warm, hypnotic vintage amplifier pulse that never drops the perceived volume of the track in the mix.

5.6 Automatic Multi-Mic Phase & Polarity Alignment (ALIGN ALL & Φ)

When recording a drum kit, guitar cabinet, acoustic piano, or horn section with multiple microphones, physical distance differences between capsules create acoustic arrival-time offsets. Sound travels through studio air at approximately 343 meters per second (1,125 feet per second), meaning every 34.3 cm (13.5 inches) of extra distance between a close microphone and an overhead or room microphone delays the acoustic wavefront by 1.0 millisecond (1000 microseconds, or 48 samples at 48.0 kHz).

5.6.1 Why Multi-Mic Time Offsets Cause Comb Filtering & Hollow Low End

When two microphones capture the same instrument at slightly different arrival times and are summed together on DRUM BUS or MST MASTER BUS:

- Frequencies whose half-wavelength matches the time offset arrive 180° out of phase and cancel each other out, carving deep, hollow notches across the frequency spectrum (known as **comb filtering**).
- A snare drum captured by both 02SnrTp (close dynamic mic) and 03OvrhdL (condenser mic 63 cm / 25 inches above the snare) suffers a 1.85 ms (1850 us) arrival delay at the overhead capsule. Without phase alignment, summing the overheads with the close snare mic thins out the 200 Hz snare body and smears the initial stick transient into a double flam.
- Similarly, a snare bottom microphone pointed upward at the snare wires moves in the opposite mechanical direction from the top snare microphone, requiring a 180° polarity inversion (Ø) in addition to time alignment.

5.6.2 Running ALIGN ALL & Auditioning with PHASE: ALIGN / PHASE A/B

Cave Recorder eliminates manual waveform sliding and guesswork through its built-in **Sub-Sample Multi-Mic Phase & Polarity Alignment** engine:

1. **Record a Short Pass or Play Back an Existing Take:** Have the drummer play the full kit for a few bars (or load any recorded Take on the timeline).
2. **Click ALIGN ALL in the Top Deck (or Φ ALIGN ALL on the Console Bar):** Cave Recorder analyzes the active Take's recorded multi-track Broadcast Wave .wav files on disk (or the live rolling capture memory if no recorded Take files exist yet) using sub-sample bandpass (150 Hz to 6 kHz) cross-correlation within a +-25 ms acoustic window, delaying closer microphones forward in time to match the furthest reference microphone. The entire multi-track phase and polarity alignment operation is captured as a single undoable step so you can revert or reapply it at any time with **CMD** / **CTRL** + **Z**.
3. **Inspect Per-Channel Ø and Φ Readouts:** Within milliseconds, each aligned channel strip updates its header telemetry and phase buttons:
   - **Ø (Cyan Polarity Badge, SHIFT + P on inspected strip):** Engages automatically on any channel whose acoustic waveform is inverted 180° relative to the reference microphone (such as CH02 02SnrTp or a bottom snare mic).

   - **Φ (Emerald Phase Align Badge) & Microsecond Readout:** Engages sub-sample time alignment (0.00 ms to +50.00 ms, displayed in microseconds from 0us to 50000us, such as 420us on CH02 and 1850us on CH03) so all wavefronts arrive at the subgroup summing bus at the exact same sub-sample instant.
4. **A/B Audition Globally (PHASE: ALIGN / PHASE A/B):** Click the emerald PHASE: ALIGN button in the Top Deck (or the PHASE A/B button on the Mix Console toolbar) to toggle all phase and polarity corrections in and out across the entire console simultaneously. You will immediately hear the kick and snare low-end punch tighten up and the stereo center image snap into focus. Both Run Multi-Mic Phase Alignment and Toggle Global Phase A/B can also be bound to custom keyboard shortcuts in the KEYMAP view (CMD / CTRL + 6).

> **INDIVIDUAL CHANNEL OVERRIDE (Ø & Φ)**
> If you want to keep distant ambient room microphones unaligned to preserve natural acoustic room size while keeping close drum mics and overheads tightly phase-locked, simply click the green Φ button on the room mic channel strip to disengage phase delay on that specific track while leaving PHASE: ALIGN active on the rest of the kit.

5.6.3 Why Delay Alone Cannot Fix Ported Cabinets, Bass DI + Amp, or Dispersive Drum Kits

Sub-sample time-of-flight delay (Φ) and 180° polarity inversion (Ø) solve pure physical distance offsets through air. However, in real rehearsal rooms and tracking studios, multi-mic sources also suffer from **frequency-dependent phase dispersion (non-linear group delay)**:

- **Bass DI + Miked Ported Bass Cabinet:** A tuned bass-reflex cabinet port acts as a high-order acoustic high-pass resonator around 45 Hz to 85 Hz, rotating the phase of the low E and A strings by 90° to 180° relative to the upper harmonics at 400 Hz. Even after you align the initial pick transient in time (Φ), the 60 Hz fundamental from the cabinet mic can sit 140° out of phase with the DI box, stealing 6 dB to 12 dB of sub-bass weight when both faders are up.
- **Multi-Mic Drum Kits & Dual-Mic Guitar Cabinets:** Drum shell resonances, off-axis cardioid proximity filters, and loudspeaker crossover networks rotate low-frequency phase differently from midrange transient phase. No single time delay or 180° polarity switch can bring both 65 Hz kick chest thump and 2.5 kHz beater attack into 0° constructive alignment simultaneously.

5.6.4 Zero-Slot Top-of-Strip Phase Align Network & Mode 1 Instant Binary Polarity Maximizer (< 50 ms)

To solve both broadband polarity conflicts and frequency-dependent phase dispersion without consuming a single slot in your 8-slot insert rack, every Input Track strip in Cave Recorder includes a dedicated, zero-slot **Phase Align Network** wired directly into the top of the channel strip alongside Ø and Φ:

```
+=================================================================+
|     TOP-OF-STRIP ZERO-SLOT PHASE & POLARITY ALIGNMENT NETWORK   |
+=================================================================+
  Channel Input (Post-Trim TRM)
         |
         v
  [1] Broadband Polarity Switch [Ø] (+1 or -1 / 180°)
         |
         v
  [2] Sub-Sample Time-of-Flight Delay [Φ] (0 .. 50,000 us)
         |
         v
  [3] Zero-Slot 5-Stage Cascaded Allpass Phase Align Network:
      * Stage 0: 1st-Order Low-Frequency Rotator (f_0 in 20..500 Hz)
      * Stages 1..4: 4x 2nd-Order Biquad Allpass (f_k in 20..12,000 Hz,
                     Q_k in 0.30..3.50, default 0.707)
      * Unified 10 ms Dual-Branch Linear Crossfader for click-free
        live auditioning between Candidates A / B / C / D and Bypass!
+=================================================================+
```

Cave Recorder's **Polarity & Phase Maximizer** operates in two distinct modes depending on whether you are in the middle of a fast rehearsal or dialing in a final studio mix:

Mode 1: Instant Binary Polarity Maximizer (< 50 ms)

When you need an immediate, zero-filter polarity check across all 18 channels right after tracking:

1. **Connected Acoustic Group Partitioning:** Cave Recorder builds the `18 x 18` pairwise broadband cross-correlation matrix across active performance regions and partitions channels into independent acoustic groups (for example, grouping the 8 drum microphones into Group 1 and the bass DI + mic'd cabinet pair into Group 2 while ignoring uncorrelated synth DI tracks).
2. **Exact Global Polarity Maximizer:** Within each group, the solver evaluates every possible binary polarity combination in under `50 milliseconds` to maximize total constructive summed energy across four selectable studio weighting criteria:
   - **Low-End Weight (30 Hz - 200 Hz Punch):** Applies a band-limited `30 Hz` to `200 Hz` low-end punch weighting so kick, snare body, and bass fundamentals never cancel out.
   - **Broadband Loudness (ITU-R BS.1770-4 K-Weighted LUFS):** Maximizes full-spectrum perceived K-weighted loudness across the summed mix.
   - **Transient Crest & Impact (60 Hz - 6 kHz + Top-K Transient Crest Re-Scoring):** Optimizes `60 Hz` to `6 kHz` punch and re-scores the top-energy transient windows so drum stick crack and guitar pick attack hit with maximum crest factor.
   - **Anti-Comb Spectral Smoothness:** Minimizes deep phase-cancellation notches across the frequency spectrum so multi-mic sources sum smoothly without hollow comb filtering.
3. **Safe Audition Backup:** Auditioning `Mode 1` or `Mode 2` candidates automatically backs up your complete per-channel phase state (`Ø`, delay `Φ`, and allpass settings) so canceling or closing the popover restores your exact original console state.

5.6.5 Mode 2: Deep Multicore Spectral Phase & Allpass Solver (Anytime 30 s .. 30 min)

When you want to lock a multi-mic drum kit, a bass DI + mic'd bass cabinet rig, or a dual-mic guitar cabinet into 0° constructive phase coherence across the entire frequency spectrum, open the **Polarity & Phase Maximizer** and run **Mode 2: Deep Multicore Spectral Phase & Allpass Solver**:

1. **Compact 256-Bin Cross-Spectral Profile:** Cave Recorder scans the highest-energy active performance windows of the Take once and condenses them into a compact 256-bin log-spaced cross-spectral profile (20 Hz to 20 kHz, with biquad allpass pole centers spanning 20 Hz to 12 kHz). Because all subsequent optimization runs purely against this in-memory spectral profile (< 500 KB), all CPU cores can evaluate millions of allpass filter configurations per second without touching the disk.
2. **Multicore Spectral Phase Optimization (Anytime 30 s .. 30 min Budget):**
   - **Theoretical Coherence Upper Bound & Efficiency Gauge:** Computes the exact mathematical upper bound on the maximum energy achievable if every channel had an ideal phase rotator at every frequency bin. The UI displays your current solution as a live **Coherence Efficiency % gauge** against this theoretical ceiling (for example, 98.7% of theoretical maximum, +4.8 dB low-end gain recovered).
   - **Global Trajectory Search & Continuous Pole Polish:** Searches across log-frequency bins to find globally optimal allpass filter trajectories and refines continuous pole frequencies f_k and bandwidths Q_k to maximize summed phase coherence across all microphones.
   - **Multicore Parallel Search Across Phase Barriers:** Runs parallel worker threads across CPU cores to explore both broad polarity inversions, low-frequency rotators, multi-band biquad allpass stages, and microsecond time-of-flight alignment without getting trapped in local phase cancellations.
3. **Live Candidates A, B, C, D & [GAIN-MATCH AUDITION]:**
   - You can choose a quick **30 s**, **2 m**, **5 m**, or **30 m** time budget (or click **[STOP & KEEP BEST]** at any instant). As the solver runs, it continuously populates four distinct studio candidate slots ordered from zero-dispersion polarity to full wavefront + allpass alignment that you can click to audition live with a click-free 10 ms crossfade:
   - **[CANDIDATE A: PURE Ø - ZERO DISPERSION]:** Optimal binary 180° polarity inversions only (0 active allpass stages, 0.0 us time shift) for zero group-delay dispersion.
   - **[CANDIDATE B: TIGHT LOW-END LOCK - MINIMAL POLES]:** Broadband polarity plus at most the 1st-order LF rotator and 1 low-frequency biquad allpass stage (<= 250 Hz, group delay < 2.5 ms, 0.0 us time shift) to lock kick and bass fundamentals while keeping midrange transients untouched.
   - **[CANDIDATE C: DEEP MULTI-BAND ALLPASS]:** Broadband polarity plus up to the full 5-stage allpass cascade (1st-order LF rotator + 4 biquads, group delay <= 6.0 ms, 0.0 us time shift) for maximum multi-band phase coherence.
   - **[CANDIDATE D: FULL WAVEFRONT + ALLPASS]:** Jointly optimizes causal microsecond time-of-flight alignment (0..25,000 us) on unlocked spot microphones alongside broadband polarity and the sparse 5-stage allpass cascade.

- **[GAIN-MATCH AUDITION] Toggle:** Because constructive phase alignment naturally increases summed bus volume by 3 dB to 8 dB, enabling [GAIN-MATCH AUDITION] automatically normalizes the summed output loudness (clamped to ±12.0 dB) to match the unaligned bypass signal. This lets you verify that the low end is genuinely tighter, punchier, and more articulate rather than merely louder before clicking **[APPLY SELECTED TO CONSOLE]**.

5.7 Hosting External CLAP & Audio Unit Plugins (1:CLAP, PDC & Low-Latency Monitoring)

While Cave Recorder's built-in TAPE, 12B EQ, VCA CMP, DE-ESS, GATE, REPAIR, RIDER, AUX 1 REVERB, and AUX 2 TAPE DLY processors cover 95% of studio tracking and mixing tasks with zero added latency, any of the eight insert slots (Slots 1..8) on any Input Track (CH01..CH18) or Subgroup Bus (SUB 1..SUB 16) can also host third-party **CLAP (Clever Audio Plug-in, .clap)** effects across macOS, Windows, and Linux, as well as native **macOS Audio Unit (AUv3 and AUv2)** audio effects ('aufx' and 'aumf').

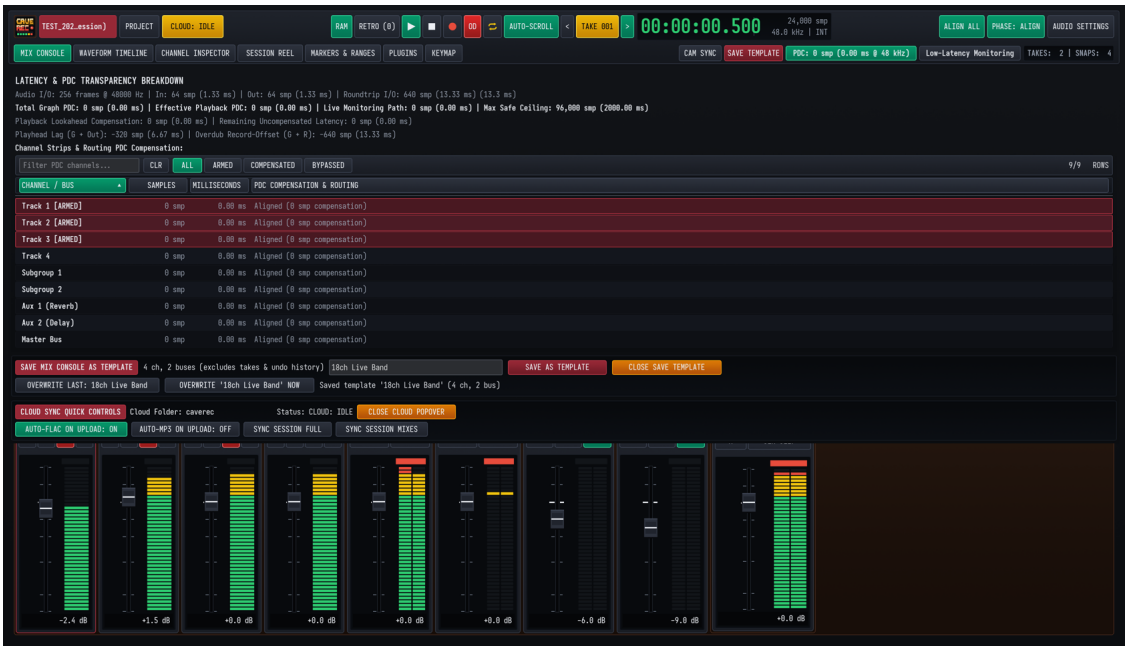


Figure 5.4: *Plugin Delay Compensation (PDC) and Low-Latency Monitoring indicators in the Top Deck.*

5.7.1 Scanning & Loading .clap & Audio Unit Plugins into the 8-Slot Rack

Cave Recorder maintains a persistent, machine-local plugin index (clap_plugin_index.json inside your user configuration directory) covering two studio plugin architectures:

Plugin Format	Supported Platforms	Standard System Discovery Locations & Identifiers

Plugin Format	Supported Platforms	Standard System Discovery Locations & Identifiers
CLAP (.clap)	macOS, Windows, Linux	**macOS:** /Library/Audio/Plug-Ins/CLAP and ~/Library/Audio/Plug-Ins/CLAP **Windows:** C:\Program Files\Common Files\CLAP and %LOCALAPPDATA%\Programs\Common\CLAP **Linux:** ~/.clap and /usr/lib/clap **Custom:** Any folder added via + ADD SEARCH FOLDER..., any file added via + ADD .CLAP FILE..., or directories listed in the CLAP_PATH environment variable. Indexed by canonical reverse-URI identifier (such as org.surge-synth-team.surge-xt-fx).
Audio Unit (AUv3 & AUv2)	macOS ('aufx' Audio Effects & 'aumf' Music Effects)	Synchronized automatically at macOS startup via the Audio Component registry across /Library/Audio/Plug-Ins/Components, ~/Library/Audio/Plug-Ins/Components, and registered macOS AUv3 application extensions. Indexed with canonical identifiers (au.<type>.<subtype>.<manufacturer>, such as au.aufx.dely.appl for AUDelay, or au.hex.<type>.<subtype>.<manufacturer> for non-alphanumeric four-character codes) and virtual audiounit://<type>/<subtype>/<manufacturer> bundle paths.

Fast Startup Validation & Out-of-Process Sandbox Protection on macOS

- **Zero-Delay Startup Validation & Auto-Discovery:** On application startup, Cave Recorder verifies the file modification timestamps of indexed .clap bundles on disk and automatically synchronizes all registered macOS Audio Unit ('aufx' and 'aumf') components via the system Audio Component registry without loading third-party dynamic libraries into memory. If a .clap bundle was moved or uninstalled while Cave Recorder was closed, its catalog entry is marked ! MISSING rather than deleted, preserving its metadata for non-destructive session recovery.
- **Out-of-Process Sandbox Protection (macOS Audio Units):** On macOS, Cave Recorder instantiates both modern AUv3 extensions and sandbox-safe AUv2 components in isolated background processes by default. Isolating third-party Audio Units in separate OS processes ensures that a faulty or unstable external plugin cannot crash your active multitrack recording session. When you open the floating editor window (PLUGIN GUI) of a legacy AUv2 plugin whose custom AUv2 graphical editor view requires direct in-process memory access, Cave Recorder automatically transitions that instance to in-process execution on the main thread while preserving its parameter state.

5.7.2 Indexing & Managing Plugins in the PLUGINS Workspace View (CMD / CTRL + 7)

Pressing **CMD** + **7** on macOS (**CTRL** + **7** on Windows and Linux) or clicking **PLUGINS** in Row 2 of the Top Transport Deck opens the dedicated **PLUGIN MANAGER & INDEXED CATALOG (CLAP & AU)** workspace view.

```
+=============================================================================================+
| PLUGIN MANAGER & INDEXED CATALOG (CLAP &
AU)                                          |
| [INDEXED: 14 | READY: 12 | MISSING: 1 | ERRORS: 1 | IN USE:
3]                        |
| [RESCAN DIRECTORIES]   [+ ADD SEARCH FOLDER...]   [+ ADD .CLAP
FILE...]              |
+---------------------------------------------------------------------------------------------
+
| STANDARD OS PATHS: /Library/Audio/Plug-Ins/CLAP  |  /Library/Audio/Plug-Ins/
Comp...  |
| CUSTOM FOLDERS: [/Volumes/StudioDSP/CLAP
[X]]                                         |
+---------------------------------------------------------------------------------------------
+
| [Search plugins...       ] [CLR] | [ALL] [AVAILABLE] [CLAP] [AUDIO UNIT]
[MISSING]   |
+---------------------------------------------------------------------------------------------
+
| STATUS    | NAME                | VENDOR | CATEGORY   | VER   | PLUGIN ID  |
ACTIONS |
+-----------+---------------------+--------+------------+-------+------------
+---------+
| [READY]   | AUDelay [IN USE x2] | Apple  | AU · Delay | 1.6.0 | au.aufx... |
[LOC..] |
| ! MISSING | VintageVerb         | Studio | CLAP · Rvb | 2.1.0 | com.stu... |
[LOC..] |
+=============================================================================================+
```

1. **Header Telemetry & Guided Discovery Actions:**
   - **Live Status Telemetry Pill:** Reports `INDEXED: N | READY: N | MISSING: N | ERRORS: N | IN USE: N` at the top of the panel (illuminated emerald when all indexed plugins are ready, or amber when any entry is missing or failed to probe).
   - **`RESCAN DIRECTORIES`:** Walks all standard OS `.clap` directories and configured custom search folders using an incremental modification-time check (probing only newly added or updated `.clap` bundles and reusing cached descriptors for unchanged bundles) and synchronizes all registered macOS `'aufx'` and `'aumf'` Audio Unit effects. Upon completion, an inline status banner summarizes `SCAN COMPLETE: N DIR(S) WALKED | N BUNDLE(S) DISCOVERED (N PROBED, N CACHED) | N READY, N MISSING, N ERROR(S)`.
   - **`+ ADD SEARCH FOLDER...`:** Opens a folder selection dialog (with an inline text-path fallback modal on headless systems) to register an additional `.clap` directory. Each custom folder appears in the search paths strip below the header as a recessed pill with an `X` button to remove the folder and prune its unreferenced entries.
   - **`+ ADD .CLAP FILE...`:** Prompts you to select and probe an individual `.clap` bundle file anywhere on disk without adding its entire parent directory.
2. **Searching, Filtering & Sorting the 8-Column Catalog Table:**
   - **Live Search (`Search plugins...`):** Filters the catalog in real time across plugin name, vendor, reverse-URI or `au.` plugin ID, version, description, primary category (`EQ`, `Compressor`, `Limiter`, `Gate / Expander`, `De-Esser`, `Transient`, `Reverb`, `Delay`, `Chorus`, `Flanger`, `Phaser`, `Saturation`, `Filter`, `Mastering`, `Analyzer`, `Restoration`, `Utility`, `Music Effect`), feature tags, bundle path, status (`ready`, `missing`, `error`), and Audio Unit format tags (`audio unit`, `audio-unit`, `auv2`, `auv3`).

- **Quick-Filter Pills:** Isolate **ALL**, **AVAILABLE** ([READY] plugins only), **CLAP** (.clap bundles only), **AUDIO UNIT** (macOS AUv2/AUv3 effects only), **MISSING / NEEDS RESOLUTION** (! MISSING and X ERROR rows), or **IN USE IN SESSION** (plugins currently assigned to one or more rack slots in the loaded session).
- **Sortable Columns & Hover Expansion:** Click any of the seven sortable column headers (**STATUS**, **NAME**, **VENDOR**, **CATEGORY**, **VERSION**, **PLUGIN ID**, **PATH**) to sort ascending or descending. Dwelling the mouse pointer over any truncated cell or folder pill for 180 ms expands the full identifier, path, or error reason in place.

3. **Per-Row Status Badges & Recovery Actions (ACTIONS Column):**
   - **[READY] (Emerald):** The .clap bundle or macOS Audio Unit is verified and ready to load into any rack slot. Plugins active in the current session display an emerald **[IN USE xN]** badge next to their name showing how many rack slots reference that plugin.
   - **! MISSING (Amber) & X ERROR (Crimson):** Indicates a .clap bundle that was moved or renamed on disk (Missing on disk (click LOCATE)), a macOS Audio Unit that is not currently registered (Missing Audio Unit (click RESCAN)), or a bundle that failed factory validation (with the diagnostic reason shown beneath the plugin name).
   - **LOCATE..., RESCAN & REMOVE:** Click **LOCATE...** to point Cave Recorder to a moved .clap bundle on disk (verifying that the selected bundle exports the expected plugin_id and immediately relinking any Ghost Slots in the active session), click **RESCAN** to re-probe that specific plugin entry, or click **REMOVE** to delete the entry from clap_plugin_index.json.

5.7.3 Selecting, Loading & Opening Plugins in the 8-Slot Rack (SLOT N PROCESSOR & PLUGIN GUI)

To load a built-in processor, third-party .clap plugin, or macOS Audio Unit effect into any channel or Subgroup Bus insert slot:

1. **Open the SLOT N PROCESSOR Selector Overlay:** Select the target Input Track (CH01..CH18) or click SEL on the target Subgroup Bus (SUB 1..SUB 16), then click any slot pill (Slots 1..8) in Upper Well 1 (MIX CONSOLE, **CMD / CTRL** + **1**) or in the left column of CHANNEL INSPECTOR (**CMD / CTRL** + **3**).
2. **Choose Between Built-In Studio Processors or CLAP / AU Plugin:**
   - Clicking any of the built-in studio processor rows (12-Band Dynamic EQ, VCA Compressor, Tape Saturation, Vocal De-Esser, Context-Aware Gate, Signal Repair, Guitar/Bass Amp Rig, 2P/4P Ladder LPF, Vocal Rider, Transient Shaper, Micro-Pitch Doubler, Gain & Polarity Utility, Studio Modulation, or Aux Send Tap) assigns that zero-latency native module to the slot, while Empty Slot clears the slot (---).
   - Clicking **CLAP / AU Plugin** (or clicking a slot pill that already hosts a plugin) expands the floating card to 348 px and reveals the **INDEXED PLUGIN CATALOG (CLAP & AU READY)** subpanel at the bottom of the selector menu.
3. **Search & Assign from the Indexed Catalog Subpanel:**
   - Type into the **Search indexed CLAP & AU plugins...** filter box (or click **CLR** to clear) to narrow the list by plugin name, vendor, category, or audio unit format tag.
   - If you have not indexed your plugins yet, click **MANAGE...** beside the search box to jump directly to the **PLUGINS** workspace tab (**CMD / CTRL** + **7**).

- Click any **[ ] [AU] Name (Vendor) [Category]** or **[ ] [CLAP] Name (Vendor) [Category]** row in the scrollable list ([x] indicates the plugin currently loaded in that slot). Cave Recorder instantiates the .clap or macOS Audio Unit effect, negotiates stereo 32-bit floating-point audio buses at the active hardware sample rate, and swaps the processor into the rack slot without interrupting transport playback.

4. **Open & Close Floating Native Plugin Editor Windows (PLUGIN GUI):**
   - Whenever a slot has an available .clap or Audio Unit plugin assigned, the header of the SLOT N PROCESSOR overlay displays a **PLUGIN GUI** toggle button (illuminated emerald when the editor window is open on screen).
   - **Cross-Platform CLAP Floating Windows:** For .clap plugins that implement the native plugin editor extension, clicking **PLUGIN GUI** opens a native floating desktop window on macOS, Windows, or Linux with live bidirectional parameter synchronization and plugin-initiated resize support.
   - **Three-Tier Native Floating GUI Hosting on macOS (Audio Units):** For hosted macOS Audio Units (AUv3 and AUv2), clicking **PLUGIN GUI** resolves the plugin's native editor view on the macOS main thread using a three-tier fallback chain and embeds it in a resizable native floating desktop window:
   - **Tier 1 (Modern AUv3 Custom Editor View):** Opens the modern AUv3 custom editor view, including sandboxed remote plugin views and bridged AUv2 view controllers.
   - **Tier 2 (Legacy AUv2 Custom Editor View):** Instantiates the custom AUv2 graphical editor view provided by legacy .component bundles.
   - **Tier 3 (Generic Parameter Editor Fallback):** If an Audio Unit does not provide a custom graphical view, Cave Recorder automatically constructs a macOS generic Audio Unit parameter editor so every Audio Unit parameter remains accessible.
5. **External Side-Chain Key Routing (SIDE-CHAIN KEY SOURCE):** When a hosted plugin declares a secondary side-chain input port (or dynamics/side-chain feature tags), you can open the **SIDE-CHAIN KEY SOURCE** dropdown to route either INTERNAL (Self) or any Input Track, Subgroup Bus, or Aux Return strip into the plugin's key input with selectable PRE, POST-EQ, or POST-FDR tap point and -24.0 dB to +24.0 dB side-chain gain trim.

5.7.4 Preset State Persistence & Non-Destructive Ghost Slot Recovery (! MISSING)

When you save a Session (*_session.json) or capture a Console Snapshot (SNAPS, **CMD** / **CTRL** + **S**), Cave Recorder serializes each populated plugin slot's complete state directly inside the session manifest:

- The canonical plugin_id (org.vendor.plugin or au.aufx.subt.manu), display label, vendor, version, and last-known bundle path hint.
- The slot's complete **saved plugin preset state** alongside the slot's parallel wet/dry mix ratio and bypass setting.

If you open a shared session on a second rehearsal or studio computer where a referenced .clap bundle or macOS Audio Unit is not yet installed (or was moved to a different disk folder), Cave Recorder **never strips the plugin or discards its preset settings**. Instead, it preserves the slot as a non-destructive **Ghost Slot**:

1. **Top-Level Missing Plugin Warning Banner:** A high-contrast crimson-and-amber warning banner appears at the top of the workspace (`! MISSING PLUGIN(S) | N PLUGIN(S) ACROSS M GHOST SLOT(S) BYPASSED (STATE PRESERVED)`), identifying the first missing plugin, vendor, `plugin_id`, strip name, and slot number, paired with three action buttons:
   - **LOCATE...:** Opens a file picker to locate the missing `.clap` bundle on disk and immediately relinks and activates every Ghost Slot referencing that `plugin_id`.
   - **OPEN PLUGINS:** Switches directly to the **PLUGINS** workspace view (**CMD** / **CTRL** + **7**) pre-filtered to **MISSING / NEEDS RESOLUTION**.
   - **X (Dismiss):** Hides the top warning banner for the remainder of the session while keeping the Ghost Slots safely bypassed in the rack.
2. **In-Slot Ghost Banner & Safe Audio Bypass:** Opening `SLOT N PROCESSOR` on a Ghost Slot displays an amber diagnostic banner (`! MISSING: <Name> (<Vendor>)` and `ID: <Plugin ID> | BYPASSED (STATE SAVED)`) with a header **LOCATE...** button. Audio passes cleanly through the bypassed Ghost Slot during playback and recording, and the saved preset state is written back untouched whenever the session or a new Snapshot is saved.
3. **One-Click Live Restoration:** The moment you relink the `.clap` bundle via **LOCATE...** or install the missing plugin and click **RESCAN DIRECTORIES** in the `PLUGINS` view, Cave Recorder instantiates the plugin, restores its saved preset state and original user bypass setting, and resumes live processing across all matching slots in the session.

5.7.5 Automatic Plugin Delay Compensation (PDC) & Transparency Breakdown

All built-in Cave Recorder processors (`TAPE`, `12B EQ`, `VCA CMP`, `DE-ESS`, `GATE`, `REPAIR`, `RIDER`, `AUX 1 REVERB`, `AUX 2 TAPE DLY`) operate with **zero internal sample latency (0 smp / 0.00 ms)**. However, when you insert a third-party `.clap` plugin or macOS Audio Unit (`AUv3` / `AUv2`) effect that uses internal lookahead buffering (such as linear-phase equalizers, spectral processors, or lookahead limiters), that plugin communicates its reported plugin latency in samples to Cave Recorder.

To prevent phase cancellation or timing drift between parallel tracks and subgroups, Cave Recorder performs **Automatic Plugin Delay Compensation (PDC)** across the entire mixing graph:

- The Top Deck displays a live **PDC: N smp (X.XX ms @ 48 kHz)** status pill at all times.
- Clicking the `PDC` pill in the Top Deck opens the **Latency & PDC Transparency Breakdown** popover (shown in Figure 5.4 above), which reports every latency stage in your studio system in both **Samples (smp)** and **Milliseconds (ms)**:
  - **Audio I/O:** Current hardware buffer size (`256 frames @ 48000 Hz`), input buffer latency (`In: 64 smp / 1.33 ms`), output buffer latency (`Out: 64 smp / 1.33 ms`), and calibrated hardware `Roundtrip I/O` (`640 smp / 13.33 ms`).
  - **Total Graph PDC** & **Effective Playback PDC:** Total delay compensation applied across parallel tracks and buses (`Max Safe Ceiling: 96,000 smp / 2000.00 ms`).
  - **Playhead Lag** & **Overdub Record-Offset:** Exact negative sample offset (`-640 smp / 13.33 ms`) applied automatically when writing new Overdub Passes (`Rec002-OD`) to disk so that everything a musician plays against playback lands sample-accurately on the timeline.

  - **Filterable Per-Channel Table (ALL, ARMED, COMPENSATED, BYPASSED):** Lists every Input Track (Track 1 [ARMED]..Track 18), Subgroup 1..2, Aux 1 (Reverb), Aux 2 (Delay), and Master Bus with its individual sample and millisecond compensation state.

5.7.6 Low-Latency Monitoring During Overdubs

When you are late in a mix and have inserted a high-latency lookahead mastering plugin or linear-phase EQ (.clap or macOS Audio Unit) on a track or Subgroup Bus, recording a last-minute vocal or guitar overdub through that high-latency plugin would introduce a distracting echo in the musician's headphones.

Clicking **Low-Latency Monitoring** in the Top Deck solves this immediately:

- Cave Recorder temporarily bypasses high-latency CLAP and Audio Unit plugins **only on record-armed (R) live monitoring paths** and their downstream buses, dropping the musician's headphone monitoring latency back down to pure zero-latency built-in console processing (0 smp graph PDC).
- All unarmed playback tracks continue playing with full Plugin Delay Compensation.
- As soon as you finish recording the Overdub Pass and disarm the track (or toggle Low-Latency Monitoring off in the Top Deck), the high-latency plugins re-engage automatically without losing a single plugin parameter setting.

5.8 Built-In Non-Destructive Audio Restoration (ADC De-Clipper & 12-Harmonic Hum Canceller)

In a high-energy rehearsal or live tracking session, a drummer will occasionally hit a rimshot 6 dB harder than during soundcheck and flat-top the hardware preamp's analog-to-digital converter (0.0 dBFS), or a vintage tube combo amp will pick up 50 Hz or 60 Hz power-line buzz in an older basement venue.

Instead of forcing you to export stems to an external audio restoration editor, Cave Recorder includes two non-destructive studio restoration tools built directly into the console and timeline:

1. **[REPAIR CLIPPING] (Two-Tier Smooth Spline & Spectral Waveform De-Clipper):**
   - Automatically detects contiguous flat-topped digital saturation bursts where the waveform hits the converter ceiling (default -0.1 dBFS threshold, user-adjustable from -12.0 dBFS to 0.0 dBFS for hard-clipped external preamps).
   - **Short Clips (1 to 8 samples):** Reconstructs the missing transient peak using smooth spline interpolation anchored to the unclipped amplitude, velocity, and curvature immediately preceding and following the clip burst.
   - **Medium & Long Clips (9 to 256 samples):** Fits a forward/backward harmonic waveform prediction model to the 512 clean context samples surrounding the clipped plateau, predicts the natural harmonic waveform trajectory forward and backward across the gap with raised-cosine crossfade blending, and restores the lost 3 dB to 6 dB transient dome cleanly.
2. **50 Hz / 60 Hz 12-Harmonic Comb Hum Canceller (Hum Canceller):**

- Power-line ground loops and neon/dimmer transformer buzz never sit solely at 50 Hz or 60 Hz. They generate a sharp comb of odd and even harmonics (60, 120, 180, 240, 300, 360, 420, 480, 540, 600, 660, 720 Hz) that sounds like an angry hornet nest inside guitar and bass tracks. Trying to cut all 12 harmonics with regular EQ bands wastes your entire parametric equalizer and hollows out the low-mid tone of the instrument.
- The dedicated **Hum Canceller** cascades **12 ultra-narrow constant-Q biquad notch filters (Q = 20.0 to 100.0, default Q = 60.0, bandwidth < 1.0 Hz at the fundamental)** locked to f_0, 2f_0, ..., 12f_0 (50.0 Hz European or 60.0 Hz North American mains, with fundamental fine-tuning from 45.0 Hz to 65.0 Hz and selectable Odd Harmonics Only vs. Odd + Even Harmonics). Because each notch is less than a couple of Hertz wide, 50/60 Hz buzz vanishes completely while 99% of the surrounding bass and guitar note warmth passes through untouched.

For step-by-step instructions on applying [REPAIR CLIPPING], range-bounded Pass comping, [PATCH FROM SOURCE...], and [SMOOTH OVER ARTIFACT] spectral spot-healing directly on the Waveform Timeline, see Section 6.5 in Chapter 6.

PART III | CHAPTER 6

Waveform Timeline, Pass Lanes, Markers & Ranges

2D waveform scrub-zooming, expandable Pass Lanes, non-destructive live-take mistake surgery, ADC clip repair, 50/60 Hz hum cancellation, range-bounded pass comping, clip gain, spectral spot-heal, markers, Take splitting, and the Multi-Camera Video Monitor and keyframe-snapped Camera Filmstrip (.mp4).

Every multitrack recording in Cave Recorder is a self-contained **Take** with its own timeline starting at `00:00:00.000` (`t = 0`). Rather than stringing an entire evening of rehearsals along a multi-hour horizontal scroll bar or stacking separate songs on top of one another, the timeline always focuses on **one active Take at a time**. Within that active Take, you can scrub and zoom sample-accurate waveforms, expand **Pass Lanes** to comp track-level overdubs, drop timestamped cue markers on the fly, loop rehearsal sections, and split long jams into individual song Takes.

6.1 Navigating the `WAVEFORM TIMELINE` View & Upper Mini-Timeline Well

Cave Recorder provides two synchronized timeline viewports for the currently active Take:

1. **Upper Mini-Timeline Well (`WAVEFORM TIMELINE` on the Mix Console):** Located in the right-hand well of the Upper Inspector Row on the `MIX CONSOLE` view (`CMD` / `CTRL` + `1`). It displays the waveform of the currently inspected track alongside quick-access transport and zoom keys so you can scrub, zoom, and switch Passes without leaving the faders.
2. **Full-Screen `WAVEFORM TIMELINE` Workspace (`CMD` / `CTRL` + `2`):** Displays all active tracks simultaneously with dedicated Marker (`MKR`) and Range (`RNG`) lanes, per-track headers, expandable Pass Lanes, and a two-row editing toolbar.

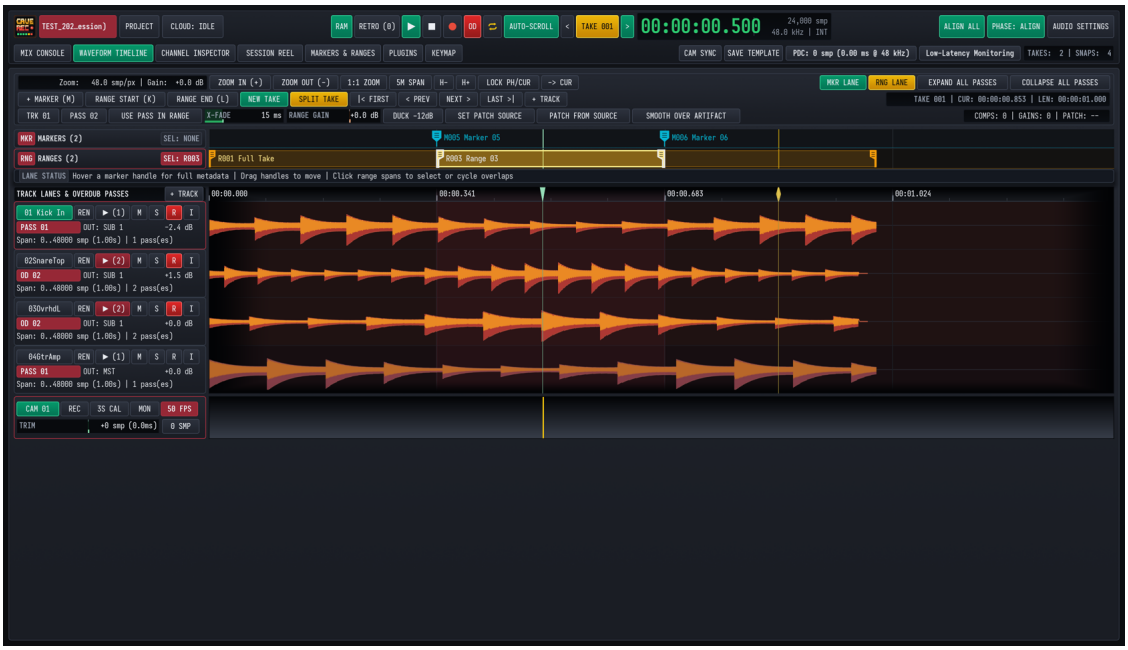


Figure 6.1: *The full-screen Waveform Timeline view showing multi-track waveforms, ruler markers, and transport controls.*

6.1.1 Multi-Resolution Waveform Detail & Viewport Readouts

Both the Upper Mini-Timeline Well and the full-screen WAVEFORM TIMELINE view render audio using multi-resolution waveform detail, covering full-song overview envelopes, medium phrase zoom, and individual sample-level waveform display. As you zoom from a 30-minute rehearsal jam down to a single snare transient, the display transitions across pre-computed resolution tiers so transient peaks remain sharp at every scale without frame drops.

The upper-left readout of the WAVEFORM TIMELINE toolbar displays the current horizontal resolution in samples per pixel (Zoom: 48.0 smp/px) and the visual waveform gain offset (Gain: +0.0 dB), while the right-hand status readout confirms the active Take number, cursor timestamp, and total Take duration (TAKE 001 | CUR: 00:00:00.853 | LEN: 00:00:01.000).

6.1.2 Upper Mini-Timeline Well Controls

At the bottom of the Upper Mini-Timeline well on the Mix Console (Chapter 4: Mix Console, Signal Flow & Channel Strips), six dedicated hardware buttons control the inspected track's timeline:

Control	Keyboard Shortcut	Function
PASSES	**E**	Expands or collapses the **Pass Lanes** for the currently inspected channel so you can switch or mute individual overdub passes directly from the console.

Control	Keyboard Shortcut	Function
LOCK PH/CUR	Configurable in KEYMAP	Toggles locking the **Playhead** (PH, green triangle) and the **Position Cursor** (CUR, amber diamond) together. When locked, clicking to place the cursor moves the playback start position simultaneously. When unlocked, the Position Cursor can be placed independently as an edit or zoom anchor while playback continues.
-> CUR	`.`	Snaps the transport Playhead immediately to the current Position Cursor location.
ZOOM +	`=`	Zooms horizontally into the waveform by one octave (halving the smp/px span).
ZOOM -	`-`	Zooms horizontally out of the waveform by one octave (doubling the smp/px span).
ZOOM 1:1	`0`	Fits the active Take to the viewport width (or resets to the default 1:1 overview span).

6.1.3 Full-Screen WAVEFORM TIMELINE Toolbar & Track Headers

Pressing `CMD` / `CTRL` + `2` (or clicking WAVEFORM TIMELINE in the navigation bar) opens the multi-track timeline workspace. Above the waveform canvas, two toolbar rows provide direct access to zoom, lane visibility, marker placement, and Take splitting:

- **Row 1 (Viewport & Lane Controls):**
  - **ZOOM IN (+) / ZOOM OUT (-) / 1:1 ZOOM:** Step horizontal zoom in (`=`), step out (`-`), or reset horizontal zoom to the default 1:1 span (`0`).
  - **5M SPAN:** Sets the visible horizontal window to a standard 5-minute (300.0 s) rehearsal span. New sessions and live recordings initialize to this 5-minute span automatically.
  - **H- / H+ (Track Lane Height):** Steps the vertical height of all track waveform lanes between 56 px (compact 18-channel overview) and 220 px (tall surgical transient inspection). Track header heights on the left stay locked 1:1 with their waveform lanes.
  - **LOCK PH/CUR & -> CUR:** Identical to the Mini-Timeline well controls, allowing coupled or decoupled Playhead and Position Cursor operation.
  - **MKR LANE & RNG LANE:** Toggle visibility of the dedicated MARKERS and RANGES ruler lanes above the audio tracks.
  - **EXPAND ALL PASSES & COLLAPSE ALL PASSES:** Unrolls (`E`) or collapses (`SHIFT` + `E`) the overdub **Pass Lanes** across every track in the active Take simultaneously.
- **Row 2 (Markers, Ranges & Take Split Bar):**
  - **+ MARKER (M) / RANGE START (K) / RANGE END (L):** Drops a cue marker at the Playhead (`M`) or defines the start (`K`) and end (`L`) boundaries of a loop or song range.
  - **NEW TAKE & SPLIT TAKE:** Extracts the active range as a new self-contained Take (NEW TAKE) or divides the active Take into two consecutive Takes at the current Playhead/Cursor position (SPLIT TAKE).

- **I< FIRST / < PREV / NEXT > / LAST >I:** Jumps the Playhead to the first marker (**SHIFT** + **[** or **HOME**), previous marker (**[**), next marker (**]**), or final marker (**SHIFT** + **]** or **END**) in the active Take.

Each row in the left-hand **TRACK LANES & OVERDUB PASSES** column displays the track number and scribble-strip name (01 Kick In), the Pass Lane disclosure button (▶ (1) or ▶ (2) showing the number of recorded Passes on that track in the current Take), **Mute (M)**, **Solo (S)**, **Record Arm (R)**, and **Input Monitor (I)** switches, the active Pass badge (PASS 01 or OD 02), the output bus assignment (OUT: SUB 1 or OUT: MST), the channel fader readout (-2.4 dB), and the sample span summary (Span: 0..48000 smp (1.00s) | 2 pass(es)).

6.1.4 Complete Reference of Timeline Navigation & Zoom Gestures

Cave Recorder supports fast one-handed **2D Scrub-Zooming** directly on the timeline ruler and waveform canvas, alongside standard mouse wheel, trackpad, and keyboard navigation:

Gesture / Action	Input / Shortcut	Behavior
Reposition Playhead / Cursor	**Single-Click** (< 4 px travel) on ruler or waveform canvas	Places the crisp 1 px Playhead or Position Cursor at the clicked timestamp. Automatically snaps to Take start (00:00:00.000) and Take end boundaries within 8 px.
Bypass Boundary Snapping	Hold **SHIFT** or **CMD** / **CTRL** + **Click**	Places the Playhead or Position Cursor at the exact sub-sample mouse coordinate without snapping to Take boundaries.
2D Ruler / Canvas Scrub-Zoom	**Click + Drag Vertically** (up/down) on ruler or canvas	Continuously zooms the timeline exponentially in or out anchored directly to the sample beneath the cursor (or around the Playhead when LOCK PH/CUR is active).
2D Ruler Scrub / Canvas Pan	**Click + Drag Horizontally** (left/right) on ruler or canvas	Dragging horizontally on the ruler scrubs the Playhead across the Take. Dragging horizontally on the waveform canvas pans the visible time window left or right.
Nudge Timeline Cursor	**LEFT** / **RIGHT** (hold **SHIFT** for 10x step)	Nudges the Position Cursor (and the Playhead when LOCK PH/CUR is active) backward or forward by a fine or coarse sample step.
Horizontal Scroll Pan	**Two-Finger Horizontal Trackpad Scroll** or **SHIFT** + **Mouse Wheel**	Smoothly pans the visible viewport backward or forward across the active Take.
Modifier Scroll Zoom	**CMD** / **CTRL** + **Mouse Wheel**	Zooms horizontally in or out anchored at the current mouse pointer position.
Step Zoom In / Out	ZOOM IN (+) (**=**) / ZOOM OUT (-) (**-**)	Zooms horizontally in or out by one octave per press.

Gesture / Action	Input / Shortcut	Behavior
Reset Zoom (1:1)	1:1 ZOOM / ZOOM 1:1 (**0**)	Resets horizontal zoom and scroll to the default 1:1 overview span.
Toggle Timeline Auto-Scroll	AUTO-SCROLL in Top Deck (configurable in KEYMAP)	Follows the advancing Playhead automatically during playback and live recording, paging the viewport forward whenever the Playhead reaches the right edge. Manually panning during playback temporarily suspends following until re-engaged.
Snap Playhead to Cursor	-> CUR (**.**)	Moves the green transport Playhead to the amber Position Cursor immediately.

> **ZOOMING AROUND THE PLAYHEAD VS. MOUSE CURSOR (LOCK PH/CUR)**
> Enable **LOCK PH/CUR** when you want keyboard zoom keys (**=** / **-**) and vertical scrub-zoom drags to stay centered on the rolling Playhead during playback. Disable **LOCK PH/CUR** when editing stationary waveforms so zooming stays pinned to the exact transient under your mouse cursor or Position Cursor.

6.2 Recording Overdubs (OD) & Comping in Expandable Pass Lanes (PASSES)

In Cave Recorder, session-level songs and track-level overdubs are strictly separated so your timeline never becomes cluttered:

- **Base Pass (Rec001, labeled PASS 01):** When **Overdub Mode (OD)** is off (the default), pressing **Record** (**R**) creates a brand-new session **Take** starting at 00:00:00.000. Every armed channel strip records its initial multitrack stem as Rec001 (PASS 01).
- **Overdub Pass (Rec002-OD, Rec003-OD, ..., labeled OD 02, OD 03, ...):** When you want to layer a second guitar solo, punch in a vocal fix, or try three alternate bass lines inside the **currently active Take**, latch the red **OD** button in the Top Deck transport bar (**O**) and arm only the target track(s). Pressing **Record** (**R**) plays back the existing tracks of the active Take from 00:00:00.000 while recording a new **Overdub Pass** (OD 02, OD 03, ...) into the active Take's rec/ folder and appending a new **Pass Lane** under each armed track.



Figure 6.2: *The Waveform Timeline with Pass Lanes expanded on Snare Top and Overhead L tracks for multi-pass comping.*

6.2.1 Step-by-Step: Recording Multi-Pass Overdubs on a Track

1. **Load the Target Take:** Use the < / > buttons beside TAKE 001 in the Top Deck (**ALT** + **[** / **ALT** + **]**) or click LOAD TAKE in the SESSION REEL view (**CMD** / **CTRL** + **4**) to load the Take you want to overdub onto.
2. **Disarm Backing Tracks & Arm the Overdub Track:** On the Mix Console or in the left track header column of the WAVEFORM TIMELINE view, disarm (R off) the backing band tracks (drums, rhythm section) so they stream from disk, and arm (R lit crimson) only the channel(s) you are overdubbing. Enable input monitoring (I / MON) if you are monitoring through Cave Recorder's console EQ and compression.
3. **Latch Overdub Mode (OD):** Click **OD** in the Top Deck transport bar (or press **O**) so the OD indicator illuminates red.
4. **Optional Pre-Roll or Cycle Range:** Position the Playhead a few bars before the section you want to overdub, or enable Cycle Loop (⇄, **C**) across a defined range (Section 6.3).
5. **Record the Overdub Pass:** Press **Record** (**R**) to begin playback and capture OD 02. Press **Stop** (**SPACE** or **SHIFT** + **SPACE**) when finished. Cave Recorder automatically promotes the newly recorded Overdub Pass (★ OD 02) as the active playback Pass on that track and captures a post-recording Snapshot.

> **AUTOMATIC OVERDUB LATENCY COMPENSATION**
> When you record an Overdub Pass (OD 02), Cave Recorder automatically compensates the recorded Pass Lane using your calibrated hardware loopback offset and live monitoring latency (displayed in the Pass Lane header as -640 smp (-13.3ms)). Backing tracks stream from disk with negative read-ahead compensation so your new overdub lands phase-aligned with the original PASS 01 drum and rhythm stems. See Chapter 2: Hardware Setup, Audio Connections & First Session for loopback calibration details.

6.2.2 Expanding Pass Lanes, Auditioning Passes & Comping Phrases

When a track within the active Take has more than one recorded Pass, the disclosure badge in the track header displays the total Pass count (for example, ▶ (2) on 02 Snare Top and 03 Overhead L).

- **Expand or Collapse a Single Track's Pass Lanes:**
  - Click the **▶ (N)** / **▼ (N)** badge in the track header on the WAVEFORM TIMELINE view.
  - Or select the track and click **PASSES** in the Upper Mini-Timeline Well (or press **E** while in the CHANNEL INSPECTOR view).
- **Expand or Collapse All Tracks' Pass Lanes:**
  - Click **EXPAND ALL PASSES** (**E**) or **COLLAPSE ALL PASSES** (**SHIFT** + **E**) in the upper-right WAVEFORM TIMELINE toolbar.

When expanded, each track unrolls its vertical stack of **Pass Lanes** in chronological order (BASE PASS PASS 01 at the top, followed by OVERDUB OD 02, OD 03, and so on):

- **Visual Active vs. Standby Distinction:** The currently active Pass Lane (labeled ACTIVE COMP LANE with a green **★ PASS 01** or **★ OD 02** badge) renders its waveform in bright amber-copper. Inactive Pass Lanes (labeled STANDBY COMP LANE with a **Use PASS 01** or **Use OD 02** button) render in muted slate-grey so you can immediately see which Pass is feeding the Mix Console channel strip.
- **Auditioning & Promoting Passes:**
  - **Promoting a Pass on the WAVEFORM TIMELINE View:** Click **Use PASS 01** (or **Use OD 02**, **Use OD 03**) in any standby Pass Lane header to promote that Pass as the active playback lane (ACTIVE COMP LANE) immediately. Because playback readers pre-buffer ahead of the Playhead, switching Passes during playback crossfades cleanly without clicks.
  - **Promoting a Pass from the MIX CONSOLE or CHANNEL INSPECTOR View:** Click **PASSES** in the Upper Mini-Timeline Well (or press **E** in the CHANNEL INSPECTOR view) to expand the inspected channel's Pass Lanes, and click **Use PASS <NN>** or **Use OD <NN>** on the Pass you want to hear.
- **Per-Pass Mute (M) & Soft Delete (Del):**
  - Click **M** on a Pass Lane to silence that Pass without changing your channel strip Mute state.
  - Click **Del** on a flawed Pass Lane to mark that Pass as deleted in the session history. Because deletion is non-destructive, pressing **CMD** / **CTRL** + **Z** restores the Pass immediately.
- **Comping Phrases Across Multiple Passes:**
  - When an overdub was punched in over a specific chorus or solo section (for example, Span: 240000..480000 smp), promoting that Overdub Pass plays your new performance across its recorded span while allowing you to compare it directly against the underlying PASS 01 Base Pass or alternate OD 03 passes. Every Pass promotion is recorded in the Undo history and stored in Console Snapshots (Chapter 7: Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync) so you can save and compare multiple comp combinations across the session.

6.3 Dropping Cue Markers & Defining Cycle Loop Ranges (MARKERS & RANGES)

During a live rehearsal jam or playback review, you can drop timestamped **Markers** and define bounded **Ranges** without stopping the transport. Markers and Ranges appear both as interactive flags in the MKR and RNG lanes of the WAVEFORM TIMELINE view and as sortable, searchable rows in the dedicated **MARKERS & RANGES** workspace (**CMD** / **CTRL** + **5**).

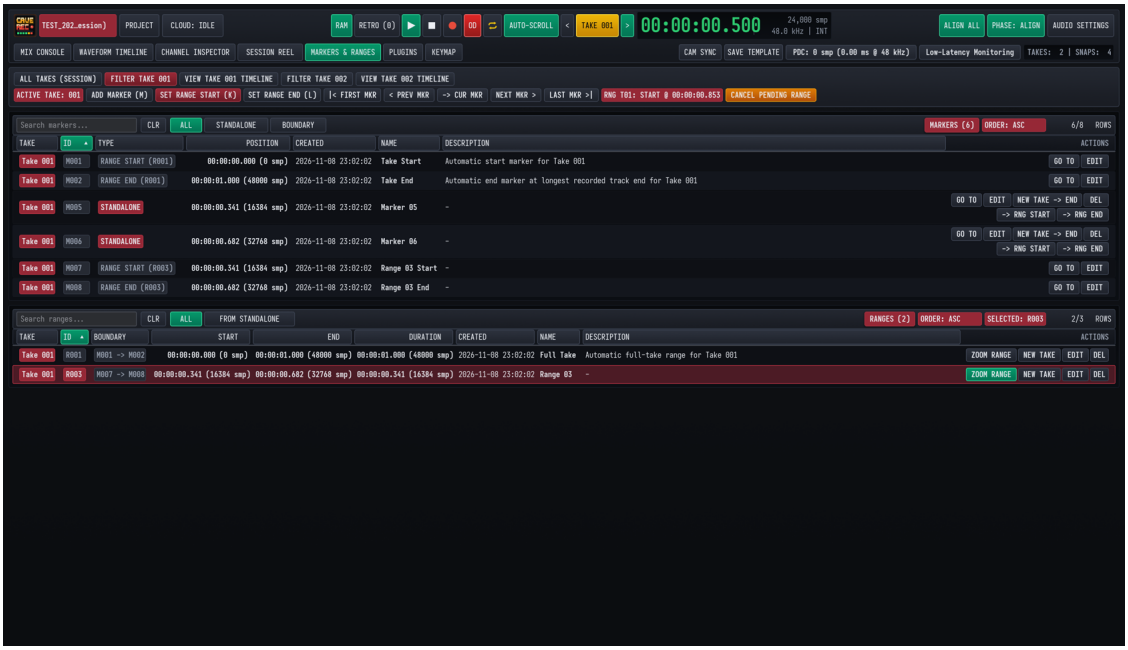


Figure 6.3: *The Markers and Ranges management view showing timestamped cue markers, loop range boundaries, and Take Split tools.*

6.3.1 Dropping & Navigating Cue Markers on the Fly (M)

Every Take automatically initializes with two boundary markers (M001 Take Start at 00:00:00.000 and M002 Take End at the end of the longest recorded stem, forming the default R001 Full Take range). To mark song transitions, solo entrances, or mix notes:

1. **Drop a Marker (`M`):** Press **`M`** at any time during live recording or playback (or click **+ MARKER (M)** in the WAVEFORM TIMELINE toolbar or **ADD MARKER (M)** in the MARKERS & RANGES toolbar). Cave Recorder places a cyan standalone marker flag (M005 Marker 05, M006 Marker 06) at the exact sample position of the rolling Playhead (or at the Position Cursor when transport is stopped).
2. **Reposition a Marker by Dragging:** In the MKR MARKERS lane of the WAVEFORM TIMELINE view, hover over any cyan marker flag to inspect its metadata, or click and drag the marker handle left or right to fine-tune its sample timestamp.
3. **Jump Between Markers:** Use the toolbar buttons or keyboard shortcuts to navigate between cue points instantly:
   - **Previous Marker (< PREV / < PREV MKR):** Press **`[`**.
   - **Next Marker (NEXT > / NEXT MKR >):** Press **`]`**.
   - **First Marker in Take (|< FIRST / |< FIRST MKR):** Press **SHIFT** + **`[`** or **HOME**.
   - **Last Marker in Take (LAST >| / LAST MKR >|):** Press **SHIFT** + **`]`** or **END**.

6.3.2 Defining Ranges (K / L) & Cycle Loop Playback (⇄)

A **Range** (R001, R003, …) is a named time span bounded by a RANGE START marker and a RANGE END marker. Ranges are used for **Cycle Loop playback**, **overdub punch/loop rehearsal**, **zooming to a song section**, and **extracting a song section as a new Take**.

1. **Mark Range Start (RANGE START (K)):** Move the Playhead or Position Cursor to the beginning of the phrase (or press it live as the band hits the downbeat) and press **K** (or click **RANGE START (K)** / **SET RANGE START (K)**). An amber status pill (RNG T01: START @ 00:00:00.853) appears in the toolbar indicating that a range start boundary is pending.
2. **Mark Range End (RANGE END (L)):** When playback reaches the end of the section, press **L** (or click **RANGE END (L)** / **SET RANGE END (L)**). Cave Recorder pairs the start and end boundaries, creates a new numbered Range span (such as R003 Range 03) in the amber RNG RANGES lane, and selects it as the active loop range.
3. **Cancel a Pending Boundary (ESC):** If you pressed **K** by mistake, press **ESC** or click **CANCEL PENDING RANGE** to discard the uncompleted boundary.
4. **Select, Drag, or Cycle Overlapping Ranges:** In the RNG RANGES lane above the waveforms, click any amber range bar to make it the active range (clicking repeatedly on overlapping spans cycles through them). Drag the left [or right] handle of any range bar in the RNG lane to adjust its start or end sample directly on the timeline.
5. **Toggle Cycle Loop Playback (⇄):** Click the Cycle Loop button ⇄ in the Top Deck transport bar (or press **C**). When ⇄ is illuminated amber, playback loops continuously between the start and end boundaries of the active Range (or across R001 Full Take if no sub-range is selected).

6.3.3 Managing, Renaming & Filtering in the MARKERS & RANGES View

Pressing **CMD** / **CTRL** + **5** (or clicking **MARKERS & RANGES** in the workspace navigation bar) opens the **MARKERS & RANGES** management workspace, divided into two synchronized tables: **MARKERS** (top) and **RANGES** (bottom).

- **Session-Wide vs. Per-Take Filtering:** Use the top filter bar (ALL TAKES (SESSION), FILTER TAKE 001, FILTER TAKE 002) to view cue points across every Take in the evening's Session at once, or isolate markers belonging to a single Take. Clicking **VIEW TAKE 001 TIMELINE** jumps directly to WAVEFORM TIMELINE with that Take loaded.
- **Marker Type Filtering & Automatic Color Badges:** Filter the upper table by **ALL**, **STANDALONE** (crimson STANDALONE badges, such as M005), or **BOUNDARY** (slate RANGE START and RANGE END badges paired to a Range, such as RANGE START (R003)), or type into Search markers... to filter by marker name or description note.
- **Per-Marker Actions (GO TO, EDIT, -> RNG START, -> RNG END, DEL):**
  - **GO TO:** Loads the owning Take (if not already active) and snaps the transport Playhead directly to that marker's exact sample position.
  - **EDIT:** Opens inline editing fields to update the marker's **Name** (for example, Verse 2 Harmonies) and free-form **Description** note (SAVE M<NNN> / CANCEL), or click **PRESET MARKER HIGHLIGHT M<NNN>** to emphasize key cue points.

  - **-> RNG START / -> RNG END:** Promotes an existing standalone cue marker into the start or end boundary of a Range so you can drop single-key **M** markers during a live jam and turn them into bounded song ranges afterward.
  - **DEL:** Deletes the marker (reversible via Undo **CMD / CTRL** + **Z**).
- **Range Filtering (ALL / FROM STANDALONE) & Per-Range Actions (ZOOM RANGE, NEW TAKE, EDIT, DEL):**
  - **FROM STANDALONE Filter Pill:** Filters the RANGES table (alongside **ALL**) to show only Ranges whose start or end boundaries were promoted from standalone cue markers via -> RNG START / -> RNG END.
  - **ZOOM RANGE:** Switches to the WAVEFORM TIMELINE view and zooms the horizontal viewport to frame that exact Range from edge to edge.
  - **EDIT:** Opens inline fields to rename the Range (for example, R003 Chorus Loop or Song 2 - Midnight Drive) and update its description.

6.4 Non-Destructive Take Splitting (Split Take at Playhead / Split by Markers)

During a band rehearsal, musicians often hit **Record** once and play a continuous 25-minute block containing warm-up banter, three separate songs, and an improvised jam. Cave Recorder lets you carve that long continuous capture into independent, self-contained **t = 0 Takes** (Take 002, Take 003, Take 004) so each song gets its own 00:00:00.000 timeline, its own song tag, and its own stereo mixdown.

6.4.1 Three Ways to Split or Extract Independent Takes

Cave Recorder provides three non-destructive workflows depending on whether you want to divide a Take in two or extract a specific marker/range span:

Split / Extract Method	Where to Trigger	How It Works
1. Split Take at Playhead / Cursor (SPLIT TAKE)	Click **SPLIT TAKE** in the WAVEFORM TIMELINE toolbar	Divides the active Take at the current Playhead/Cursor timestamp into two consecutive segments: the first part (00:00:00.000 to the split point) remains in the current Take, and the second part (from the split point to the end) becomes a brand-new t = 0 Take (Take<NNN>).
2. Extract from Marker to End (NEW TAKE -> END)	Click **NEW TAKE -> END** on any standalone marker row in the MARKERS & RANGES view (**CMD / CTRL** + **5**)	Extracts everything from that marker's sample timestamp through the end of the Take into a brand-new t = 0 Take (Take<NNN>), leaving the source Take's full duration untouched.

Split / Extract Method	Where to Trigger	How It Works
3. Extract Bounded Range as New Take (NEW TAKE)	Click **NEW TAKE** in the WAVEFORM TIMELINE toolbar (with a Range selected) or click **NEW TAKE** on any row in the RANGES table (**CMD** / **CTRL** + **5**)	Copies the exact sample span between the Range's START and END boundaries (R003) into a brand-new t = 0 Take (Take<NNN>) whose 00:00:00.000 origin corresponds to the start of that Range, leaving the source Take's span untouched.

6.4.2 How Non-Destructive Splitting Preserves Stems, Passes & Phase Offsets

When you divide a Take with SPLIT TAKE or extract a span with NEW TAKE:

1. **Zero Destructive File Overwrites & Full Undo:** None of your original .wav files on disk are ever overwritten or deleted. Both SPLIT TAKE and NEW TAKE are tracked in the Session's Undo history, so pressing **CMD** / **CTRL** + **Z** reverses the split or extraction immediately.
2. **Sample-Zero (t = 0) Alignment:** Every multitrack stem in the newly created Take (Take<NNN>) starts at its own 00:00:00.000 timeline zero aligned to the split or range start point, with Broadcast Wave (bext) timecode metadata preserved.
3. **Pass Structure & Console Routing Preserved:** All recorded **Base Passes** (Rec001) and **Overdub Passes** (Rec002-OD) that overlap the extracted span are preserved in their respective **Pass Lanes**, along with per-track phase alignment offsets, latency compensation metadata, and internal markers shifted relative to the new 00:00:00.000 origin.
4. **Immediate Session Reel Registration:** The newly created Take appears immediately in the SESSION REEL catalog (Chapter 7: Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync) ready for tagging, live Print conversion, or offline Mixdown rendering.

6.5 Non-Destructive Live-Take Salvage & Timeline Mistake Surgery

When a full band plays live in one room and captures a magic Take where the groove, tempo push, and collective energy are unrepeatable, you should never have to throw away that performance because of a minor technical or musical blemish: a snare rimshot that clipped the preamp converter for 14 samples, a vintage guitar combo that buzzed at 60 Hz, a lead vocal phrase that jumped out 6 dB too loud in the bridge, a flubbed chord in bar 38, or a vocalist bumping the microphone stand during an acoustic intro.

Cave Recorder provides six **100% non-destructive Live-Take Salvage and Mistake Surgery tools** directly inside the WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**) and CHANNEL INSPECTOR (**CMD** / **CTRL** + **3**). Every operation preserves your original 32-bit floating-point .wav files untouched on disk and is tracked in the Session's Undo/Redo history (**CMD** / **CTRL** + **Z**):

```
+=================================================================+
|     NON-DESTRUCTIVE LIVE-TAKE SALVAGE & MISTAKE SURGERY SUITE   |
+=================================================================+
  [1] [REPAIR CLIPPING] (Two-Tier Smooth Spline & Spectral De-Clipper)
      * Reconstructs flat-topped ADC converter clips (1..256 samples)
  -----------------------------------------------------------------
  [2] 50/60 Hz 12-HARMONIC COMB HUM CANCELLER (Q = 20..100 Notch)
      * Strips f_0 .. 12*f_0 mains & dimmer buzz without dulling tone
  -----------------------------------------------------------------
  [3] RANGE-BOUNDED PASS COMPING ([USE OD 02 IN RANGE])
      * Punches an Overdub Pass into a selected Range with 5..50 ms
        equal-power sin/cos crossfades (zero clicks, zero file edits)
  -----------------------------------------------------------------
  [4] PRE-INSERT CLIP GAIN ENVELOPE (-24.0 dB to +24.0 dB)
      * Normalizes a single loud phrase or quiet intro BEFORE the
        8-slot rack so compressors & tape saturators hit evenly
  -----------------------------------------------------------------
  [5] [PATCH FROM SOURCE...] (Same-Take or Cross-Take Bar Graft)
      * Copies a clean chord or drum fill from another bar (or another
        Take) into the selected Range with phase-locked crossfades
  -----------------------------------------------------------------
  [6] [SMOOTH OVER ARTIFACT] (5..150 ms Spectral Spot-Heal)
      * Erases a mic stand bump, cable pop, lip smack, or stick click
        using bidirectional waveform & spectral synthesis
+=================================================================+
```

6.5.1 [REPAIR CLIPPING]: Reconstructing Clipped Preamp Transients

If an enthusiastic snare hit or vocal shout overloaded the hardware audio interface's analog-to-digital converter (0.0 dBFS) during tracking, red clip markers appear along the top edge of the waveform in WAVEFORM TIMELINE:

1. **Select the Clipped Region (or Full Track):** Frame the clipped transient using a Range (K and L) or leave no sub-range selected to scan the entire active Pass on the selected track.
2. **Click [REPAIR CLIPPING]:** Cave Recorder locates every contiguous flat-topped sample plateau where |x[n]| >= Clip Threshold (default -0.1 dBFS, adjustable from -12.0 dBFS to 0.0 dBFS if an external analog preamp or pedal clipped below digital full scale).
3. **Two-Tier Waveform Reconstruction:**
   - **Short Clips (1 to 8 samples):** Rebuilds the rounded peak using smooth spline interpolation matched to the entry and exit slopes of the waveform.
   - **Medium & Long Clips (9 to 256 samples):** Fits a forward/backward harmonic waveform prediction model to the 512 clean context samples before and after the clip, predicts the harmonic waveform forward and backward across the clipped gap with a raised-cosine blend, and writes the repaired region to a companion repair layer while keeping the original Rec001.wav stem intact.

6.5.2 50 Hz / 60 Hz 12-Harmonic Comb Hum Canceller

When a single-coil guitar pickup, vintage bass amplifier, or unshielded stage cable injects 50 Hz or 60 Hz power-line buzz into a great live Take, enable the **Hum Canceller** on that track (50 Hz or 60 Hz fundamental selector, 45.0..65.0 Hz fine tune, 12 Harmonics, Q = 20.0..100.0, default Q = 60.0). Twelve ultra-narrow constant-Q notch filters remove the fundamental and its first 11 overtones (60 Hz through 720 Hz) without carving a wide hole out of your bass or guitar low-mids.

6.5.3 Range-Bounded Pass Comping ([USE OD 02 IN RANGE])

In Section 6.2, clicking Use OD 02 switches the entire track from start to finish to Overdub Pass OD 02. Frequently, however, the original live PASS 01 (Rec001) is magical for 95% of the song and you only recorded a quick Overdub Pass (OD 02) to fix a single missed lyric or guitar clam in Verse 2:

1. Expand the track's Pass Lanes with **E** in WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**).
2. Mark the start and end of the phrase to replace using **K** (RANGE START) and **L** (RANGE END).
3. Click **[USE OD 02 IN RANGE]** (or the desired standby Pass row inside the bounded Range).
4. Cave Recorder keeps PASS 01 active across the rest of the song and cleanly splices OD 02 into that exact Range using a phase-accurate **5 ms** to **50 ms** **equal-power (sin/cos) crossfade** at both boundaries.

6.5.4 Pre-Insert Clip Gain Envelope (-24.0 dB to +24.0 dB)

If a guitarist kicked on a solo boost pedal that jumped out +7 dB too loud in the bridge, lowering the channel fader *after* the 8-slot rack will not stop the channel's TAPE saturator and VCA CMP compressor from being slammed into heavy distortion during that bridge.

Instead, select the Range (K .. L) on the waveform and drag the **Pre-Insert Clip Gain Handle** (-24.0 dB to +24.0 dB, with smooth 5 ms entry/exit ramps). Because Clip Gain is applied directly to the disk playback reader **before** the 8-slot processor rack, pulling the loud bridge down by -6.0 dB on the waveform ensures the track hits your channel EQ, tape drive, and compressor at the exact same sweet-spot level as the rest of the song.

6.5.5 [PATCH FROM SOURCE...]: Grafting a Clean Bar or Chord Across Bars or Takes

What if the bass player or rhythm guitarist hit one wrong chord in the second chorus of a live Take, and nobody noticed until mixing day when the band has already packed up their gear?

Because the band played the exact same chord progression cleanly in Chorus 1 (or in another Take from the same rehearsal session), you can repair the mistake using **[PATCH FROM SOURCE...]**:

1. **Select the Flawed Target Range:** In WAVEFORM TIMELINE, mark the 1-bar or 1-chord mistake span using **K** and **L** on the target track.

2. **Click [PATCH FROM SOURCE…] and Choose the Clean Source Bar:** Select an earlier or later timestamp from the same Take (for example, Chorus 1) or from another Take in the Session.
3. **Automatic Transient & Phase-Correlation Micro-Alignment (±50 ms):** Even when a band plays without a click track, Cave Recorder automatically slides the source audio snippet within a ±50 ms window to maximize waveform cross-correlation and zero-crossing phase alignment at both the entry and exit boundaries, welding the clean chord into place with adjustable 10 ms to 100 ms equal-power crossfades.

6.5.6 [SMOOTH OVER ARTIFACT]: Spectral & Waveform Spot-Heal (5 ms to 150 ms)

During an intimate acoustic intro or quiet vocal breakdown, a stray mechanical noise (a vocalist bumping the mic stand, a guitar cable pop, a loud lip smack, or an accidental drumstick click on a rim) can momentarily distract from the performance. Simple muting leaves an unnatural hole in the room ambience.

To erase a short transient blemish (5 ms to 150 ms) without leaving a gap:

1. Zoom in (ZOOM + or vertical drag) and bracket the 5 ms to 150 ms artifact with RANGE START (K) and RANGE END (L).
2. Click **[SMOOTH OVER ARTIFACT]** in the timeline toolbar.
3. Cave Recorder analyzes the 250 ms of clean room tone and sustained note harmonics immediately before and after the artifact:
   - For short clicks and pops (5 ms to 25 ms), it performs **bidirectional harmonic waveform interpolation** across the gap.
   - For longer thumps or stand bumps (25 ms to 150 ms), it performs **Windowed Spectral Magnitude & Phase Coherence Interpolation**, replacing only the transient anomaly with continuous harmonic sustain and natural room ambience while leaving the rest of the track untouched.

6.6 Multi-Camera Video Monitor & Timeline Filmstrip (video/*.mp4 & .vididx)

When one or more studio cameras or wireless **Cave Companion** mobile camera nodes are armed during a Take, Cave Recorder displays each camera angle as a sample-synchronized **Video Timeline Strip** lane inside WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**) paired with a real-time **Multi-Camera Video Monitor**.

6.6.1 48.0 kHz Audio-Clock Synchronization & the .vididx Frame Index

To prevent audio/video drift across 30-minute rehearsal jams and ensure instant frame-accurate scrubbing without scanning multi-gigabyte .mp4 container files, every recorded camera file (<TakeFolder>/video/<SessionID>_Take<NNN>_Cam<CC>-<Name>_<Stamp>.mp4) is accompanied by a deterministic frame index file (.vididx):

- **Exact Integer 48.0 kHz Audio-Clock Timestamps:** Rather than trusting the operating system wall clock or USB bus arrival jitter, Cave Recorder stamps every captured video frame against the hardware audio interface's monotonic sample counter (960 samples per frame at 50 FPS, or 1,920 samples per frame at 25 FPS).

- **.vididx Frame Index Structure:** Each .vididx file stores the camera's nominal frame rate, 48.0 kHz sample rate, calibrated optical latency offset, user camera trim offset, and a per-frame index of t = 0 audio sample positions and keyframe offsets.
- **Instant Keyframe Lookup & Scrubbing:** During timeline playback or rapid ruler scrubbing, Cave Recorder looks up the target video frame directly from the .vididx index, jumps to the nearest preceding 0.5 s (25-frame) keyframe, and decodes only the minimal forward frames required to display the exact target image.

6.6.2 BT.709 Video Monitor & Sample-Locked Playback

The **Video Monitor** renders camera frames in BT.709 studio color space. While transport playback is rolling, Cave Recorder continuously keeps all active camera angles locked to the 48.0 kHz audio playhead:

Sync State	Audio-to-Video Offset (|Δt|)	Playback Synchronizer Action
Locked Cadence	|Δt| <= 0.5 frame (<= 10.0 ms @ 50 FPS)	Presents every decoded CFR frame in natural 1:1 cadence with zero frame drops or repeats.
Smooth Catch-Up	0.5 frame < |Δt| <= 2.5 frames (10..50 ms)	Advances or holds a single frame step within the current 0.5 s GOP so visual sync converges smoothly without decoder reset.
Keyframe Seek	|Δt| > 2.5 frames (> 50.0 ms, or after a timeline scrub jump)	Dispatches an immediate .vididx keyframe seek to the target GOP and pre-rolls to the exact playhead sample.

6.6.3 Operating the Video Timeline Filmstrip & Camera Trim Offset Slider

Each camera lane in WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**) renders a **time-anchored thumbnail filmstrip** aligned 1:1 with the horizontal audio sample ruler:

1. **Camera Angle Selection & Split-Screen Preview:** Click any camera lane header (Cam01-Drums, Cam02-GtrMobile, Cam03-RoomMobile) to focus that angle in the **Video Monitor**, or switch the monitor to **Multi-Camera Split-Screen** mode (Side-by-Side Crop, Side-by-Side 16:9, 3-Cam Band, or Quad 2x2) to inspect all angles simultaneously.
2. **Time-Anchored Filmstrip Span & 35 mm Sprocket Tick Rails:** Every recorded video clip spans its exact start-to-end sample range on the timeline (incorporating its recorded start time and **Camera Trim Offset**) with frame-accurate thumbnail decoding and top/bottom 35 mm filmstrip sprocket tick rails marking exact frame boundaries when zoomed in.
3. **Interactive Pointer Skimming & Floating 16:9 Frame Loupe Popover:** Moving your pointer across any camera filmstrip lane reveals a cyan skimmer hairline, a floating high-resolution 16:9 BT.709 **Frame Loupe Popover** displaying the exact frame number and timestamp (#F<frame> · <MM:SS.mmm>), and a vertical transient alignment guide extending down across the audio tracks so you can visually line up drumstick impact or pick attack against the audio waveform.

4. **Per-Clip Fine Trim Nudge (`Camera Trim Offset`):** Although the **3-Second Camera Latency Calibration** modal (Section 2.5.2) automatically compensates for hardware sensor and capture-card delay, you can fine-tune any individual video clip's alignment on the timeline using its **Camera Trim Offset** slider (adjustable in exact audio samples and milliseconds, with `Double-Click` / `Alt+Click` reset to `0 smp`). Adjusting the camera trim offset updates the `.vididx` and `take.json` metadata non-destructively and propagates automatically into offline split-screen `.mp4` mixdowns and `.fcpxml` v1.9 multicam exports.

PART III | CHAPTER 7

Session Reel, Mixdowns, Snapshots, Disk Layout & Cloud Sync

Managing Takes, converting live Prints to FLAC and MP3, rendering offline Mixdowns, Stem Bounces, and Multi-Camera Split-Screen .mp4 Video Mixdowns, exporting Takes to external workstation project formats (.rpp, .als, Session XML, .dawproject, .fcpxml multicam and stacked split-screen timelines, and BWF/iXML bundles), comparing Snapshots with Live Mix A/B recall, disk naming rules, resumable Google Drive video and mix sync, and the Cave Companion Mobile ecosystem (Android .apk and iOS .ipa).

A typical three-hour band rehearsal or studio tracking night produces anywhere from five to twenty recorded **Takes** alongside evolving console mixes, overdub passes, and stereo mixes for the band group chat. Cave Recorder centralizes all session management inside the **SESSION REEL** workspace (**CMD** / **CTRL** + **4**), combining a searchable **Takes Catalog**, one-click **Print-to-FLAC** and **Print-to-MP3** conversion, offline **Master Mixdowns**, **Stem Bounces**, and **External Workstation Project & Bundle Exports (EXPORT TAKE)**, non-destructive **Console Snapshots** with **Live Mix** protection, a self-documenting disk layout, and built-in two-way **Google Drive Cloud Sync**.

7.1 Managing Takes in the SESSION REEL View

Pressing **CMD** / **CTRL** + **4** (or clicking **SESSION REEL** in the workspace navigation bar) opens the Session Reel ledger for the currently open Session.

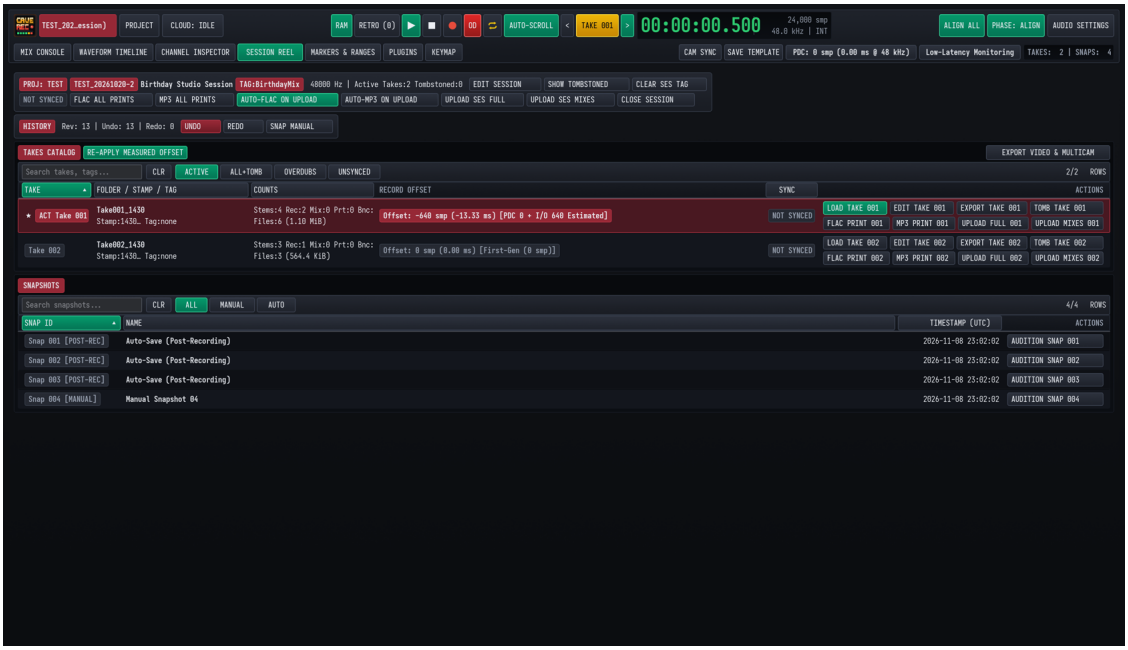


Figure 7.1: *The Session Reel view listing recorded Takes, durations, star ratings, Print conversion buttons, and Snapshot controls.*

7.1.1 The Session Header & Undo/Redo History Bar

At the top of the SESSION REEL view, two horizontal control bars summarize the active Session and its edit history:

- **Session Metadata & Global Actions Bar:**
  - Displays the parent Project badge (PROJ: TEST), the permanent Session ID (TEST_20261020-2), the human-readable Session display title (Birthday Studio Session), the optional folder tag (TAG:BirthdayMix), the hardware sample rate (48000 Hz), and the count of active and soft-deleted Takes.
  - **EDIT SESSION & CLEAR SES TAG:** Rename the Session's display title or edit/clear its <SessionTag>. Editing a Session tag cleanly renames the Session folder on disk and updates all relative links in <ProjectTag>/_Mixes/ without renaming any audio files inside.
  - **Deleted Takes Toggle & EMPTY TRASH:** Toggle visibility of soft-deleted Takes in the catalog (paired with the **ALL+TOMB** filter pill below) or click the red **EMPTY TRASH** button when soft-deleted Takes are present to permanently remove their audio files from disk.
  - **FLAC ALL PRINTS, MP3 ALL PRINTS, AUTO-FLAC ON UPLOAD & AUTO-MP3 ON UPLOAD:** Batch-convert all uncompressed live stereo Prints across the Session to lossless 24-bit .flac or 320 kbps CBR .mp3 files, or automatically convert Prints to .flac and/or .mp3 whenever uploading to cloud storage.
  - **UPLOAD SES FULL / UPLOAD SES MIXES / CLOSE SESSION:** Trigger a full or mixes-only cloud sync for the entire Session, or close the Session and return to the PROJECT browser modal (**CMD** / **CTRL** + **O**).
- **HISTORY Bar (Reversible Undo/Redo & Manual Snapshot Capture):**
  - Every fader drag, EQ node move, phase alignment, marker edit, Pass promotion, and Take deletion is tracked as a reversible step in the Session's command history (Rev:13 | Undo:13 | Redo:0). Continuous knob or fader gestures automatically group into a single undo step when you release the mouse button.
  - Click **UNDO** (**CMD** / **CTRL** + **Z**) or **REDO** (**CMD** / **CTRL** + **SHIFT** + **Z**) to step backward or forward through your session adjustments.
  - Click **SNAP MANUAL** (**CMD** / **CTRL** + **S**) to capture an immediate point-in-time Console Snapshot (Section 7.3).

7.1.2 Switching, Starring, Renaming & Deleting Takes in the TAKES CATALOG

The **TAKES CATALOG** table lists every Take recorded or split during the Session. You can filter the catalog using the search box (Search takes, tags...) or the filter pills: **ACTIVE** (default, showing non-deleted Takes), **ALL+TOMB** (showing both active Takes and soft-deleted Takes), **OVERDUBS** (showing only Takes that contain Rec002-OD overdub passes), and **UNSYNCED** (showing Takes with local changes not yet uploaded to Google Drive).

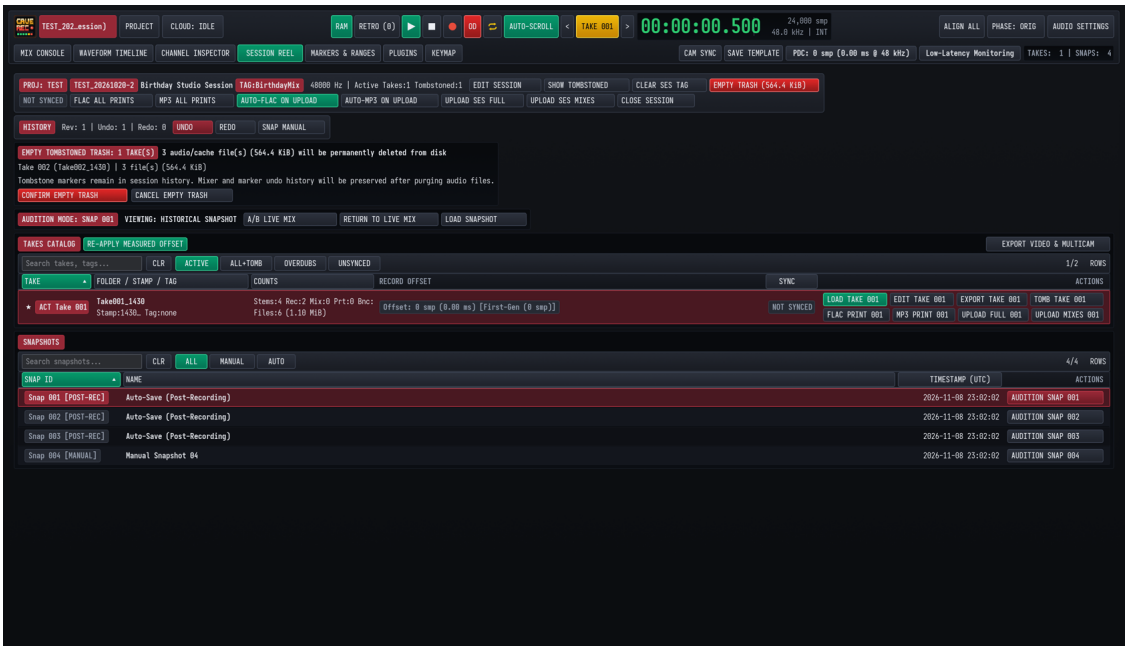


Figure 7.2: *The Session Reel view showing the Empty Trash confirmation banner and the burgundy Audition Mode banner with Live Mix controls.*

Each row in the TAKES CATALOG provides the following status columns and actions:

Column / Action	Operation
Favorite Star (★)	Click the star icon on the far left of any row to mark a keeper Take. Starring a Take lets bandmates and engineers spot the best performance of a song at a glance.
Active Take Badge (ACT Take 001)	Highlights the Take currently loaded into the Mix Console and Waveform Timeline. To switch the active Take, click **LOAD TAKE <NNN>** in the ACTIONS column, or use the **<** and **>** buttons beside TAKE 001 in the Top Deck (**ALT** + **[** and **ALT** + **]**).
FOLDER / STAMP / TAG	Shows the on-disk Take folder name (Take001_1430), capture timestamp (Stamp:1430), and optional song tag (Tag:none or Tag:SongA).
COUNTS	Summarizes the audio assets inside the Take: multitrack stems (Stems:4), recording passes (Rec:2), rendered offline mixdowns (Mix:0), live stereo prints (Prt:0), and stem bounces (Bnc:0), plus total disk footprint.
RECORD OFFSET & RE-APPLY MEASURED OFFSET	Displays the roundtrip latency offset applied to overdub passes in that Take (Offset: -640 smp (-13.33 ms)). If you recalibrate hardware loopback latency in AUDIO SETTINGS (**CMD** / **CTRL** + **,**), clicking **RE-APPLY MEASURED OFFSET** in the catalog header updates the alignment offset on the active Take.
EDIT TAKE <NNN>	Opens the Take metadata editor to rename the Take's song title, assign or change its alphanumeric <TakeTag> (for example, changing Take001_1947 to Take001_1947_MidnightDrive), and record multi-line rehearsal or mix notes.

Column / Action	Operation
CONVERT TO REAL <NNN> & PURGE RETRO <NNN>	Shown on Takes captured from the 30-minute rolling memory buffer (RETRO Takes, **SHIFT** + **R**). Click **CONVERT TO REAL <NNN>** to promote a retroactive memory capture into a permanent standard Session Take, or click **PURGE RETRO <NNN>** to discard an unwanted retroactive capture and reclaim disk space immediately.
TOMB TAKE <NNN> (Soft Delete)	Marks an unwanted false-start Take as deleted in the session history and hides it from the ACTIVE view. Soft-deleted Takes are never overwritten or re-downloaded during cloud sync, and can be restored at any time with **UNDO** (**CMD** / **CTRL** + **Z**) or by viewing **ALL+TOMB**.
EMPTY TRASH (CONFIRM EMPTY TRASH)	When one or more Takes have been soft-deleted, a red **EMPTY TRASH** button appears in the Session Reel header showing the exact byte count to be reclaimed. Clicking **EMPTY TRASH** opens a confirmation banner listing the exact Take folders and file counts to be removed from disk. Click **CONFIRM EMPTY TRASH** to permanently delete those audio files from your drive, or **CANCEL EMPTY TRASH** to keep them. Even after purging audio files via Empty Trash, the Take remains marked as deleted in the session history so deleted Takes are not re-downloaded during cloud sync.

7.2 Live Stereo Prints vs. Offline Mixdowns & Stem Bounces

Cave Recorder provides four distinct stereo, multitrack, and external project interchange workflows so you can share rough rehearsal recordings with the band in seconds, export studio-grade stems for external mastering, or hand off complete multitrack project files to an external mixing or video editing suite.

7.2.1 Comparison of Export & Bounce Outputs

Output Type	Capture Mode	Signal Path & Destination	Studio Use Case
1. Live Stereo Print (Print<NNN>)	**Real-time** during every Take. Convert to .flac or .mp3 via **FLAC PRINT <NNN>**, **MP3 PRINT <NNN>**, **FLAC ALL PRINTS**, or **MP3 ALL PRINTS**.	Live post-Master-Bus stereo mix saved to <TakeFolder>/mix/ and linked into <ProjectTag>/_Mixes/.	Instant post-rehearsal listening on phones, web browsers, or band group chats without waiting for an offline render.

Output Type	Capture Mode	Signal Path & Destination	Studio Use Case
2. Offline Stereo Mixdown (Mix<NNN>)	**Faster-than-realtime** offline render (triggered via the Master Bus **FLAC**, **MP3**, and **BOUNCE** controls).	Full Mix Console signal path (trims, phase delays, active Pass Lanes, 8-slot inserts, EQ, VCA compression, Subgroup/ Aux buses, Master limiter) saved to <TakeFolder>/mix/ and linked into _Mixes/.	Sharing polished stereo mixes in 24-bit FLAC or 320 kbps CBR MP3 after refining faders, EQ, dynamics, or overdub comps.
3. Offline Stem Bounce (Bounce<NNN>)	**Faster-than-realtime** offline multi-stem render.	Individual post-DSP Track channels (<CC>-<TrackName>) and/or Subgroup Buses (Bus<BB>-<BusName>) rendered from 00:00:00.000 into <TakeFolder>/bounce/.	Delivering phase-aligned, delay-compensated instrument or bus stems (WAV or FLAC) for external remixing or mastering.
4. Multi-Camera Split-Screen Video Mixdown (VideoMix<NNN>.mp4)	**Offline hardware-accelerated** multi-camera video + console audio render.	Multi-camera split-screen compositor (Single, Side-by-Side Crop, Side-by-Side 16:9, 3-Cam Band, Quad 2x2) paired with console-mixed 48.0 kHz stereo AAC-LC audio saved to <TakeFolder>/mix/ and linked into <ProjectTag>/ _Mixes/.	Sharing complete multi-angle band performance videos with studio-mixed console audio immediately after rehearsal.
5. External Project, .fcpxml v1.9 Multicam & Bundle Export (EXPORT TAKE)	**Instant project interchange**, **multicam video timeline**, or **stem bundle export** from the SESSION REEL or Export modal.	Generates .rpp project files, .als Live Set project archives, <Session version="7003"> XML session folders, .dawproject open interchange archives, .fcpxml v1.9 Final Cut XML timelines (both stacked split-screen and native <mc-clip> multicam), or universal BWF/iXML/MIDI/IR stem bundles in <TakeFolder>/ export/ (linked into <ProjectTag>/_Exports/).	Handing off a Take (with faders, pans, polarity, phase delays, Pass Lanes, markers, calibrated multi-camera video angles, and optional .nam neural amp models) to an external mixing or video post-production workstation.

7.2.2 Converting Live Stereo Prints to Lossless FLAC (FLAC PRINT) & 320 kbps MP3 (MP3 PRINT)

While you are tracking a Take, Cave Recorder writes the live 2-track Master Bus print as an uncompressed .wav file (Print001_<Stamp>.wav) with automatic Plugin Delay Compensation (PDC) latency trimming so the audio engine expends zero processor time on compression during live recording. Once the transport stops:

1. **Convert a Single Take's Print:** In the SESSION REEL view (**CMD** / **CTRL** + **4**), click **FLAC PRINT 001** (for a lossless 24-bit .flac file) or **MP3 PRINT 001** (for a 320 kbps CBR .mp3 file with a Xing/Info seek header for instant web-player scrubbing) in the ACTIONS column of that Take row.
2. **Convert All Prints in the Session:** Click **FLAC ALL PRINTS** or **MP3 ALL PRINTS** in the Session Reel header bar to convert every unconverted .wav Print across all active Takes in one batch.
3. **Safe Write & Project _Mixes/ Linking:** Cave Recorder encodes the 24-bit .flac counterpart (50-60% smaller than the source .wav file) or 320 kbps CBR .mp3 counterpart in the background before finalizing the file on disk. If local disk space ever runs out during encoding, any incomplete file is removed immediately so corrupt files never appear on disk. Once converted, the **FLAC PRINT 001** or **MP3 PRINT 001** button illuminates in its active state and Cave Recorder automatically places a relative link to the converted file inside the top-level <ProjectTag>/_Mixes/ folder.

7.2.3 Rendering Offline Stereo Mixdowns & Stem Bounces

After you refine channel EQ, VCA compression, multi-mic phase alignment (Chapter 5: Built-In Processors, Phase Alignment & External Plugins (CLAP & AU)), or overdub Pass selections (Chapter 6: Waveform Timeline, Pass Lanes, Markers & Ranges), you can render an updated stereo Mixdown (Mix<NNN>) or multitrack Stem Bounce (Bounce<NNN>) faster than real time:

1. **Load and Verify the Take:** Load the target Take into the console (LOAD TAKE <NNN>) and verify your mute, solo, active Pass Lane, and Master Bus limiter settings.
2. **Render an Offline Stereo Mixdown (Mix<NNN>):** On the Master Bus strip (Chapter 4: Mix Console, Signal Flow & Channel Strips), toggle **FLAC** (24-bit lossless .flac), **MP3** (320 kbps CBR .mp3 with a Xing/Info seek header), or both simultaneously, and click **BOUNCE**. Cave Recorder processes the entire Take from 00:00:00.000 through the full console signal path, increments the sequential mix number (Mix001, Mix002, …), writes the rendered file(s) into <TakeFolder>/mix/, and links them into <ProjectTag>/_Mixes/. When both FLAC and MP3 are selected, a single BOUNCE click renders both files under the same Mix<MMM> number.
3. **Render Processed Track or Subgroup Stems (Bounce<NNN>):** When exporting stems for an external mix or mastering session, Cave Recorder renders **Individual Track Stems** (01-Kick-In, 02-Snare-Top, …) and/or **Stereo Subgroup Bus Stems** (Bus01-Drum-Bus, Bus02-Guitar-Bus, …) into <TakeFolder>/bounce/ in WAV or FLAC. Every bounced stem starts at exact sample 0 (00:00:00.000) with full Plugin Delay Compensation applied so all stems line up sample-accurately when dragged into any external workstation.

7.2.4 Exporting Takes to External Workstation Project Formats (EXPORT TAKE)

When an outside mixing engineer or video editor wants to open a recorded Take in an external digital audio workstation or non-linear video editor without rebuilding track names, fader balances, subgroup routing, or marker flags by hand, click **EXPORT TAKE** on any Take row in the SESSION REEL view (**CMD** / **CTRL** + **4**).

Cave Recorder writes the selected interchange package into <TakeFolder>/export/ and automatically maintains a flat relative link inside the project-level <ProjectTag>/_Exports/ directory:

- **.rpp Project File (EXPORT TAKE -> .rpp):** Writes a plain-text .rpp project file referencing the Take's audio stems with sample-accurate microsecond phase offsets, active and muted Pass Lanes, channel faders, stereo pans, polarity switches (Ø), Subgroup Bus folder routing, and timeline markers and loop ranges.
- **.als Live Set Project (EXPORT TAKE -> .als):** Writes a gzip-compressed XML .als project file with AudioTrack lanes, sample-zero AudioClip entries (IsWarped=false so natural band tempo fluctuations remain untouched), linear mixer volume and pan values, Subgroup GroupTrack routing, and Timeline locators.
- **Open-Source Workstation Session XML (EXPORT TAKE -> Session XML 7003):** Generates a complete <Session version="7003"> XML directory structure with Sources, Regions, Playlists, Route strips, GainControl faders, Pannable stereo positions, Subgroup Buses, and Locations markers.
- **.dawproject Open Interchange Archive (EXPORT TAKE -> .dawproject):** Packages a standard ZIP container (project.xml + metadata.xml + optional audio stems) preserving tracks, channel gains, pans, Subgroup Bus routing, Pass clips, and timeline markers for any workstation supporting the open .dawproject specification.
- **.fcpxml v1.9 Video Timeline XML (EXPORT TAKE -> .fcpxml):** Writes a Final Cut XML v1.9 timeline (<fcpxml version="1.9">) with <asset-clip> audio lanes tagged by track role (dialogue.<TrackName>), rational frame-aligned sample timing, decibel volume adjustments, and chapter markers for immediate sync with multi-camera rehearsal or live-session video footage.
- **Universal BWF / iXML / MIDI / IR Stem Bundle (EXPORT TAKE -> BWF Bundle):** Exports Broadcast Wave (bext + <BWFXML> iXML track and pass metadata) audio stems paired with a Type 1 Standard MIDI File (_markers.mid carrying marker cues and tempo map metadata), a README_SESSION_NOTES.txt console sheet, and 24-bit 48 kHz minimum-phase impulse response .wav files capturing each channel's 5-stage Phase Alignment Network + 12-Band static EQ curve so any workstation can reproduce Cave Recorder's exact phase and tonal alignment via a standard convolution loader.

When exporting a Take, you can choose **Zero-Copy Relative References** (which links directly to your existing rec/*.wav stems on disk in milliseconds with zero duplicated audio data) or **Self-Contained Copy / Render** (which copies or renders stems alongside any active .nam neural amplifier models into the export package for portable delivery on an external drive).

7.2.5 Rendering Multi-Camera Split-Screen .mp4 Video Mixdowns & .fcpxml v1.9 Multicam Timelines

When a Take contains one or more recorded studio or wireless **Cave Companion** camera angles in <TakeFolder>/video/, click **EXPORT VIDEO & MULTICAM** in the SESSION REEL view (**CMD** / **CTRL** + **4**) to open the **Export Video Mixdown & Multicam FCPXML** modal:

1. **Choose a Split-Screen Compositor Layout:** Cave Recorder composites up to four camera angles directly in BT.709 color space at 1080p50 (1920×1080 @ 50 FPS) or 1080p25 (25 FPS) with 2 px dark studio dividers:
   - **Single (Full-Frame 16:9):** Renders a single selected camera angle across the full 1920×1080 frame.
   - **Side-by-Side Crop (2-Camera Vertical 8:9 Split with Horizontal Pan):** Places Camera 1 on the left half (960×1080) and Camera 2 on the right half (960×1080), with independent **Horizontal Camera Pan (-100% to +100%)** sliders so you can center each musician cleanly within their vertical half-frame.
   - **Side-by-Side 16:9 (2-Camera Letterboxed 16:9 Split):** Displays two full-width 16:9 camera frames side by side (960×540 each) with zero cropping.
   - **3-Cam Band (3-Camera Triptych / Band Layout):** Combines a wide or featured upper/left hero angle with two companion bandmate panes and per-pane crop/pan framing.
   - **Quad 2x2 (4-Camera 2×2 Studio Grid):** Displays four synchronized 960×540 16:9 quadrants (Cam 1 Top-Left, Cam 2 Top-Right, Cam 3 Bottom-Left, Cam 4 Bottom-Right).
2. **Offline Split-Screen .mp4 Mixdown Engine (960/1920-Sample Video Cadence to 1024-Sample AAC-LC Bridge):**
   - Clicking **RENDER VIDEO MIXDOWN (.MP4)** runs the Take from 00:00:00.000 through the full Mix Console audio engine while decoding and compositing every calibrated camera stream using its .vididx calibrated latency and clip trim offset.
   - Because 50 FPS (960 samples/frame) and 25 FPS (1920 samples/frame) video cadences differ from the 1024-sample frame size required by hardware AAC-LC audio encoders, Cave Recorder bridges the 48.0 kHz stereo console bus through an exact sample-accurate accumulator into the hardware H.264 + AAC-LC (320 kbps) .mp4 muxer alongside gapless 48.0 kHz .flac and .mp3 post-take audio prints.
   - The finished video is saved to <TakeFolder>/mix/<SessionID>_Take<NNN>_VideoMix<MMM>_<Stamp>.mp4 and linked into <ProjectTag>/_Mixes/ for instant local playback, LAN companion streaming, or Google Drive sharing.
3. **Dual-Mode .fcpxml v1.9 Video Timeline Export (Stacked Split-Screen & Native <mc-clip> Multicam):**
   - Clicking **EXPORT MULTICAM FCPXML (.FCPXML)** generates a Final Cut XML v1.9 package in <TakeFolder>/export/ supporting two non-linear editing workflows:
   - **Stacked Split-Screen Spine (<adjust-crop> & <adjust-transform>):** Translates your selected Cave Recorder split-screen layout (Side-by-Side Crop, 3-Cam Band, Quad 2x2) directly into stacked <asset-clip> video lanes with exact rational <adjust-crop>, <adjust-transform> pan/scale coordinates, and <timeMap> clock-drift tables derived from each camera's .vididx index.
   - **Native <mc-clip> Multi-Camera Switching Angle Assembly:** Builds a standard Final Cut <mc-clip> multicam resource combining all calibrated camera angles (<mc-angle>) and post-trim audio stems so a video editor can cut live between camera angles on the timeline with zero manual clap-syncing.

7.3 Non-Destructive Console Snapshots & Live Mix (SNAPS)

A **Console Snapshot** (Snap 001, Snap 002, …) is a complete, point-in-time capture of your entire mixing console and timeline comp state stored as a compact JSON file inside <SessionFolder>/snapshots/. Because Snapshots reference your existing audio stems rather than duplicating .wav files on disk, you can save dozens of mix variations per Session with near-zero disk footprint.

7.3.1 What a Snapshot Captures & Automatic Snapshot Policies

Every Snapshot records:

- All input channel strip parameters (input trim, 180-degree polarity invert, microsecond acoustic phase delay, reorderable 8-slot processing rack order, 12-band Parametric EQ nodes, VCA Compressor settings, Tape Saturation, CLAP plugin states, Aux send levels and tap points, pan, fader gain, mute, solo, and output bus routing).
- All stereo Subgroup Bus (SUB 1..16), Aux Return (AUX 1 Convolution Reverb, AUX 2 Stereo Tape Delay), and Master Bus parameters.
- Active **Pass Lane** selections (PASS 01, OD 02, …) and per-pass mute states across every Take in the Session.
- The complete compact Undo/Redo history stack at that moment in time, so restoring a Snapshot also restores its undo timeline.

Snapshots are created in three ways and can be filtered in the **SNAPSHOTS** table at the bottom of the SESSION REEL view using the **ALL**, **MANUAL**, and **AUTO** pills:

1. **Manual Snapshot ([MANUAL]):** Click **SNAP MANUAL** in the Session Reel history bar or press **CMD** / **CTRL** + **S** from any view to save a named milestone (for example, Manual Snapshot 04 or Pre-Vocal-Overdub Balance).
2. **Post-Recording Auto-Save ([POST-REC]):** Automatically captured immediately after every normal recording stop (Auto-Save (Post-Recording)), preserving the exact monitor mix and Pass state that was active when the band finished the Take. (If you abort a false start using the clean-restart discard stop command, the post-recording auto-save is suppressed.)
3. **Periodic 20-Minute Auto-Save ([AUTO]):** Automatically captured every 20 minutes while a Session is open, provided at least one console or timeline action has occurred since the last saved Snapshot.

7.3.2 Auditioning Historical Snapshots with Live Mix Protection

In traditional mixing consoles, loading an old scene or snapshot overwrites your current unsaved fader moves unless you remember to save a temporary file first. Cave Recorder eliminates this hazard by automatically preserving your **Live Mix**:

1. **Click AUDITION SNAP <NNN> to Enter Audition Mode:** The moment you click **AUDITION SNAP 001** on any row in the SNAPSHOTS table while working on a live mix, Cave Recorder automatically saves your current unsaved working console state as your **Live Mix** in memory and illuminates the burgundy **AUDITION MODE: SNAP 001 | VIEWING: HISTORICAL SNAPSHOT** banner (see Figure 7.2).

2. **Click Through Multiple Historical Snapshots Freely:** While in Audition Mode, you can click **AUDITION SNAP 002**, **AUDITION SNAP 003**, **AUDITION SNAP 004**, and so on across five or more historical Snapshots to compare balances. Your original working **Live Mix** remains safely preserved in memory the entire time.
3. **Instant Zero-Latency Comparison (A/B LIVE MIX):** Click **A/B LIVE MIX** in the Audition Mode banner to toggle directly between the historical Snapshot and your preserved **Live Mix** while audio is playing. All fader, EQ, and dynamics changes transition via sample-accurate smoothing ramps (5 to 10 ms) without audio clicks.
4. **Restore Your Working Mix (RETURN TO LIVE MIX or BACKSPACE):** Click **RETURN TO LIVE MIX** in the Audition Mode banner (or press **BACKSPACE** while in the SESSION REEL view) to exit Audition Mode and restore your exact working console state with zero lost fader or EQ moves.
5. **Adopt the Historical Snapshot (LOAD SNAPSHOT):** If you decide you prefer the historical Snapshot over your previous working state, click **LOAD SNAPSHOT** to promote that Snapshot as your active live session mix.

> **ZERO LOST FADER MOVES DURING SNAPSHOT COMPARISON**
> You never need to manually save a throwaway snapshot before checking how the mix sounded an hour ago. **Live Mix** protection engages automatically on the first AUDITION SNAP click and guards your working console balance until you explicitly click **RETURN TO LIVE MIX** (**BACKSPACE**) or **LOAD SNAPSHOT**.

7.4 Self-Documenting Disk Folder Structure & File Naming Rules

Cave Recorder stores all Projects, Sessions, Takes, Passes, Prints, Mixdowns, Stem Bounces, and Snapshots in an open, human-readable directory hierarchy on your drive (defaulting to `~/Music/caverec/` on macOS and Linux, and `%USERPROFILE%\Music\caverec\` on Windows). Every folder and audio filename states which Project, Session, Take, Channel, and Pass it belongs to, so any engineer can open the folder in macOS Finder, Windows Explorer, Linux file managers, or any external audio editor ten years from now with zero proprietary database lock-in.

7.4.1 Complete Project & Session Directory Tree on Disk

Conceptual studio hierarchy (Project -> Session -> Take -> Pass / Prints / Mixdowns / Snapshots):

```
ProjectName/
└── SessionName/
    ├── session.json
    ├── Snapshots/
    ├── Mixdowns/
    ├── Prints/
    │   └── Take001_LiveMix.flac
    └── Takes/
        └── Take001_SongName/
            ├── Pass001_Base/
            │   └── Ch01_Kick.wav
            └── Pass002_Overdub/
                └── Ch09_LeadGtr.wav
```

On disk inside your Cave Recorder Library Root (`~/Music/caverec/`), this hierarchy is organized on disk using self-identifying prefixed filenames and a project-wide `_Mixes/` index folder:

```
~/Music/caverec/                          # Library Root folder
├── .caverec/
│   ├── projects.json                     # Local Project registry
│   ├── drive_sync_state.json             # Cloud sync state ledger
│   └── templates/
│       ├── MicAndGuitar001.json          # Default 2-ch session template
│       └── GuitarBassDrumTrio001.json    # 11-ch band session template
├── GF/                                   # Project ("Grottfolk")
│   ├── .caverec_project.json             # Cloud Project index
│   ├── _Mixes/                           # Flat links to Mix* & Print*
│   │   ├── GF_20261015-2_Take001_Print001_1947.flac
│   │   ├── GF_20261015-2_Take001_Print001_1947.mp3
│   │   ├── GF_20261015-2_Take001_Mix001_2315.flac
│   │   └── GF_20261015-2_Take001_Mix001_2315.mp3
│   ├── _Exports/                         # Flat links to EXPORT TAKE packages
│   └── GF_20261015-2_Jam/                # Session folder (tag "Jam")
│       ├── GF_20261015-2_session.json    # Console & marker manifest
│       ├── models/                       # Self-contained .nam neural models
│       ├── snapshots/                    # Saved Console Snapshots
│       │   └── GF_20261015-2_001_2026-10-15_19-34-12.json
│       ├── Take001_1947_Improv/          # Take 001 folder
│       └── Take002_0015_SongB/           # Late-night Take (same day)
└── MISC/                                 # Built-in default project
    └── _Mixes/
```

Inside each individual Take folder (`Take001_1947_Improv/`), raw multitrack passes, multi-camera video recordings and `.vididx` sidecar indices, stereo prints and audio/video mixdowns, stem bounces, external project exports, and Take-level neural amp models are separated into clean subfolders:

```
Take001_1947_Improv/                      # Take 001 (tag "Improv")
├── GF_20261015-2_Take001_take.json       # Take metadata & notes
├── rec/                                  # Base (Rec001) & Overdub (-OD)
│   ├── GF_20261015-2_Take001_01-Kick-In_Rec001_1947.wav
│   ├── GF_20261015-2_Take001_02-Snare-Top_Rec001_1947.wav
│   ├── GF_20261015-2_Take001_09-LeadGtr_Rec001_1947.wav
│   ├── GF_20261015-2_Take001_09-LeadGtr_Rec002-OD_2210.wav
│   └── GF_20261015-2_Take001_09-LeadGtr_Rec003-OD_20261019-1922.wav
├── video/                                # 50/25 FPS CFR Camera Angles & .vididx
Sidecars
│   ├── GF_20261015-2_Take001_Cam01-Drums_1947.mp4
│   ├── GF_20261015-2_Take001_Cam01-Drums_1947.vididx
│   ├── GF_20261015-2_Take001_Cam02-RoomMobile_1947.mp4
│   └── GF_20261015-2_Take001_Cam02-RoomMobile_1947.vididx
├── mix/                                  # Live Prints, Audio & Split-Screen
Video Mixdowns
│   ├── GF_20261015-2_Take001_Print001_1947.wav
│   ├── GF_20261015-2_Take001_Print001_1947.flac
│   ├── GF_20261015-2_Take001_Print001_1947.mp3
│   ├── GF_20261015-2_Take001_Mix001_2315.flac
│   ├── GF_20261015-2_Take001_Mix001_2315.mp3
│   └── GF_20261015-2_Take001_VideoMix001_2320.mp4
├── bounce/                               # Track & Subgroup Bounces
│   ├── GF_20261015-2_Take001_01-Kick-In_Bounce001_2315.wav
│   └── GF_20261015-2_Take001_Bus01-Drum-Bus_Bounce001_2315.wav
├── export/                               # External workstation exports
(.rpp, .als, .dawproject, .fcpxml, BWF)
├── models/                               # Take-level .nam neural amplifier
models
└── .cache/                               # Waveform display cache
```

7.4.2 Canonical File & Folder Naming Reference

Every file created by Cave Recorder follows consistent, predictable naming rules so files sort chronologically and never collide when dragged into an external folder or shared across operating systems:

Item	Naming Pattern	Concrete Example
Project Tag & Folder	1 to 8 uppercase letters or digits (A-Z, 0-9)	GF, INSOF, MISC
Project _Mixes/ Folder	<ProjectTag>/_Mixes/ (relative links to active Mix*, Print*, and VideoMix* .flac/.mp3/.wav/.mp4 files)	GF/_Mixes/ GF_20261015-2_Take001_Print001_1947.flac
Project _Exports/ Folder	<ProjectTag>/_Exports/ (relative links to exported external workstation packages)	GF/_Exports/ GF_20261015-2_Take001.rpp
Session ID (Permanent)	<PROJECT_TAG>_<YYYYMMDD> (plus -2, -3, ... for additional sessions on the same session day)	GF_20261015, GF_20261015-2
Session Folder	<ProjectTag>/<SessionID>[_<SessionTag>]	GF/GF_20261015-2_Jam

Item	Naming Pattern	Concrete Example
Take Folder	Take<NNN>_<Stamp>[_<TakeTag>]	Take001_1947_Improv
Base Pass Stem (rec/)	<SessionID>_Take<NNN>_<CC>-<TrackName>_Rec001_<Stamp>.wav	GF_20261015-2_Take001_01-Kick-In_Rec001_1947.wav
Overdub Pass Stem (rec/)	<SessionID>_Take<NNN>_<CC>-<TrackName>_Rec<MMM>-OD_<Stamp>.wav	GF_20261015-2_Take001_09-LeadGtr_Rec002-OD_2210.wav
Camera Video & Index (video/)	<SessionID>_Take<NNN>_Cam<CC>-<CamName>_<Stamp>.<mp4|vididx>	GF_20261015-2_Take001_Cam01-Drums_1947.mp4
Live Stereo Print (mix/)	<SessionID>_Take<NNN>_Print<NNN>_<Stamp>.w	GF_20261015-2_Take001_Print001_1947.flac
Offline Mixdown (mix/)	<SessionID>_Take<NNN>_Mix<NNN>_<Stamp>.fla	GF_20261015-2_Take001_Mix001_2315.flac
Split-Screen Video Mix (mix/)	<SessionID>_Take<NNN>_VideoMix<NNN>_<Stam	GF_20261015-2_Take001_VideoMix001_2320.mp
Track Stem Bounce (bounce/)	<SessionID>_Take<NNN>_<CC>-<TrackName>_Bounce<NNN>_<Stamp>.wav / .flac	GF_20261015-2_Take001_09-LeadGtr_Bounce001_2315.wav
Subgroup Stem Bounce (bounce/)	<SessionID>_Take<NNN>_Bus<BB>-<BusName>_Bounce<NNN>_<Stamp>.wav / .flac	GF_20261015-2_Take001_Bus01-Drum-Bus_Bounce001_2315.wav
External Take Export (export/)	<TakeFolder>/export/<SessionID>_Take<NNN>.<rpp|als|dawproject|fcpxml>	Take001_1947_Improv/export/GF_20261015-2_Take001.rpp
Neural Amp Model (models/)	<SessionFolder>/models/<ModelName>_<ShortHash>.nam	GF_20261015-2_Jam/models/StudioCrunch_a1b2c3d4.nam
Console Snapshot (snapshots/)	<SessionID>_<NNN>_<YYYY-MM-DD_HH-MM-SS>.json	GF_20261015-2_001_2026-10-15_19-34-12.json

Key studio rules governing disk files:

- **08:00 Local Time Session-Day Rollover:** A studio session day runs from **08:00 to 07:59 local time** the following morning. If your band starts rehearsing on October 15 at 21:00 and records a Take after midnight at 01:30 on October 16, that late-night recording still belongs to session date 20261015 and receives a four-digit <Stamp> (0130). If you open the same Session four days later on October 19 to record a guitar overdub, <Stamp> automatically expands to include the calendar date (20261019-1922).
- **Permanent File IDs vs. Editable Folder Tags:** The <SessionID>, Take numbers (Take001, Take002), Pass numbers (Rec001, Rec002-OD), and timestamps are assigned once at creation and never change, and numbers are never reused after a deletion. Editable tags (<SessionTag> like Jam and <TakeTag> like Improv) appear **only in folder names**, never inside .wav or .flac filenames. As a result, you can rename song tags and display titles freely at any time without breaking file references or re-uploading gigabytes of audio to the cloud.
- **Broadcast Wave (bext) Headers & Sample-Zero Alignment:** Every .wav stem is written with standard Broadcast Wave (bext) metadata embedding the wall-clock origin timecode while aligning the start of the Take to sample 0 (00:00:00.000).

7.5 Band Collaboration & Google Drive Cloud Sync (CLOUD)

Cave Recorder includes built-in two-way **Google Drive Cloud Sync** designed specifically for bands and production teams. Bandmates can pull lightweight stereo .flac and .mp3 Prints, Mixdowns, and Console Snapshots to their laptops or stream .mp3 files directly in the Google Drive web player immediately after rehearsal, while selectively downloading multi-gigabyte raw multitrack stems only for the Takes they want to remix or overdub at home.

7.5.1 Connecting a Project to Google Drive & Reading the CLOUD Badge

1. **Authenticate with Google Drive:** Click **PROJECT** in the Top Deck (**CMD**/**CTRL** + **O**) to open the Project & Session Browser modal (or click the **CLOUD: IDLE** status badge in the Top Deck). In the top Cloud Folder row, click **Check /Connect Drive**. When a Google OAuth Client ID is configured, Cave Recorder launches your default web browser for the Google account sign-in flow and stores your refresh credentials inside your operating system's user configuration folder. If no OAuth Client ID is configured yet on your workstation, Cave Recorder displays DRIVE: OAUTH NOT CONFIGURED with inline setup guidance for placing gdrive_oauth_client.json in your Cave Recorder configuration directory (~/Library/Application Support/caverec/ on macOS, ~/.config/caverec/ on Linux, or %APPDATA%\caverec\ on Windows) or setting the CAVEREC_GDRIVE_CLIENT_ID environment variable.
2. **Monitor the Top Deck CLOUD Status Pill:**
   - **CLOUD: IDLE:** Connected and ready, with no transfer currently running.
   - **CLOUD: SYNCING:** Actively uploading or downloading session manifests, mixes, or stems in the background.
   - **CLOUD: SYNCED:** All local Session changes match the remote Google Drive project index.

3. **Automatic Real-Time Recording Guard:** If a background cloud upload or download is running when you press **Record** (`R`) on the transport bar, Cave Recorder automatically pauses and terminates the background transfer immediately so disk and network activity never compete with live multitrack audio capture.

7.5.2 Selective Sync Tiers: MIXES vs. FULL, AUTO-FLAC ON UPLOAD & AUTO-MP3 ON UPLOAD

Because an evening of 18-channel 48 kHz / 24-bit multitrack stems can total 10 to 30 GB while stereo .flac and .mp3 Prints and Console Snapshots take only a few megabytes, Cave Recorder supports two explicit transfer modes for both uploading and downloading:

Sync Tier	Upload / Download Buttons	What Is Transferred	When to Use
Mixes & Metadata Only (MIXES)	UPLOAD SES MIXES, UPLOAD MIXES <NNN>, Sync Library (Mixes), DOWNLOAD MIXES	Session & Take JSON manifests, Markers & Ranges, Console Snapshots (snapshots/), .flac/.mp3/.wav Live Prints and Mixdowns (mix/), Stem Bounces (bounce/), and <ProjectTag>/_Mixes/. **Skips raw rec/ multitrack WAV stems.**	Right after rehearsal over studio Wi-Fi or mobile hotspot so bandmates can listen to every Take, stream .mp3 mixes in Google Drive, inspect markers, and audition mixes within seconds.
Full Multitrack Backup (FULL)	UPLOAD SES FULL, UPLOAD FULL <NNN>, Sync Library (Full), DOWNLOAD FULL	Everything in MIXES **plus** all raw multitrack Base Pass (Rec001) and Overdub Pass (Rec002-OD) .wav stems in rec/.	Backing up the complete multitrack session to cloud storage, or pulling raw stems of a keeper Take (UPLOAD FULL 001 / DOWNLOAD FULL) to overdub or remix on another computer.

> **AUTOMATIC PRINT CONVERSION ON UPLOAD (AUTO-FLAC ON UPLOAD & AUTO-MP3 ON UPLOAD)**
> Both the SESSION REEL header bar and the Top Deck CLOUD popover provide independent **AUTO-FLAC ON UPLOAD** and **AUTO-MP3 ON UPLOAD** toggles (AUTO-FLAC ON UPLOAD is enabled by default). When enabled, clicking UPLOAD SES MIXES or UPLOAD MIXES <NNN> automatically converts any uncompressed Print<NNN>.wav files to 24-bit .flac and/or 320 kbps CBR .mp3 before uploading so bandmates get compact lossless files and browser-streamable MP3s without uploading huge uncompressed stereo WAV prints. If your local drive ever runs completely out of free space during pre-conversion, Cave Recorder presents a three-choice recovery banner (**PROCEED WITHOUT FLAC PRINTS**, **UPLOAD PRINT.WAV INSTEAD**, or **CANCEL**) so you can still back up your session safely.

7.5.3 Sharing a Project with Bandmates (+ Link Remote Project...)

Each Project in Cave Recorder is an independent sync root, allowing you to keep personal projects in your own Google Drive while linking band projects hosted in a bandmate's Google Drive:

1. **Share the Project Folder:** The bandmate who created the Project shares its <ProjectTag> folder in Google Drive with the band's Google accounts (or clicks **COPY SHARE LINK** in the PROJECT modal to copy a caverec:// project link to the clipboard).
2. **Link the Remote Project:** On each bandmate's computer, open the **PROJECT** modal (**CMD** / **CTRL** + **O**) and click **+ Link Remote Project...**.
3. **Auto-Discovery or Link Paste:** Cave Recorder automatically lists all Cave Recorder projects found in your Google Drive "Shared with me" view for one-click linking, or lets you paste a Google Drive folder URL or caverec:// share URI directly.
4. **Automatic Tag Collision Resolution:** Because every Project in your local library must have a unique <ProjectTag>, if a linked remote project uses a tag (such as GF) that already exists on your machine, Cave Recorder prompts you with three clear choices:
   - **RENAME LOCAL PROJECT TAG...:** Re-tags your existing local project (for example, GF to GF2) to free GF for the shared band project.
   - **MERGE SAME PROJECT (ADOPT REMOTE):** Used when two bandmates independently created the GF project before linking, adopting the shared cloud project identity cleanly.
   - **CANCEL:** Aborts linking without changing local files.

7.5.4 Exclusive Session Write-Locking (Read-Only Preview) & Zero-Loss Conflict Merging

When multiple bandmates collaborate on the same Session across different computers, Cave Recorder prevents accidental overwrites using **Exclusive Session Write-Locking** paired with **Snapshot-Backed Conflict Merging**:

- **Automatic Write Lock on Open & Release on Close:**
  - When you open a Session while online, Cave Recorder acquires an exclusive edit lock for your computer and automatically pulls any newer cloud metadata if your local copy has no unsaved edits.
  - When you close the Session (CLOSE SESSION) or quit Cave Recorder while online, the edit lock is released automatically so another bandmate can open the Session for editing.
- **Opening a Session Locked by a Bandmate (Read-Only Preview Mode):**
  - If a bandmate currently has the Session open on another computer, Cave Recorder alerts you and offers **OPEN READ-ONLY PREVIEW** (recommended), **BREAK LOCK & TAKE WRITE ACCESS**, or **CANCEL**.
  - In **Read-Only Preview** mode, an amber **READ-ONLY (LOCKED BY <MACHINE>)** badge appears in the header. You have full playback, timeline scrubbing, and live in-memory console mixing (faders, solos, 12-band EQ, VCA compression, and Snapshot auditioning), while transport Record / Overdub and background disk writes are disabled so you never conflict with your bandmate's live recording session.

  - If you dial in a great mix while in Read-Only Preview mode, click **SAVE AS LOCAL SNAPSHOT** in the header bar to preserve your mix settings locally, or click **TAKE LOCK** once your bandmate finishes.
- **Zero-Loss Merging of Offline Edits, Takes & Deletions:**
  - Audio files in rec/, mix/, and bounce/ are never overwritten, and new recordings and bounces are always saved as new numbered files. Pulling a MIXES update onto the studio computer that holds the raw rec/ multitrack WAV stems adds new mixes and snapshots without ever deleting local multitrack stems.
  - Takes or Passes that were marked as deleted in the session history on one machine remain marked as deleted across syncs so purged false starts are never re-downloaded.
  - If two bandmates both edit a Session's mix while offline and later sync to Google Drive, Cave Recorder detects that both computers edited the Session while offline, combines all newly recorded Takes, Passes, Mixdowns, and Snapshots from both machines, and prompts you to choose either **KEEP LOCAL MIX & SAVE CLOUD MIX AS SNAPSHOT** or **ADOPT CLOUD MIX & SAVE LOCAL MIX AS SNAPSHOT**. The alternate mix is automatically saved into snapshots/ so zero fader or EQ work is ever lost, and you can immediately compare both bandmates' mixes side by side using **Live Mix** (Section 7.3.2).

7.5.5 Unified Google Drive Resumable Video/Print Sync & Cross-Device Deduplication

When a Session includes multi-camera .mp4 + .vididx recordings alongside .flac and .mp3 audio prints, Cave Recorder coordinates desktop and **Cave Companion** mobile transfers through a unified **Google Drive Resumable Sync** pipeline:

1. **Automatic Cross-Device Deduplication:** Every uploaded .mp4, .vididx, .flac, .mp3, and .wav asset is tagged in Google Drive metadata with its content fingerprint. If a mobile camera node has already transferred a 1080p50 .mp4 angle to the desktop workstation over studio Wi-Fi (or uploaded it directly from the phone), Google Drive confirms the matching file fingerprint so the second device never re-uploads the same multi-hundred-megabyte video file.
2. **Chunked Resumable Uploads:** Large .mp4 video angles, split-screen VideoMix<NNN>.mp4 renders, and .wav/.flac stems upload via resumable upload sessions in 8 MB chunks. If a rehearsal basement Wi-Fi connection drops mid-upload, Cave Recorder automatically resumes from the last committed byte offset when connectivity returns.
3. **Verified Downloads & Pairing QR Folder Propagation:** Downloads stream into a temporary .part staging file and finalize into place only after the integrity check passes. When you pair a **Cave Companion** phone via the desktop pairing QR code, the project's Google Drive root folder link propagates automatically to the phone so mobile sign-in connects directly to the shared band project folder.

7.6 Cave Companion Mobile Ecosystem (Android .apk & iOS .ipa)

The **Cave Companion** mobile application for Android (caverec-v0.4.2-android-arm64.apk) and iOS (caverec-v0.4.2-ios-arm64.ipa) turns every bandmate's phone or tablet into a low-latency **Wireless Remote Transport & 16-Channel Meter Bridge**, a sample-locked **Wireless 1080p50 CFR Studio Camera Node**, and an offline-capable **Multi-Angle & Multi-Print Take Player** styled in the same anodized studio console aesthetic as the desktop workstation (complete with the top-bar CaveLogoBadge, unified **Studio Status Bar**, and adaptive 2-pane landscape split console on iPad and Android tablets).

7.6.1 Desktop Pair Mobile Companion Modal (Studio QR Code, 6-Digit PIN & Device Access Roles)

Click **PAIR MOBILE** in the Top Transport Deck to open the **Pair Mobile Companion** modal on your desktop workstation:

- **Studio Pairing QR Code & 6-Digit PIN Fallback:** Displays a high-contrast QR code encoding the workstation's local network address, workstation identity, a secure pairing credential, and the optional Google Drive project folder link. Inside the mobile app's **Pair** tab, switch between **SCAN STUDIO QR** and **6-DIGIT PIN** entry at any time. If a device camera is already mounted on a tripod across the room, bandmates can type the workstation's IP address and the 6-digit numeric PIN displayed below the QR code instead.
- **Automatic Studio Reconnection & Safe Unpair Sheet:** Upon pairing, the mobile app securely saves the studio pairing credential on the device so subsequent studio rehearsals reconnect automatically when the phone joins the studio Wi-Fi network. Tapping the connected studio badge in the top **Studio Status Bar** opens the paired connection sheet with live clock-lock telemetry and a two-step **Unpair Device** confirmation.
- **Paired Device Access Roles (Access Role):** The desktop modal lists every paired phone with live connection status, clock-lock offset, and a per-device **Access Role** selector:
  - **Full Control + Camera:** Can trigger Play, Stop, Record, Overdub, RETRO, safe two-step Discard Take confirmation, marker drops (**M**), 60 Hz timeline scrubbing, and record/transfer wireless 1080p50 camera angles.
  - **Camera + Player:** Can record synchronized 1080p50 camera angles and play back Takes and mixes, while transport Record/Stop triggers are locked out to prevent accidental bumps during a live song.
  - **Player & Meters Only:** Read-only 16-channel metering, session browsing, and multi-angle Take playback.
- **Companion App Download QR Code:** Includes a toggleable download QR code pointing directly to the release packages (caverec-v0.4.2-android-arm64.apk and caverec-v0.4.2-ios-arm64.ipa).

7.6.2 Wireless Studio Link Architecture (Real-Time Control & Background Media Sync)

Cave Companion separates sub-millisecond studio telemetry and transport control from bulk .mp4 / .flac media transfers across two dedicated local Wi-Fi channels:

Studio Channel	Capability	Cadence & Protection	Studio Function
Real-Time Studio Control Link	**Sub-Millisecond Clock Lock**	Continuous drift-compensated clock synchronization	Maps the phone's camera and player clock directly onto the desktop audio interface's 48.0 kHz sample timeline (< 0.5 ms lock).
Real-Time Studio Control Link	**Remote Transport Commands**	Authenticated studio session commands	Triggers Play, Stop, Record, Overdub, RETRO, Take switching, and live Marker drops (**M**) from behind the drum kit or amp stack.
Real-Time Studio Control Link	**Remote Timeline Scrubbing**	**60 Hz** smooth playhead scrubbing	Lets a musician scrub the desktop Waveform Timeline smoothly by dragging across the mobile timeline ruler.
Real-Time Studio Control Link	**Live Console Telemetry & Meters**	**25 Hz** broadcast from desktop to paired phones	Streams active Take timecode, transport state (PLAY, REC, OD), and 16-channel Peak/RMS + Master Bus LED meter levels.
High-Speed Media Transfer Link	**Manifest, Video & Audio Sync**	Authenticated transfer with automatic .part resume and file integrity verification	Transfers session/take manifests, sends wireless 1080p50 .mp4 + .vididx camera recordings to <TakeFolder>/video/, and downloads .mp3, .flac, and VideoMix<NNN>.mp4 files to phones at full Wi-Fi speed.

7.6.3 Wireless 1080p50 CFR Mobile Camera Node & Automatic Desktop Timeline Ingest

Switching to the **Camera** tab inside the Cave Companion app configures the phone's hardware camera and H.264/HEVC encoder as a sample-locked wireless studio camera:

1. **Locked 1080p50 Constant Frame Rate (CFR) Capture:** Locks sensor exposure duration and frame delivery to an exact 50 FPS (20.000 ms / 960 audio samples per frame at 48.0 kHz) with zero B-frames and 0.5 s closed GOP keyframes, stamping every frame's presentation timestamp into a local .vididx sidecar locked to the desktop audio clock.

2. **Live Studio HUD (Clock Lock Badge, 16-Channel Mini-Meters & Tally Border):** Displays a green **CLOCK LOCKED (±0.2 ms)** badge, a compact 16-channel console meter bridge so the performer can verify mic levels from across the room, and a bright crimson **Recording Tally Border** whenever the desktop transport enters Record.
3. **Rear LED Torch 4-Pulse Optical Calibration:** Tapping **CALIBRATE TORCH LATENCY** fires the phone's rear LED Torch in a 4-pulse sequence (80 ms optical pulse + acoustic click) to calibrate the phone camera's rolling-shutter and hardware encoder pipeline offset.
4. **Automatic Wi-Fi Camera Transfer & Desktop Timeline Auto-Ingest:** The instant recording stops, the phone finalizes its .mp4 and .vididx files and automatically transfers both over studio Wi-Fi into the desktop workstation's <TakeFolder>/video/ folder. As soon as the verified transfer completes, the desktop workstation automatically attaches the mobile camera angle to the active Take and renders its **Video Timeline Strip** thumbnail filmstrip on the WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**).

7.6.4 Multi-Angle & Multi-Print Take Player & Mobile Storage Quota Tiering

In the **Takes & Player** tab, bandmates can review every recorded Take either in the studio over local Wi-Fi or on the road via **Google Drive Cloud Sync**:

- **Instant Multi-Angle & Multi-Print Switching:** While a Take is playing on the phone, tap any camera angle pill (Cam01-Drums, Cam02-GtrMobile, Split-Screen Mix) or switch audio sources on the fly between the **Live Stereo Print (Print001.mp3 / .flac)** and subsequent **Offline Mixdowns (Mix001.flac, Mix002.flac)**. Continuous drift-free audio/video synchronization keeps video frame presentation locked to the active audio print with zero timeline restart.
- **Smart Mobile Storage Quota Cache Tiering:** To prevent multi-gigabyte 1080p50 camera files from filling up a bandmate's phone storage, Cave Companion enforces a three-tier local cache policy:
  - **Pinned Offline Reference Tier (.mp3 Prints & Mixes + Manifests):** Compact 320 kbps .mp3 Prints/Mixdowns and Session/Take JSON manifests are retained locally for instant offline listening in the car or on a flight.
  - **On-Demand High-Resolution Tier (24-Bit .flac & Multi-Angle .mp4):** Lossless .flac mixes and multi-angle .mp4 videos are fetched on demand (over studio Wi-Fi or Google Drive) and managed within a configurable mobile storage ceiling (2 GB to 16 GB automatic cache).
  - **Automatic Camera Outbox Eviction:** Raw 1080p50 .mp4 files recorded by the phone's **Mobile Camera** are automatically removed from the phone's temporary camera outbox once studio Wi-Fi or Google Drive confirms that the desktop workstation or cloud project holds a verified copy.

PART IV | CHAPTER 8

Action, Mouse Gesture & Keyboard Shortcut Reference

Complete reference for all 59 keyboard actions across macOS, Windows, and Linux, mouse interaction gestures, the Keymap View, and command-line options.

Cave Recorder is engineered for fast, eyes-on-the-band operation during live rehearsals and studio tracking sessions. Every transport command, workspace switch, marker placement, channel strip toggle, timeline zoom action, and snapshot operation can be triggered from the keyboard or with direct mouse and trackpad gestures.

This chapter provides the complete reference for all **59 keyboard-assignable studio actions** across macOS, Windows, and Linux, all **mouse and trackpad modifiers** across the console and timeline, the interactive **KEYMAP configuration workspace** (`CMD` + `6` on macOS, `CTRL` + `6` on Windows and Linux), and the **command-line options** for automated studio verification and visual inspection.

8.1 Complete Keyboard Shortcut Reference (All 59 Actions by Scope)

Cave Recorder organizes its 59 bindable actions into **seven execution scopes** so that single-key shortcuts can be reused across different workspace views without colliding:

- **Platform Modifier Mapping:** Cave Recorder supports four modifier keys across platforms: the primary command modifier maps to `CMD` on macOS and `CTRL` on Windows and Linux, the alt modifier maps to `OPT` (`ALT`) on macOS and `ALT` on Windows and Linux, the shift modifier maps to `SHIFT` on all platforms, and the secondary control modifier maps to `CTRL` on macOS and `SUPER` (`WIN`) on Windows and Linux. Keymap profiles saved on one operating system automatically translate to the native modifiers on the other platforms.
- **Modal Priority (PROJECT MODAL):** While the PROJECT (Project & Session Browser) modal is open, only PROJECT MODAL actions (`ENTER` and `ESC`) respond to keyboard input.
- **View-Specific Priority & Global Fallback:** When a workspace view tab (MIX CONSOLE, WAVEFORM TIMELINE, CHANNEL INSPECTOR, SESSION REEL, or MARKERS & RANGES) is active, Cave Recorder checks that view's dedicated shortcuts first and then falls back to GLOBAL actions. Unmodified single-letter keys (R, O, C, M, K, L) remain reserved for global transport and cue marker actions, while channel strip toggles on the Mix Console use `SHIFT` combinations (`SHIFT` + `M`, `SHIFT` + `S`, `SHIFT` + `A`, `SHIFT` + `I`, `SHIFT` + `P`) so you never accidentally mute or disarm a track while dropping a marker.

8.1.1 Global Transport & Record (GLOBAL TRANSPORT)

These 10 actions belong to the GLOBAL / TRANSPORT category (GLOBAL scope) and are active from every workspace view whenever the Project Browser modal is closed.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Play / Pause Transport	**SPACE**	**SPACE**	Starts or pauses Master Transport playback from the active Take playhead.
Stop Transport	**SHIFT** + **SPACE**	**SHIFT** + **SPACE**	Stops Master Transport playback or active recording immediately and finalizes open audio files.
Toggle Record	**R**	**R**	Starts or stops multitrack recording (creates a new t = 0 Take when OD is off, or records an Overdub Pass when OD is latched).
Toggle Overdub Mode (OD)	**O**	**O**	Latches Overdub Mode (OD) on the Top Deck so pressing Record captures a track-level Overdub Pass (Rec002-OD) inside the active Take.
Toggle Cycle / Loop	**C**	**C**	Toggles continuous loop playback and loop overdubbing (⇄) across the active Take range.
Capture Retroactive Take	**SHIFT** + **R**	**SHIFT** + **R**	Flushes the rolling 30-minute capture memory (RETRO) to disk as a new numbered Session Take.
Toggle Retroactive RAM Ring	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Enables or disables continuous background 30-minute RAM capture across record-armed inputs.
Toggle Timeline Auto-Scroll	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Toggles AUTO-SCROLL so the Waveform Timeline viewport follows the advancing playhead automatically.
Activate Previous Session Take	**OPT** + **[** (ALT + [)	**ALT** + **[**	Loads the previous non-deleted Session Take (<) into the Waveform Timeline and Mix Console.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Activate Next Session Take	OPT +] (ALT +])	ALT +]	Loads the next non-deleted Session Take (>) into the Waveform Timeline and Mix Console.

8.1.2 Global Navigation & Modals (GLOBAL NAVIGATION & MODALS)

This group includes the 10 GLOBAL / NAVIGATION actions (GLOBAL scope) used to switch workspace tabs and open studio configuration dialogs, plus the 2 MODAL / PROJECT actions (PROJECT MODAL scope) active inside the Project & Session Browser modal.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Switch to Mix Console View	CMD + 1	CTRL + 1	Activates the large-format MIX CONSOLE workspace view (Chapter 4).
Switch to Waveform Timeline View	CMD + 2	CTRL + 2	Activates the multi-track WAVEFORM TIMELINE and Pass-Lane comping view (Chapter 6).
Switch to Channel Inspector View	CMD + 3	CTRL + 3	Activates the full-screen single-channel CHANNEL INSPECTOR view (Chapter 5).
Switch to Session Reel View	CMD + 4	CTRL + 4	Activates the SESSION REEL Takes Catalog, Snapshots, and Cloud Sync view (Chapter 7).
Switch to Markers & Ranges View	CMD + 5	CTRL + 5	Activates the sortable MARKERS & RANGES table view (Chapter 6).
Switch to Keymap View	CMD + 6	CTRL + 6	Activates the interactive KEYMAP reference and shortcut configuration view (Section 8.3).
Switch to Plugins View	CMD + 7	CTRL + 7	Activates the PLUGINS CLAP and macOS Audio Unit catalog and search directory management view (Section 5.7.2).
Open Project & Session Browser	CMD + O	CTRL + O	Opens the PROJECT modal to switch Projects, create Sessions from templates, or open recent Sessions.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Toggle Audio Device & Latency Settings	**CMD** + **,**	**CTRL** + **,**	Opens or closes the AUDIO SETTINGS modal for device selection, buffer size, and loopback calibration.
Toggle PDC Latency Breakdown	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Opens or closes the per-channel Plugin Delay Compensation (PDC) transparency popover in the Top Deck.
Confirm / Open Selected Session	**ENTER**	**ENTER**	Scoped to PROJECT MODAL: opens the selected Session or confirms new Session creation in the Project Browser.
Close Project Browser / Dismiss Error	**ESC**	**ESC**	Scoped to PROJECT MODAL: closes the Project Browser modal or clears an inline validation message.

8.1.3 Global Undo, Snapshots & Markers (GLOBAL UNDO / SNAPSHOTS / MARKERS)

This group covers the 5 GLOBAL / SESSION actions (Undo, Redo, Capture Manual Console Snapshot, Template popover, and Cloud Sync popover), the 9 GLOBAL / MARKERS cue and range hotkeys, and the 3 VIEW / MARKERS TAB table filter actions scoped to the MARKERS & RANGES view.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Undo Last Console Action	**CMD** + **Z**	**CTRL** + **Z**	Reverses the most recent console parameter, routing, Pass selection, or marker edit.
Redo Console Action	**CMD** + **SHIFT** + **Z**	**CTRL** + **SHIFT** + **Z**	Re-applies the most recently undone console parameter, routing, Pass selection, or marker edit.
Capture Manual Console Snapshot	**CMD** + **S**	**CTRL** + **S**	Saves an immutable point-in-time Console Snapshot ([MANUAL]) in <SessionFolder>/snapshots/.
Toggle Save Console Template Popover	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Opens or closes the SAVE TEMPLATE popover in the Top Deck to store a reusable room template.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Toggle Cloud Sync Status Popover	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Opens or closes the Google Drive CLOUD sync status and quick-upload popover in the Top Deck.
Add Standalone Marker at Playhead	**M**	**M**	Drops a standalone cue marker (M005, M006, ...) at the active playhead or position cursor without stopping transport.
Set Range Start Boundary	**K**	**K**	Marks the RANGE START sample for a new Take range (or completes a pending RANGE END boundary).
Set Range End Boundary	**L**	**L**	Marks the RANGE END sample for a new Take range (or completes a pending RANGE START boundary).
Abort Pending Range / Dismiss Overlay	**ESC**	**ESC**	Cancels an in-progress range boundary definition (K or L) or closes open Top Deck popovers.
Jump to Previous Marker	**[**	**[**	Moves the transport playhead to the preceding marker or cursor stop in the active Take (< PREV).
Jump to Next Marker	**]**	**]**	Moves the transport playhead to the next marker or cursor stop in the active Take (NEXT >).
Jump Playhead to Position Cursor	**.**	**.**	Snaps the green transport playhead directly to the amber timeline position cursor (-> CUR).
Jump to First Marker in Take	**SHIFT** + **[** (or **HOME**)	**SHIFT** + **[** (or **HOME**)	Moves the transport playhead to the earliest marker in the active Take (I< FIRST).
Jump to Last Marker in Take	**SHIFT** + **]** (or **END**)	**SHIFT** + **]** (or **END**)	Moves the transport playhead to the final marker in the active Take (LAST >I).
Show All Markers in Table	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Scoped to MARKERS & RANGES: shows both STANDALONE and BOUNDARY markers (ALL filter pill).

Action Title	macOS Default	Windows / Linux Default	Studio Function
Filter Standalone Markers Only	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Scoped to MARKERS & RANGES: filters the markers table to standalone cue markers only (STANDALONE).
Filter Range Boundary Markers Only	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Scoped to MARKERS & RANGES: filters the markers table to range start/end boundary markers only (BOUNDARY).

8.1.4 Mix Console View (MIX CONSOLE VIEW)

These 7 actions belong to the VIEW / MIX CONSOLE category (MIX CONSOLE scope) and operate on the currently inspected Input Track or Subgroup Bus when the MIX CONSOLE tab (**CMD** / **CTRL** + **1**) is active.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Toggle Mute on Inspected Strip	**SHIFT** + **M**	**SHIFT** + **M**	Toggles post-fader Mute (M) on the currently inspected Input Track or Subgroup Bus strip.
Toggle Solo on Inspected Strip	**SHIFT** + **S**	**SHIFT** + **S**	Toggles Solo-in-Place (S) on the currently inspected Input Track or Subgroup Bus strip.
Toggle Record Arm on Inspected Track	**SHIFT** + **A**	**SHIFT** + **A**	Toggles Record Arm (R) on the currently inspected Input Track (CH01..CH18).
Toggle Input Monitor (MON) on Inspected Track	**SHIFT** + **I**	**SHIFT** + **I**	Toggles live hardware Input Monitoring (I / MON) on the currently inspected Input Track.
Toggle 180 deg Polarity Invert on Inspected Strip	**SHIFT** + **P**	**SHIFT** + **P**	Toggles 180-degree Polarity Invert (Ø) on the currently inspected channel strip.
Trigger Phase Align on Inspected Track	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Runs sub-sample acoustic Phase Alignment (Φ) on the currently inspected Input Track relative to the reference track.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Trigger Global Multi-Mic Phase Align	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Runs automatic multi-microphone Phase Alignment (ALIGN ALL) across all active tracks in the Session.

8.1.5 Waveform Timeline View (WAVEFORM TIMELINE VIEW)

These 6 bindable actions belong to the VIEW / TIMELINE category (WAVEFORM TIMELINE scope) and are active when the WAVEFORM TIMELINE tab (**CMD** / **CTRL** + **2**) is selected, supplemented by the direct left/right arrow cursor step keys.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Timeline Zoom In	**=**	**=**	Zooms the horizontal waveform viewport in by one octave (ZOOM IN (+)), halving the smp/px span.
Timeline Zoom Out	**-**	**-**	Zooms the horizontal waveform viewport out by one octave (ZOOM OUT (-)), doubling the smp/px span.
Timeline Zoom Reset (1:1)	**0**	**0**	Resets horizontal zoom and scroll position to the default 1:1 overview span (1:1 ZOOM).
Toggle Playhead & Cursor Lock	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Toggles LOCK PH/CUR to couple or decouple the green transport Playhead and amber Position Cursor.
Expand All Track Pass Lanes	**E**	**E**	Expands overdub **Pass Lanes** (PASS 01, OD 02, ...) across all tracks in the active Take (EXPAND ALL PASSES).
Collapse All Track Pass Lanes	**SHIFT** + **E**	**SHIFT** + **E**	Collapses expanded overdub **Pass Lanes** across all tracks back to compact single-lane rows (COLLAPSE ALL PASSES).

Action Title	macOS Default	Windows / Linux Default	Studio Function
Step Position Cursor (Built-In Navigation)	`LEFT` / `RIGHT`	`LEFT` / `RIGHT`	Nudges the timeline Position Cursor (and Playhead when LOCK PH/CUR is active) backward or forward by one fine step.
Coarse Step Position Cursor (Built-In Navigation)	`SHIFT` + `LEFT` / `RIGHT`	`SHIFT` + `LEFT` / `RIGHT`	Nudges the timeline Position Cursor (and Playhead when LOCK PH/CUR is active) backward or forward by a 10x coarse step.

8.1.6 Channel Inspector View (CHANNEL INSPECTOR VIEW)

These 4 actions belong to the VIEW / INSPECTOR category (CHANNEL INSPECTOR scope) and are active when the CHANNEL INSPECTOR tab (`CMD` / `CTRL` + `3`) is selected.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Toggle VCA Compressor Sidechain HPF	`H`	`H`	Engages or bypasses the 90 Hz Sidechain High-Pass Filter (SC HPF) on the inspected channel's VCA Compressor.
Toggle VCA Compressor Stereo Link	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Engages or bypasses stereo sidechain detector linking (LINK) on the inspected channel or Subgroup Bus.
Reset Channel Meter Clip Hold	`X`	`X`	Clears the latched red 0.0 dBFS overload indicator (CLR CLIP) and yellow peak-hold marker on the inspected meter.
Toggle Pass Lanes on Inspected Channel	`E`	`E`	Expands or collapses the overdub **Pass Lanes** (PASSES) for the currently inspected Input Track.

8.1.7 Session Reel View (SESSION REEL VIEW)

These 3 actions belong to the VIEW / SESSION REEL category (SESSION REEL scope) and are active when the SESSION REEL tab (`CMD` / `CTRL` + `4`) is selected.

Action Title	macOS Default	Windows / Linux Default	Studio Function
Return from Snapshot Audition to Live Session	**BACKSPACE**	**BACKSPACE**	Exits historical Snapshot Audition Mode (RETURN TO LIVE MIX) and restores the working **Live Mix**.
Show All Session Takes	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Switches the TAKES CATALOG filter to show all active Session Takes.
Filter Takes with Overdubs	Unbound (Assignable in KEYMAP)	Unbound (Assignable in KEYMAP)	Filters the TAKES CATALOG (OVERDUBS) to display only Takes containing Rec002-OD Overdub Passes.

8.2 Complete Mouse & Trackpad Gesture Reference

All faders, rotary knobs, horizontal drag bars, meter overload lamps, processor rack slots, interactive EQ and compressor graphs, and timeline rulers share consistent mouse and trackpad gestures across Cave Recorder.

Control / Surface	Mouse / Modifier Gesture	Behavior
Faders, Rotary Knobs & Value Bars	Click + Drag (Vertical on faders and knobs, Horizontal on strip bars)	Sweeps the parameter smoothly along its logarithmic dB, frequency, or percentage taper.
Faders, Rotary Knobs & Value Bars	**SHIFT** + Drag (can be pressed or released mid-drag)	Engages 10x fine precision adjustment (0.1 dB steps on faders, trims, sends, and thresholds, sub-Hz on EQ, 0.01 on Q).
Faders, Rotary Knobs & Value Bars	Double-Click or **OPT** / **ALT** + Click	Resets any fader or trim to 0.0 dB unity, any PAN or BAL bar to center (C), and any processor knob to its factory default.
Faders, Rotary Knobs & Value Bars	Mouse Scroll Wheel	Steps the hovered fader, knob, or horizontal bar up or down in calibrated increments.
Numeric dB Fader Readout Box	Click readout box below fader, type dB value, press **ENTER**	Opens an inline numeric entry field at the foot of the fader well. Type an exact value (such as -3.5 or 0) and press **ENTER** to apply, or **ESC** to cancel.

Control / Surface	Mouse / Modifier Gesture	Behavior
Peak/RMS Meters & Clip Lamps	Click red clip indicator or CLR CLIP button	Resets latched red 0.0 dBFS overload lamps and floating yellow peak-hold markers on the channel or Master Bus meter.
8-Slot Processor Rack (Slots 1..8)	Click slot label (1:CLAP, 2:12B EQ, 3:VCA CMP, ! MISSING)	Selects that rack slot and opens the inline SLOT N PROCESSOR picker to swap processor types, assign a discovered .clap or macOS Audio Unit (AUv3 / AUv2) plugin, open a floating PLUGIN GUI window, or relink a Ghost Slot via LOCATE....
8-Slot Processor Rack (Slots 1..8)	Drag :: handle vertically or Click ^ / v arrows	Reorders processor slots (1:CLAP, 2:12B EQ, 3:VCA CMP, 4:AUX SND, or loaded third-party CLAP and macOS Audio Unit plugins) live in real time with clickless crossfading and zero audio dropouts.
8-Slot Processor Rack (Slots 1..8)	Click B button on any rack slot	Bypasses or re-engages that individual processor slot in place (illuminates amber when bypassed) for instant A/B listening.
8-Slot Processor Rack Header	Click SWAP EQ/CMP or RESET	SWAP EQ/CMP exchanges the slot order of 12B EQ and VCA CMP in one click. RESET restores all built-in channel processors to flat defaults.
12-Band Parametric EQ Graph (2:12B EQ)	Click + Drag band node horizontally / vertically	Sweeps the band's center/corner frequency (20 Hz to 20 kHz) and gain (-24.0 dB to +24.0 dB) simultaneously over the live Dual-Trace RTA.
12-Band Parametric EQ Graph (2:12B EQ)	Mouse Scroll Wheel over node or **SHIFT** + Drag	Scrolling over a node narrows or widens filter bandwidth (Q, 0.10 to 18.00). Holding **SHIFT** while dragging engages 10x fine frequency and gain resolution.
12-Band Parametric EQ Graph (2:12B EQ)	Double-Click band node (or **OPT** / **ALT** + Click)	Resets the band's gain to 0.0 dB (or toggles band bypass when double-clicking a band selector pill).

Control / Surface	Mouse / Modifier Gesture	Behavior
VCA Compressor Graph (3:VCA CMP)	Click + Drag threshold or ratio handle on transfer curve	Adjusts compressor THRESH (-60.0 dB to 0.0 dB) and compression RATIO slope (1.0:1 to 20.0:1) directly against the scrolling GR trace.
Waveform Timeline Ruler & Canvas	Single-Click (< 4 px travel) on ruler or waveform canvas	Repositions the green Playhead or amber Position Cursor to the clicked timestamp, snapping to Take start (00:00:00.000) and Take end within 8 px.
Waveform Timeline Ruler & Canvas	**SHIFT** + Click or **SHIFT** + Drag	Bypasses Take-boundary snapping so you can place or scrub the Playhead and Position Cursor at exact sub-sample coordinates.
Waveform Timeline Ruler & Canvas	Click + Drag Horizontally (left / right)	Dragging horizontally on the ruler scrubs the Playhead across the Take. Dragging horizontally on the waveform canvas pans the visible time window.
Waveform Timeline Ruler & Canvas	Click + Drag Vertically (up / down)	Performs continuous exponential **2D Scrub-Zoom** anchored at the sample under the cursor (or around the Playhead when LOCK PH/CUR is active).
Waveform Timeline Viewport	Two-Finger Horizontal Trackpad Scroll or **SHIFT** + Mouse Wheel	Smoothly pans the horizontal waveform viewport backward or forward across the active Take.
Waveform Timeline Viewport	**CMD** + Mouse Wheel (macOS) / **CTRL** + Mouse Wheel (Win/Linux)	Zooms the waveform timeline horizontally in or out anchored at the mouse pointer position.
Marker (MKR) & Range (RNG) Lanes	Click + Drag cyan marker flag or amber range [/] handles	Moves a standalone cue marker or adjusts the start and end sample boundaries of a loop range directly on the timeline.

8.3 Customizing Shortcuts in the KEYMAP View (CMD/CTRL + 6)

Pressing **CMD** + **6** on macOS (**CTRL** + **6** on Windows and Linux) or clicking **KEYMAP** in Row 2 of the Top Transport Deck opens the **KEYMAP & SHORTCUT CONFIGURATION** workspace.

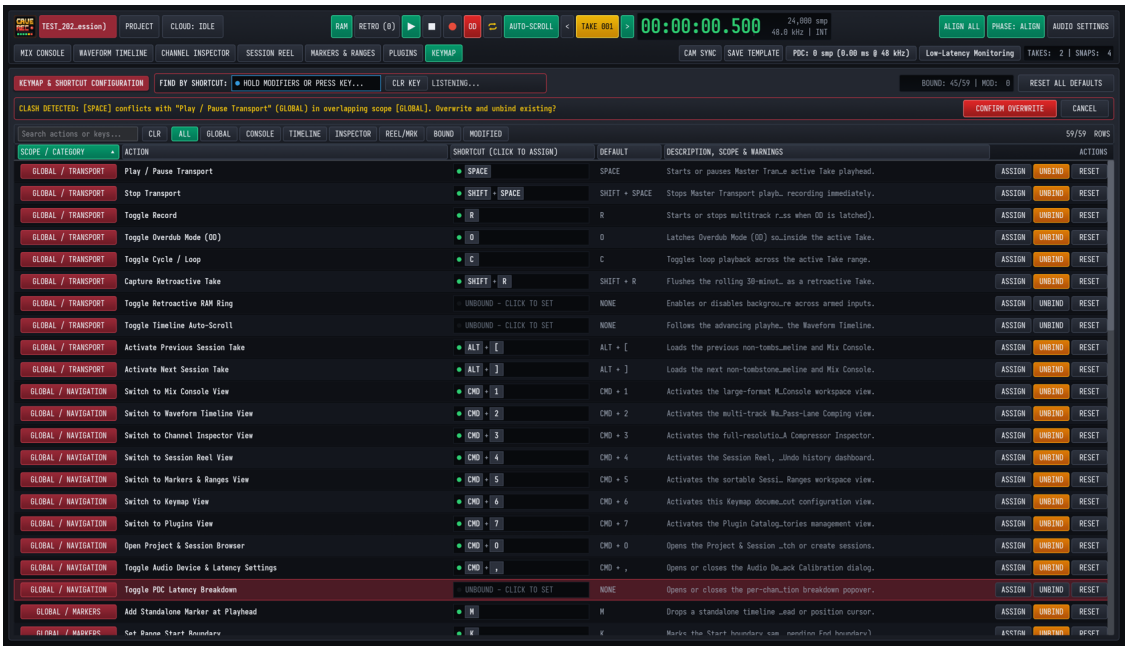


Figure 8.1: *The Keymap View showing keyboard shortcut scopes, search filtering, conflict warnings, and custom shortcut recording.*

8.3.1 Searching by Action Name, Reverse-Searching by Key Chord & Filtering by Scope

The KEYMAP view provides three complementary ways to inspect bindings across all 59 studio actions:

1. **Text Search (Search actions or keys...):** Type any action name (Phase Align, Snapshot, Pass Lanes), action identifier (transport_record), scope name (MIX CONSOLE), or key chord (SHIFT + M, CMD + 1) into the left search box to filter the table in real time. Click CLR to clear the text query.
2. **Interactive Reverse-Search by Shortcut (FIND BY SHORTCUT:):** Click inside the CLICK & PRESS KEYS TO FIND... capture well in the top header bar. Its status indicator switches from IDLE to LISTENING... and the prompt changes to HOLD MODIFIERS OR PRESS KEY...:
   - **Hold Modifier Keys (MODS: [...]):** Hold down any combination of modifiers (**CMD** / **CTRL** , **OPT** / **ALT** , **SHIFT**) without pressing a letter key to filter the table immediately to every action whose shortcut includes those held modifiers.
   - **Press a Trigger Key (EXACT: [...]):** Press any non-modifier key (such as **SPACE** , **R** , or **1**) to lock the complete key chord and isolate the exact action(s) bound to that shortcut across all scopes.
   - **Clear Reverse Search (CLR KEY):** Click CLR KEY beside the capture box to clear the key filter and restore all 59 rows.
3. **Scope & Status Quick-Filter Pills:** Click any of the eight filter pills in the toolbar (59/59 ROWS) to isolate a specific subset of actions:
   - **ALL:** Displays all 59 bindable studio actions.
   - **GLOBAL:** Displays the 34 global actions (GLOBAL / TRANSPORT, GLOBAL / NAVIGATION, GLOBAL / MARKERS, and GLOBAL / SESSION).
   - **CONSOLE:** Displays the 7 VIEW / MIX CONSOLE channel strip actions.
   - **TIMELINE:** Displays the 6 VIEW / TIMELINE zoom and Pass Lane actions.
   - **INSPECTOR:** Displays the 4 VIEW / INSPECTOR processor and clip-reset actions.

- **REEL/MRK:** Displays the VIEW / SESSION REEL, VIEW / MARKERS TAB, and MODAL / PROJECT actions.
- **BOUND:** Displays only actions that currently have an active keyboard shortcut assigned (45/59 by factory default).
- **MODIFIED:** Displays only actions whose current shortcut differs from the factory default (marked with an amber * in the DEFAULT column and counted in the MOD: telemetry badge at the top right).

8.3.2 Recording Custom Shortcuts, Unbinding & Resetting Defaults

Every row in the KEYMAP table includes an interactive **SHORTCUT (CLICK TO ASSIGN)** capture well and row action buttons on the far right:

1. **Enter Capture Mode (ASSIGN or Click Capture Well):** Click the **ASSIGN** button on the target row, or click directly inside its SHORTCUT (CLICK TO ASSIGN) box (UNBOUND - CLICK TO SET). The row's capture well highlights and prompts PRESS KEYS, THEN APPLY, while the right-hand action column switches to **APPLY**, **CANCEL**, and **UNBIND**.
2. **Press Your Desired Key Combination:** Hold any desired modifier keys (**CMD** / **CTRL**, **OPT** / **ALT**, **SHIFT**) and press a character key (A..Z, 0..9, [,], -, =, ,, .), function key (**F1** .. **F12**), or named navigation key (**SPACE**, **ENTER**, **TAB**, **BACKSPACE**, **DELETE**, **HOME**, **END**, **PGUP**, **PGDN**, **UP**, **DOWN**, **LEFT**, **RIGHT**). Shifted punctuation symbols (such as { or +) are automatically normalized to their canonical physical base key (**SHIFT** + **[** or **SHIFT** + **=**) for consistent cross-platform matching.
3. **Commit the Binding (APPLY):** Click **APPLY** to validate and save the captured key chord, or click **CANCEL** (or click outside the capture well) to exit without changing the shortcut.
4. **Unbind or Reset Individual Actions (UNBIND / RESET):** Click **UNBIND** on any row to clear its shortcut to UNBOUND, or click **RESET** to restore that single action to its factory default binding.
5. **Restore Factory Defaults Across the Entire Studio (RESET ALL DEFAULTS):** Click **RESET ALL DEFAULTS** at the top right of the KEYMAP header to restore all 59 actions to their original factory keybindings in one step.

8.3.3 Live Scope Conflict Detection, Shared Non-Conflicting Scopes & OS Warnings

When you assign a custom shortcut, Cave Recorder validates the key chord against two safety layers before writing to disk:

1. **Scope Clash Detection (CLASH DETECTED Confirmation Banner):**
   - **Overlapping Scopes (Blocked Without Confirmation):** Because GLOBAL shortcuts are active across every workspace view, assigning the same key chord to two GLOBAL actions, to two actions in the **same** workspace view, or to a GLOBAL action and a workspace view action creates an ambiguous collision. When you click APPLY on a colliding chord, Cave Recorder pauses and displays the amber **Clash Detection Banner** at the top of the KEYMAP view (CLASH DETECTED: [SPACE] conflicts with "Play / Pause Transport" (GLOBAL) in overlapping scope [GLOBAL]. Overwrite and unbind existing?). Click **CONFIRM OVERWRITE** to assign the shortcut and automatically unbind the conflicting action, or click **CANCEL** to keep your existing bindings untouched.

  - **Non-Conflicting Sibling Views (Allowed & Highlighted in Green):** Because only one workspace view tab is visible at a time, sibling view scopes (MIX CONSOLE, WAVEFORM TIMELINE, CHANNEL INSPECTOR, SESSION REEL, and MARKERS & RANGES) never conflict with each other, and PROJECT MODAL only runs while the modal browser is open. When two actions in separate views share the same key (such as **E** bound to Expand All Track Pass Lanes in WAVEFORM TIMELINE and Toggle Pass Lanes on Inspected Channel in CHANNEL INSPECTOR), the DESCRIPTION, SCOPE & WARNINGS column highlights SHARED IN NON-CONFLICTING SCOPE: CHANNEL INSPECTOR in emerald green.

2. **Operating System Shortcut Conflict Warnings (OS WARN):**
  - If you assign a key combination commonly intercepted by the host desktop environment, Cave Recorder displays an amber OS WARN (<Platform>): <System Function> badge in the row description so you know the operating system may capture the key press first:
  - **macOS Warnings:** **CMD** + **Q** (Quit Application), **CMD** + **H** (Hide Window), **CMD** + **M** (Minimize Window), **CMD** + **W** (Close Window), **CMD** + **SPACE** (Spotlight Search), **CMD** + **TAB** (Application Switcher), **CMD** + **SHIFT** + **3** / **4** / **5** (System Screenshot), and **CTRL** + Arrow keys (Mission Control / Desktop Spaces).
  - **Windows & Linux Warnings:** **ALT** + **F4** (Close Window), **ALT** + **TAB** (Window Switcher), **ALT** + **SPACE** (Window Menu), **CTRL** + **W** / **CTRL** + **Q** (Close / Quit), **CTRL** + **SHIFT** + **ESC** (Task Manager), **CTRL** + **ALT** + **T** / **DELETE** (Terminal / Session Dialog), and **SUPER** (**WIN**) + **L** / **D** / **E** / **R** / **I** / **TAB** / **SPACE** /Arrow keys.

8.3.4 Persisting, Backing Up & Sharing Studio Keymap Profiles (keymap.json)

Every change you make in the KEYMAP view is saved immediately to a human-readable JSON profile (keymap.json) inside your user configuration folder (stored alongside preferences.json and recent.json):

- **macOS:** ~/Library/Application Support/caverec/keymap.json
- **Linux:** $XDG_CONFIG_HOME/caverec/keymap.json (or ~/.config/caverec/keymap.json when XDG_CONFIG_HOME is unset)
- **Windows:** %APPDATA%\caverec\keymap.json (or %USERPROFILE%\AppData\Roaming\caverec\keymap.json)

```
{
  "schema_version": 1,
  "bindings": {
    "console_trigger_group_phase_align": {
      "modifiers": {
        "command": false,
        "control": false,
        "alt": false,
        "shift": true
      },
      "key": {
        "kind": "character",
        "value": "G"
      }
    }
  }
}
```

To transfer your custom studio keymap to a second rehearsal computer or back it up to version control:

1. Copy `keymap.json` from the source workstation's configuration directory into the target workstation's `caverec` configuration directory while Cave Recorder is closed, and launch `caverec`.
2. On startup, Cave Recorder automatically normalizes character symbols, maps `command: true` to **CMD** on macOS and **CTRL** on Windows and Linux, populates any omitted actions with their factory defaults, and sanitizes the table so no two overlapping scopes can ever load with a conflicting shortcut.

8.4 Command-Line & Automated Studio Verification Reference

In addition to launching Cave Recorder from your desktop application menu, studio engineers and system integrators can run `caverec` from a terminal shell to execute automated 18-channel signal-path verification or generate offscreen workspace captures for studio documentation and rig sign-off.

```
# 1. Standard interactive desktop studio launch
caverec

# 2. Run the 18-channel studio self-test & signal-path verification
caverec --self-test
caverec --stress-test

# 3. Render all 16 workspace & modal inspection screenshots
caverec --screenshot-dir .tmp/caverec-shots

# 4. Render a single workspace preset at custom dimensions
caverec --screenshot-dir .tmp/caverec-shots \
  --screenshot-view console-18ch \
  --screenshot-size 1680x960
```

8.4.1 Command-Line Arguments Reference

Command-Line Flag	Arguments / Values	Studio Function
(No arguments)	None	Launches the interactive Cave Recorder desktop workstation (1680x960 default window), loads user keybindings (keymap.json) and studio preferences (preferences.json), and opens the PROJECT browser modal to select or create a Session.
--self-test --stress-test	None	Runs the automated **18-Channel Studio Self-Test & Hardware Verification Suite** and exits with code 0 on pass or 1 on failure. Verifies 18-in/20-out routing (CH01..CH08 analog preamps, CH09..CH10 S/PDIF, CH11..CH18 ADAT optical), 48.0 kHz / 44.1 kHz sample-rate clock switching, multi-mic drum phase alignment, 30-minute retroactive memory capture, simultaneous 14-track playback + 4-track overdub pass recording, and offline mixdown and stem rendering.
--screenshot-dir	<dir> (Output directory path)	Renders offscreen PNG captures of Cave Recorder's workspace views and detail regions into <dir> for studio documentation and visual rig inspection without opening an interactive desktop window.

Command-Line Flag	Arguments / Values	Studio Function
--screenshot-view	<slug|all> (requires --screenshot-dir)	Selects which workspace preset to render: all (default), console, console-18ch, console-routing, timeline, timeline-expanded, inspector-track, inspector-subgroup, session-reel, session-reel-modals, markers-ranges, keymap, plugins, device-settings, top-deck-popovers, or the PROJECT Session Browser modal presets (welcome and welcome-forms).
--screenshot-size	<WxH> (320x240 to 4096x4096, default 1680x960)	Sets the logical viewport resolution used when rendering offscreen workspace captures with --screenshot-dir. Both space-separated (--screenshot-size 1920x1080) and =-separated (--screenshot-size=1920x1080) syntax are supported.

8.4.2 Studio Environment Variables

Cave Recorder also recognizes the following optional environment variables on startup:

Environment Variable	Platform / Scope	Studio Function
CLAP_PATH	macOS, Windows, Linux	Colon-separated (macOS/Linux) or semicolon-separated (Windows) list of additional directories to scan on startup for third-party .clap audio plugins alongside the standard system plugin folders (Section 5.7.1).
CAVEREC_GPU_POWER	Linux (low, igpu, low_power, low-power or high, dgpu, high_performance, high-performance)	Selects the graphics adapter power preference on multi-GPU Linux studio workstations. Low-power values (low, igpu, low_power, low-power, the default) select integrated low-noise graphics so cooling fans never bleed into live room microphones, while high-performance values (high, dgpu, high_performance, high-performance) select a discrete GPU.

Environment Variable	Platform / Scope	Studio Function
CAVEREC_GDRIVE_CLIENT_ID	macOS, Windows, Linux	Optional Google OAuth 2.0 Desktop Client ID used by Check / Connect Drive for the Google account browser sign-in flow when gdrive_oauth_client.json is not present in the user configuration directory (Section 7.5.1).
CAVEREC_GDRIVE_CLIENT_SECRET	macOS, Windows, Linux	Optional Google OAuth 2.0 Client Secret paired with CAVEREC_GDRIVE_CLIENT_ID when required by a custom Desktop OAuth client credential.
CAVEREC_GDRIVE_ACCESS_TOKEN	macOS, Windows, Linux	Optional OAuth 2.0 access token override for unattended Google Drive cloud synchronization in automated studio backup scripts when interactive browser sign-in is not available.

PART IV | CHAPTER 9

Studio Glossary & Alphabetical Index

Canonical definitions of Cave Recorder terminology, Take vs. Pass disambiguation, multi-camera video and Cave Companion mobile terms, and an alphabetical subject index.

This chapter defines the canonical studio terminology used across every console faceplate, timeline lane, session file, and workflow in Cave Recorder (caverec), followed by the complete Alphabetical Index of the manual.

9.1 Take vs. Pass Disambiguation (Quick Guide)

Traditional desktop digital audio workstations use the word "Take" for both an entire song recording and an overdubbed layer on a single track, causing constant confusion during band rehearsals. Cave Recorder enforces a strict four-tier studio hierarchy where **Session-level Takes** and **Track-level Passes** are completely distinct concepts:

```
Project (Band / Artist Workspace, e.g., GF)
 └── Session (Rehearsal Night or Tracking Day, e.g., GF_20261020-1)
      └── Take (Complete Song Run-Through Starting at 00:00:00.000)
           └── Pass (Single-Track Stem Inside That Take: PASS 01 / OD 02)
```

Studio Term & Level	UI Badge & Location	On-Disk Pattern	How It Is Created & Used
Project (Level 1: Band or venture)	PROJECT modal (**CMD** / **CTRL** + **O**)	<LibraryRoot>/<ProjectTag>/	Top-level band folder containing all Sessions plus a unified _Mixes/ folder for rapid listening and cloud sharing.
Session (Level 2: Rehearsal or studio day)	Session pill in Top Deck & SESSION REEL (**CMD** / **CTRL** + **4**)	<ProjectTag>/<SessionID>_<Tag>/	Defines the room's 18-channel microphone patch, strip names, bus routing, and console processing shared across all Takes recorded that day.

Studio Term & Level	UI Badge & Location	On-Disk Pattern	How It Is Created & Used
Take (Level 3: Complete song or jam at t = 0)	< TAKE 001 > in Top Deck & TAKES CATALOG	Take<NNN>_<Stamp>[_<TakeTag>]/	Created whenever you press Record (**R**) with Overdub Mode (OD) **off**. Every Take starts at its own 00:00:00.000 origin and is never stacked on top of another Take.
Base Pass (Level 4A: Initial track stem)	PASS 01 (BASE PASS) on strip & Pass Lane	rec/*_<CC>-<Name>_Rec001_<Stamp>.wav	The initial 32-bit float pre-fader mono stem captured on a record-armed channel strip when the Take was first recorded from 00:00:00.000.
Overdub Pass (Level 4B: Layered track stem)	OD 02, OD 03 (OVERDUB) on strip & Pass Lane	rec/*_<CC>-<Name>_Rec<MMM>-OD_<Stamp>.wav	A subsequent track-level recording captured inside the currently active Take when Overdub Mode (OD, **O**) is latched on the Top Deck.
Pass Lanes (Track sub-lanes inside a Take)	PASSES (**E** / **SHIFT** + **E**) in WAVEFORM TIMELINE	Stored in *_Take<NNN>_take.json	Expandable vertical sub-lanes under a track displaying its PASS 01 Base Pass and OD 02..N Overdub Passes for one-click comp selection (Use PASS 01 / ★ OD 02).

> **PASS LANES ARE NEVER CALLED TAKE LANES**
> In Cave Recorder, track-level overdub lanes under a channel are always called **Pass Lanes** (PASS 01, OD 02, OD 03) and are never called "Take Lanes", and the project and session browser modal is always labeled **PROJECT** and is never called a "Welcome Screen" or "Welcome Dialog". A **Take** (TAKE 001, TAKE 002) is always an entire multitrack song performance starting at 00:00:00.000, while a **Pass** is a single track's recording layer inside that Take.

9.2 Studio Glossary

The alphabetical definitions below cover every core studio concept, signal-processing stage, console switch, and workflow term in Cave Recorder, with direct links to the chapters where each topic is covered in depth.

12-Band Parametric EQ (2:12B EQ)

Zero-latency 12-band precision state-variable parametric equalizer (B1 through B12) available in the 8-slot processor rack of every Input Track (CH01..CH32) and Subgroup Bus (SUB 1..SUB 16). Each band provides -24.0 dB to +24.0 dB gain across 20 Hz to 20 kHz, variable Q (0.10 to 18.00), and seven selectable filter shapes: peaking Bell (BELL), Low Shelf (LSHF), High Shelf (HSHF), High-Pass Filter (HPF), Low-Pass Filter (LPF), surgical Notch (NTCH), and Spectral Tilt (TILT). See Section 5.3.

30-Minute Retroactive Capture (RAM / RETRO)

Continuous background audio capture system that retains the most recent 30 minutes of live 32-bit floating-point audio across all record-armed (R) input channels in system memory whenever the green RAM button is lit in the Top Transport Deck. Clicking RETRO (or pressing **SHIFT** + **R**) flushes the rolling 30-minute memory to disk as a new numbered Session Take (Take<NNN>) so spontaneous jams played while the transport was stopped are never lost. See Section 1.1.2, Section 2.2.3, and Section 3.2.

8-Slot Processor Rack

Multi-instance reorderable eight-slot insert processing chain (Slots 1..8) on every Input Track and Subgroup Bus. Any built-in zero-latency studio processor (2:12B EQ, 3:VCA CMP, 4:AUX SND movable send tap, Amp Rig, standalone Cabinet & IR Loader, Tape Saturation, Context-Aware Gate, De-Esser, Vocal Rider, Transient Shaper, Ladder Filter, Modulation, Micro-Pitch Doubler, Gain Utility, or Signal Repair) or third-party .clap and macOS Audio Unit (AUv3 / AUv2) plugin can be inserted in one or more slots across the same strip with independent per-slot state and telemetry. Clicking any slot opens the inline SLOT N PROCESSOR picker to swap processor types, assign an installed plugin, open a floating PLUGIN GUI window, or preserve an uninstalled plugin as a non-destructive Ghost Slot (! MISSING). Slots can be reordered live during playback by dragging their :: handle or clicking ^ / v with clickless crossfading, or bypassed individually via B. See Section 4.1, Section 5.1.3, and Section 5.7.3.

Active Comp Lane (ACTIVE COMP LANE)

The single **Pass Lane** currently selected for disk playback on a track inside the active Take. Highlighted in amber-copper waveform color with a green star badge (★ PASS 01 or ★ OD 02) in WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**) and reflected on the channel strip's gold pass indicator badge on the Mix Console. See Section 3.4 and Section 6.2.2.

Amp Dynamic Gate (AMP DYNAMIC)

High-gain guitar and bass variant of the Context-Aware 3-Variant Noise Gate engineered for stoner rock, doom, psychedelic rock, and tube/fuzz rigs. Combines an autocorrelation pitch periodicity detector with a spectral flatness detector so whisper-quiet -34 dBFS clean fingerpicked notes (Whisper Clean Lock) and intentional 1.5 s to 5.0 s (default 2.5 s) harmonic cabinet feedback swells (Feedback Bloom Lock) remain wide open, while 50/60 Hz hum and fuzz hiss clamp shut in 15 ms to 40 ms (default 18 ms to 22 ms) on a palm-mute stop (Palm-Mute Clamp) along a click-free S-curve with split-band HF Tame de-hissing. See Section 5.4.5.

Amp Rig & Neural Amp Capture ([CIR] / [NAM] / HEAD ON/BYP / CAP)

Zero-latency guitar and bass amplifier, cabinet, and neural capture insert processor available in any of the 8 rack slots on Input Tracks and Subgroup Buses. Operates in either **[CIR] Circuit Mode** (modeling 8 vacuum-tube preamp and power-amp topologies with 2x / 4x anti-aliasing oversampling) or **[NAM] Neural Capture Mode** (running open .nam neural amplifier models alongside 8 modeled speaker cabinets and custom .wav/.nam linear impulse responses), with independent **HEAD ON/BYP** and **CAB BYP** switches for pairing an amplifier head with downstream stereo effects and a standalone Cabinet & IR Loader slot. Includes a guided 3-step **Hardware Amp & Pedal Capture Wizard (CAP)** supporting three training quality modes (**Standard** 150 training passes, **Mastering** up to 1,000 training passes with held-out validation checkpointing, adaptive step-size refinement, automatic convergence early stopping, and interactive **STOP & KEEP BEST**, and **Turbo** 60 training passes), live training and validation ESR metering (STUDIO < 1% ESR), and automatic replication into <SessionFolder>/models/ and <TakeFolder>/models/ for portable cloud sync. See Section 5.4.6.

Analog VCA Compressor (3:VCA CMP)

Zero-latency voltage-controlled amplifier dynamics processor included in every Input Track and Subgroup Bus 8-slot rack. Provides adjustable Threshold (-60.0 dB to 0.0 dB), Ratio (1.0:1 to 20.0:1 with 2:1, 4:1, 8:1, and 20:1 presets), Attack (0.1 ms to 100.0 ms), Release (10.0 ms to 1000.0 ms), Makeup Gain (0.0 dB to +24.0 dB), a 90 Hz Sidechain High-Pass Filter (SC HPF, shortcut **H** in Channel Inspector), Stereo Link (LINK), New York parallel blend (MIX 50%), and a real-time Gain Reduction (GR) trace. See Section 5.4.

Audio Settings Modal (AUDIO SETTINGS)

Hardware configuration dialog opened via AUDIO SETTINGS in the Top Deck (**CMD** + **,** on macOS, **CTRL** + **,** on Windows and Linux). Selects the physical input and output audio interface, sample rate (44.1, 48.0, 88.2, or 96.0 kHz), hardware buffer size (32 to 2048 frames), checks macOS Microphone permission status with one-click permission request and macOS permission reset guidance, and runs the automated hardware impulse loopback latency calibration (Measure Latency). See Section 2.2 and Section 2.3.

Audio Unit (AUv2 & AUv3 on macOS)

Native macOS audio effect ('aufx') and music effect ('aumf') plugins discovered across /Library/Audio/Plug-Ins/Components, ~/Library/Audio/Plug-Ins/Components, and registered AUv3 application extensions. Hosted directly in any of the eight reorderable slots (Slots 1..8) on Input Tracks and Subgroup Buses with canonical au.<type>.<subtype>.<manufacturer> identifiers, isolated process crash protection (AUv3 and bridged AUv2), three-tier native floating GUI hosting (PLUGIN GUI), binary Audio Unit preset state persistence, side-chain bus routing, automatic Plugin Delay Compensation (PDC), and Low-Latency Monitoring bypass. See Section 5.7.

Aux Return (AUX 1 REVERB & AUX 2 TAPE DLY .. AUX 8)

Up to eight dedicated, solo-safe stereo effect return strips (AUX 1 through AUX 8, dynamically added via **+ AUX**) on the right side of the Mix Console feeding the Master Bus:

- **Reverb Return (AUX 1 REVERB):** Zero-latency acoustic convolution and FDN plate reverb (EMT 140 Plate, Studio Room, Chamber, Spring) with Input Low-Cut (LC), Pre-Delay (PD, 0 to 200 ms), Decay Time (DCY, 0.20 to 12.00 s), and WET blend toggle.
- **Delay Return (AUX 2 TAPE DLY):** Capstan magnetic tape, BBD analog, or pristine digital echo processor with Wow & Flutter (WOW), High-Frequency Damping (DMP), Delay Time (TM, 1 to 2000 ms), Feedback (FB), and Stereo Ping-Pong (PP) widening.

See Section 4.3.2 and Section 5.5.

Aux Send (RVB & DLY / PRE, EQ, PST)

Per-channel and per-subgroup stereo send level controls (-inf to +12.0 dB) feeding AUX 1 REVERB (RVB) and AUX 2 TAPE DLY (DLY), paired with a 3-way pickoff tap selector: PRE (pre-insert and pre-fader for independent cue sends), EQ (post-EQ and pre-fader), or PST (post-fader and post-mute, scaling wet effects naturally with fader moves). See Section 4.2.3.

Base Pass (Rec001 / PASS 01)

The initial multitrack audio stem captured on a record-armed channel strip when a Session **Take** is originally recorded from 00:00:00.000 with Overdub Mode (OD) turned off. Stored on disk in <TakeFolder>/rec/<SessionID>_Take<NNN>_<CC>-<TrackName>_Rec001_<Stamp>.wav and displayed as PASS 01 (BASE PASS) in the channel strip header and **Pass Lanes**. See Section 1.2 and Section 6.2.

Cab IR (Speaker Cabinet & Linear Impulse Response Loader)

Standalone reorderable speaker cabinet and linear impulse response (IR) convolution rack processor (SPEAKER CABINET & IR LOADER) available in any of the 8 rack slots on Input Tracks and Subgroup Buses. Provides 8 modeled physical speaker enclosures (2x12, 4x12 25, 4x12 75, 4x12 BR, 4x12 TG, 4x12 OV, 4x12 15, and 8x10), 5 studio microphone capsule models (DYN-C, DYN-L, RBN-8, CND-L, DYN-B), continuous MIC POS and ROOM AIR, an interactive MIC HPF × MIC LPF (20..500 Hz × 2..20 kHz) 2D XY spectral contour pad, and zero-allocation time-domain FIR convolution (EXT IR, up to 2048 taps per channel) for custom mono/stereo .wav and .nam ("Linear") impulse responses managed through the three-tier (Take -> Session -> Global) **[CAB IR]** browser. Enables placing stereo modulation, delay, or reverb rack slots between an Amp Rig head (CAB BYP enabled) and the speaker cabinet. See Section 5.4.7.

Camera Latency Calibration (3-Second 4-Pulse Optical & Acoustic Calibration)

Automated 3-second (34,320-sample at 48.0 kHz) optical and acoustic calibration workflow (CALIBRATE CAMERA LATENCY) that measures the physical sensor, capture-card, and rolling-shutter pipeline delay of connected studio cameras and wireless **Cave Companion** mobile camera nodes to sub-frame microsecond accuracy. Emits 4 staggered pulses combining an 80 ms full-white screen flash (or mobile rear LED Torch flash) with a studio monitor click, correlates the direct-path acoustic arrival sample against partial-exposure luminance ramps across frames, and stores the calibrated camera latency in session.json so every recorded .vididx sidecar index locks visually to the 48.0 kHz audio timeline. See Section 2.5.2.

Cave Companion Mobile App (Android .apk & iOS .ipa)

Native mobile studio companion application (caverec-v0.4.2-android-arm64.apk and caverec-v0.4.2-ios-arm64.ipa) paired with the desktop workstation via the **Pair Mobile Companion** modal (studio pairing QR code or 6-digit PIN fallback with per-device access role permissions). Combines a low-jitter wireless studio control and telemetry link (sub-millisecond clock lock, authenticated remote transport control with safe two-step Discard Take confirmation, 60 Hz playhead scrubbing, and 25 Hz 16-channel segmented LED meter broadcast), a high-speed Wi-Fi media sync client (with .part resume and file integrity verification), a **Wireless 1080p50 CFR Mobile Camera Node** (16:9 Cinema HUD with ULTRA-WIDE / WIDE / TELEPHOTO / FRONT lens selection and automatic desktop timeline ingest), and a **Video-First Multi-Angle & Multi-Print Take Player** with pinned transport, continuous audio/video synchronization, Google Drive Resumable Sync, and storage quota cache tiering. See Section 3.7 and Section 7.6.

Channel Inspector (CHANNEL INSPECTOR)

Full-screen single-channel, Subgroup Bus, Aux Return, and Master Bus processing workbench (**CMD** + **3** on macOS, **CTRL** + **3** on Windows and Linux). Combines a full-height channel strip on the left with the top **Signal-Flow Chain & Layout Mode Bar** (SPLIT, CHAIN LANE, EQ+CMP, and HIDE TL / SHOW TL), an expanded 12-band Parametric EQ and Dual-Trace RTA, an expanded Analog VCA Compressor with scrolling Gain Reduction trace, full-height **Large Processor Views** featuring interactive **2D XY Control Pads** and real-time visualizers across all 15 built-in processors and the Master Bus Brickwall Limiter, a multi-bus output routing popover (OUTPUT BUSES), and a collapsible single-track Mini-Timeline with Pass Lane comping controls. See Section 5.1.2.

CLAP Plugin (1:CLAP)

Third-party audio effect plugin conforming to the open **CLAP (Clever Audio Plug-in, .clap)** standard on macOS, Windows, and Linux, hostable in any of the eight reorderable slots (Slots 1..8) on any Input Track or Subgroup Bus. Supports full catalog indexing and management in the PLUGINS view (**CMD** / **CTRL** + **7**), inline rack assignment (SLOT N PROCESSOR), native floating desktop windows (PLUGIN GUI), binary CLAP preset state persistence with non-destructive Ghost Slot recovery, automatic Plugin Delay Compensation (PDC), and one-click Low-Latency Monitoring bypass. See Section 5.7.

Cloud Sync (CLOUD)

Built-in two-way Google Drive synchronization engine for backing up Sessions and collaborating across bandmates' computers. Supports two transfer tiers (MIXES for fast synchronization of manifests, Snapshots, and .flac/.mp3/.wav Prints and Mixdowns, and FULL for complete raw rec/*.wav multitrack stems), automatic Print conversion (AUTO-FLAC ON UPLOAD and AUTO-MP3 ON UPLOAD), federated shared project linking (+ Link Remote Project...), exclusive session write-locking (Read-Only Preview), and zero-loss offline conflict merging via Snapshots. See Section 7.5.

Control Room Monitor Bar & AUTO-LVL

Speaker-check and loudness-matched audition strip wired strictly downstream of the Master Bus 2-TRK live stereo Print tap and offline BOUNCE engine. Includes 4-band Linkwitz-Riley (24 dB/oct) frequency isolation buttons (SUB < 90 Hz, LOW 90-350 Hz, MID 350 Hz - 3.5 kHz, HIGH > 3.5 kHz), unity-gain MONO (M⊕) fold-down, equal-power DIFF ((L - R) × 0.7071) stereo difference solo, DIM (-20 dB default, configurable -40 dB to -6 dB) speaker attenuation, and AUTO-LVL loudness-matched processor/rack bypass (300 ms RMS tracker gated below -60 dBFS and clamped to ±18 dB). See Section 4.3.4.

Cycle Loop (⇄)

Master Transport mode (**C** or clicking ⇄ in the Top Deck) that loops playback or overdub recording continuously between the START and END boundaries of the active Take Range (R001, R003, ...). See Section 3.4 and Section 6.3.2.

Drum Bleed Gate (DRUM BLEED)

Percussion-focused variant of the Context-Aware 3-Variant Noise Gate designed to isolate rack and floor toms from loud snare and cymbal spill. Combines a resonant fundamental sidechain filter (Q = 4.0) with one-click [LRN] pitch auto-detection (55 Hz to 350 Hz, default 110 Hz), differential snare/cymbal spill rejection (E_key - α * E_ref), 1.5 ms lookahead with < 0.25 ms attack, and an envelope-driven dynamic decay low-pass filter (20 kHz sweeping down to an adjustable 250 Hz to 800 Hz cutoff, default 380 Hz) that removes hi-hat wash during tom ring-outs. See Section 5.4.5.

Dual-Trace Spectrum Analyzer (RTA)

High-resolution real-time acoustic spectrum analyzer displayed behind the 12-Band Parametric EQ curve in both the Mix Console Upper Well 2 and the CHANNEL INSPECTOR view. Plots the pre-EQ input spectrum (darker shaded trace) and post-EQ output spectrum (bright emerald trace) simultaneously from 20 Hz to 20 kHz. See Section 5.3.2.

Export Take (EXPORT TAKE / _Exports/)

Multi-format external project and stem interchange workflow inside the SESSION REEL view (**CMD** / **CTRL** + **4**) that packages any recorded Take into .rpp project files, .als Live Set projects, <Session version="7003"> XML session folders, .dawproject open interchange archives, .fcpxml v1.9 video timelines, or universal Broadcast Wave (bext / iXML / MIDI marker / 24-bit 48 kHz minimum-phase impulse response) bundles. Writes to <TakeFolder>/export/ and links into <ProjectTag>/_Exports/ with either zero-copy relative stem references or self-contained copied/rendered audio stems and .nam neural amplifier models. See Section 7.2.4.

Ghost Slot (! MISSING Plugin Recovery)

Non-destructive rack placeholder activated automatically when a Session, Snapshot, or Session Template references a CLAP or Audio Unit plugin that is not currently installed on the workstation or was moved on disk. Marked in amber as ! MISSING (! <Name>) in the 8-Slot Processor Rack, a Ghost Slot passes unity-gain dry audio so the session plays cleanly while preserving the original plugin_id, plugin display name, and base64-encoded binary preset state untouched across saves until the plugin is relinked via LOCATE..., reinstalled and rescanned via RESCAN DIRECTORIES, or replaced in SLOT N PROCESSOR. See Section 5.7.4.

Input Monitor (I)

Per-track console switch (I on channel strips, MON in Channel Inspector, shortcut **SHIFT** + **I** on the inspected track in Mix Console) that routes the live physical hardware input (IN 01..IN 18) through the channel's 8-slot processor rack and fader even while transport playback is rolling, allowing musicians to rehearse or monitor live overdubs through console EQ, compression, and reverb. See Section 4.2.4.

Input Trim (TRM)

Digital pre-insert gain control (-24.0 dB to +24.0 dB, default 0.0 dB) located at the top of every Input Track and Subgroup Bus strip. On Input Tracks, TRM sits upstream of both the raw multitrack stem recorder (rec/) and the 8-slot processor rack so you can fine-tune preamp gain staging before stems hit disk. On Subgroup Buses, TRM gain-stages the summed multi-channel signal entering the subgroup's SUB RACK. See Section 4.1 and Section 4.2.2.

Keymap View (KEYMAP)

Interactive keyboard shortcut reference and customization workspace (**CMD** + **6** on macOS, **CTRL** + **6** on Windows and Linux). Lists all 59 studio actions across seven scopes with live substring search, interactive Reverse-Search by Shortcut (FIND BY SHORTCUT:), scope filter pills, inline key-chord capture (ASSIGN / APPLY), scope clash detection, operating-system conflict warnings (OS WARN), and automatic persistence to keymap.json. See Section 8.3.

Live Mix (A/B LIVE MIX & RETURN TO LIVE MIX)

Automatic working-memory safeguard that preserves your live, unsaved Mix Console state the instant you click AUDITION SNAP <NNN> in the SESSION REEL view (**CMD** / **CTRL** + **4**). While in Audition Mode, you can step across any number of historical Snapshots, click A/B LIVE MIX for clickless real-time comparison against your working mix, or press **BACKSPACE** (RETURN TO LIVE MIX) to restore your exact working console state with zero lost fader moves. See Section 7.3.2.

Live Stereo Print (2-TRK / Print001)

Real-time 2-track post-Master-Bus stereo recording captured concurrently alongside raw pre-fader multitrack stems whenever a Take is recorded with 2-TRK enabled on the MST MASTER BUS strip. Written to <TakeFolder>/mix/<SessionID>_Take<NNN>_Print<NNN>_<Stamp>.wav with automatic Plugin Delay Compensation (PDC) latency trimming, and convertible in one click (FLAC PRINT, MP3 PRINT, FLAC ALL PRINTS, or MP3 ALL PRINTS) to a lossless 24-bit .flac file or a 320 kbps CBR .mp3 file linked into <ProjectTag>/_Mixes/ for immediate post-rehearsal band listening. See Section 1.1.3, Section 4.1.1, and Section 7.2.

Low-Latency Monitoring

Top Deck monitoring mode (Low-Latency Monitoring in Row 2 of the Top Deck) that temporarily bypasses high-latency lookahead CLAP and Audio Unit plugins on record-armed (R) live monitoring paths and their downstream buses during overdubbing, dropping foldback latency to zero internal samples while keeping full Plugin Delay Compensation active on unarmed playback tracks. See Section 5.7.6.

Markers & Ranges (M001 / R001)

Sample-accurate timeline cue flags and bounded time spans stored per Take and managed in both the WAVEFORM TIMELINE (MKR and RNG lanes) and the dedicated MARKERS & RANGES workspace (**CMD** / **CTRL** + **5**). Standalone cue markers (M005, …) are dropped on the fly with **M** and navigated via **[**, **]**, **SHIFT** + **[** (**HOME**), and **SHIFT** + **]** (**END**). Ranges (R001, R003, …) are bounded by start (**K**) and end (**L**) markers for Cycle Loop playback (⇄) and non-destructive Take extraction (NEW TAKE). See Section 6.3.

Master Bus (MST MASTER BUS)

The final stereo summing strip on the right edge of the Mix Console, summing all Subgroup Buses, direct Input Tracks, and Stereo Aux Returns. Incorporates a built-in Master Tape Saturation stage (TAPE with DRIVE and WARM), a Master Bus Glue VCA Compressor (COMP), a Lookahead Brickwall Limiter (LIM with THR and CEIL), the live 2-TRK Print recorder arm switch, independent FLAC and MP3 format toggles with a one-click BOUNCE offline Mixdown trigger, CLR CLIP, Master Fader, and 40-segment stereo Peak/RMS meter. See Section 4.3.3.

Mix Console (MIX CONSOLE)

Primary studio workspace view (**CMD** + **1** on macOS, **CTRL** + **1** on Windows and Linux) combining the four Upper Inspector Wells (8-Slot Rack & Tape, 12-Band EQ & RTA, VCA Compressor & GR, and Mini-Timeline) with a dynamically configurable lower bank of up to 32 mono or stereo-linked Input Track strips (CH01..CH32, added via + TRK), up to 16 stereo Subgroup Buses (SUB 1..SUB 16, added via + SUB), up to 8 Stereo Aux Returns (AUX 1..AUX 8, added via + AUX), and the MST MASTER BUS. See Chapter 4.

Multi-Camera Video Suite (50 FPS CFR, .vididx Sidecar & Video Timeline Strip)

Sample-accurate multi-camera studio video recording, monitoring, and editing system supporting desktop USB/HDMI cameras (macOS and Linux) and wireless **Cave Companion** 1080p50 mobile camera nodes. Captures Constant Frame Rate (50 FPS = 960 samples/frame or 25 FPS = 1920 samples/frame at 48.0 kHz, 50 Hz mains anti-flicker, zero B-frames, 0.5 s closed GOP) .mp4 streams into <TakeFolder>/video/ paired with a binary .vididx sidecar index (locked to 48.0 kHz audio-clock timing for instant frame lookup and drift-free playback), rendered in BT.709 on the **Video Monitor** and as a keyframe-snapped **Video Timeline Strip** filmstrip with a per-clip timing nudge slider in WAVEFORM TIMELINE. See Section 2.5 and Section 6.6.

Multi-Mic Phase Alignment (ALIGN ALL & Φ)

Automatic acoustic arrival-time and polarity correlation processor that analyzes phase relationships across multi-miked instruments (such as a drum kit or dual-miked guitar/bass cabinet) relative to a timing anchor channel (0us) using sub-sample bandpass (150 Hz to 6 kHz) GCC-PHAT cross-correlation on recorded .wav stems or rolling capture memory. Clicking ALIGN ALL in the Top Deck (or Φ on an individual strip) calculates and engages sub-sample time delay (0 to 50,000 us / 0.00 to +50.00 ms) and 180-degree polarity inversion (Ø) to eliminate comb filtering, with single-step Undo/Redo support and instant global comparison via PHASE: ALIGN / PHASE: BYP. See Section 3.3 and Section 5.6.

Offline Stereo Mixdown (Mix001)

Faster-than-real-time stereo render of an entire Take processed from 00:00:00.000 through the complete Mix Console signal chain (including updated fader balances, 8-slot inserts, Subgroup/Aux buses, Master Bus processing, and active Pass Lane selections). Triggered by selecting FLAC and/or MP3 and clicking BOUNCE on the Master Bus, saved to <TakeFolder>/mix/<SessionID>_Take<NNN>_Mix<NNN>_<Stamp>.<flac|mp3> (with a Xing/Info seek header on .mp3 files for web browser scrubbing), and linked into <ProjectTag>/_Mixes/. See Section 7.2.3.

Overdub Mode (OD)

Transport recording mode toggled via the red OD button in the Top Deck (or pressing **O**). When OD is off, pressing Record (**R**) creates a brand-new t = 0 Session **Take**. When OD is latched (illuminated red), pressing Record (**R**) plays back the existing tracks of the currently active Take while recording a new track-level **Overdub Pass** (Rec002-OD, OD 02) onto armed channels with automatic roundtrip loopback latency compensation. See Section 3.4 and Section 6.2.

Overdub Pass (Rec002-OD / OD 02)

A layered or replacement track-level recording captured inside an existing Session **Take** when Overdub Mode (OD) is enabled on the transport bar. Saved as <TakeFolder>/rec/<SessionID>_Take<NNN>_<CC>-<TrackName>_Rec<MMM>-OD_<Stamp>.wav and displayed as OD 02, OD 03, and higher inside that track's expandable **Pass Lanes**. See Section 1.2 and Section 6.2.

Pass

Any individual audio recording on a single channel strip belonging to a specific Session **Take**, encompassing both the initial **Base Pass** (Rec001 / PASS 01) captured when the Take was first created and any subsequent **Overdub Passes** (Rec002-OD / OD 02, OD 03, ...) recorded into that Take. See Section 9.1.

Pass Lanes (PASSES)

Expandable vertical sub-lanes under an Input Track in WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**, expanded with **E** and collapsed with **SHIFT** + **E**) and CHANNEL INSPECTOR (**CMD** / **CTRL** + **3**, toggled with **E** or PASSES). Pass Lanes display the track's PASS 01 Base Pass and all OD 02..N Overdub Passes inside the active Take for one-click comp promotion (Use PASS 01 / ★ OD 02), per-pass muting (M), and non-destructive deletion (Del). Track-level overdub lanes are always called **Pass Lanes** and are never called "Take Lanes". See Section 6.2.2.

Phase Alignment Network (Zero-Slot Allpass Rotator)

Dedicated zero-slot 5-stage allpass phase rotation network (one 1st-order 20-500 Hz low-frequency rotator plus four 2nd-order 20-12,000 Hz biquad allpass stages with Q in 0.30..3.50, default 0.707, and a clickless 10 ms dual-branch preset crossfade) built into the top of every Input Track strip alongside Ø and Φ. Programmed automatically by Mode 2 of the Polarity & Phase Maximizer while leaving all 8 rack slots free. See Section 5.6.4.

Plugin Delay Compensation (PDC)

Automatic sample-accurate delay compensation engine that aligns parallel tracks, Subgroup Buses, and Aux paths whenever a third-party CLAP or Audio Unit plugin introduces internal lookahead buffering. Displayed live in Row 2 of the Top Deck (PDC: N smp (X.XX ms @ 48 kHz)) and detailed per channel in the **Latency & PDC Transparency Breakdown** popover. See Section 5.7.5.

Plugins View (PLUGINS)

Dedicated plugin catalog and management workspace (**CMD** + **7** on macOS, **CTRL** + **7** on Windows and Linux). Displays every discovered .clap and macOS Audio Unit (AUv3 / AUv2) plugin with live text search, status and usage filter pills (ALL, AVAILABLE, MISSING / NEEDS RESOLUTION, IN USE IN SESSION), custom folder registration (+ ADD SEARCH FOLDER..., + ADD .CLAP FILE...), hot directory rescanning (RESCAN DIRECTORIES), and one-click bundle relinking (LOCATE...) backed by clap_plugin_index.json. See Section 5.7.2.

Polarity & Phase Maximizer (Mode 1 & Mode 2)

Two-tier multi-channel phase optimization suite that maximizes constructive acoustic energy across multi-mic instruments without moving microphone stands:

- **Mode 1 (Instant Binary Polarity Maximizer, < 50 ms):** Partitions channels into connected acoustic groups and evaluates all polarity combinations (Ø) across four weighting criteria (Low-End Weight 30-200 Hz Punch, Broadband Loudness ITU-R BS.1770-4 K-Weighted LUFS, Transient Crest & Impact 60 Hz-6 kHz, and Anti-Comb Spectral Smoothness).
- **Mode 2 (Deep Multicore Spectral Phase & Allpass Solver, Anytime 30 s .. 30 min):** Analyzes active performance regions across 256 frequency bands (20 Hz to 20 kHz) and runs multicore global phase and allpass optimization across all CPU cores to populate Candidates A (Pure Ø - Zero Dispersion), B (Tight Low-End Lock - Minimal Poles), C (Deep Multi-Band Allpass), and D (Full Wavefront + Allpass) with [GAIN-MATCH AUDITION].

See Section 5.6.4 and Section 5.6.5.

Polarity Audition Memory

Automatic session memory safeguard that preserves your complete per-channel phase state (polarity, microsecond delay, and 5-stage allpass settings) before you audition Mode 1 or Mode 2 Polarity & Phase Maximizer candidates, restoring your exact prior polarity, microsecond delay, and allpass state if you cancel or revert. See Section 5.6.4.

Polarity Invert (Ø)

Console switch (Ø, shortcut **SHIFT** + **P** on the inspected strip in Mix Console) that flips positive and negative waveform excursions by 180° (illuminates cyan when active). Used to correct phase cancellation on bottom snare microphones, rear open-back guitar cabinet microphones, or inverted balanced XLR cables, and set automatically during ALIGN ALL. See Section 4.2.2 and Section 5.6.

Project & Project Tag (PROJECT / _Mixes/)

Top-level workspace representing a band, artist, or musical venture, identified by a unique 1-to-8 character uppercase alphanumeric **Project Tag** (GF, TEST, MISC) and stored at <LibraryRoot>/<ProjectTag>/. Managed in the **PROJECT** modal (**CMD** / **CTRL** + **O**), each Project contains all of that band's Sessions plus a flat _Mixes/ index folder containing relative links to all active .flac, .mp3, and .wav Prints and Mixdowns across every Session. See Section 1.2, Section 2.4, and Section 7.4.

Roundtrip Loopback Calibration (Measure Latency)

Built-in hardware impulse measurement tool inside the AUDIO SETTINGS modal (**CMD** / **CTRL** + **,**) that transmits a calibration pulse out a selected physical output jack (Out 20 or Out 1) over a patch cable into an analog input jack to measure exact physical converter, ADAT, and USB/Thunderbolt roundtrip delay in samples. The measured offset is automatically applied as negative record-offset compensation on all subsequent Overdub Passes (OD). See Section 2.3.

Session & 08:00 Rollover Rule

A single rehearsal evening, jam night, or studio tracking day within a Project (<ProjectTag>/<SessionID>[_<SessionTag>]/), defining the shared 18-channel hardware patch, strip names, bus topology, and console settings across all Takes recorded during that gathering. Cave Recorder rolls the session calendar date over at **08:00 local time** so Takes recorded after midnight (01:30) remain grouped under the same evening session date (YYYYMMDD) as recordings started at 21:00. See Section 1.2 and Section 7.4.2.

Session Reel (SESSION REEL)

Session management dashboard (**CMD** + **4** on macOS, **CTRL** + **4** on Windows and Linux) containing the session metadata and cloud sync header, the reversible Undo/Redo history bar (UNDO, REDO, SNAP MANUAL), the searchable **TAKES CATALOG** (Take loading, starring, tagging, FLAC PRINT and MP3 PRINT encoding, soft-delete trash management, and latency offset re-application), and the **SNAPSHOTS** table. See Chapter 7.

Session Template (SAVE TEMPLATE / MicAndGuitar001)

Reusable console configuration file (<LibraryRoot>/.caverec/templates/<Name>.json) saved via the SAVE TEMPLATE button in Row 2 of the Top Deck and selectable when creating a new Session in the PROJECT modal. Cave Recorder seeds two built-in studio templates on first launch: **MicAndGuitar001** (the default 2-channel singer-songwriter template with Channel 1 Mic feeding an -8.0 dB Post-Fader Reverb send and Channel 2 Guitar, both with 80 Hz high-pass filters) and **GuitarBassDrumTrio001** (an 11-channel band template with Guitar Bus, Drum Bus, and parallel Drum Crush Bus subgroups). Templates store all channel strip names, physical input bindings, Subgroup Bus assignments, 8-slot processor racks, EQ curves, and compressor settings while excluding recorded audio Takes and undo history. See Section 2.4.3.

Snapshot (SNAPS)

Immutable point-in-time capture of the entire Mix Console state, active Pass Lane selections, and compact undo history stored in <SessionFolder>/snapshots/<SessionID>_<NNN>_<Timestamp>.json. Created manually via SNAP MANUAL (**CMD**/**CTRL** + **S**), automatically after every normal recording stop ([POST-REC]), or every 20 minutes during active editing ([AUTO]), and auditioned losslessly via **Live Mix**. See Section 7.3.

Spectral Spot-Heal & Live-Take Mistake Surgery

Suite of non-destructive timeline repair operations in WAVEFORM TIMELINE (**CMD**/**CTRL** + **2**) designed to salvage an unrepeatable live band Take without overwriting raw .wav stems on disk: [REPAIR CLIPPING] (smooth waveform interpolation for 1..8 samples and harmonic waveform prediction for 9..256 samples of clipped ADC transients), 50/60 Hz (45..65 Hz fine-tune) 12-harmonic Q = 20..100 (default Q = 60) comb hum cancellation, range-bounded Pass comping ([USE OD 02 IN RANGE]), pre-insert Clip Gain, [PATCH FROM SOURCE...] bar/chord grafting with ±50 ms cross-correlation phase lock, and [SMOOTH OVER ARTIFACT] (5 ms to 150 ms waveform and spectral spot-healing of mic stand bumps, pops, and clicks). See Section 6.5.

Split-Screen Video Mixdown (VideoMix001.mp4) & .fcpxml v1.9 Multicam Export

Offline multi-camera video rendering and Final Cut XML v1.9 export workflow opened from the **Export Video Mixdown & Multicam FCPXML** modal in SESSION REEL (**CMD** / **CTRL** + **4**). Composites up to four calibrated camera angles in BT.709 color space across five layouts (Single, Side-by-Side Crop with horizontal camera pan, Side-by-Side 16:9, 3-Cam Band, and Quad 2x2), bridges 960/1920-sample CFR video cadence to 1024-sample AAC-LC stereo console audio to write <TakeFolder>/mix/<SessionID>_Take<NNN>_VideoMix<MMM>_<Stamp>.mp4 (linked into <ProjectTag>/_Mixes/) alongside gapless 48.0 kHz .flac/.mp3 audio prints, and exports .fcpxml v1.9 timelines with both stacked split-screen lanes (<adjust-crop>, <adjust-transform>, <timeMap>) and native <mc-clip> multicam switching angles. See Section 7.2.5.

Standby Comp Lane (STANDBY COMP LANE)

Any recorded **Pass Lane** (PASS 01 Base Pass or OD 02..N Overdub Pass) on a track that is not currently selected as the ACTIVE COMP LANE. Rendered in slate-grey waveform color with a clickable Use PASS <NN> or Use OD <NN> button that promotes the standby pass to active playback immediately without audio clicks. See Section 6.2.2.

Stem Bounce (Bounce001)

Faster-than-real-time offline post-processor render of individual Input Track channels (<CC>-<TrackName>_Bounce<NNN>.wav / .flac) and/or stereo Subgroup Buses (Bus<BB>-<BusName>_Bounce<NNN>.wav / .flac) written into <TakeFolder>/bounce/. Bounced stems start at exact sample 0 (00:00:00.000) with all insert processing, phase alignment, and Plugin Delay Compensation applied for delivery to external remixing or mastering engineers. See Section 7.2.3.

Subgroup Bus (SUB 1 DRUM BUS & SUB 2 INST BUS)

Stereo summing bus strip (SUB 1 through SUB 16) that combines routed Input Tracks into a shared stereo processing chain with its own Input Trim (TRM), Stereo Link (LINK), SEL inspection button, reorderable **8-Slot SUB RACK** (Tape Saturation, 12-Band EQ with Dual-Trace RTA, Analog VCA Bus Compressor, and third-party CLAP or macOS Audio Unit plugins), stereo Aux Sends (RVB and DLY), stereo Balance (BAL), Mute (M), Solo (S), and logarithmic stereo fader. See Section 4.3.1.

Take (TAKE 001)

A complete, self-contained multitrack song recording or rehearsal jam captured within a Session (<SessionFolder>/Take<NNN>_<Stamp>[_<TakeTag>]/), starting at its own independent timeline origin (00:00:00.000, sample 0). Pressing Record (**R**) with Overdub Mode (OD) off always creates a new numbered Take (Take 001, Take 002, …) and switches the active timeline to that Take. Two different Takes are never stacked on top of one another in the same timeline view. See Section 1.1.1 and Section 9.1.

Take Splitting (SPLIT TAKE & NEW TAKE)

Non-destructive editing operations in WAVEFORM TIMELINE (**CMD** / **CTRL** + **2**) and MARKERS & RANGES (**CMD** / **CTRL** + **5**) that divide a continuous recording at the Playhead (SPLIT TAKE), extract from a standalone marker to the end (NEW TAKE -> END), or extract a bounded range (NEW TAKE) into a brand-new t = 0 Session Take while preserving all original .wav files, Pass Lanes, and phase alignment offsets with full Undo support. See Section 6.4.

Tape Saturation (TAPE)

Zero-latency analog magnetic tape modeling stage built into every Input Track, Subgroup Bus, and the Master Bus. Models transformer core warmth, magnetic hysteresis soft-clipping, AC bias asymmetry, and high-frequency head self-erasure via six continuous controls: Input Drive (DRV), Harmonic Warmth (WRM), Tape Bias (BIAS), High-Cut Roll-Off (HC), Output Trim (OUT), and Parallel Wet/Dry Mix (MIX). See Section 5.2.

Vocal Rider (VOCAL RIDER)

Zero-latency automatic vocal and instrument leveler (±6.0 dB range around a selectable -36 to -6 dBFS target) driven by a speech-weighted (250 Hz to 4.0 kHz) detector and protected by a **Breath & Bleed Freeze Gate** (Freeze Threshold, default -48 dBFS) that holds the fader steady during vocal pauses so live drum and amplifier room bleed is never boosted before the compressor. See Section 5.3.4.

Waveform Timeline (WAVEFORM TIMELINE)

Multi-track waveform editing and overdub comping workspace (**CMD** + **2** on macOS, **CTRL** + **2** on Windows and Linux) focused on the currently active t = 0 Session Take. Features multi-resolution waveform detail, one-handed 2D ruler scrub-zooming, expandable **Pass Lanes** (PASSES, **E** / **SHIFT** + **E**), dedicated Marker (MKR) and Range (RNG) lanes, Playhead/Cursor lock (LOCK PH/CUR), and non-destructive Take splitting (SPLIT TAKE / NEW TAKE). See Chapter 6.

9.3 Alphabetical Index

The alphabetical index below lists every studio topic, console control, processor parameter, keyboard shortcut, workflow procedure, and glossary definition across Chapters 1 through 9, with direct section links and exact PDF page numbers.

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